

Cellular Expert Express User Guide

Version 7.3

2026



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About Cellular Expert

Cellular Expert UAB (CE) developed ultra-fast wave propagation, communication systems deployment planning and radio/optical visibility calculation software for ESRI's ArcGIS mapping environment, which is widely used within Telecom, Defense, IoT, and other companies and organizations.

CE's communication network planning, network asset management, operational support software and customer-tailored solutions enhance the intelligence and business efficiency of more than 170 communication network companies, regulators, and defense organizations in more than 50 countries.

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1. Introduction

In this User Guide, we introduce the Cellular Expert Express application. Express is the merged Inventory 3D and Cellular Expert Express solution in one product. Inventory 3D is responsible for data management. From a desktop, laptop, or tablet computer with internet access, the Webapp connects to a database located on a server and enables the user to edit database records, sites, and objects, draw diagrams and locate the assets on an integrated map. Inventory3D integrates data derived from various sources, such as sensors, Google Sheets, maps, and changes in database entries are synchronized with the server database content, keeping the database constantly up to date.

Cellular Expert Express is a telecommunication network planning, optimization, and data management solution built on Esri ArcGIS Enterprise, the most advanced GIS platform. We continually expand the functionality of Cellular Expert Express, but it is not intended to mimic and match Cellular Expert Desktop one hundred percent. The idea is to have a simplified task-oriented user interface to gather and share the data across the company. This means that the solution is dedicated across the organization and to fit various tasks and workflows.

The Cellular Expert Express application is a versatile tool for inventory planning, design, space, and assets management. It consists of several building blocks that together provide a customized and tailored solution for a specific need. Take inventory management: Error-prone databases which contain faulty inventory lists can be deleterious, especially in large industrial environments. Thus, it is essential that upon inventory changing databases are regularly updated.

2. CE Express application

To access the Cellular Expert Express web application, type its URL in the address field of a web browser. You can use any web browser, however, we recommend the up-to-date Google Chrome web browser.

The URL depends on the installation and initial configuration:

1. Web address, as an example: <https://<yourdomain>/ceexp>
2. An address can be configured by the administrator.

Please note that for security reasons the application uses only HTTPS.

2.1 Log in to the Express Network Data Management application

Users can log in with the organization account of ArcGIS Enterprise or Cellular Expert Express and are identified by a *Username* and a *Password*.



Use the Login as ArcGIS button to log in with an ArcGIS Enterprise account. It will allow to access the Network Data Management view for database management and the Map view for analysis, calculations, etc.

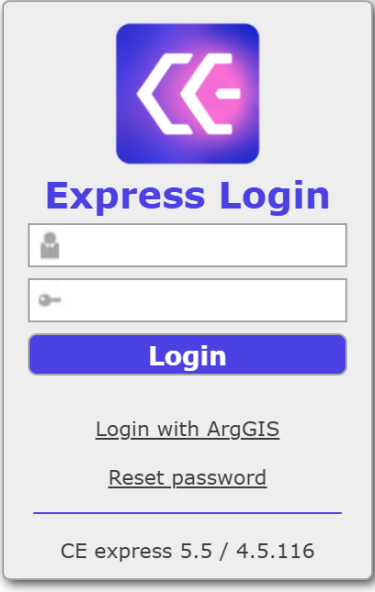
Sign in to ArcGIS Enterprise with 

ArcGIS login

☐ Keep me signed in

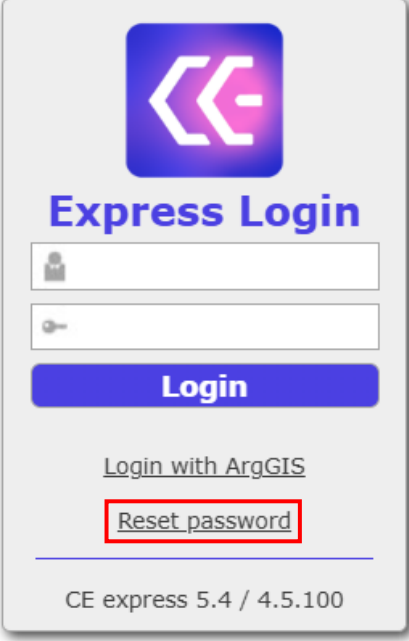
[Forgot password?](#)

Use log in with Express account to login with a Cellular Expert Express account. It will allow to access only the Network Data Management view. To reach Map view you will be able to log in with an ArcGIS Enterprise account later.

A screenshot of the 'Express Login' form. At the top is a logo consisting of two stylized 'C' shapes in blue and purple. Below the logo is the title 'Express Login' in blue. There are two input fields: the first has a person icon and the second has a key icon. Below these is a blue 'Login' button. Under the button are two links: 'Login with ArgGIS' and 'Reset password'. At the bottom, it says 'CE express 5.5 / 4.5.116'.

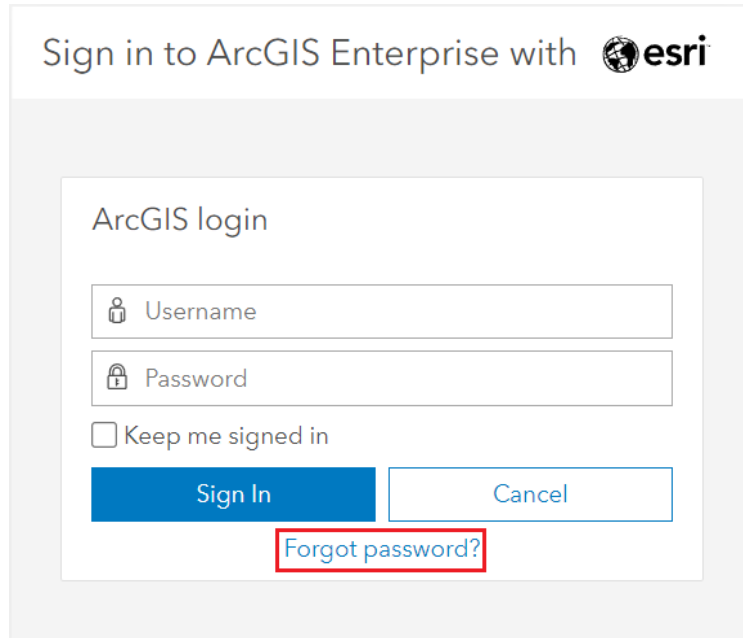
For the option Login with ArcGIS an ArcGIS Portal account is required and the user gains access to the application map widgets. The second option, Login with Express account, enables only access to view and manage tables, data and so on.

Registered users may change their Cellular Expert password any time by clicking Reset password.

A screenshot of the 'Express Login' form, similar to the one above but with version '4.5.100'. In this version, the 'Reset password' link is highlighted with a red rectangular border.

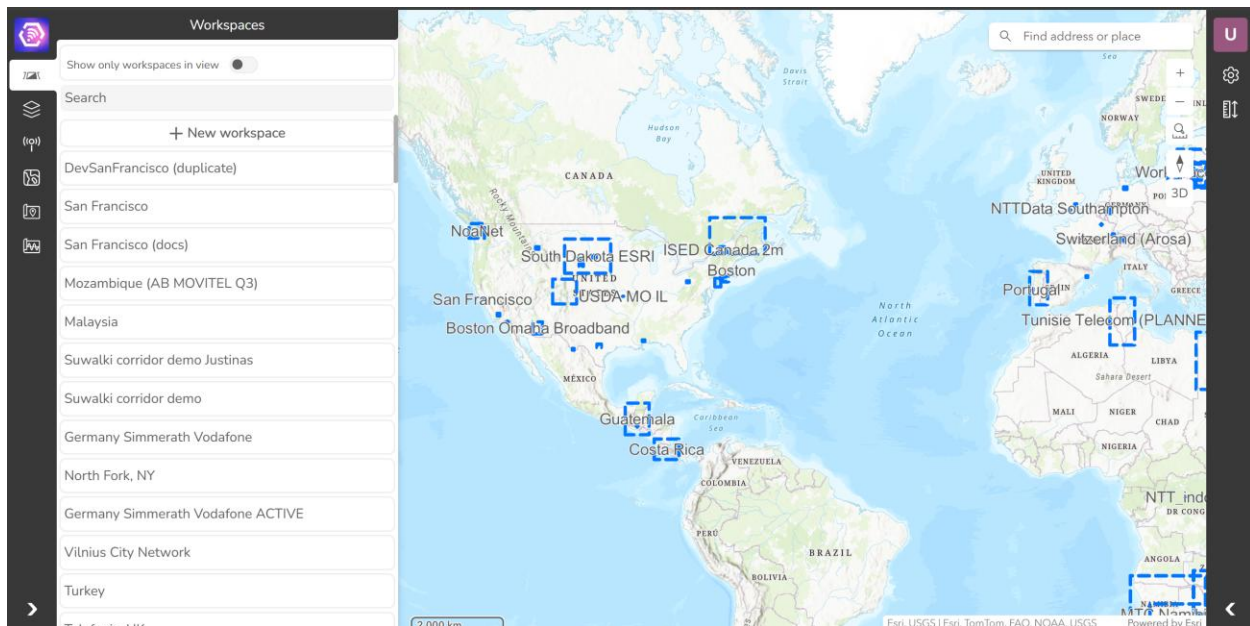
Note: before using the Reset Password feature the server should be configured as a mail server and have

the possibility to send email. The Cellular Expert Express Network Data Management view opens. To recover an ArcGIS Enterprise account, use the Forgot password button.

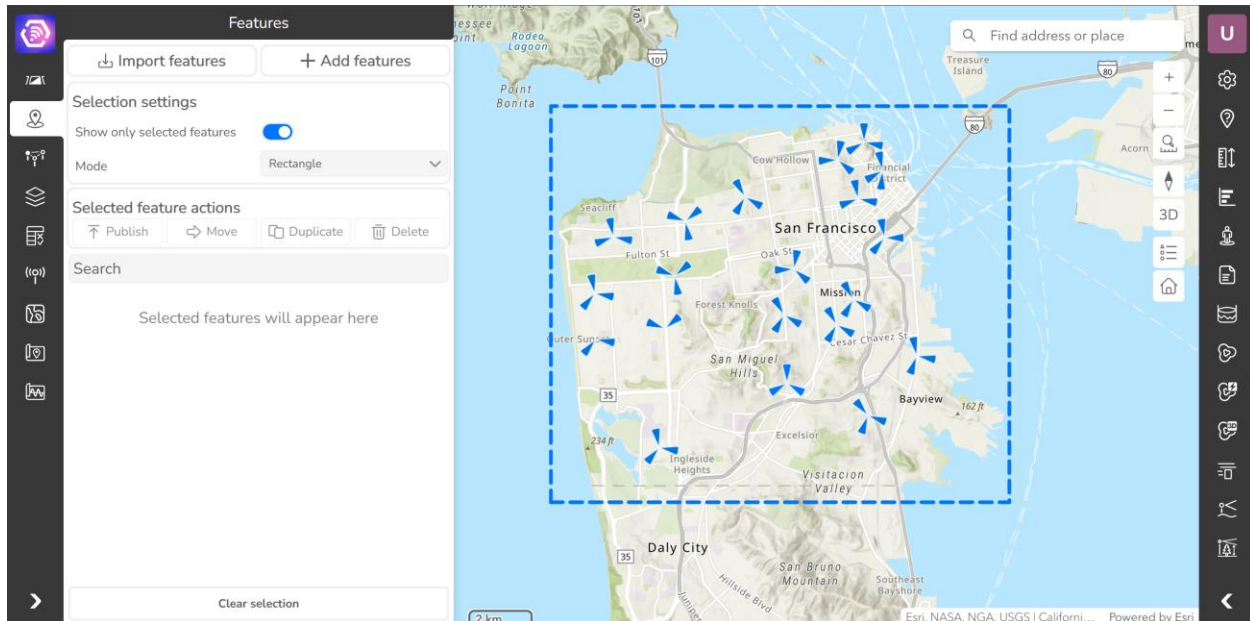
The image shows the ArcGIS login interface. At the top, it says "Sign in to ArcGIS Enterprise with" followed by the Esri logo. Below this is a white box titled "ArcGIS login". Inside the box, there are two input fields: "Username" with a person icon and "Password" with a lock icon. Below these fields is a checkbox labeled "Keep me signed in". At the bottom of the box are two buttons: "Sign In" in blue and "Cancel" in white with a blue border. Below the "Sign In" button, the text "Forgot password?" is highlighted with a red rectangular box.

2.2 Open the Express Map view

In the Network Data Management view the map and the database are connected. Features, described in the tables, are displayed on the map. Select a workspace.



Map view and features are visible only for the selected workspace.

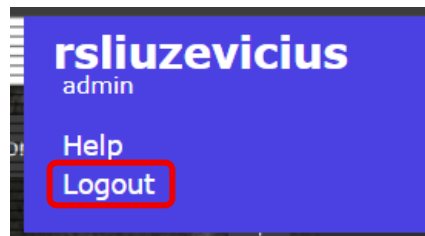


2.3 Log out

To exit the Express Network Data Management application, click on the User menu button at the top right corner.



And click on the *Logout* button.



3. Map view

The Map view has a predefined layout divided into:

1. Inventory3D window layout, settings, user
2. Data management and visualization tools
3. Map view, graphical data management
4. Table view, attribute data management

5. Data analysis tools and calculation results
6. Calculation result graphs and tables, additional analysis tools

The screenshot displays the Cellular Expert Express software interface. The interface is divided into several panels:

- 1. Inventory3D window layout, settings, user:** Located at the top right, showing a search bar and user profile.
- 2. Data management and visualization tools:** Located on the left side, showing a list of features and a search bar.
- 3. Map view, graphical data management:** The central map area showing a city street view with various data points and overlays.
- 4. Table view, attribute data management:** A table at the bottom showing cell data with columns: cell_name, site_name, x, y, height, azimuth, tilt, frequency, power, misc_loss, bandwidth, noise_figure, downlink_duplex_factor, subcarrier_spacing, tx_mimo, rx_mimo, active_antenna_effect, cell_load.
- 5. Data analysis tools and calculation results:** A panel on the right showing calculation settings, transmitter information, and a profile graph.
- 6. Calculation result graphs and tables, additional analysis tools:** A panel in the center showing a graph of signal strength and a table of calculation results.

3.1 Cellular Expert Express tools

3.1.1 Workspaces

Click this button  to open Workspaces tool.

Shows an active workspace. The tool allows to switch between available workspaces. The Map view will be automatically zoomed to the selected workspace extent and only objects defined for that workspace will be visible.

A workspace is a project-based view in which objects are displayed. Workspaces can be assigned individually for each user group.

Workspaces

Show only workspaces in view ☐

Search

+ New workspace

San Francisco

London

Lithuania

Colorado

Switzerland (Arosa)

Show only workspaces in view

When enabled, this option filters the workspace list to display only the workspaces currently visible within the map's viewport.

Upon hovering the mouse over a workspace item, options for it appear.

Boston

San Francisco

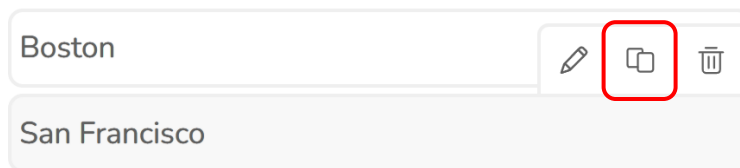
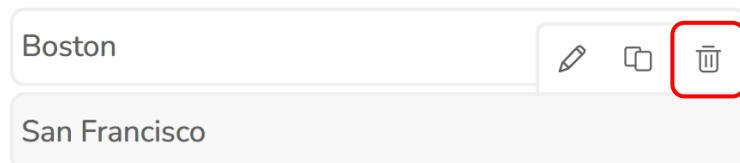
Edit Workspace

Boston

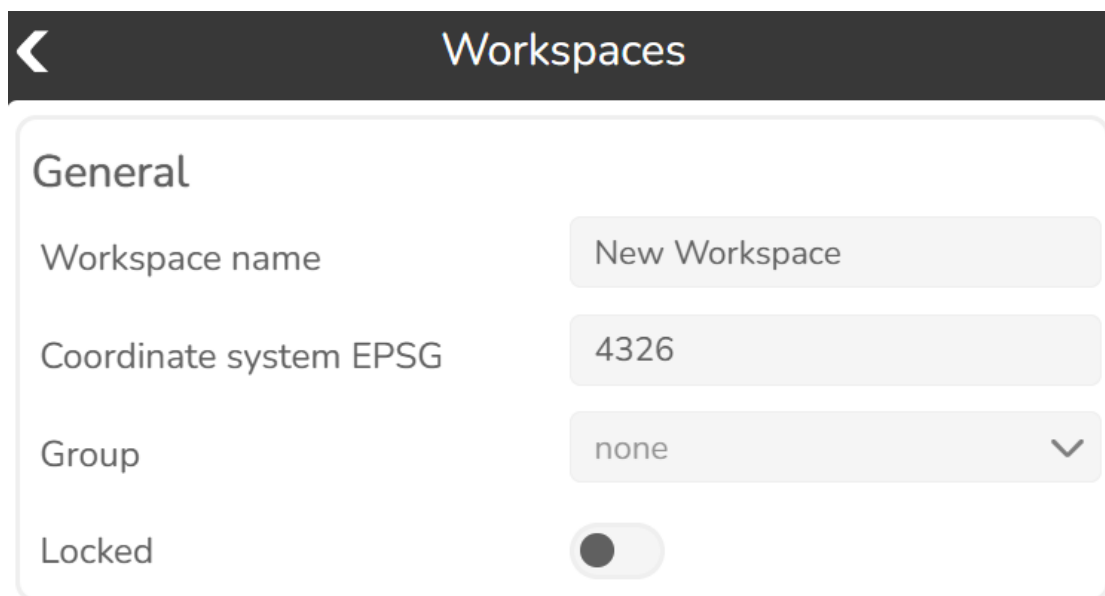
  

San Francisco

Duplicate Workspace

**Delete Workspace****3.1.1.1 Add workspace**

To create a workspace, press the *New workspace* button.

**General****Workspace name**

Workspace identification.

Coordinate system EPSG

EPSG code of the coordinate system (spatial reference) used within workspace. Feature coordinates will be saved in this coordinate system. This does not affect the geodata used within calculations. Default is 4326 (WGS84).

Group

Workspaces are grouped based on the value in this field. To group workspaces, set this field to the same value for multiple workspaces.

Locked

Locks feature editing within the workspace. Useful when you want to keep a workspace for archiving purposes. Only an admin user can disable this for a locked workspace.

Extent

Draw on map

☐

Min. X

0

Min. Y

0

Max. X

0

Max. Y

0

Extent

Workspace extent defines where the map gets zoomed to when the workspace is loaded. It is also used

as a zoom reference for the home button.

**Draw on map**

Enabling this allows for clicking on the map to draw a desired square for workspace extent.

Min. X

Minimum x (leftmost) coordinate of the workspace extent (in workspace epsg)

Min. Y

Minimum y (bottommost) coordinate of the workspace extent (in workspace epsg)

Max. X

Maximum x (rightmost) coordinate of the workspace extent (in workspace epsg)

Max. Y

Maximum y (topmost) coordinate of the workspace extent (in workspace epsg)

Coordinate origin

X	0
Y	0
Z	0

Coordinate origin

Origin point from which coordinates are calculated from in the user interface. Global coordinates are saved in the database regardless of this setting.

Calculations

Calculate EIRP	☒
Use clutter	☒
Feature height reference	Elevation ▼
Receiver height reference	Elevation ▼
Geodata set	none ▼

Calculations**Calculate EIRP**

Enabled – EIRP is calculated with the formula: power – misc. loss + antenna gain.

Disabled – power value is used as EIRP.

Use clutter

Determines whether Clutter Loss is used in prediction calculations.

Transmitter height reference

The height above which the absolute transmitter height is calculated, e.g. if “elevation” is selected, and transmitter height is set to 10 m, the absolute transmitter height is calculated as elevation + 10. This is used within CE calculations.

Available options:

- Elevation
- Clutter height (buildings only)
- Clutter height
- Absolute

Receiver height reference

The height above which the absolute receiver height is calculated, e.g. if “elevation” is selected, and receiver height is set to 10 m, the absolute receiver height is calculated as elevation + 10. This is used within CE calculations.

Available options:

- Elevation
- Clutter height (buildings only)
- Clutter height
- Absolute

Geodata set

Geodata set used in calculations launched within the workspace.

Extra layers**URL / Portal ItemID**

Specifies the source of the layer. Accepts either a direct URL or an ArcGIS Portal Item ID.

Title

The title field defines the name displayed for a layer in the layer list

Opacity

Adjusts layer transparency.

Visible

Turns the layer's display on or off.

Group

Specifies the group assignment for the layer. You can select an existing group, create a new one, or leave the field blank to place the layer in the default group Other.

Feature naming schemes	
Sites	none ▼
Candidate sites	none ▼
Site search areas	none ▼
Cells	none ▼
Repeaters	none ▼
Radars	none ▼
CPE	none ▼
Measurements	none ▼
Lights	none ▼
Sirens	none ▼
Omens	none ▼
Mesh nodes	none ▼
Links	none ▼
HCM links	none ▼

Feature naming schemes

Allows for automatic feature name setting depending on features already existent in workspace. For example: Cells feature naming scheme set to "1,2,3.." → Cells 1, 2, and 3 already exist in workspace → next placed cell in add features tool will automatically have the name set to 4.

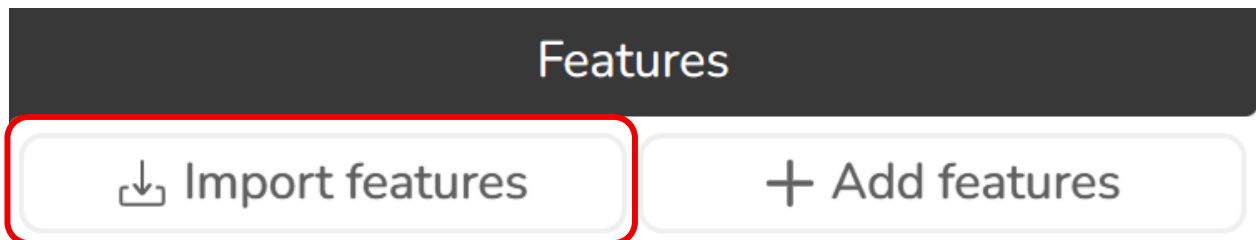
3.1.2 Features

Click this button  to open Features tool.

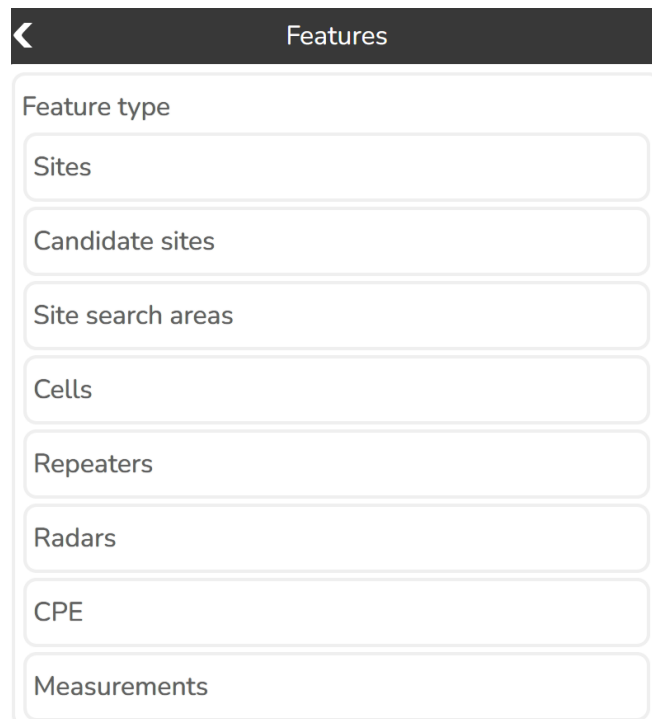
Use this tool to import or add features, select and visualize them.

3.1.2.1 Import

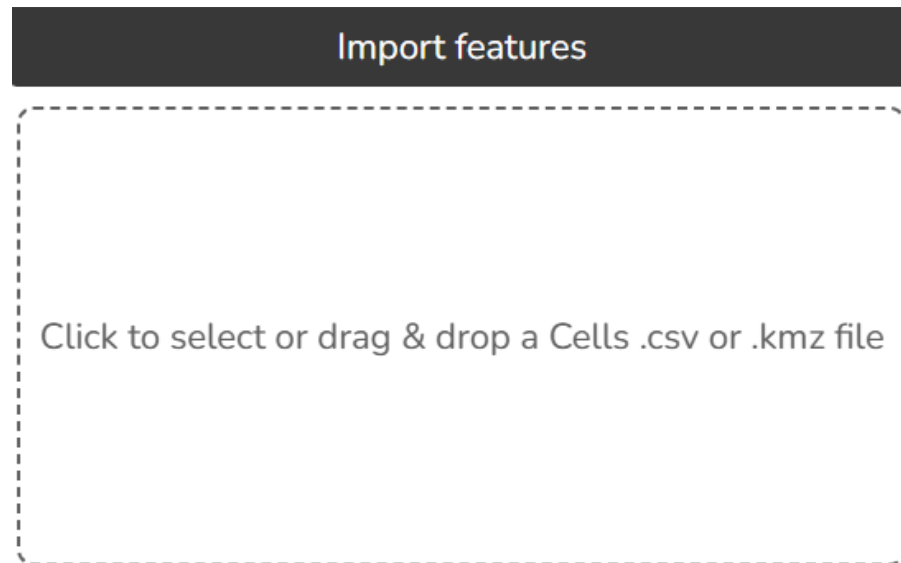
The widget allows the creation of objects from a text file. To start importing new objects press the *Import features* button.



Select the object type.



A new dialog on the right side of the window will appear. The widget imports object data to the database. The imported objects will be displayed automatically on the map. The supported file formats are CSV and KMZ.



To start the import process, select or drag and drop a CSV or KMZ file.



3.1.2.1.1 Mapping

The data in the import files may have names, values and units which do not match the data in the Cellular Expert database. To resolve such issues, check *Use mapping* button.

Import

Importing

cells3392

Use mapping☒

Create sites☐

Mapping presets

+ New mapping preset

Required mapping

	Source	Fill
Cell name	Cell name	
X	X	
Y	Y	
Azimuth, deg	none	

Optional mapping

	Source	Fill
Height, m	none	
Downtilt, deg	none	
EL Downtilt, deg	none	
Frequency, MHz	none	
Power, dBm	none	

Accept

Cancel

Source - the name of the value that is written in the data file.

Fill - value which will be used when an object in the data file has no value for a particular property. In this example, if "Azimuth" is not set, then it will by default be assigned the value of 0. Leaving the default to empty means that no default value will be applied.

Required mapping		
Parameter	Source	Fill
Cell name	name ▼	
Latitude	lat ▼	
Longitude	long ▼	
Azimuth, deg	▼	0

3.1.2.1.2 Mapping presets

It is possible to create import presets.

Mapping presets

+ New mapping preset

New preset

Apply

×

To create a preset, first define source or fill value, then press *New mapping preset* button.

+ New mapping preset

A new preset will be created.

New preset

Apply

×

Define a preset name. The defined preset will be applied the next time for importing.

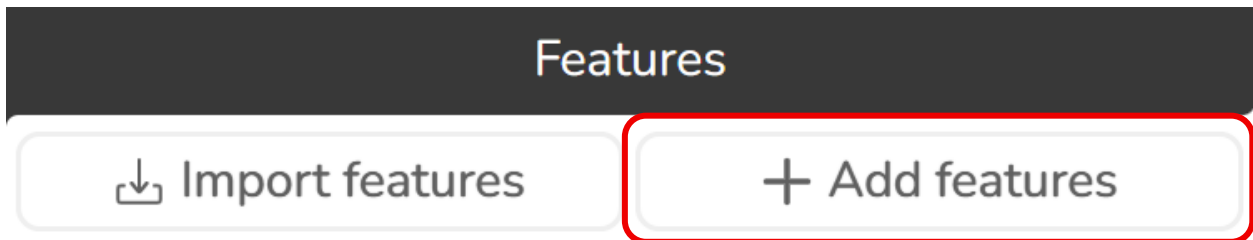
Apply

Applies a mapping preset for importing file.

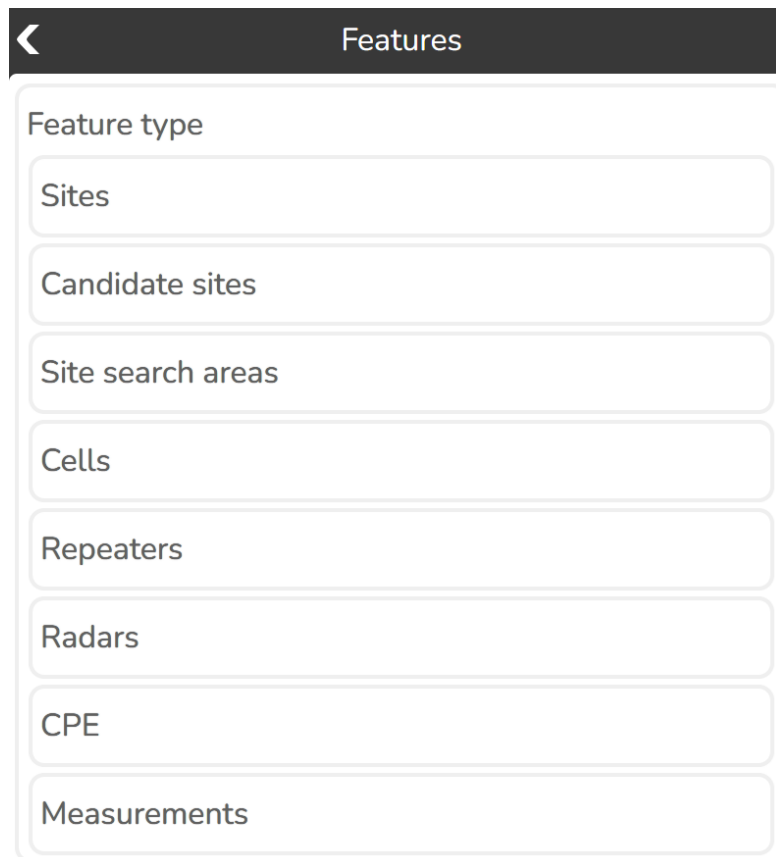
Delete

Deletes a preset.

3.1.2.2 Add features



Select the object type.



A new dialog on the right side of the window will appear.

New objects can be created in several ways. They can be:

- Created from templates
- Created with Cellular Expert tools from zero (define all parameters in the process)

Add feature

Template

none

Required parameters

Cell name

X

Y

Azimuth, deg

Optional parameters

Height, m

Downtilt, deg

El. Downtilt, deg

Frequency, MHz

Power, dBm

Misc loss, dBm

Bandwidth, MHz

Noise figure, dB

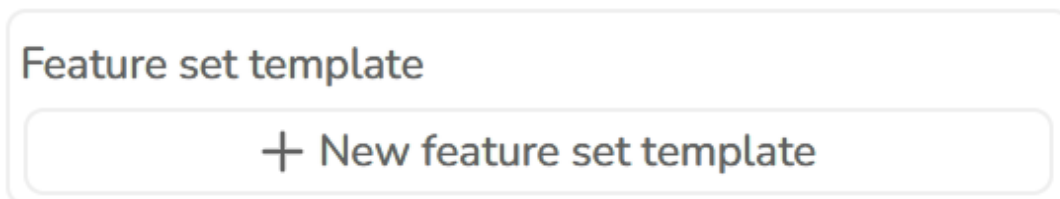
Downlink duplex factor

Accept

Cancel

3.1.2.2.1 Feature set template

Allows for saving a group of features as a “feature set template”, with one of the features acting as the origin for the feature set coordinates. The entire feature set can then be placed with a single operation via the add features tool. Coordinates relative to the origin feature are maintained by the other features in the set.



Features

General

Feature set template name

Feature set template

Features

Coordinate origin feature

none

Sites

Site_A

Cells

Cell_A

Cell_B

Cell_C

General**Feature set template name**

Name of the feature set template

Features**Coordinate origin feature**

Feature which will be used as the coordinate origin point for the feature set. When the feature set is placed, this feature will be at the mouse click point.

Feature parameters

Parameters of the features saved in the feature set template. The placed feature parameters will be automatically filled with the saved values.

3.1.2.2.2 Add Site

Add feature

Template

none

Required parameters

Site name

X

Y

Optional parameters

Height, m

Required parameters

Site name
Site identification.

X
Coordinate in the projected coordinate system.

Y
Coordinate in the projected coordinate system.

Optional parameters

Height
Height above the terrain.

3.1.2.2.3 Add Candidate sites

Add features

Required parameters

Site name

X

Y

Optional parameters

Max Site Height

m

City

Street

Status

Type

Notes

Required parameters

Site name
Site identification.

X
Coordinate in the projected coordinate system.

Y
Coordinate in the projected coordinate system.

Optional parameters

Max Site Height
Maximum site height above the terrain in meters.

City
City where the site is located.

Street
Street address of the site.

Status
Free-form text.

Type

Free-form text.

Notes

Free-form text.

3.1.2.2.4 Add Site search areas

Add features

Required parameters

Area Name	
Draw mode	Polygon 
Geometry segments	0
Group	

Required parameters**Area Name**

Area identification.

Draw mode

Method used to draw the area geometry:

- Polygon
- Circle

Circle radius

Radius of the circle, in meters.

Geometry segments

Number of segments used to approximate the geometry.

Group

Group to which the area belongs.

3.1.2.2.5 Add Cell

Add feature

Template

none

Required parameters

Cell name

X

Y

Azimuth, deg

Optional parameters

Height, m

Downtilt, deg

El. Downtilt, deg

Frequency, MHz

Power, dBm

Misc. loss, dB

Bandwidth, MHz

Noise figure, dB

Downlink duplex factor

Subcarrier spacing, kHz

Required parameters**Cell name**

Cell identification.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Azimuth

Cell direction from the North in degrees.

Optional parameters**Height, m**

Height above the terrain.

Downtilt

Mechanical tilt value.

El. Downtilt, deg

Electrical tilt value

Frequency

Frequency value in MHz.

Power

Power value in dBm.

Misc. loss, dB

Miscellaneous loss value in dB.

Bandwidth, MHz

Value in MHz. Required for 4G and 5G technologies. For other technologies define the value as 0.015.

Noise figure, dB

Value in dB. Required for 4G and 5G technologies.

Downlink duplex factor

Value range from 0 to 1. Required for Duplex mode TDD, which is applicable for 4G and 5G technologies, and used for Downlink Throughput calculations. For example, if defined value is 0.7, then 70% of available bandwidth will be dedicated to Downlink, and 30% - for Uplink.

Subcarrier spacing, kHz

Value in kHz. Required for 4G and 5G technologies. For other technologies define value 15.

Tx Mimo

Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

Rx Mimo

Receiver antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

Active antenna effect

The parameter is dedicated to smart antenna modeling. The default value is 0, but if massive MIMO is used, a smart antenna effect can be included to lower the interference and boost throughput.

Recommended values:

- For MIMO 32x32 – value 6.
- For MIMO 64x64 – value 9.

Cell load, %

The parameter is described in percentages and varies from 0 to 100. It describes how the cell is loaded.

The Cell load affects RSSI, RSRQ, and DL Throughput calculations. For example, if the Cell load is higher, the DL Throughput is lower.

Color index

Describes the cell visualization. Available values:

- None – blue color.
- 1 – red color.
- 2 – light green color.
- 3 – dark green color.
- 4 – light blue color.
- 5 – dark blue color.
- 6 – purple color.

Technology

Describes the technology of the network object. Possible values are 2G, 3G, 4G, and 5G.

Prediction model

Prediction model for Path Loss simulation.

Frequency group

Used to divide calculations into parts. If the selection range includes two or more different frequency group values, the cells won't be predicted together.

Antenna

Define antenna patterns for the Cell object.

Carriers

Describes the carrier values used for 2G calculations: C/I interference and C/A interference. The values are written in brackets, [...]. If more than one value is defined, the values are separated by a comma. If there is no carrier information, the brackets are left empty [].

Site ID

Describes to which Site the Cell belongs.

Duplex mode

Available values FDD or TDD. Required for 4G and 5G technologies. For other technologies define value FDD.

Status

Free-form text.

Type

Free-form text.

3.1.2.2.6 Add Repeater

Add feature

Template none

Required parameters

Repeater name

X

Y

Azimuth, deg

FS Threshold 1, dBm

Power 1, dBm

Optional parameters

Height, m

Downtilt, deg

El. Downtilt, deg

Frequency, MHz

FS Threshold 2, dBm

Power 2, dBm

FS Threshold 3, dBm

Power 3, dBm

Required parameters**Repeater name**

Repeater identification.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Azimuth

Direction from the North in degrees.

Optional parameters**Height**

Object's height above the terrain.

Downtilt

Mechanical tilt in telecommunications repeaters is the physical angling of the antenna to optimize signal coverage.

Electrical Tilt

Electrical tilt in a repeater refers to the electronic adjustment of an antenna's vertical radiation pattern to optimize network coverage and reduce interference.

Frequency

Frequency value in MHz.

Thresholds 1, 2, 3

The minimum field strength in dB at repeater location at which power of the corresponding index (1-3) will be applied. Values should be in ascending order. If a higher threshold is satisfied, the power corresponding to it will be used.

Power 1, 2, 3

A power that is assigned to cells based on the repeater thresholds and cell signal strength. When the cell's signal strength is categorized the power of the repeater will be assigned as well.

Misc loss

Miscellaneous loss value in dB.

Bandwidth

Value in MHz. Required for 4G and 5G technologies. For other technologies define the value as 0.015.

Subcarrier Spacing

Value in kHz. Required for 4G and 5G technologies. For other technologies define value 15.

Tx Mimo

Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

Rx Mimo

Receiver antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

Technology

Describes the technology of the network object.

Prediction Model

Lets the user select which prediction model and configuration should be used for calculations.

Frequency group

Used to divide calculations into parts. If the selection range includes two or more different frequency group values, the cells won't be predicted together.

Antenna

Antenna name for Repeater object.

3.1.2.2.7 Add Radar

Add feature

Template none 

Required parameters

Radar name

X

Y

Optional parameters

Height, m

Downtilt, deg

Frequency, MHz

Power, dBm

Misc loss, dBm

View angle, deg

Prediction model

none



Required parameters**Radar name**

Radar identification.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Optional parameters**Height**

Object's height above the terrain.

Downtilt
Mechanical tilt value.

Frequency
Frequency value in MHz.

Power
Power value in dBm.

Misc Loss
Miscellaneous loss value in dB.

View Angle
Visible field (vertical angle) of the radar in degrees.

Prediction Model
Prediction model for Path Loss simulation.

3.1.2.2.8 Add CPE

Add feature

Template

none

Required parameters

CPE name

X

Y

Optional parameters

Height, m

Azimuth, deg

Antenna

none

Power, dBm

Misc loss, dBm

Cell ID

Throughput, Mbps

Status

Notes

Required parameters**CPE name**

CPE identification.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Optional parameters**Height**

Object's height above the terrain.

Azimuth

Direction from the North in degrees.

Antenna

Antenna name for CPE location.

Power

A power value in dBm.

Misc loss

Miscellaneous loss value in dB.

Cell ID

Describes to which Cell the CPE point belongs.

Throughput

The speed at which data is transferred. Measured in Mb/s.

Status

Current status of the network object.

Notes

Additional information for network predictions can be noted here.

3.1.2.2.9 Add Measurements

Add feature

Required parameters

Field strength, dB	
X	
Y	
Cell ID	

Required parameters**Field strength, dB**

Field strength.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Cell ID

A field which binds the measurement to a cell network object.

3.1.2.2.10 Add Omen

Add feature

Required parameters

Omen name	
X	
Y	

Optional parameters

Height, m	
Notes	

Required parameters**Omen name**

Omen identification.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Optional parameters**Height, m**

Height above the terrain.

Notes

Free-form text.

3.1.2.2.11 Add Sirens

Add feature

Template none 

Required parameters

Siren name

X

Y

Azimuth, deg

Optional parameters

Height, m

Downtilt, deg

Frequency, Hz

Power, W

Misc. loss, dB

Prediction model none 

Antenna none 

Status

Type

Required parameters**Siren name**

Siren identification.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Azimuth

Cell direction from the North in degrees.

Optional parameters**Height, m**

Height above the terrain.

Downtilt

Mechanical tilt value.

Frequency

Frequency value in MHz.

Power

Power value in dBm.

Misc. loss, dB

Miscellaneous loss value in dB.

Prediction model

Only ISO9613 can be applied to calculate sound loss for the siren.

Antenna

Antenna name for Siren object.

Status

Free-form text.

Type

Free-form text.

3.1.2.2.12 Add Lights

Add feature

Template

none

Required parameters

Light name

X

Y

Azimuth, deg

Optional parameters

Height, m

Downtilt, deg

Antenna

none

Required parameters**Light name**

Light identification.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Azimuth

Cell direction from the North in degrees.

Optional parameters**Height, m**

Height above the terrain.

Downtilt

Mechanical tilt value.

Antenna

Antenna name for Light object.

3.1.2.2.13 Add Mesh nodes

Add feature

Template

none

Required parameters

Mesh node name

X

Y

Optional parameters

Height

m

Frequency

MHz

Power

dBm

Misc. loss

dB

Prediction model

none

Antenna

none

Sensitivity

dBm

Max connections

Layer

none

Group name

Status

Type

Required parameters

Mesh node name

Mesh node identification.

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Optional parameters**Height, m**

Height above the terrain.

Frequency, MHz

Frequency of the mesh node.

Power, dBm

Power value of the mesh node.

Misc Loss, dB

Miscellaneous loss value of the mesh node.

Prediction model

The prediction model that will be used for the mesh node in Mesh Connectivity and Quick Mesh Connectivity calculations.

Antenna

Define antenna patterns for the mesh node object.

Sensitivity

Sensitivity value of the mesh node.

Max connections

The maximum number of connections the mesh node can have (used for mesh connectivity calculations).

Layer

The layer of the mesh node (number value). Used for priority calculations in Automatic Frequency Planning.

Group name

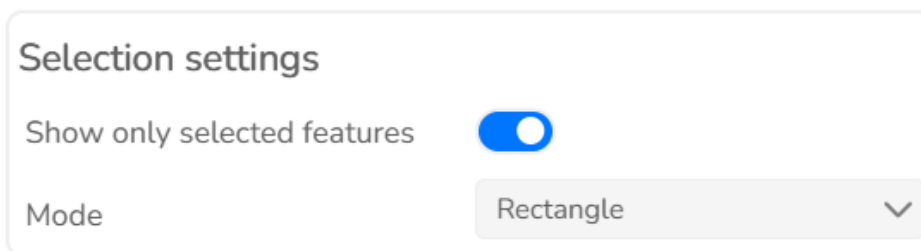
The group name of the mesh node (text description). Several mesh nodes may belong to the same group.

Status

Free-form text.

Type

Free-form text.

3.1.2.3 Selecting

The image shows a 'Selection settings' dialog box. It contains a toggle switch for 'Show only selected features' which is currently turned on (blue). Below this is a dropdown menu labeled 'Mode' with 'Rectangle' selected and a downward arrow icon.

Selection settings

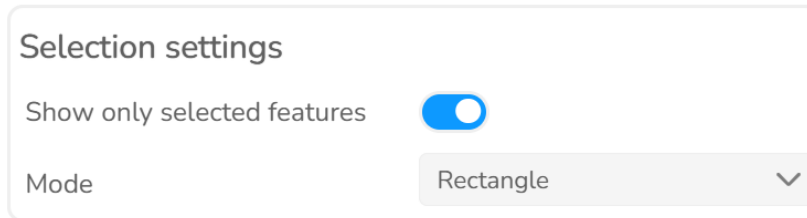
Show only selected features

Only the selected features will be displayed in the feature list. When the switch off, all features are visible regardless of selection. When it is on non-selected features are hidden, so you see only what you have selected.

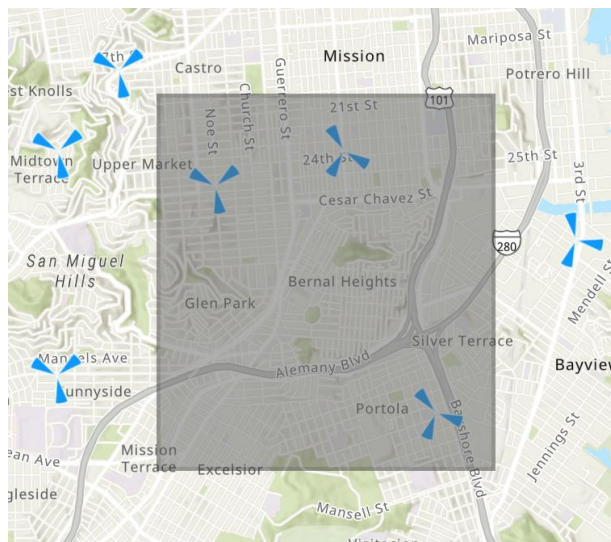
Mode

There are several different modes for selecting objects:

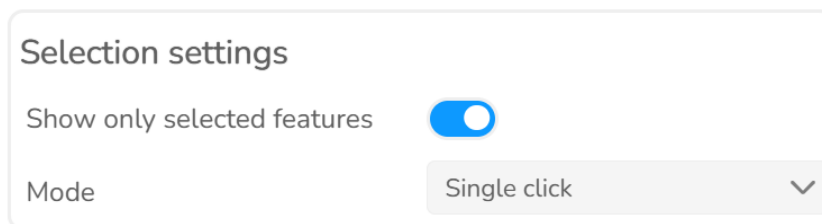
- **Rectangle.**



To start the selection process, click once on the map, then move the mouse cursor to define an area, click a second time to finish the selection process.



- **Single click.** Click once on an object to select it.



- **Radius.**

Selection settings

Show only selected features ☒

Mode

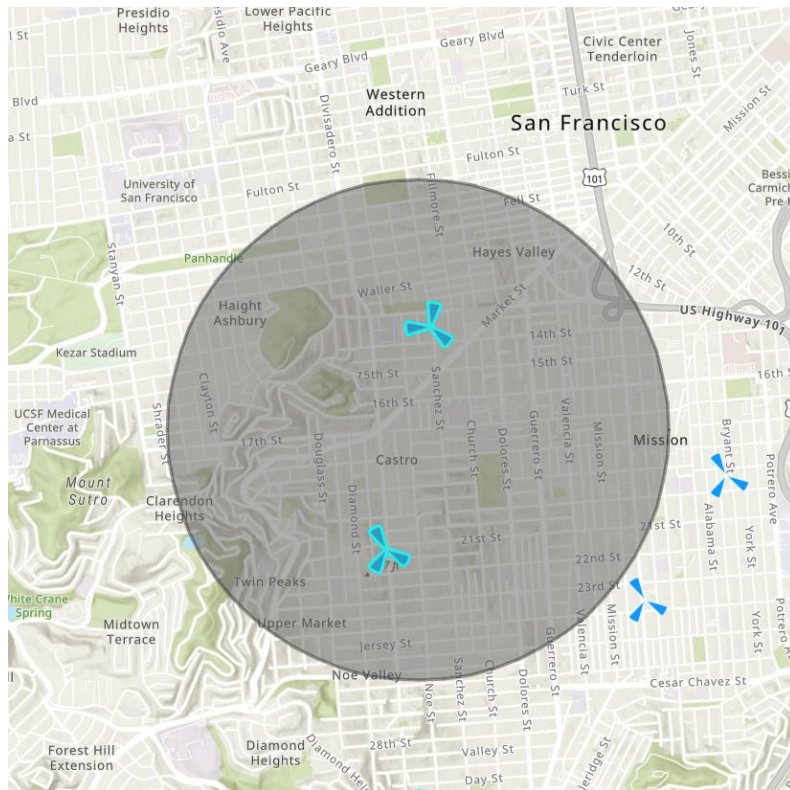
Radius

Radius

100

 m

Set the desired distance, then click on the map and the surrounding objects within the set distance will be selected.



- **Polygon.**

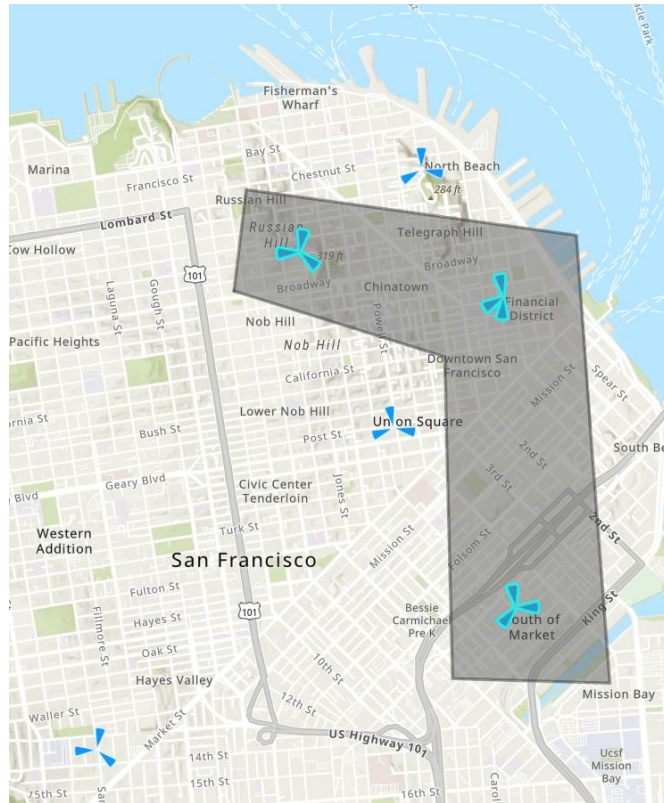
Selection settings

Show only selected features ☒

Mode

Polygon

Click once on the map to create the vertices of the polygon. Double-click to close the polygon and select the feature.



- **Polygon layer.**

Selection settings

Show only selected features ☒

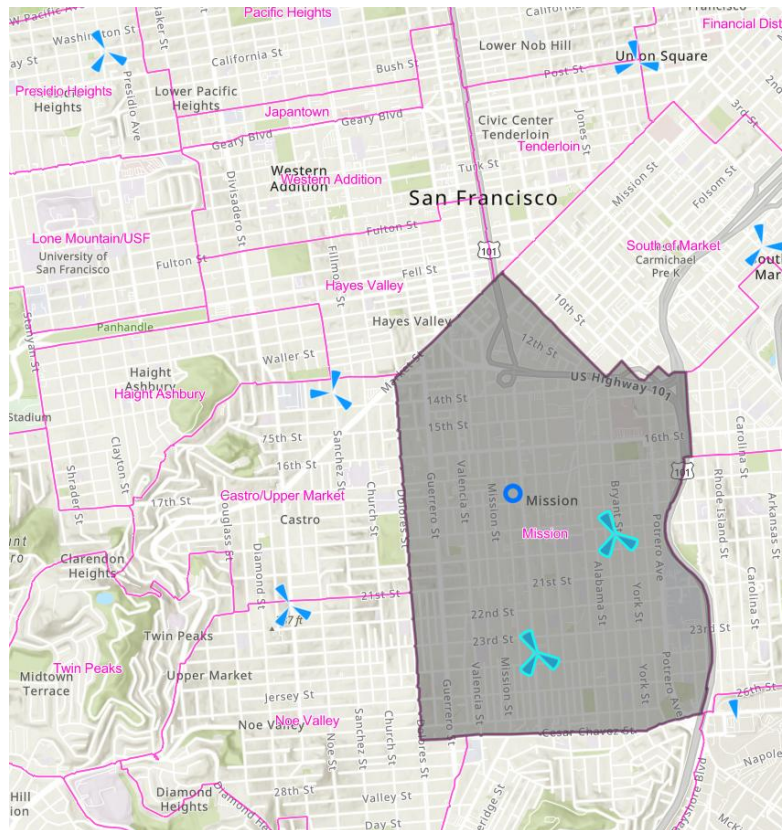
Mode

Polygon layer

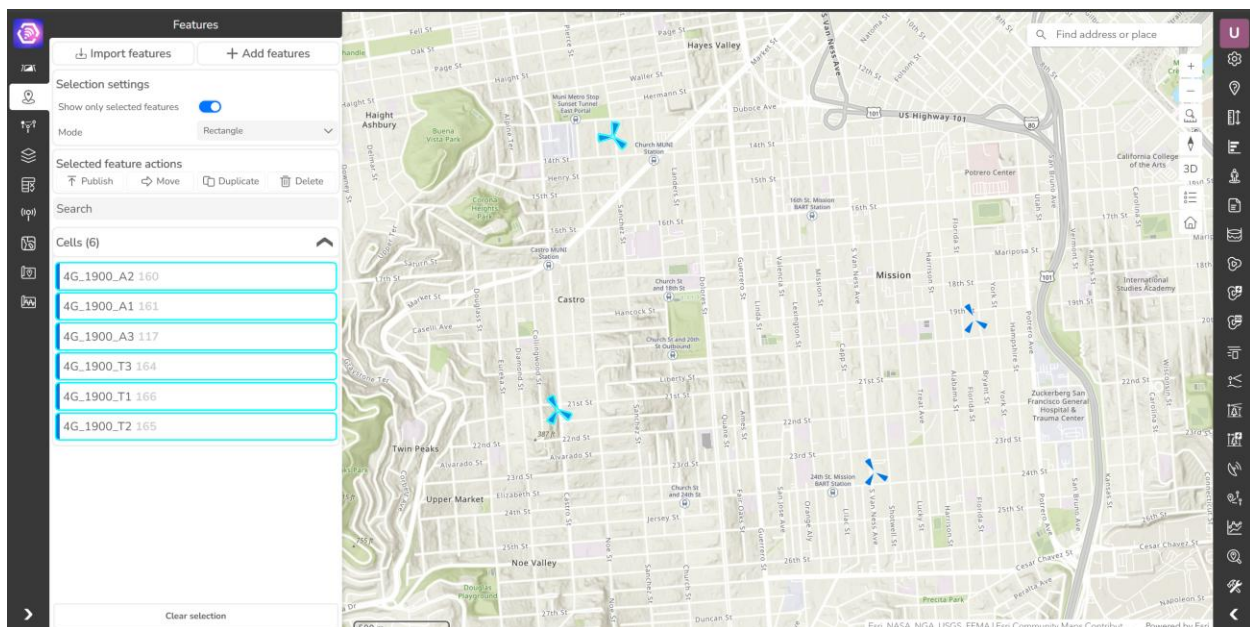
Polygon layer

SF Neighborhoods

To use the polygon layer selection, choose a predefined polygon layer from the dropdown list. Then, click on the map. All features located within the selected polygon boundary will be highlighted and displayed in the feature list.



Selected objects will be highlighted on the map and listed in the tool.



Clear selection

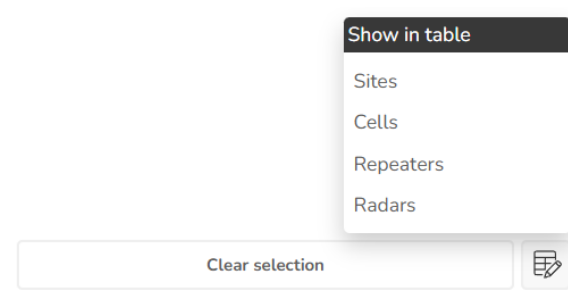
Clears a objects selection on the map.

Search

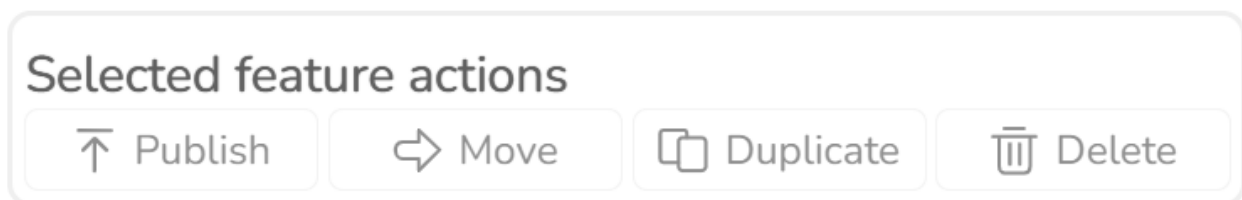
Initiates the search procedure in the selected objects list.

Show in table

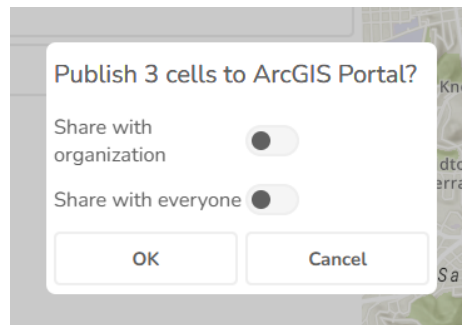
Opens the selected objects attribute table

**3.1.2.4 Move/Duplicate/Delete/Publish selected objects**

Select objects on the map to start the Publish, Move, Duplicate or Delete functionality.

**Publish**

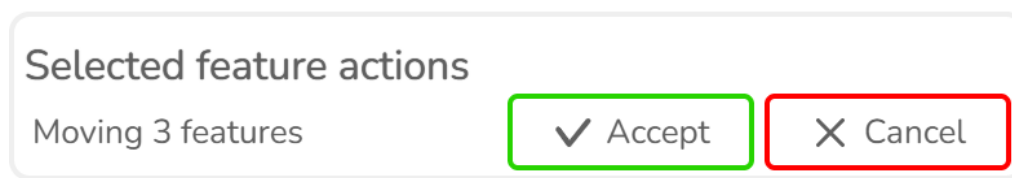
Select the features you want to publish to portal as a feature layer, then click the publish button. An option menu then appears to let you select who the newly created feature layer will be shared to:



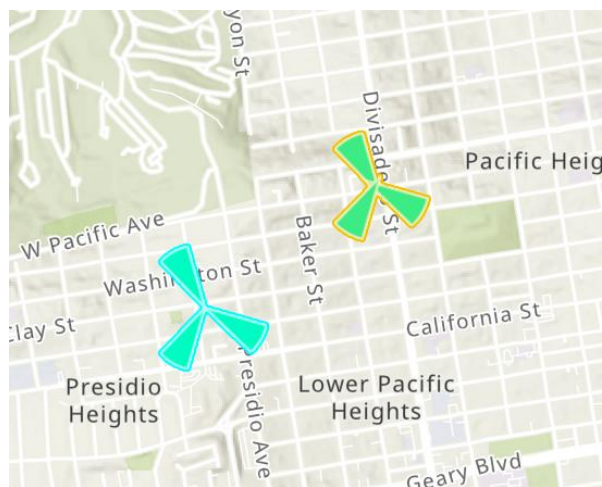
Press OK to start the publishing process.

Move

Select the objects which shall be moved and choose the Move tool. The Move selected features dialog will appear.



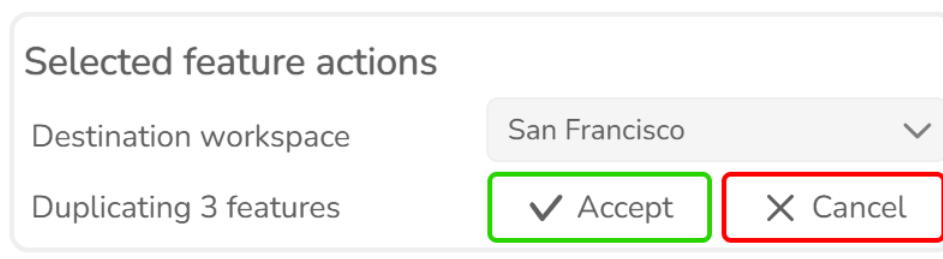
Hold the left mouse button and drag the object to the preferred location.



Press Accept to save changes.

Duplicate

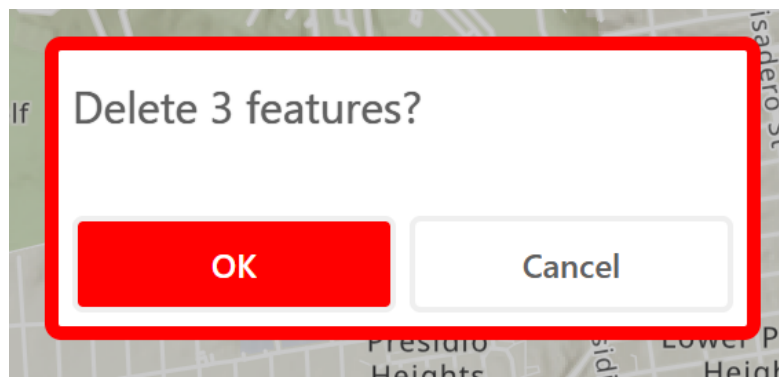
Select the objects which shall be duplicated and choose the Duplicate tool. The Duplicate features dialog will appear. Hold the left mouse button and drag the object to the preferred location. Selected objects can be duplicated to another workspace.



Press Accept to save changes.

Delete

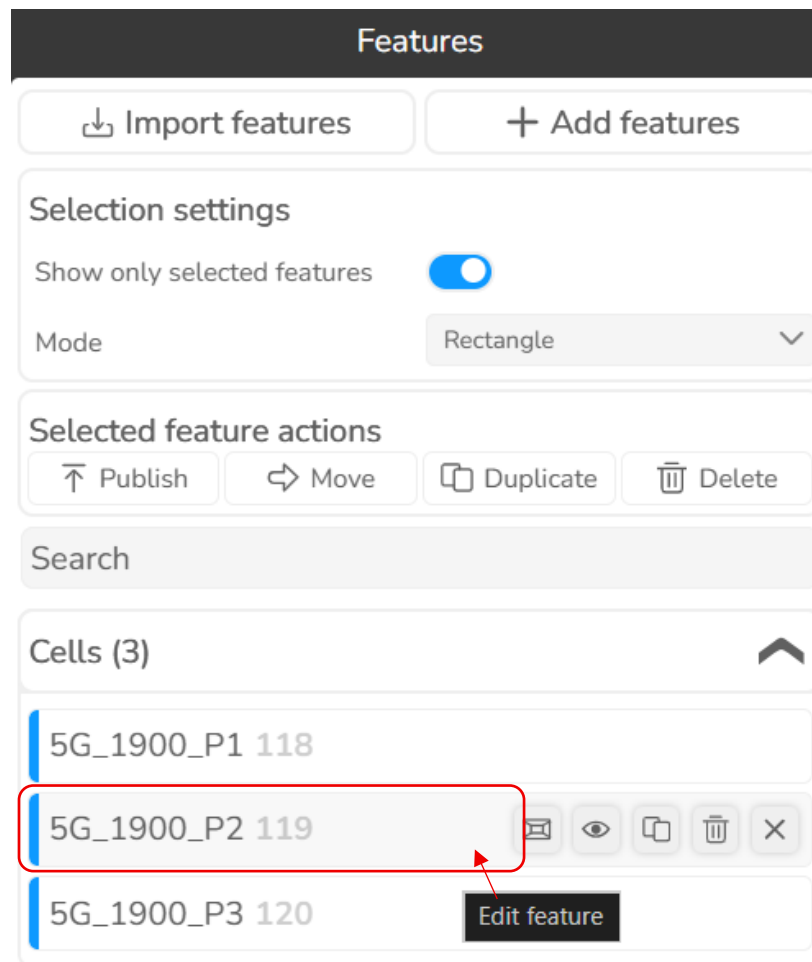
Select the objects on the map, then press the Delete button to remove them from the Map view and database tables.



Press Accept to save changes.

3.1.2.5 Edit objects

Object attributes can be edited using the Cellular Expert graphical interface. Move the mouse cursor to the required object in the Features tool and click on it to edit object attributes.



A new dialog opens on the right side with the most important attributes.

Edit feature

Required parameters

Cell name

2G_1900_U2

X

-122.43175528440705

Y

37.74952330683332

Azimuth, deg

170

Optional parameters

Height, m

20

Downtilt, deg

0

El. Downtilt, deg

0

Frequency, MHz

1900

Power, dBm

Misc loss, dBm

Bandwidth, MHz

20

Noise figure, dB

6

Downlink duplex factor

0.75

Subcarrier spacing, kHz

15

TX MIMO

4

▼

RX MIMO

4

▼

Active antenna effect

0

Cell load, %

10

Accept

Cancel

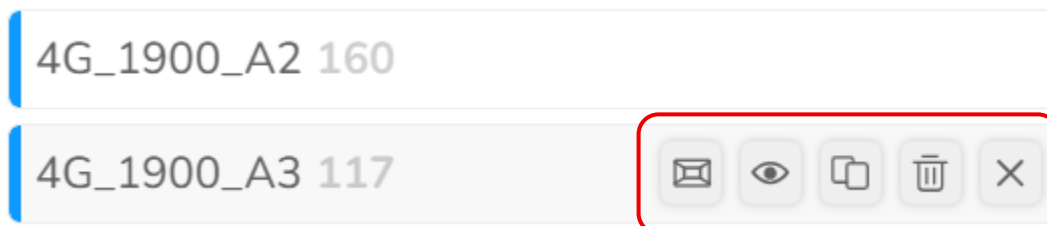
Accept

Saves all changes.

Cancel

Discards all changes and closes the dialog.

Upon hovering the mouse over a feature item, options for it appear.



View from perspective

Viewpoint from the position of cell.

Highlight feature

Highlights feature on the map.

Duplicate feature

Creates a copy of the feature.

Delete feature

Deletes feature.

Remove from selection

Removes feature from the selected features list.

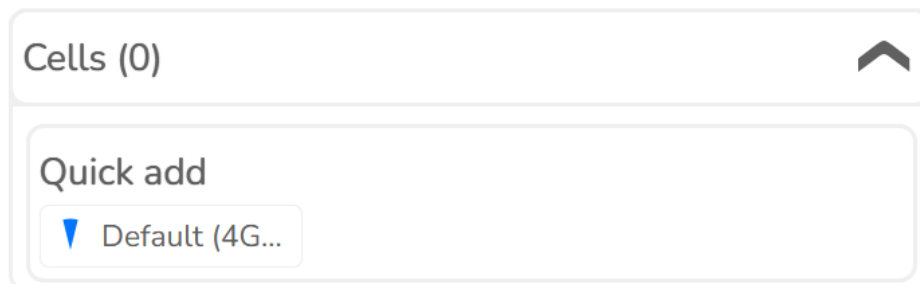
Add to selection

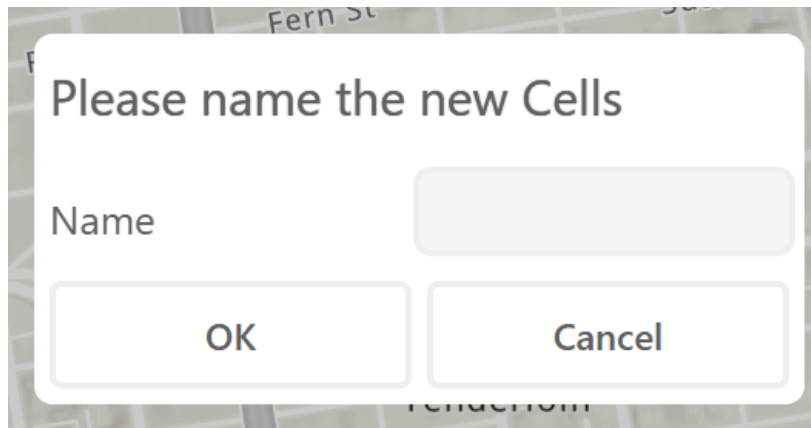
Adds feature to selected features list.

3.1.2.6 Quick add

If you have feature templates favorited, the quick add section will appear under the feature type in the features list. You may then use the listed favorite feature templates to quickly add features with the template applied. This can be done in 2 ways:

- Click the feature template name to automatically open the add features tool and have the template applied
- Drag and drop the feature template on the map to quickly add a new feature to the desired location with only having to enter the feature name in a popup.

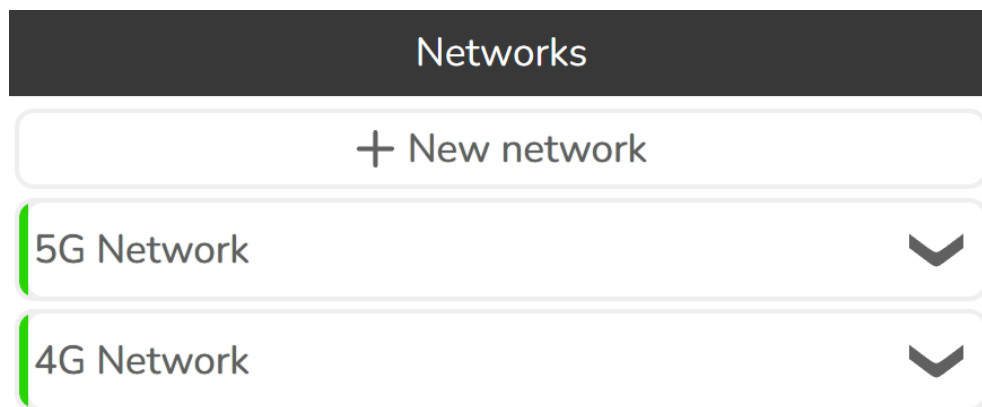




3.1.3 Networks

Click this button  to open Networks tool.

The networks tool allows for easy management of multiple objects, grouped into networks. Networks can be defined by feature attribute or via the selection of the features required. Then, feature attribute change tracking and calculations can be performed for each network without the need for selection.



Once expanded, an existing network item displays the following information:

4G Network

Feature type

cells

Feature count

45

Uncalculated features

0

Calculation status

Complete

Last calculated

4/25/2025 11:03 AM

Results

All results

export

open

4G 1800 Field Strength 1, dBm

export

open

4G 1800 Best Server 1

export

open

4G 1800 Field Strength 2, dBm

export

open

4G 1800 Best Server 2

export

open

Calculate

Feature type
The feature type associated with the network.

Feature count
Total feature count in the network.

Uncalculated features
Number of Network objects that have not yet been added to a calculation.

Calculation status
Status of the current calculation.

Last calculated
The last time the calculation was started.

Results
Select which prediction results you want to add to the Map view and manage in the layers tool.

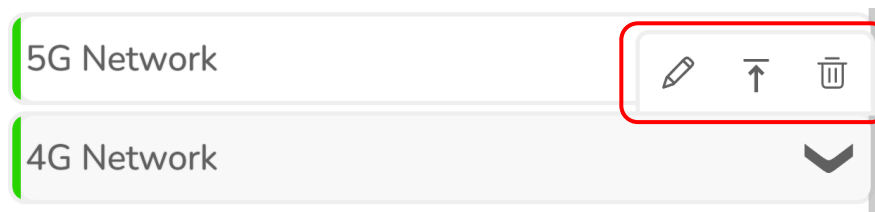
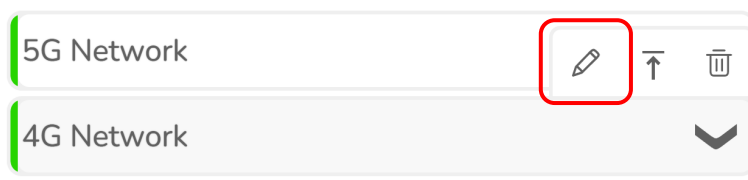
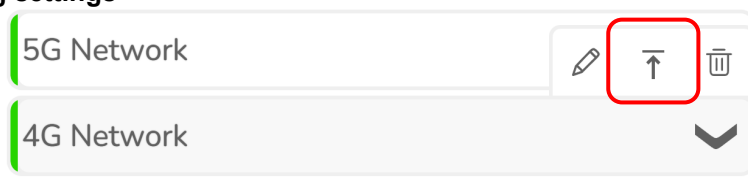
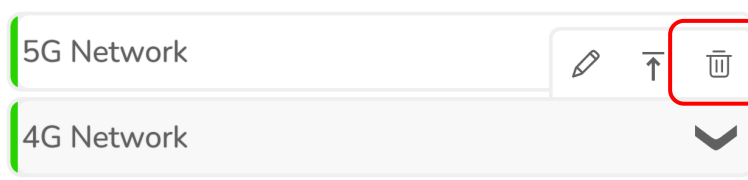
Export

Exports the selected prediction results layer as a TIF raster.

Open

Opens the selected prediction results layer.

Upon hovering the mouse over a network item, options for it appear.

**Edit Network****Network publishing settings****Delete Network****3.1.3.1 Add network**

To create a network press the *New network* button.

Networks

Network properties

Network name

New network

Feature type

none

▼

Filter

Feature filter

By attribute

▼

Filter attribute values

+

Network name

The name of the network

Feature type

The type of feature to be included in the network.

Feature filter

- By attribute – Include features in the network by selecting an attribute and inputting the required attribute values. The features with these values will then be filtered and included in the network.
- From selection – Include features in the network from the currently selected items.

Attribute

The attribute field to be filtered by.

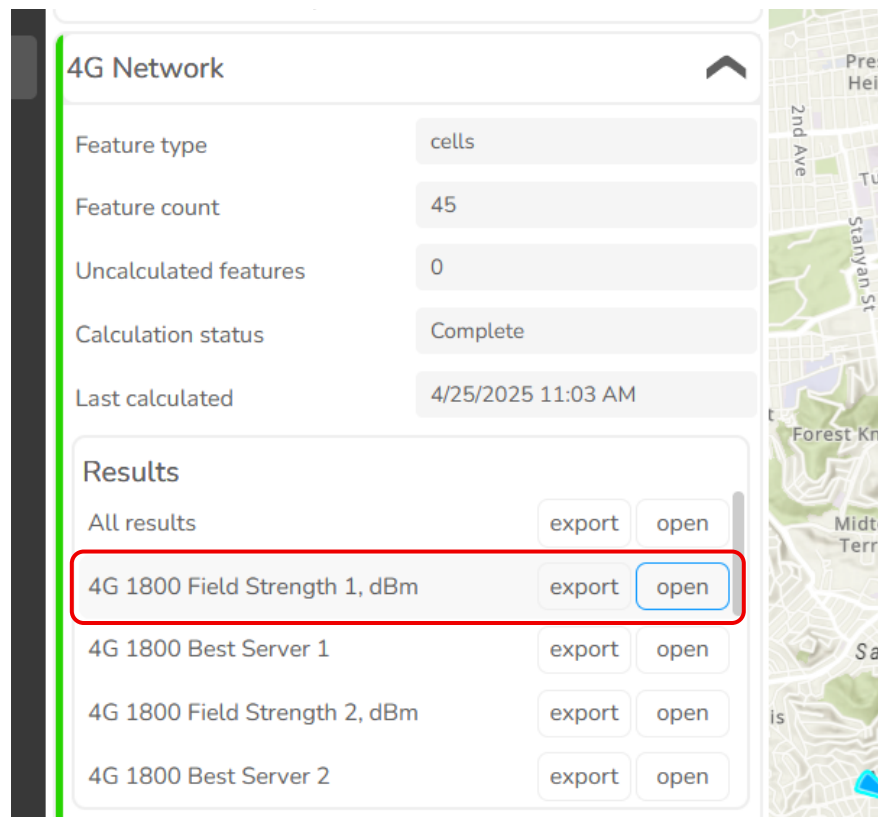
Filter attribute values

Required values for the filter attribute. Features containing ANY of the entered values will be included.

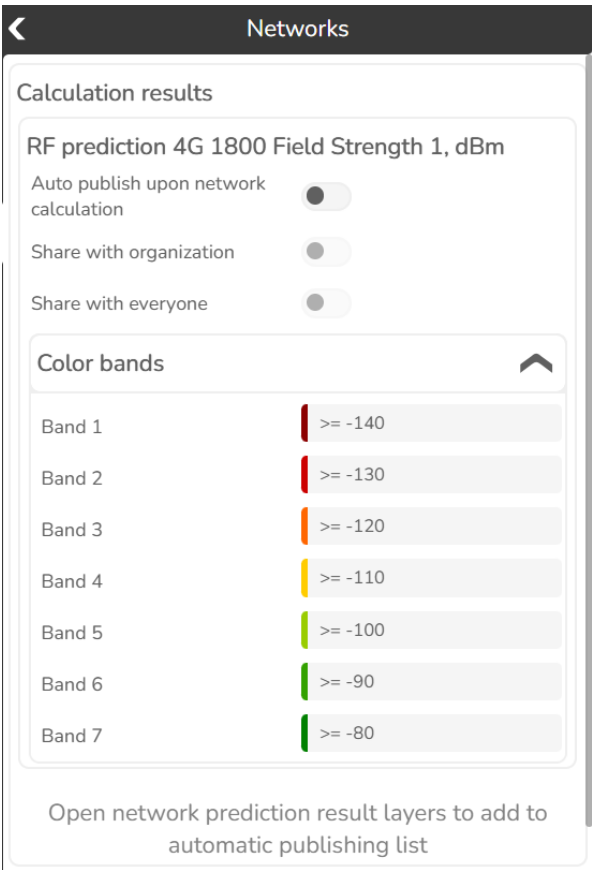
3.1.3.2 Network publishing setting

If you would like the calculation results to be automatically published when the network is re-calculated, you may set it up in the network publishing settings menu.

First open the calculation result layer you would like to be automatically published, and set up the color band values as per your preference:

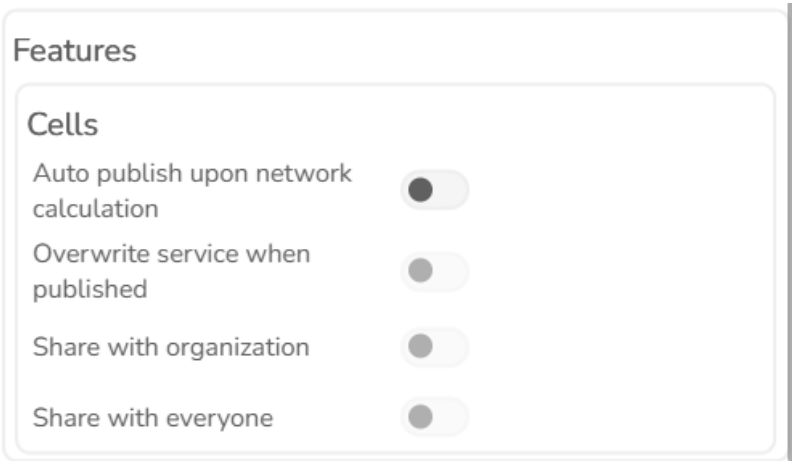


Then, options for your layer's publishing will appear under the calculation results section:



You may enable automatic publishing and select who the results are shared to for this layer.

The same options are also available for the features that are associated with the network.

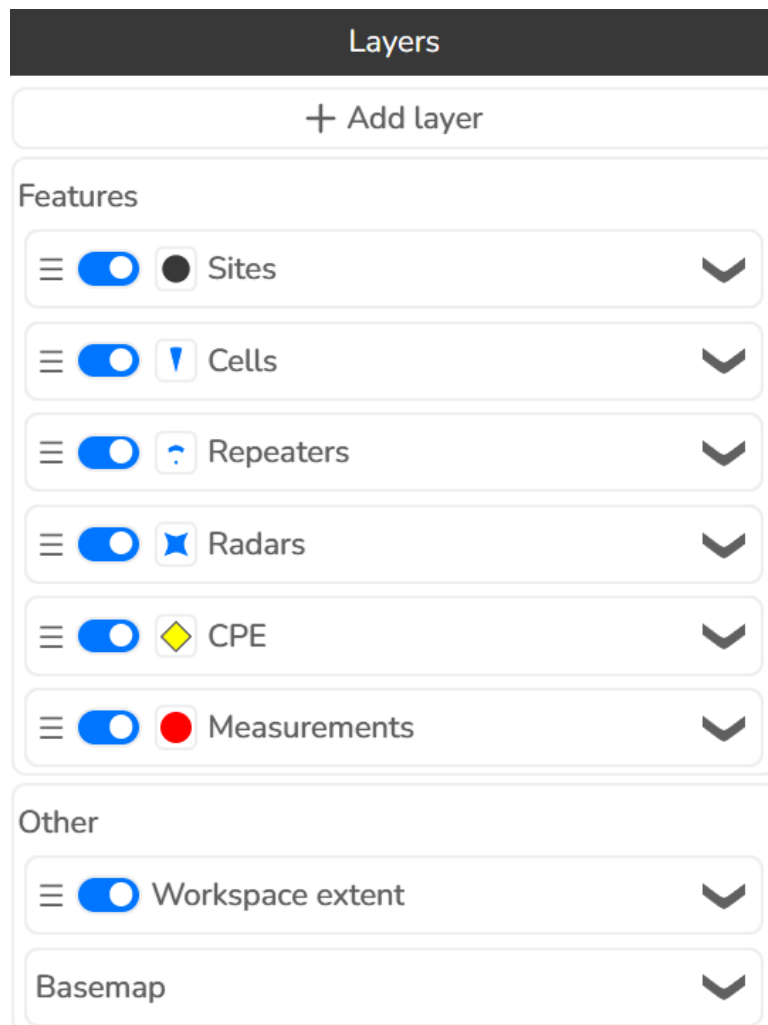


After setting up the automatic publishing options, click Accept to save. The publishing is then executed the next time the network calculation is run.

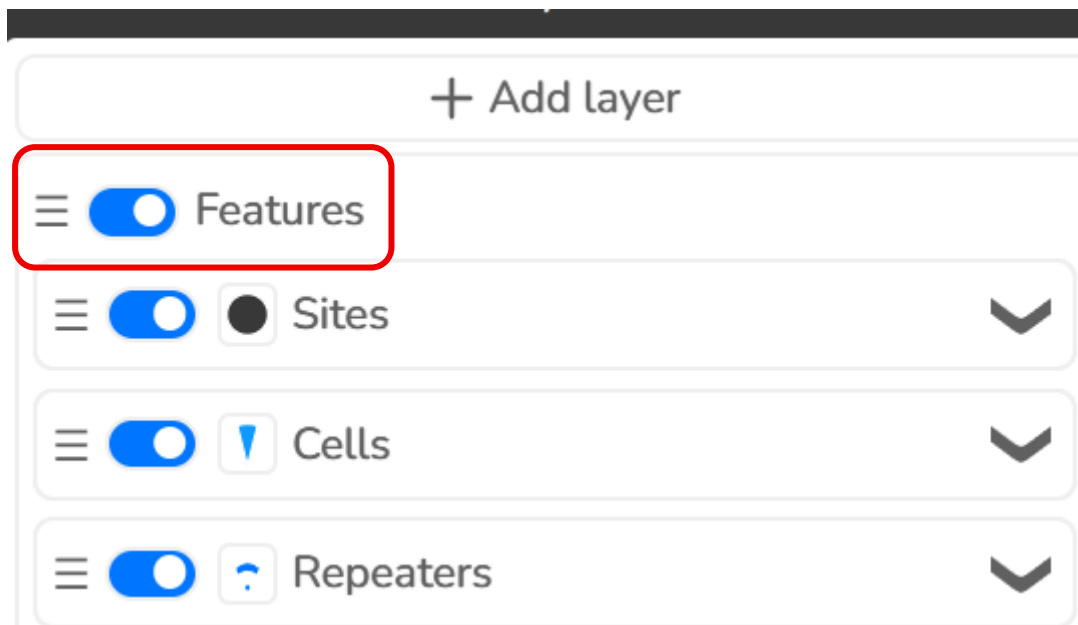
3.1.4 Layers

Click this button  to open Layers tool.

Manages the visualization of features and calculation results in the Map view. Displays and manages all layers in the workspace, grouped into categories such as Features, Geodata, Prediction results, and Other. Each layer can be shown, hidden, or reordered.



You can reorder the entire group or toggle its visibility to show or hide all layers within it at once.

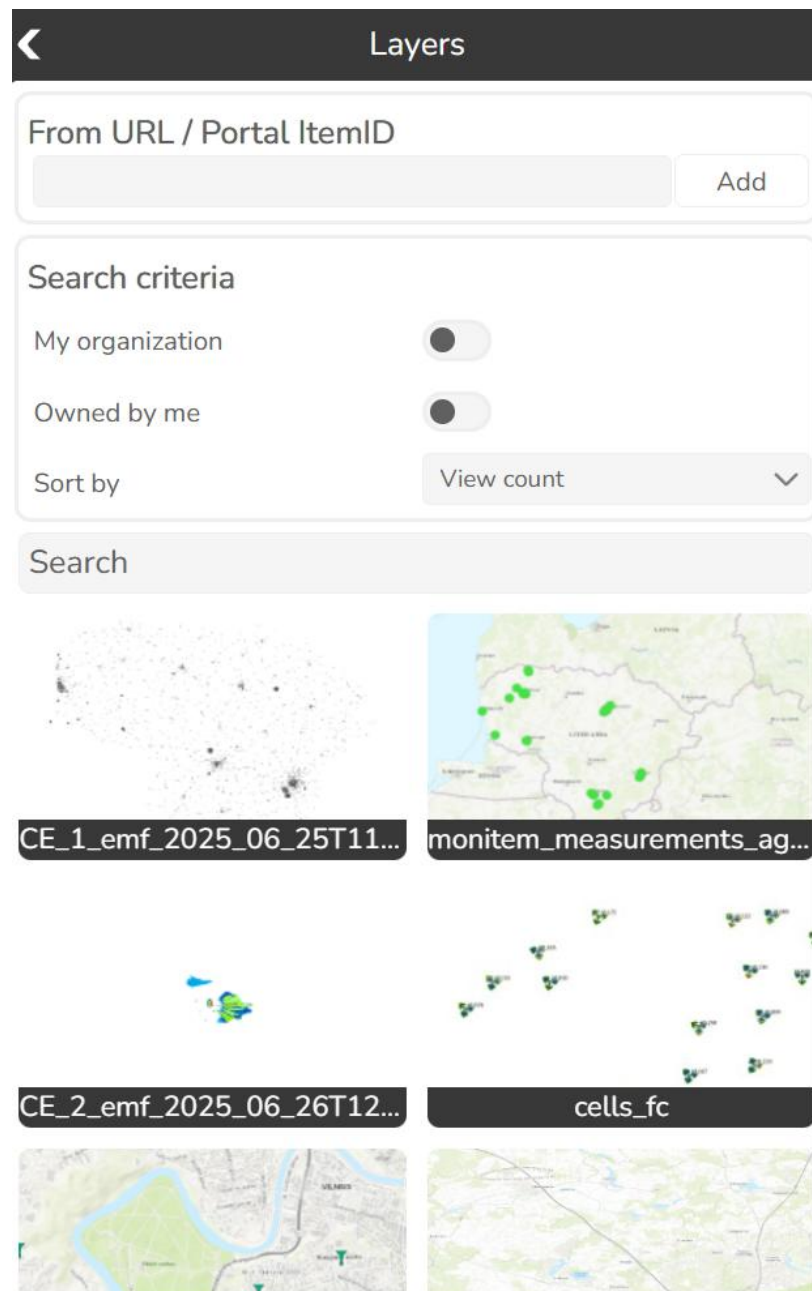


Layers can also be organized by dragging them into existing groups or dropping them into **New layer group** to create a custom group.



3.1.4.1 Add layer

Layers added from this panel are visible only to you. Other users who open the same workspace will not see these layers unless they also add them manually.

**From URL / Portal ItemID**

Allows you to add a layer directly by entering its URL or Portal Item ID, then selecting Add.

Search criteria**My organization**

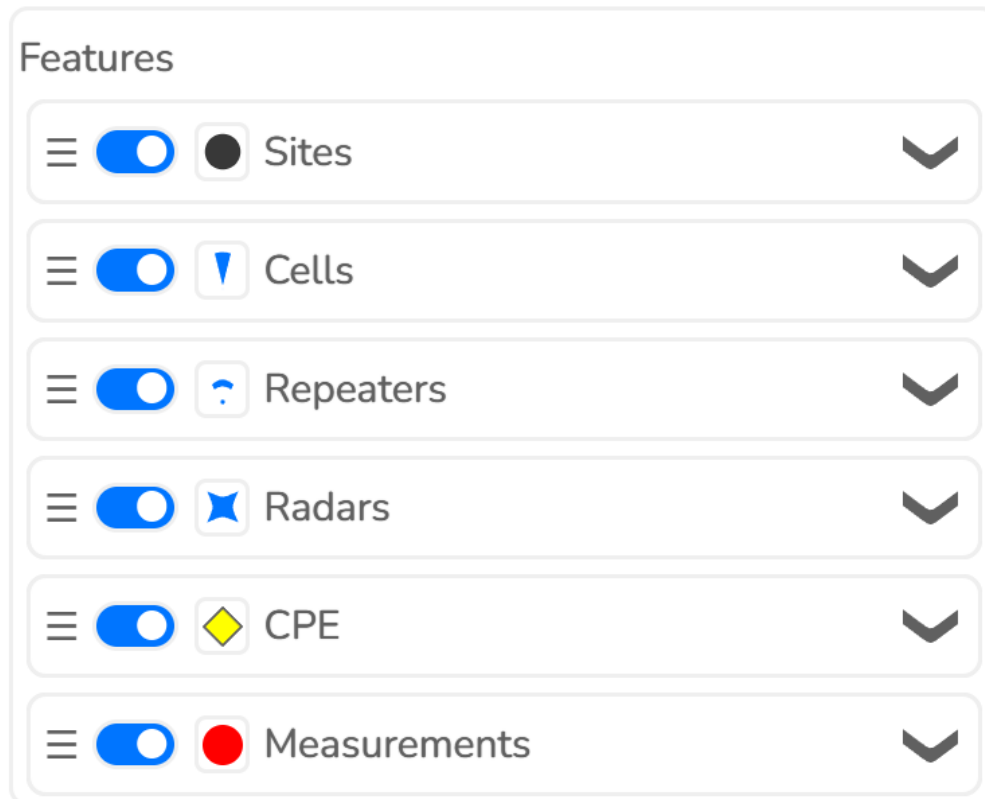
Limits search results to layers shared within your organization.

Owned by me

Shows only layers you own.

Sort by

Determines the order of search results (e.g., by view count or date).

3.1.4.2 Features**Rearrange layers**

Allows you to reorder layers by dragging.

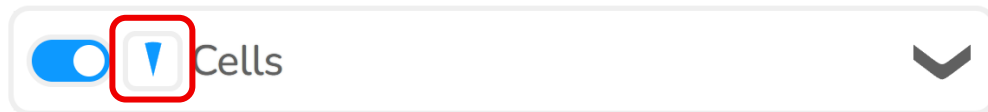
**Turn on/off the layer**

Turn on or off a layer in the Map view

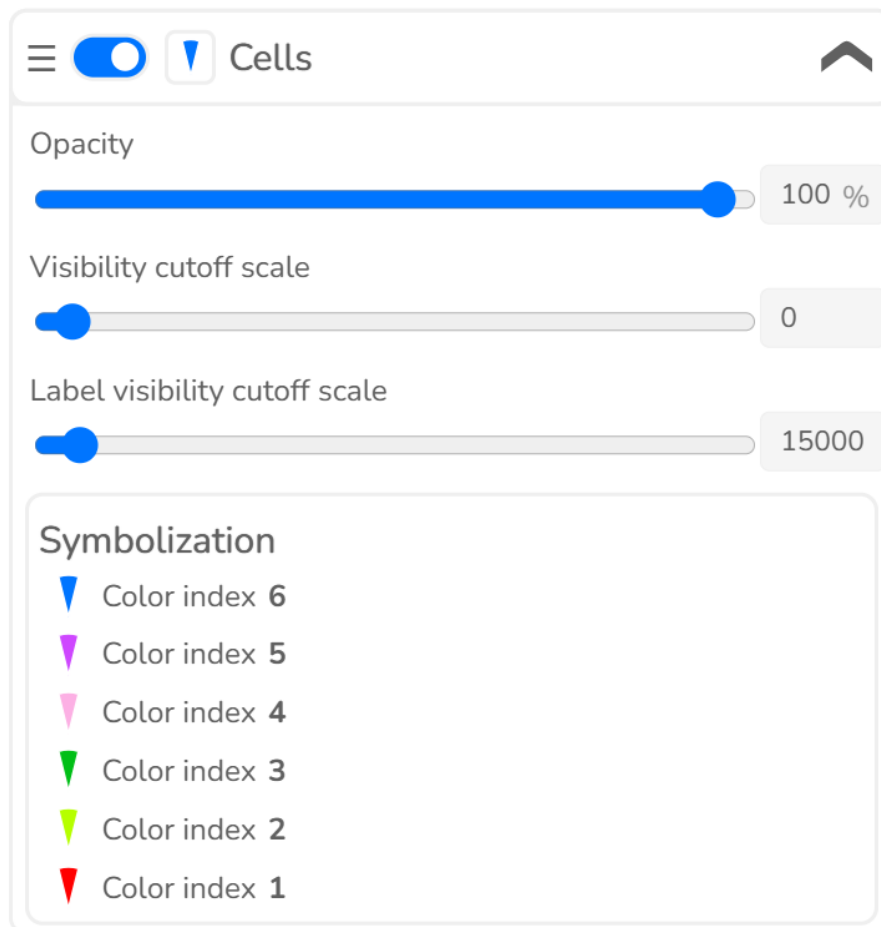
- The layer is visible - 
- The layer is not visible - 

Layer symbol

Preview layer symbol. Click on it to edit the symbol.

**Expand layer**

Previews and manages the layer opacity and map scale .

**3.1.4.2.1 Edit symbolization**

Click on the layer symbol to edit it. A new dialog opens on the right side.

Symbol



Edit 3D

☐

Shape

path ▾

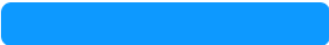
SVG Path

M 7 22 L 2 2 C 5 1 9 1 12 2 Z

Size

20

Color



Outline

☒

Outline color

Outline width

1

Offset

☒

X offset

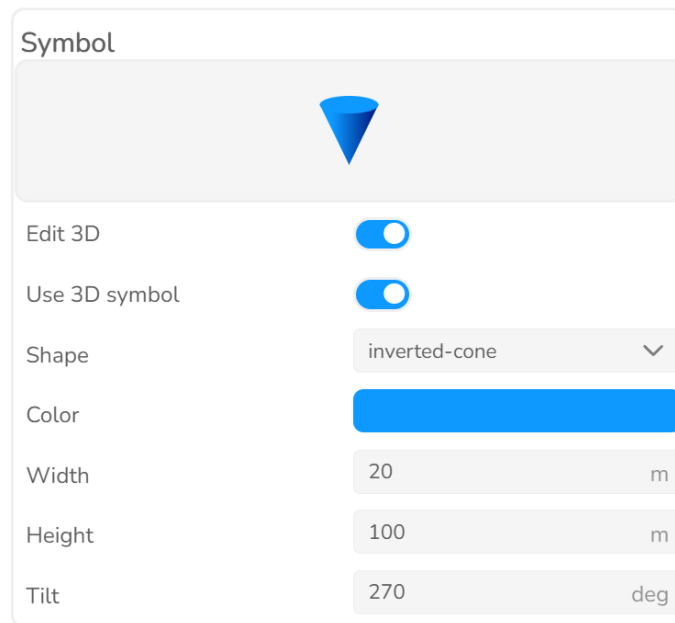
0

Y offset

10

In layer symbolization it is possible to change:

- Shape.
 - if a path is selected in shape, svg path can be changed.
- Size.
- Color.
- Turn on and off outline.
- Outline color.
- Turn on and off offset.
- X/Y offset.



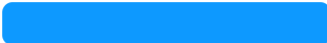
Symbol



Edit 3D ☒

Use 3D symbol ☒

Shape inverted-cone ▼

Color 

Width 20 m

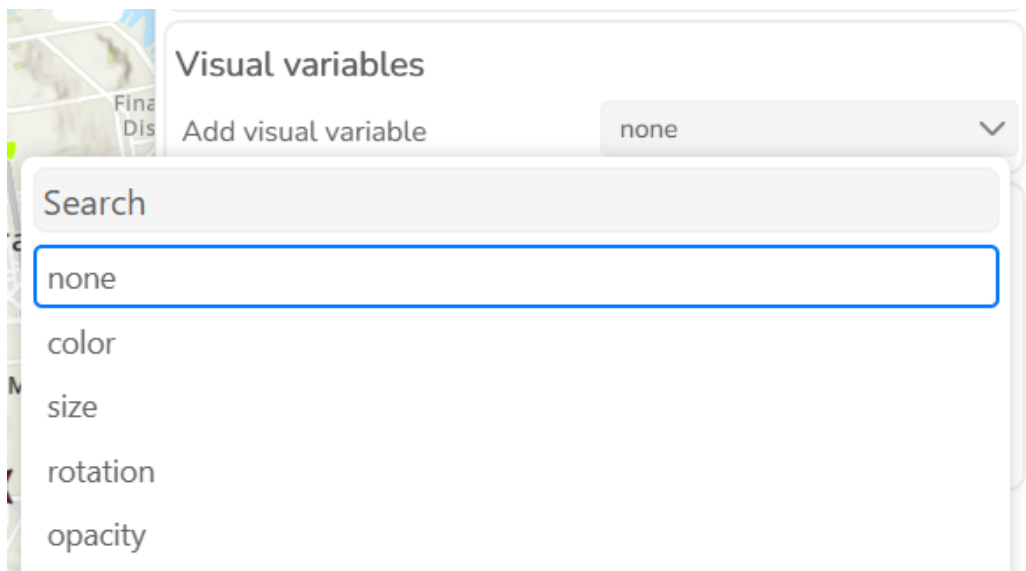
Height 100 m

Tilt 270 deg

There is an option to use 3d symbol. The following parameters can be changed in the 3d symbol:

- Shape
- Color
- Width
- Height
- Tilt

Visual variables can be added for each layer. Add visual variables to a layer using the *Add visual variable* dropdown.



Visual variables

Add visual variable none ▼

Search

none

color

size

rotation

opacity

Choose a visual variable. The choices are:

- Color.
- Size.
- Rotation.
- Opacity.

Visual variables

Add visual variable none ▼

Rotation ✕

Field azimuth ▼

Color ✕

Field color_index ▼

Type unique ▼

Stops

+ Add stop

Value	6	Color		✕
Value	5	Color		✕
Value	4	Color		✕
Value	3	Color		✕
Value	2	Color		✕
Value	1	Color		✕

Accept

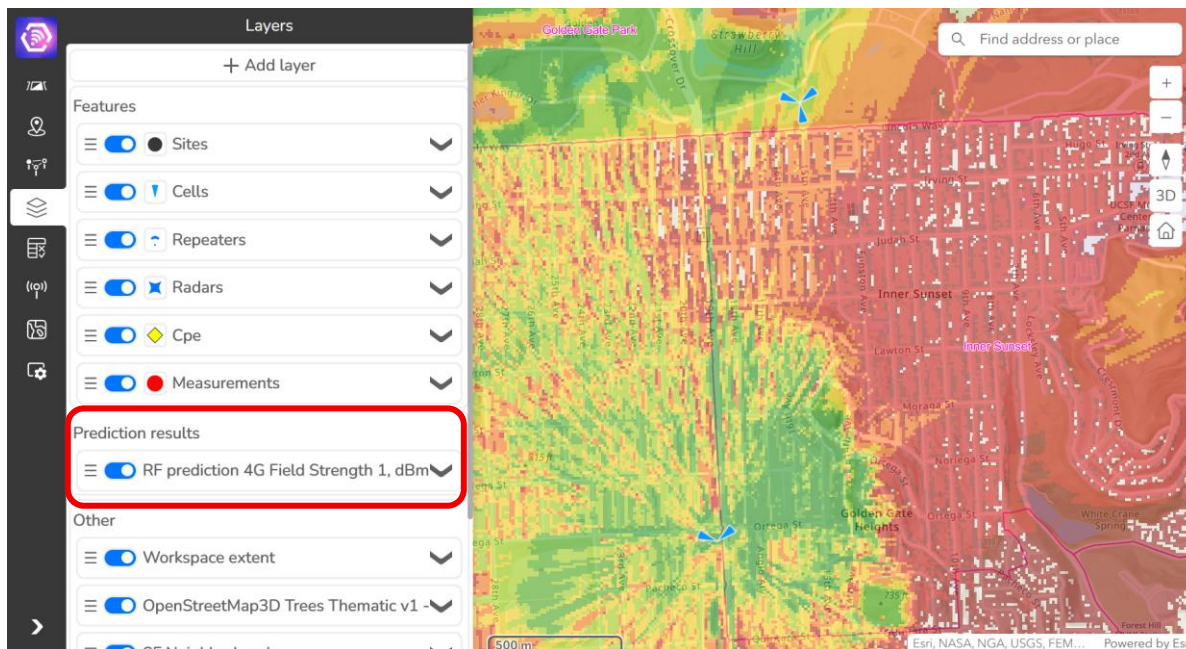
Saves all changes.

Cancel

Discards all changes and closes the dialog.

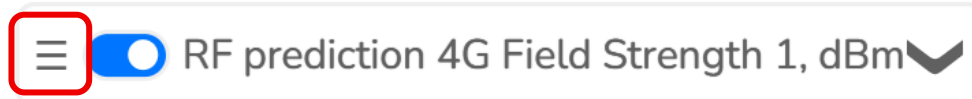
3.1.4.3 Prediction results

The prediction rasters calculated in Cellular Expert Express and loaded in the map view will appear in the Layers tool.



Rearrange layers

Allows you to reorder layers by dragging.



Turn on/off the layer

Turn on or off a layer in the Map view

- The layer is visible - 
- The layer is not visible - 

Expand layer

Expand the layer so you can see and edit the symbology.

For these types of layers, the visualization can be edited in the following ways:

- Add threshold using *Add color band* button

+ Add color band

- Adjust threshold value

Value -91

- Adjust color band

Color



- Adjust color opacity



- Delete threshold using *Delete band* button

×

Color bands

+ Add color band

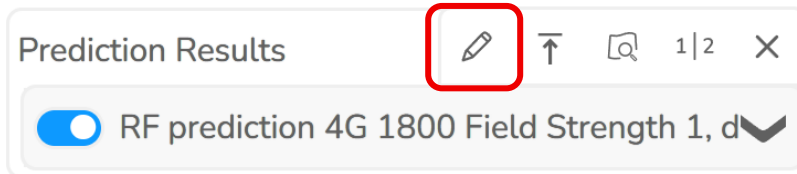
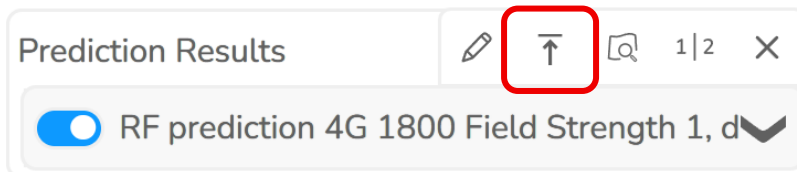
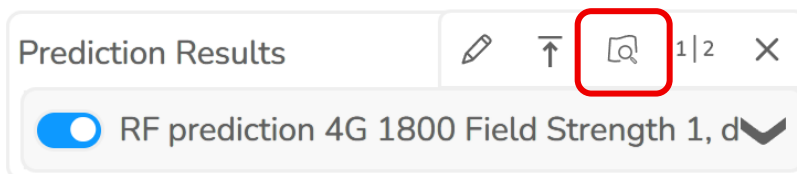
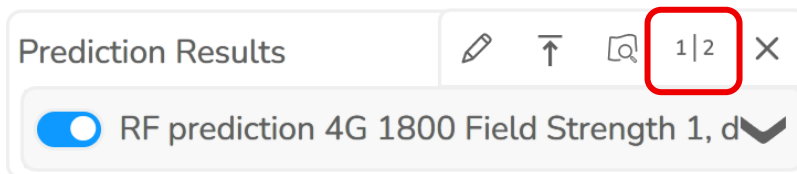
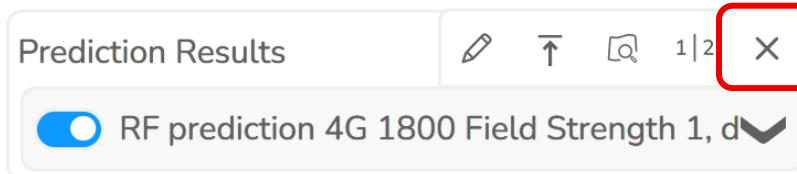
Value	-80	Color			×
Value	-90	Color			×
Value	-100	Color			×
Value	-110	Color			×
Value	-120	Color			×
Value	-130	Color			×
Value	-140	Color			×

Upon hovering the mouse over a prediction result item, options for it appear.

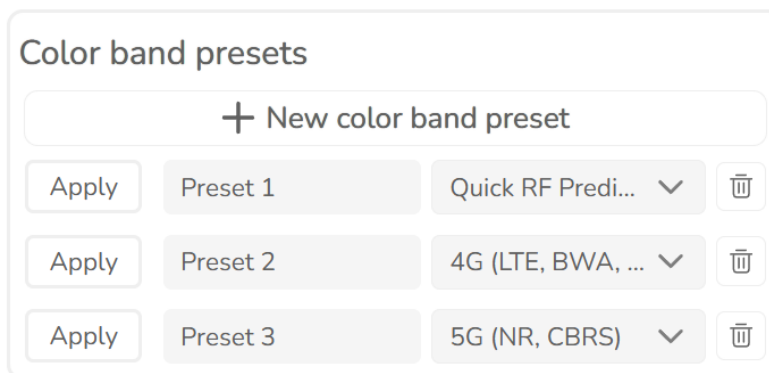
Prediction Results

1/2

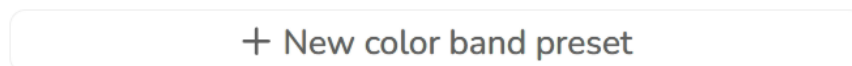
☒ RF prediction 4G 1800 Field Strength 1, d

Rename layer**Publish to portal****Zoom to layer****Compare****Remove layer****3.1.4.3.1 Presets**

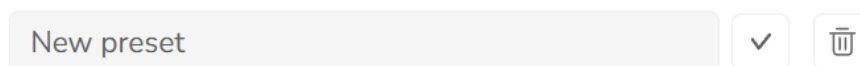
It is possible to create symbology presets. This means that when a prediction raster is loaded, a symbology preset can be used:



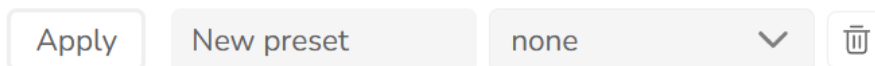
To create a preset, first define the bands and colors, then press *New color band preset* button.



Define a preset name.



A new preset will be created.



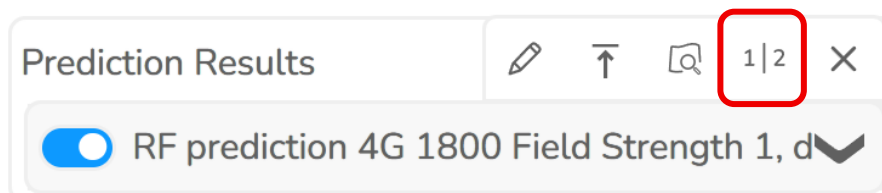
The defined preset will be applied the next time when the raster will be loaded.

Apply

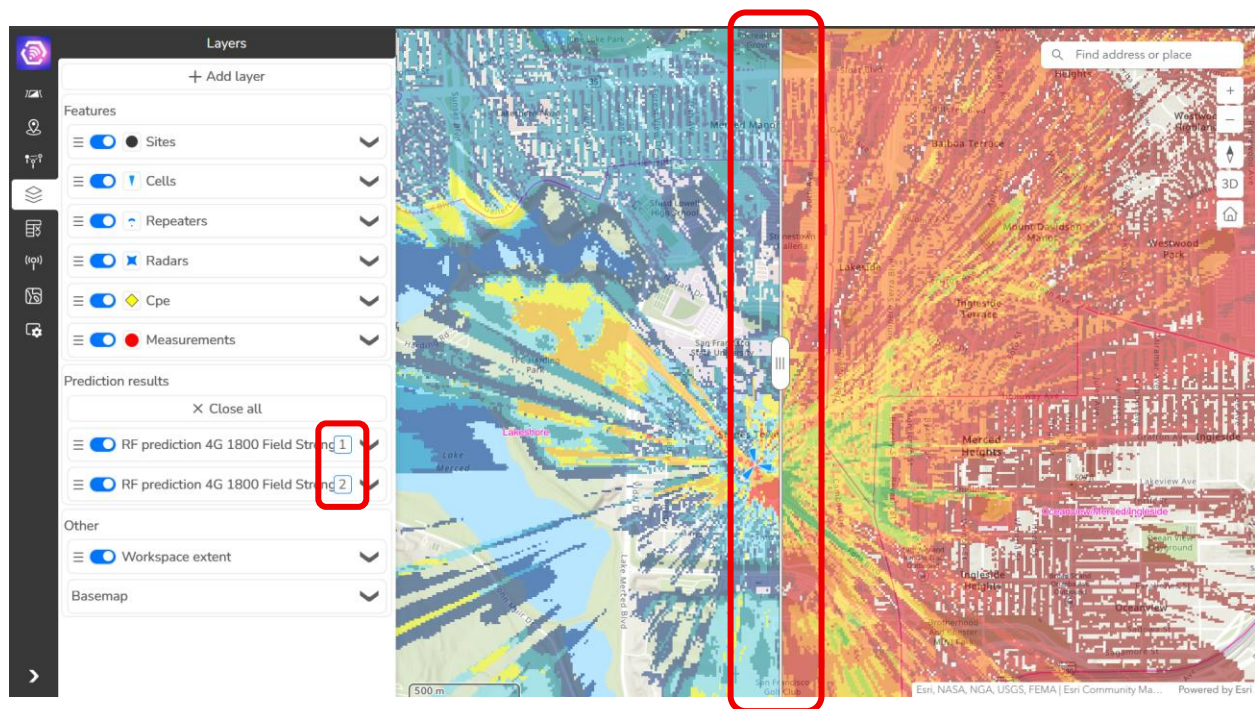
Applies a preset symbology for the current prediction raster layer.

3.1.4.3.2 Swipe

The Swipe widget enables you to easily compare the content of different layers in a map. Mouse over the layers you want to compare and press the compare button.



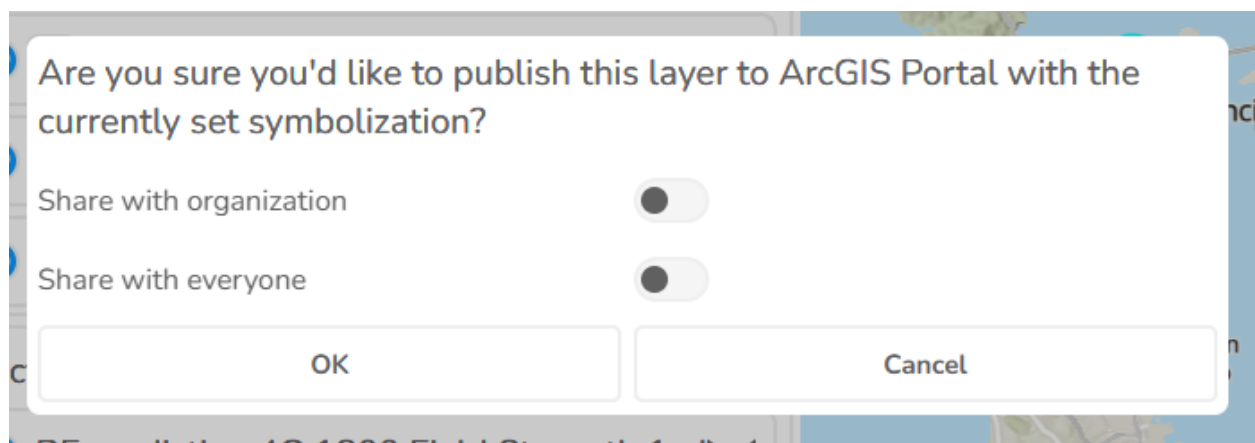
When the first 1 and second 2 layers are selected for comparison, a swipe widget will appear on the map. Slide the swipe tool to compare different layers.



3.1.4.3.3 Publish to portal

Raster prediction results may be individually published to portal by using the publish to portal button.

Upon clicking this button, publishing settings will appear:



Set up the publishing options and click OK to start the publishing process.

3.1.4.4 Geodata

This section appears when enabled in the settings.

Keep prediction results open
between sessions ☒

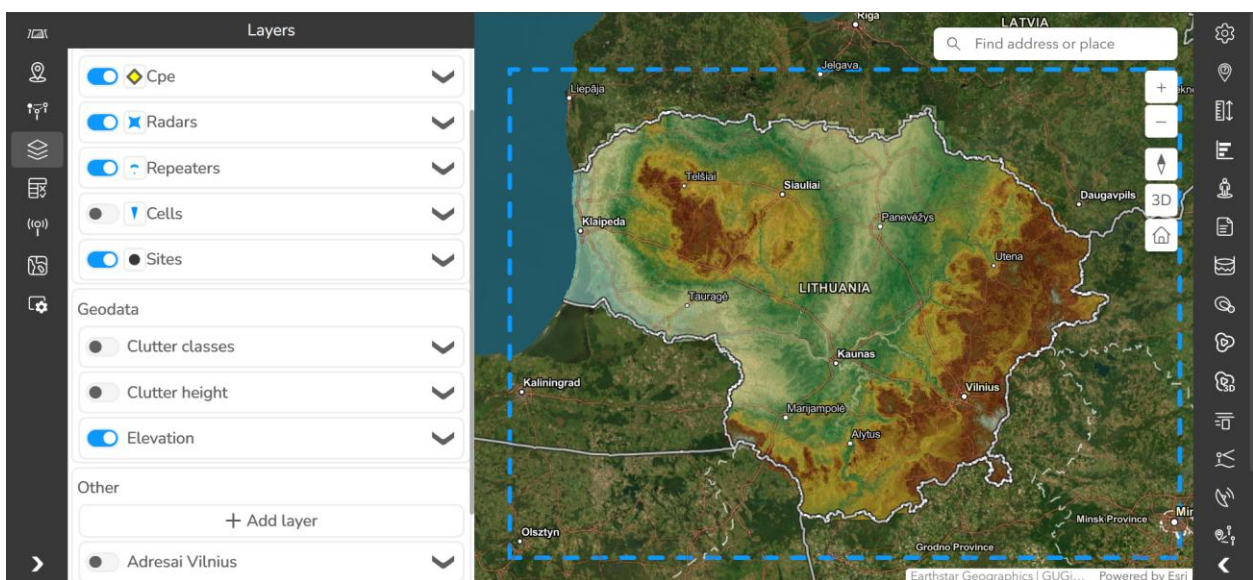
**Load geodata when opening
workspace ☒**

Show only feature layers related
to active module ☒

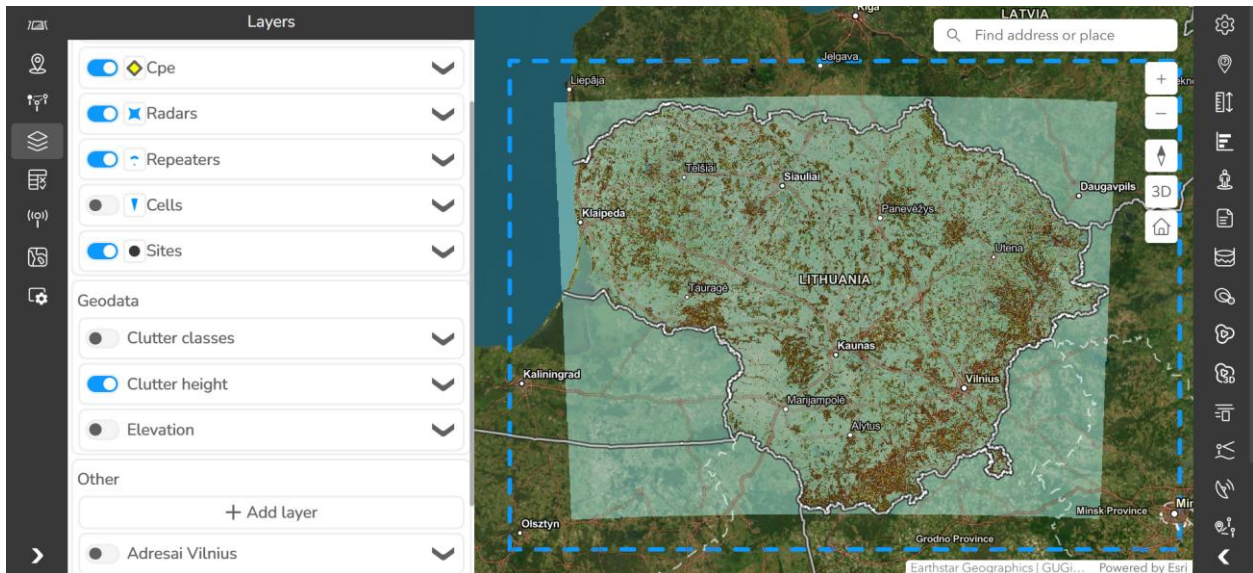
Section contains geodata raster layers from the geodata set assigned to the current workspace.



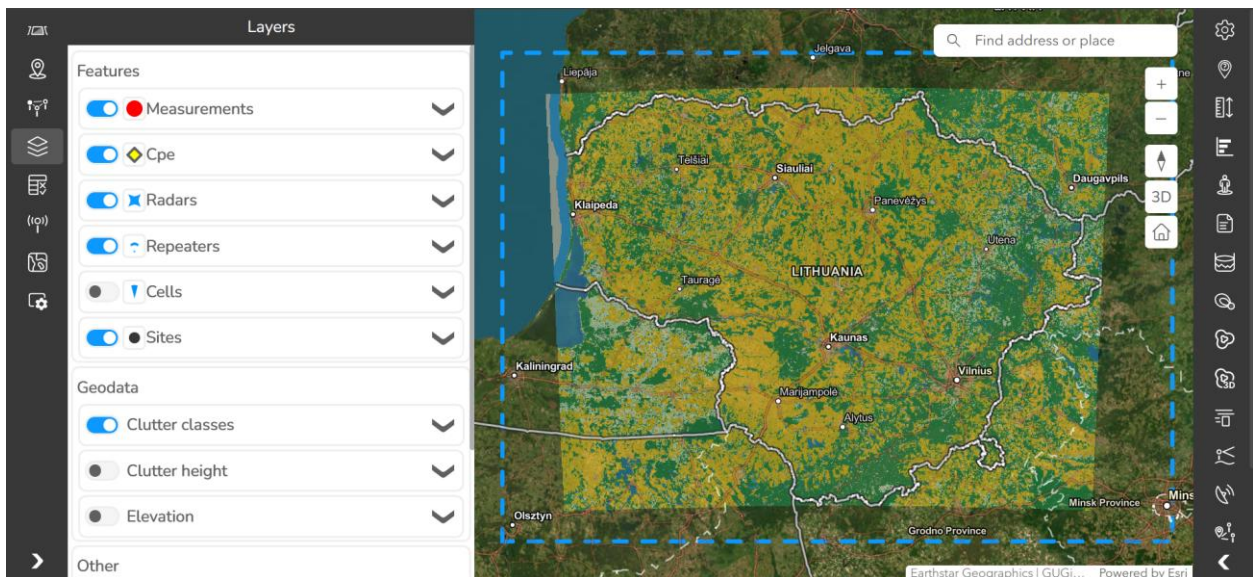
- **Elevation:**



- **Clutter height:**



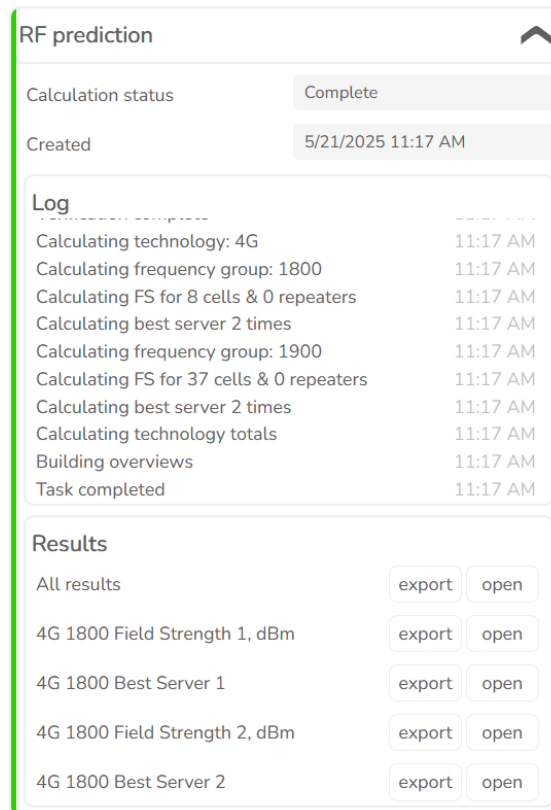
- **Clutter classes:**



3.1.5 Prediction history

Click this button  to open Prediction history tool.

All calculations performed in Cellular Expert Express are loaded in this tool. This includes failed and successful calculations. The accessibility of prediction rasters is user specific. This means that users can not view the prediction results of other users, except the owner of the prediction results shares them with the organization.



Calculation status

Status of the current calculation

Log

The tool shows the progress of a prediction in the Task log section.

Results

Select which prediction results you want to add to the Map view and manage in the layers tool.

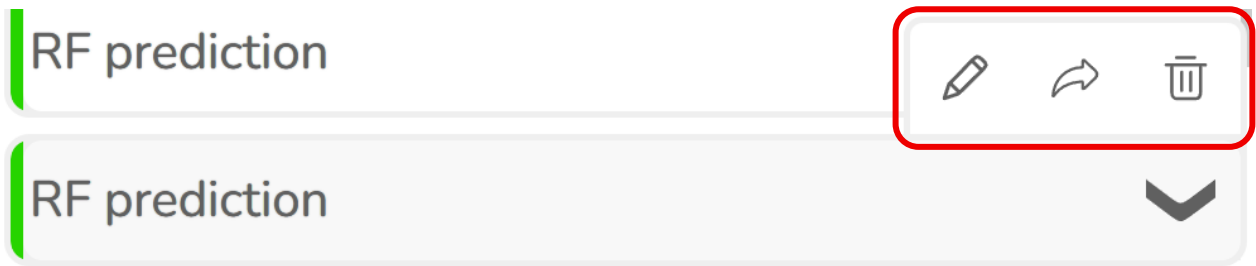
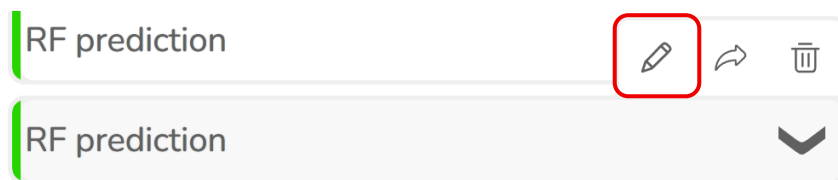
Export

Exports the selected prediction results layer as a TIF raster.

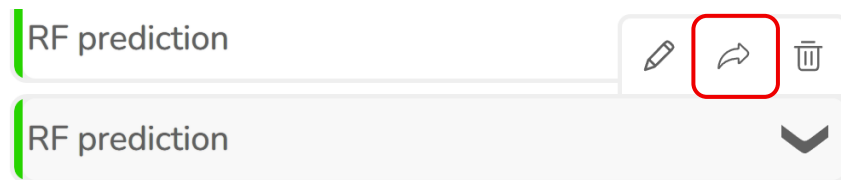
Open

Opens the selected prediction results layer.

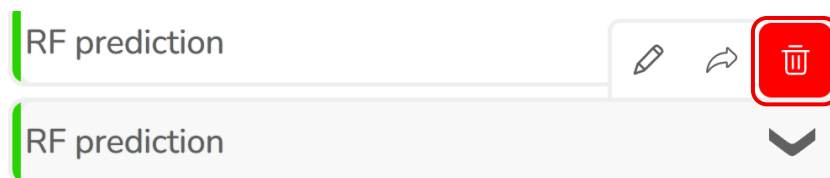
Upon hovering the mouse over a prediction result item, options for it appear.

**Rename layer****Share**

Mouse over the calculation you want to share with the organization members and click *Share calculation task* button.

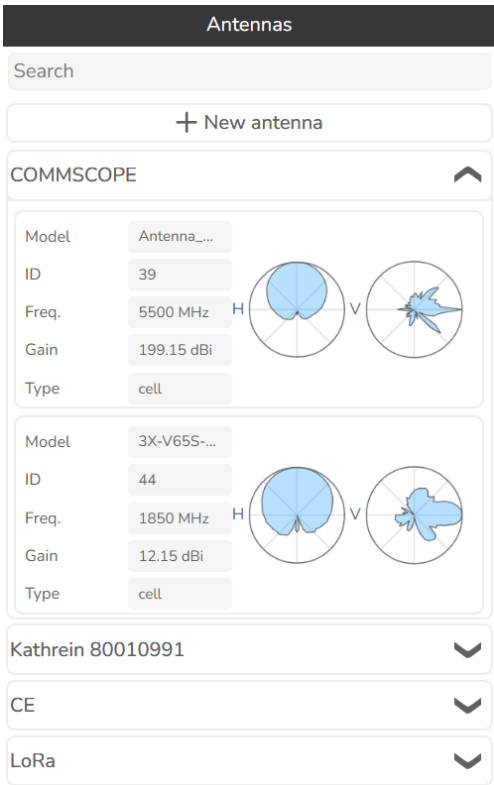
**Delete**

Deletes the calculation from the Prediction History widget.

**3.1.6 Antennas**

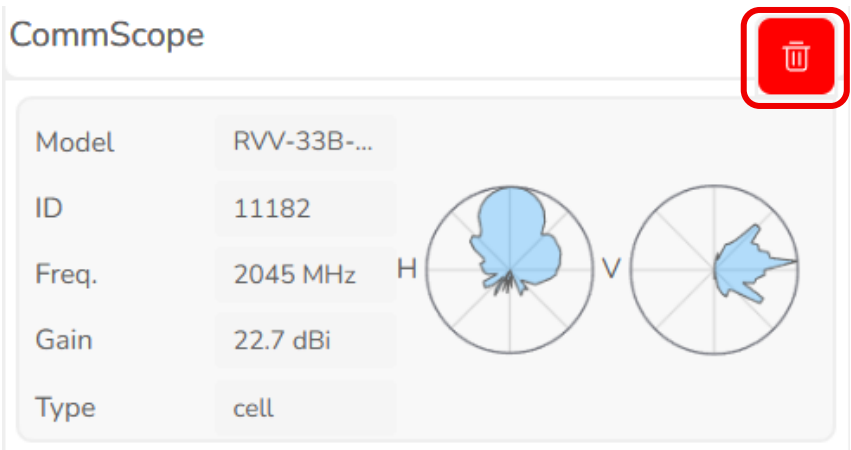
Click this button  to open Antennas tool.

Preview the list of available antennas in the project. The antennas are used for point-to-area calculations.

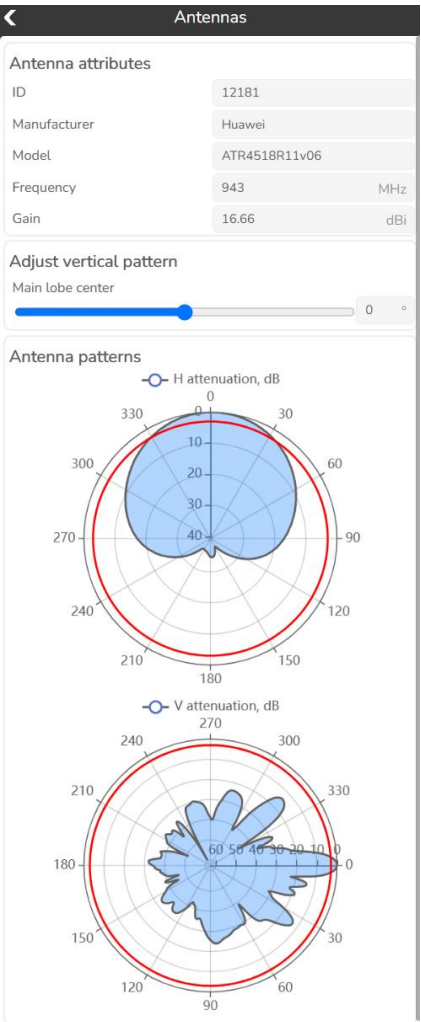


Search
Initiates the search procedure in the antennas list.

Delete antenna
Delete antenna using *Delete* button.

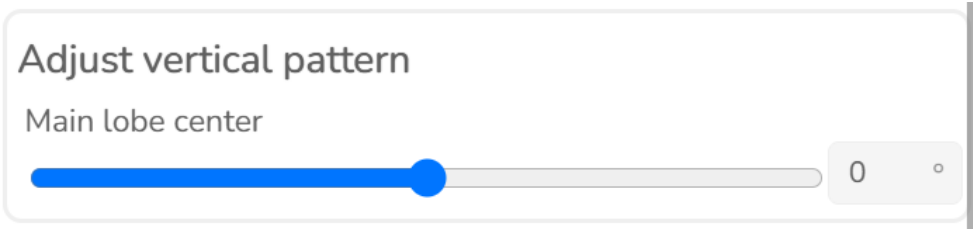


Click on an antenna to preview the antenna patterns.



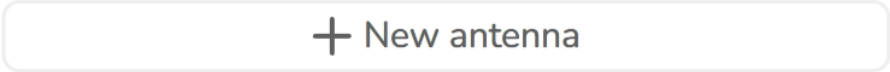
Adjust vertical pattern

In the case where you have an electrically tilted vertical antenna pattern imported, you might want to tilt it back so it is essentially a 0 tilt antenna pattern (recommended if you use electrical tilt field in cell attributes). You may use the adjust vertical pattern tool to do this. Find the center of the main lobe of the vertical antenna pattern using the “main lobe center” slider, then click accept.

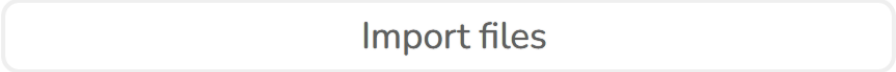


3.1.6.1 Import Antennas

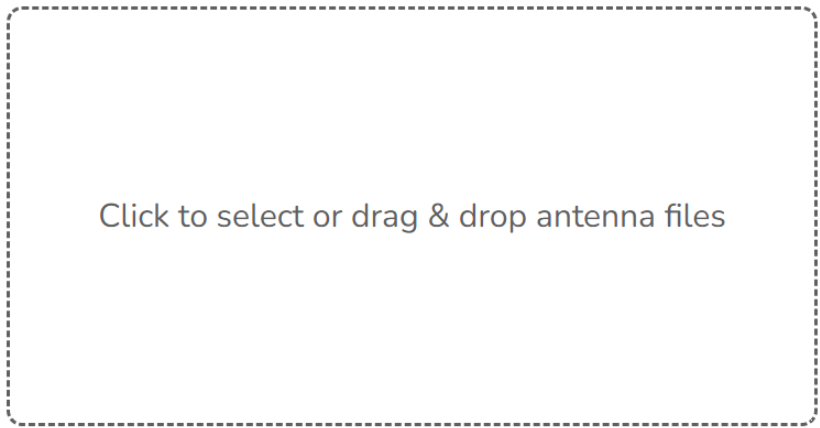
To import new antennas, click the *New Antenna* button

A rectangular button with rounded corners, containing a plus sign icon followed by the text "New antenna".

and then select *Import Files*.

A rectangular button with rounded corners, containing the text "Import files".


Antennas

A large rectangular area with a dashed border, containing the text "Click to select or drag & drop antenna files".

To start the import procedure, select or drag and drop the antenna pattern file. Planet format antenna pattern files are supported. Here is an example:

```
NAME Typical 33deg H and 7deg V TR 36 942
FREQUENCY 0
H_WIDTH 33
V_WIDTH 0
FRONT_TO_BACK 0
GAIN 0 dBi
TILT 0
HORIZONTAL 360
0.00 0.00
1.00 0.01
2.00 0.04
3.00 0.10
4.00 0.18
```

3.1.6.1 Import HCM code file

To import HCM code patterns as antennas, you may use this import option. The required file is a CSV with 3 fields, antenna gain, horizontal HCM code and vertical HCM code. These are then combined and imported as antenna patterns under the HCM manufacturer group. Multiple lines are allowed.

CSV file example:

AM	AN	AO
gain	horizontal_hcm	vertical_hcm
17dBi	060EA30	008PC00
17dBi	062EA30	007PA00
17dBi	061EA30	007PC00
17dBi	063EA30	008PF00
17dBi	070EA30	008PG00
17dBi	032EA30	008PA00

After file upload, you are prompted to assign which CSV field corresponds to the required parameters:

 **Antennas**

Antenna parameters

Horizontal HCM code

none

▼

Vertical HCM code

none

▼

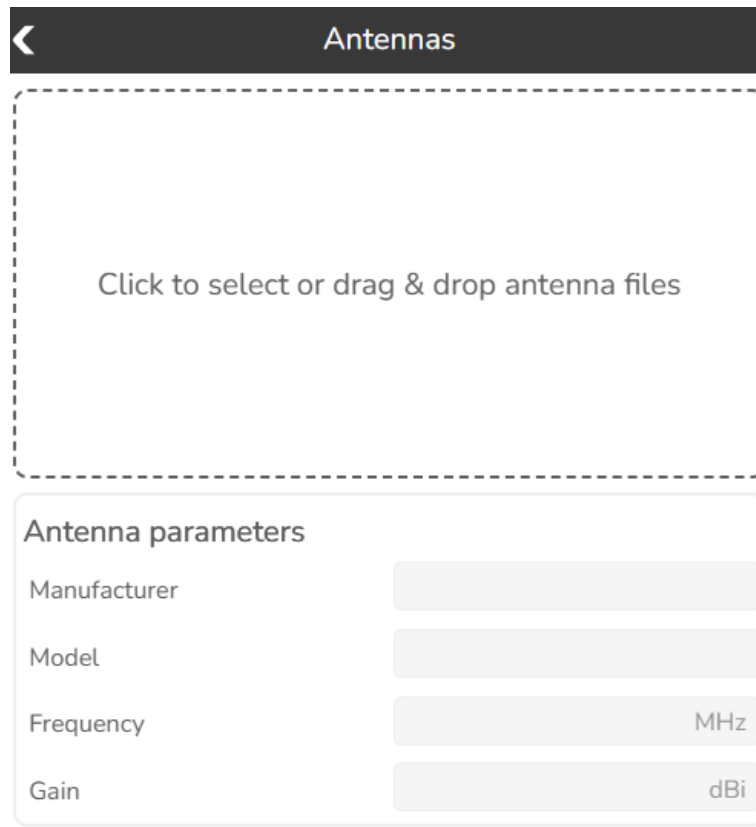
Gain

none

▼

3.1.6.2 Import cell array antenna

If you have separate pattern files for a multibeam antenna, you may use this import option to combine them into a single cell array antenna pattern.



Antennas

Click to select or drag & drop antenna files

Antenna parameters

Manufacturer

Model

Frequency MHz

Gain dBi

Antenna parameters**Manufacturer**

Antenna manufacturer.

Model

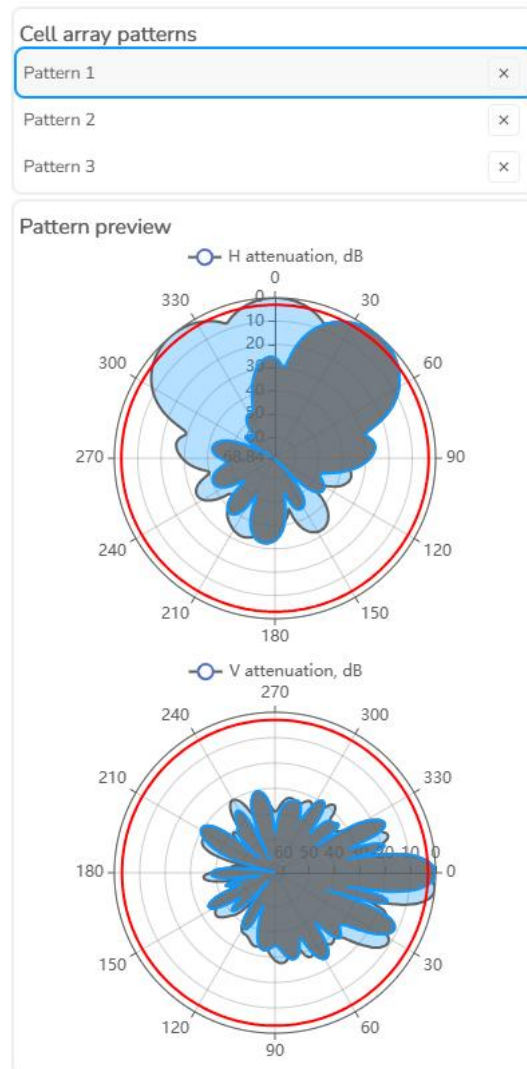
Antenna name.

Frequency

Antenna frequency value in MHz.

Gain

Antenna gain value in dBi.



3.1.6.3 Create manually

To manually create an antenna, click the *New Antenna* button

+ New antenna

and then select *Create Manually*.

Create manually

Antennas

Antenna parameters

Manufacturer

Manually created

Model

Frequency

1800

MHz

Gain

20

dBi

Max attenuation

50

dB

Horizontal pattern

Beamwidth

60

deg

Smoothing degree count

5

deg

Vertical pattern

Beamwidth

60

deg

Smoothing degree count

5

deg

Angle offset

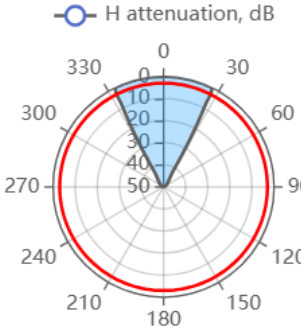
0

deg

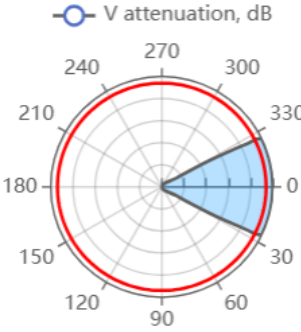
Mirror pattern

Pattern preview

H attenuation, dB



V attenuation, dB



Accept

Cancel

Antenna parameters

Manufacturer

Antenna manufacturer (company or entity).

Model

Antenna name.

Frequency

Antenna frequency value in MHz.

Gain

Antenna gain value in dBi.

Max attenuation

Attenuation value assigned to the attenuated parts of the antenna pattern

Horizontal pattern**Beamwidth**

Antenna's horizontal beamwidth value in degrees. Used for generating the antenna pattern.

Smoothing degree count

Transition width between attenuated and non-attenuated parts of the antenna pattern in degrees. The attenuation values are linearly interpolated between 0 and the max attenuation value within this transition range.

Vertical pattern**Beamwidth**

Antenna's vertical beamwidth value in degrees. Used for generating the antenna pattern.

Smoothing degree count

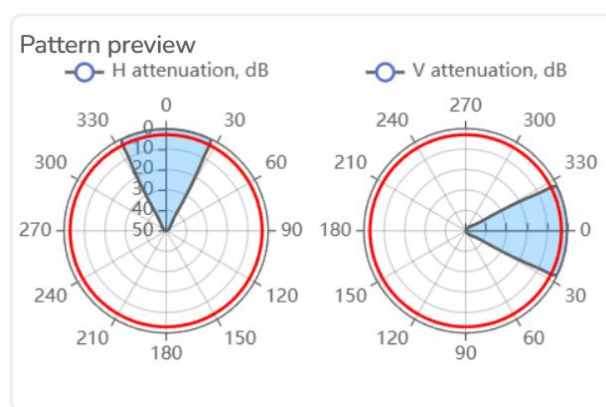
Transition width between attenuated and non-attenuated parts of the antenna pattern in degrees. The attenuation values are linearly interpolated between 0 and the max attenuation value within this transition range.

Angle offset

Transition width between attenuated and non-attenuated parts of the antenna pattern in degrees. The attenuation values are linearly interpolated between 0 and the max attenuation value within this transition range.

Mirror pattern

Ability to mirror the vertical attenuation pattern.

Pattern preview

3.1.7 Geodata sets

Click this button  to open Geodata sets tool.

Geodata sets are collections of geographical raster data that is used in CE calculations. Geodata sets are comprised of the following raster types:

Elevation – DTM (Digital terrain model), each pixel defines the absolute terrain height above sea level. No buildings or other obstacles are included in this height.

Clutter height (optional) – Relative height of obstructions (buildings, forests, etc.) above elevation.

Clutter classes (optional) – Each pixel defines the ID of the clutter class, which the area belongs to. Usually derived from land use data. If building heights are included in the clutter height raster, the clutter classes raster must have building outlines separated into their own class ID.

To edit a geodata set, click the desired geodata set name from the list. To create a new one, click “+ New geodata set”.

Geodata sets

Search

+ New geodata set

Australia

Guatemala

UK

Switzerland

Lithuania_1m

San Francisco

When creating a new geodata set, some general details, like the geodata set name, must be entered and

saved. Only then you may then upload the geodata rasters to the set after selecting it from the list.

The screenshot shows a mobile application interface titled 'Geodata sets'. It has a dark header bar with a back arrow and the title. Below the header, there are two main sections: 'General' and 'Rasters'. The 'General' section contains two input fields: 'Geodata set name' with a 'New geodata' button next to it, and 'Geodata set ID' with the value '53'. The 'Rasters' section contains three dashed rectangular boxes, each representing a file upload area. The boxes are labeled 'Elevation', 'Clutter height', and 'Clutter classes' from top to bottom.

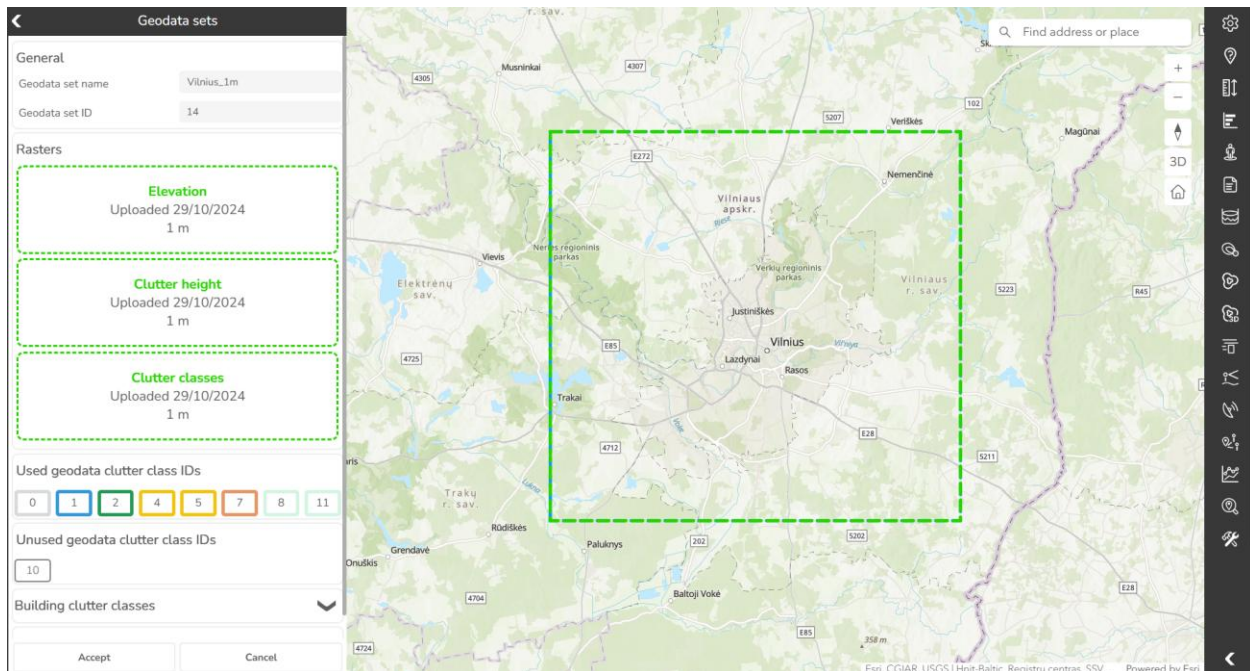
Under the rasters section, there are 3 File upload squares for uploading the rasters. Click to select a file or drag & drop to upload a raster file.

Uploaded rasters have the following requirements:

- Must be in projected coordinate system
- Coordinate system units must be meters
- All rasters must have the same coordinate system
- Raster resolution in X and Y axis must match

After uploading rasters, the file upload squares change to green, and show some metadata, like upload date and resolution, about the uploaded raster files.

The data extent is also outlined in green on the map when a geodata set is selected:



3.1.7.1 Clutter classes

If a clutter class raster is available in the geodata set, the clutter classes section should be set up to represent the data in the file.

All available clutter class raster values are listed under the “Used” and “Unused” categories.

Used geodata clutter class IDs

0

1

2

4

5

7

8

Unused geodata clutter class IDs

11

Each relevant clutter class needs these fields filled out:

IDs in geodata raster – clutter class IDs from your uploaded clutter classes raster need to be assigned to predefined clutter classes within CE. For example if your clutter classes raster has buildings outlined with the ID of 2, you would need to assign “2” to the buildings clutter class in the menu. Multiple IDs may be assigned to each class.

Height – Nominal height for clutter class if it is not represented in the clutter height raster. If you already have heights for this class in the clutter height raster, this should be set to 0.

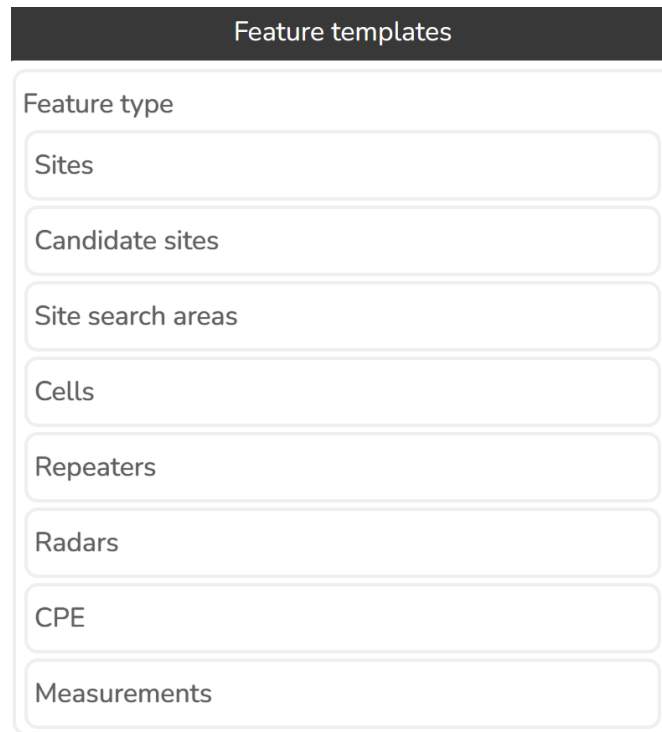
Color – the color by which the clutter class is represented in different UI elements, for example – the profile chart.

The screenshot shows a configuration panel titled "Clutter classes" with a scrollable list of four clutter classes. Each class has three configuration fields: "IDs in geodata raster", "Height", and "Color".

Clutter Class	IDs in geodata raster	Height	Color
Open / Terrain (default)	none	0	Light Green
Grassland	none	0	Bright Green
Sparse forest	none	0	Medium Green
Medium dense forest	none	0	Dark Green

3.1.8 Feature templates

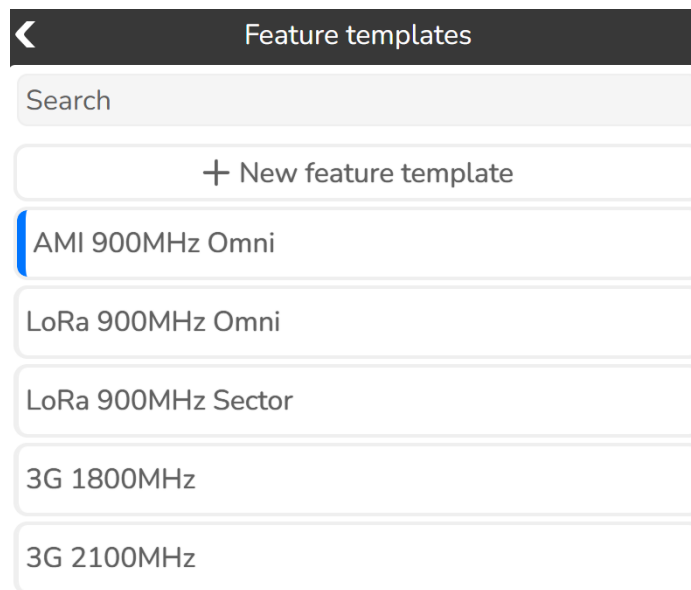
Feature templates serve the purpose of allowing the user to quickly fill out feature parameters from pre-saved data. In the case of feature creation, you can fill out the parameters by selecting a feature template in the add features tool. In the case of starting some calculations, you may pick a template from which missing feature parameters will be filled during the calculation. The feature templates tool allows you to set up these templates before using them.



A screenshot of a mobile application screen titled "Feature templates". Below the title is a list of feature types, each in a rounded rectangular button. The list includes: Sites, Candidate sites, Site search areas, Cells, Repeaters, Radars, CPE, and Measurements.

Feature type
Sites
Candidate sites
Site search areas
Cells
Repeaters
Radars
CPE
Measurements

After opening the tool, you get to choose which feature type you want to manage templates for.



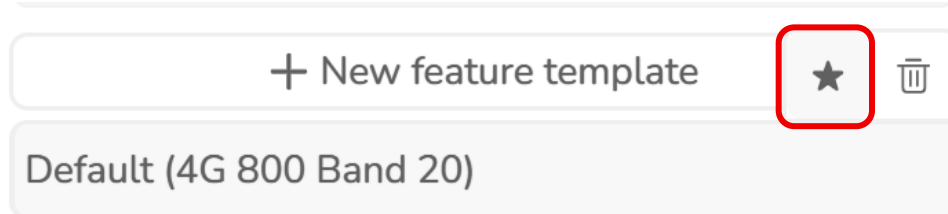
A screenshot of a mobile application screen titled "Feature templates". At the top left is a back arrow. Below the title is a search bar labeled "Search". Under the search bar is a button labeled "+ New feature template". Below that is a list of existing feature templates, each in a rounded rectangular button. The list includes: AMI 900MHz Omni (highlighted with a blue bar on the left), LoRa 900MHz Omni, LoRa 900MHz Sector, 3G 1800MHz, and 3G 2100MHz.

Feature template
AMI 900MHz Omni
LoRa 900MHz Omni
LoRa 900MHz Sector
3G 1800MHz
3G 2100MHz

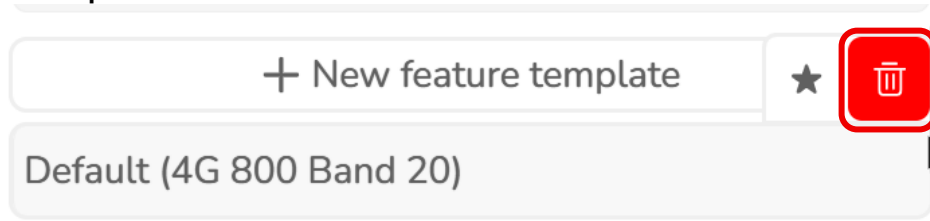
After selecting the feature type, a list of existing feature templates is shown, click an existing feature template to edit it, or click new feature template to create a new one.

Upon hovering the mouse over a feature template, options for it appear.

Mark as favorite



Delete Feature template

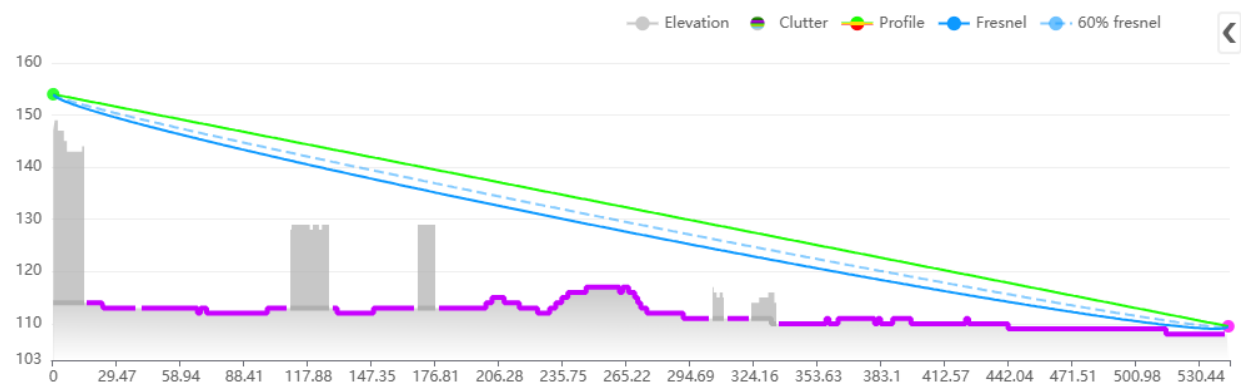


3.1.9 Prediction models

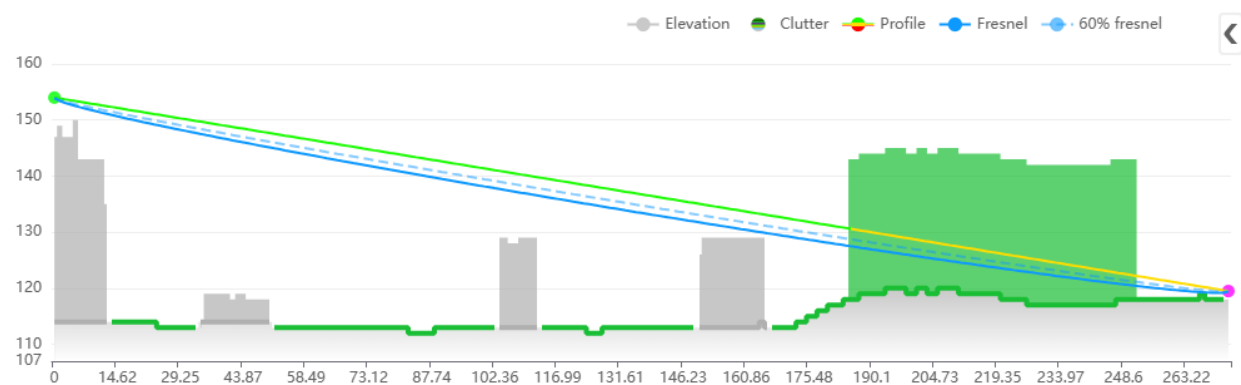
Click this button  to open Prediction models tool.

The CE Path Loss Modelling aims to perform near-deterministic calculation of received signal levels at each specific point (pixel) in the network's target coverage area by applying selective path loss model depending on the radio visibility condition between the transmitter antenna vis-à-vis a receiver antenna located at a given point in coverage area. The radio visibility is evaluated based on the DTM, Obstacles and Clutter path profile information. This verification of radio visibility will result in the receiver antenna point assigned into one of three possible radio visibility conditions:

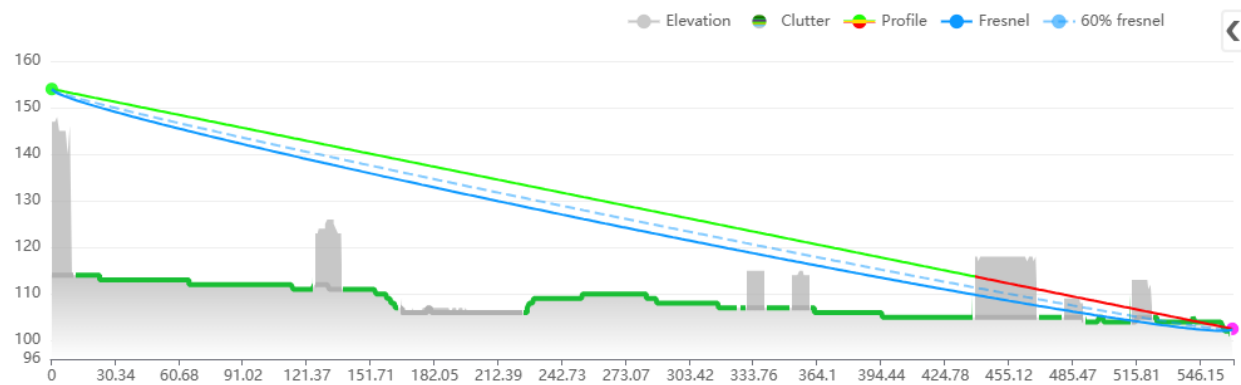
- **Line-of-Sight (LOS)** – occurs when there are neither terrain irregularities, obstacles or clutter interposing the direct radio path between the transmitter and receiver antennas. The radio path is understood to include the 1st Fresnel zone around the direct line and account for Spherical Earth effect. The LOS condition is illustrated by the path profile depicted in Fig. 3(a).
- **Obstructed LOS (OLOS)** – occurs when the direct radio propagation line is interposed by clutter, see illustration in Fig. 3(b).
- **Non-LOS (NLOS)** – occurs when the direct radio propagation line is interposed by terrain bulges or obstacles, see illustration in Fig. 3(c).



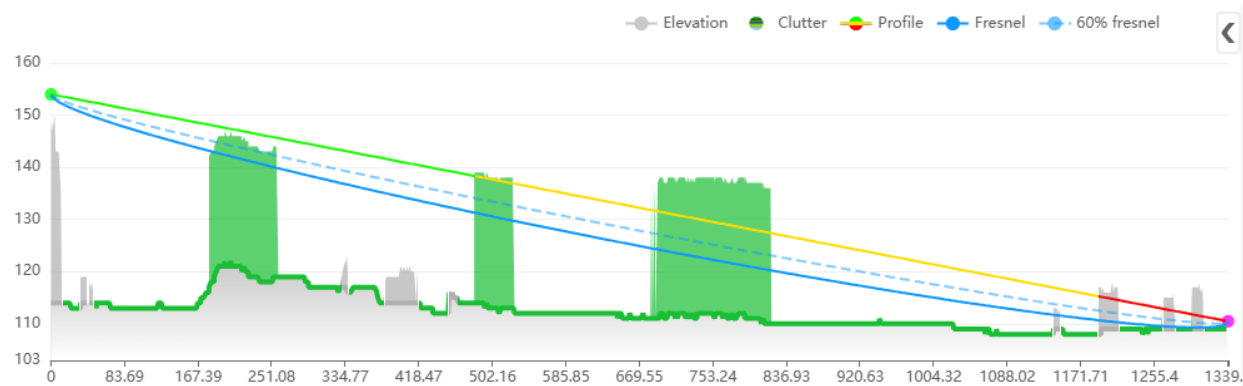
a. Example of path profile with LOS condition (green line of direct radio link)



b. Example of path profile with OLOS condition (yellow segment of radio link path)



(c) Example of path profile with NLOS condition (red segment of radio link path)



(d) Example of path profile with OLOS+NLOS condition (yellow+red segment of radio link path)

Depending on the LOS condition for the receive antenna at specific location (area map pixel), the CE tools will apply the specific sub-set of path loss prediction model, as explained in the following section.

Prediction models available in Cellular Expert support frequencies from 10kHz to 350 GHz.

Prediction models	
ITU-R P452 (6GHz - 50GHz)	
ID	1
Configurations	3
UniMacro (400MHz - 3GHz)	
ID	2
Configurations	3
CEC ITU-R (100MHz - 6GHz)	
ID	3
Configurations	20
ITU-R P525 (6GHz - 100GHz)	
ID	4
Configurations	6
ITU-R P368 (10KHz - 30MHz)	
ID	5
Configurations	1

CEC ITU-R Model (100MHz – 6GHz) is a combination model intended for use in a variety of different radiocommunication systems which is derived explicitly from ITU-R path loss modelling methods as follows:

- a. Receive antenna in LOS condition – path loss calculated as FSL based on Recommendation ITU-R P.525 ([ref URL](#)).
- b. Receive antenna in OLOS condition – total path loss modelled as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)) with included dual slope option and CE combined clutter loss modelling, which is the combination of Recommendation ITU-R P.2108 ([ref URL](#)) and ITU-R P.833 (2015) Attenuation in vegetation.
- c. Receive antenna in NLOS condition – path loss as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)) with included dual slope option, additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)).
- d. Receive antenna in OLOS+NLOS condition – path loss as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)) with included dual slope option, additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)), and CE combined clutter loss modelling, which is the combination of Recommendation ITU-R P.2108 ([ref URL](#)) and Single Knife-Edge diffraction.
- e. Receive antenna in the clutter (building, vegetation, etc) – path loss is calculated as described above based on LOS, OLOS and NLOS conditions, and additional penetration loss is added to simulate Outdoor-to-Indoor scenario which is based on ITU-R P.833 recommendation.

ITU-R P.452 Model (6GHz – 50GHz) is provided as a universally applicable model with a very wide frequency range from 0.1-50 GHz. Its implementation is based on the methodology described in the Recommendation ITU-R P.452 ([ref URL](#)). This model does not provide for definition of OLOS visibility condition; instead, it considers clutter as part of the general obstacles category and accordingly distinguishes only two radio visibility cases:

- a. Receive antenna in LOS condition – path loss model based on FSL principle.
- b. Receive antenna in NLOS condition – total path loss modelled using a combination of basic transmission losses and losses due to diffraction.

ITU-R P.1546 Model (30MHz – 4GHz) ([ref URL](#)) is a widely recognized radio propagation prediction method developed by the **International Telecommunication Union (ITU)**. It is primarily used for estimating **point-to-area radio signal coverage** in the frequency range from **30 MHz to 4000 MHz** over **terrestrial paths**. This model is especially suitable for **broadcasting, land mobile, and fixed services**.

Key Features

- **Versatile Application:** Supports predictions over **land, sea, and mixed paths**, making it adaptable to various geographic conditions.
- **Input Parameters:** Takes into account factors such as **transmitter and receiver heights, terrain profile, clutter (buildings, vegetation), climate, and time/location variability**.
- **Time and Location Variability:** Predictions can be tailored for different statistical reliability levels (e.g., 50% or 10% time availability).
- **Clutter and Terrain Handling:** The model can incorporate detailed **digital elevation models (DEM)** and **clutter data** for more accurate predictions, reflecting the influence of buildings, forests, and other surface features.

Use in Cellular Expert

In the **Cellular Expert** software, the ITU-R P.1546 model is implemented to support **real-world coverage planning** and **regulatory studies**. Users can configure environmental parameters and resolution settings to match local conditions and improve prediction accuracy.

LOS ITU-R P.525 Model (6GHz – 100GHz) is the FSL path loss calculated based on the method in Recommendation ITU-R P.525 ([ref URL](#)). As such it could be used for modelling radio links where LOS is considered a necessary condition, e.g., for Fixed (Point-to-Point) Links or Mobile Systems in mmWave bands.

UniMacro Model (400MHz – 3GHz) is the CE's proprietary combination model developed over the years of practical experience with the operational planning of cellular mobile networks in the frequency ranges from 400-3000 MHz. It had been fine-tuned to produce coverage predictions that are most closely aligned with what could be expected to be experienced by the actual mobile network users in the field. The model will model different path losses depending on radio visibility conditions as follows:

- a. Receive antenna in LOS condition – path loss model based on FSL principle and dual slope based on breakpoint distance.
- b. Receive antenna in OLOS condition – path loss modelled using Extended Hata (Open Area) model and CE combined clutter loss modelling, which is the combination of Recommendation ITU-R P.2108 ([ref URL](#)) and ITU-R P.833 (2015) Attenuation in vegetation.
- c. Receive antenna in NLOS condition – path loss modelled using Extended Hata model with additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)).
- d. Receive antenna in OLOS+NLOS condition – path loss modelled using Extended Hata (Open Area) model with additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)), and CE combined clutter loss modelling, which is the combination of Recommendation ITU-R P.2108 ([ref URL](#)) and Single Knife-Edge diffraction.
- e. Receive antenna in the clutter (building, vegetation, etc) – path loss is calculated as described above based on LOS, OLOS and NLOS conditions, and additional penetration loss is added to simulate Outdoor-to-Indoor scenario which is based on ITU-R P.833 recommendation.

ITU-R P.368 (10kHz – 30MHz) provides a standardized prediction method for assessing the ground-wave field strength of radio waves in the 10 kHz to 30 MHz frequency range. This frequency band is primarily associated with long-range communication systems using amplitude modulation (AM) and shortwave bands, often for maritime, aeronautical, military, and broadcasting services.

This model offers guidance for engineers, planners, and researchers working on system design and analysis in the MF (Medium Frequency) and HF (High Frequency) bands.

The ITU-R P.368 model calculates signal strength based on several key environmental and system parameters:

- **Frequency (f)**
Higher frequencies tend to attenuate more rapidly over ground. The attenuation rate increases significantly above 3 MHz.
- **Distance (d)**
Field strength diminishes with increasing distance due to geometrical spreading and absorption by the ground and atmosphere.
- **Surface refractivity, Surface Conductivity (σ) and Relative Permittivity (ϵ_r)**
The surface over which the wave propagates critically affects signal strength:
 - Sea water: High conductivity, minimal loss
 - Dry land or desert: Low conductivity, high loss
 - Typical values range from:
 - Conductivity: 10^{-4} to 5 S/m

- Relative permittivity: 4 to 81

ISO 9613 standard provides a validated, practical method for predicting the outdoor propagation of sound, and is increasingly applied in siren sound modeling within public security, emergency warning systems, and defense operations. This modeling ensures that acoustic alert systems (e.g. civil defense sirens, disaster warnings, military alert signals) achieve their intended coverage, intelligibility, and effectiveness across various terrain and urban environments.

Purpose in the Public Security Context

In emergency and defense scenarios, reliable audibility of sirens is critical for:

- Civil alert and evacuation systems
- Military base perimeter alarms
- Air raid or missile defense warning networks
- Disaster alert systems (e.g. earthquakes, tsunamis, nuclear incidents)

By applying ISO 9613-2, engineers can model how far a siren can be heard under specific environmental conditions, optimizing:

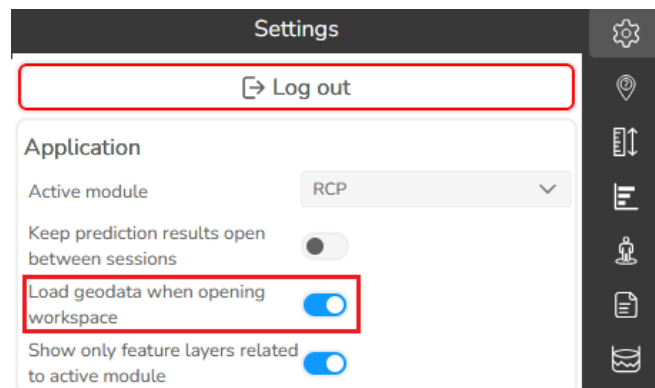
- Placement and spacing of sirens
- Sound power selection
- Minimization of acoustic shadow zones
- Compliance with national safety and civil defense regulations

The model assumes standard favorable propagation:

- Downwind or moderate inversion conditions
- Ambient temperature ~10 °C
- Relative humidity ~70%

These are conservative conditions ensuring that siren reach is never overestimated, supporting public safety margin planning.

CEC 3GPP TR Indoor is used in indoor environments (in building) for wireless communication simulations. This model is standardized by 3GPP (3rd Generation Partnership Project) and is part of the TR 38.901 technical report, which specifies channel models for frequencies from 0.5 GHz to 100 GHz. Cellular expert version requires to convert CAD drawing into a clutter height raster file which is imported to workspace. By toggle ON "Load geodata when opening workspace" this raster file becomes visible, allowing for accurate placement and planning new transmitters as indoor antennas.



In the left ribbon, navigate to Layers → Geodata, expand Clutter Heights, and set the opacity to 100% to enhance building visibility. For more accurate RF prediction, use a high-resolution grid of 0.1 meters in the prediction model to account for wall thickness and fine structural details.

3.1.9.1 CEC ITU-R Model

This deterministic model is designed for precise tracking of the main, strongest radio ray, while also empirically modeling the scattering of other rays around the receiver. It applies to all ranges of cellular mobile and public safety networks, including 2G, 3G, 4G, and 5G, within the 100 MHz to 6 GHz frequency range.

The model is recommended for accurate wide-area propagation and coverage modeling, especially when precise and up-to-date topographic data are available. This includes Digital Terrain Models (DTM), building data (including height information), and vegetation data (with height details) derived from a Digital Surface Model (DSM). Ideally, these data should be created with LiDAR or similar methods, at a resolution of at least 10 meters, though 5, 2, or even 1 meter or higher is preferable for optimal accuracy.

When building data and their heights are not available, and only DTM and clutter data at a resolution of 10 meters or lower are accessible, the UniMacro Model should be considered for wide-area propagation and coverage modeling in a slightly narrower frequency range of 400 MHz to 3 GHz.

3.1.9.1.1 CEC ITU-R Model settings

 Prediction models

Model parameters

Configuration name	
Radius	km
Receiver height	1.5 m
Effective earth radius	8500 km
Offset coefficient	37 dB
Distance coefficient	20
Distance coefficient obstructed	30
Frequency coefficient	20

Configuration name

Name of the prediction configuration name.

Radius

Maximum prediction radius in kilometers to calculate path loss.

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

Effective earth radius

Earth radius in kilometers, used for the calculations.

Offset coefficient

Represents the offset in decibels added to the path loss grid. The default value is 37 dB.

Distance coefficient

Defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

Distance coefficient obstructed

Represents the slope based on the obstructed distance between the cell and the receiver location. The default value is 30.

Frequency coefficient

Indicates the slope determined by the frequency value, with a default value of 20.

The Clutter Class option defines several predefined clutter categories, each with unique values for diffraction loss, clutter loss, penetration loss, and receiver loss coefficients. These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.

Other clutter values

Open / Terrain

Nominal distance

1

m

Diffraction loss coefficient

1

Enclosed receiver loss offset

17

dB

Enclosed receiver loss scaling coefficient

0.25

Enclosed receiver loss frequency exponent coefficient

1

Receiver point loss offset

0

dB

Receiver height

m

Sparse forest

Nominal distance

27

m

Diffraction loss coefficient

0.5

Enclosed receiver loss offset

0

dB

Enclosed receiver loss scaling coefficient

0.82

Enclosed receiver loss frequency exponent coefficient

0.65

Receiver point loss offset

0

dB

Receiver height

m

Nominal distance

The average distance between objects within the clutter class, ranging from 1 to 100 meters.

Diffraction loss coefficient

A multiplier used in diffraction calculations. Lower values result in reduced diffraction loss, while higher values increase it. Typically, this coefficient is higher for buildings compared to forests or other clutter

types. It is a multiplier for diffraction calculations. If value is lower, diffraction will be lower, if higher – then diffraction will be higher. Usually, for buildings clutter class this parameter is higher than forest or other clutter classes.

Enclosed receiver loss offset

The initial entry loss into an obstacle within the clutter class, expressed as an offset in dB, which is added to the path loss grid. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Enclosed receiver loss scaling coefficient

Represents additional signal loss as a function of the distance travelled in an obstacle within the clutter class. Higher values increase path loss. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Enclosed receiver loss frequency exponent coefficient

Reflects additional loss inside an obstacle within the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Receiver point loss offset

An additional loss offset in dB applied to the path loss grid, representing user equipment (UE) losses.

Receiver height

Receiver height value used when receiver lands on this clutter class. If this value is not defined, the main receiver height value defined in the prediction model is used.

Clutter Classes default values

	Nominal Distance	Diffraction loss coefficient	Enclosed receiver loss offset	Enclosed receiver loss scaling coefficient	Enclosed receiver loss frequency exponent coefficient	Receiver point loss offset
Open / Terrain	1	1	17	0.25	1	0
Grassland	27	0.5	0	0.82	0.65	0
Sparse forest	27	0.5	0	0.82	0.65	0
Medium dense forest	27	0.5	0	0.89	0.65	0
Very dense forest	15	0.5	0	0.95	0.65	0
Low density urban (Low buildings)	27	0.5	0	0.89	0.65	0
Low density urban (High buildings)	27	0.5	0	0.89	0.65	0
Medium density urban (Low buildings)	15	0.6	0	0.89	0.65	0
Medium density urban (High buildings)	15	0.6	0	0.89	0.65	0
High density urban	10	0.7	0	0.89	0.65	0

(Low buildings)						
High density urban (High buildings)	10	0.7	0	0.89	0.65	0
High density urban (Very high buildings)	10	0.7	0	0.89	0.65	0
Building blocks	15	0.6	0	0.89	0.65	0
Transportation	27	0.5	0	0.89	0.65	0
Agriculture	27	0.5	0	0.89	0.65	0
Plantation	27	0.5	0	0.89	0.65	0
Parks	27	0.5	0	0.89	0.65	0
Airport	27	0.6	0	0.89	0.65	0
Sea	27	0.5	0	0.89	0.65	0
Inland water	27	0.5	0	0.89	0.65	0
Concrete building	1	1	5	0.25	1	0
Glass building	1	1	2	0.25	1	0
Wood building	1	0.6	2	0.25	1	0
Low loss building	1	1	8.5	0.25	1	0
High loss building	1	1	17	0.25	1	0

3.1.9.2 ITU-R P.452 Model

This model is designed to estimate radio signal propagation over long distances, including terrestrial paths. It is specifically designed to predict signal attenuation caused by various mechanisms, such as diffraction, tropospheric scatter, ducting, and reflections from the Earth's surface, across frequencies ranging from 0.1 GHz to 50 GHz. While other prediction model covers frequencies from 100MHz to 6GHz, we recommend to use it from 6GHz to 50GHz frequencies.

This model is particularly well-suited for microwave links and is widely used for planning and interference analysis in fixed and mobile radio communication systems. By accounting for the effects of terrain, atmospheric conditions, and other factors, it helps engineers assess link reliability and optimize network performance in a variety of environmental scenarios.

3.1.9.2.1 ITU R. P452 Model settings

 Prediction models

Model parameters

Configuration name	
Radius	km
Receiver height	1.5 m
Effective earth radius	8500 km
Offset coefficient	37 dB
Distance coefficient	20
Frequency coefficient	20

Configuration name

Name of the prediction configuration name.

Radius

Maximum prediction radius in kilometers to calculate path loss.

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

Effective earth radius

Earth radius in kilometers, used for the calculations.

Offset coefficient

Represents the offset in decibels added to the path loss grid. The default value is 37 dB.

Distance coefficient

Defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

Frequency coefficient

Indicates the slope determined by the frequency value, with a default value of 20.

Clutter class option describes several fixed Clutter names with penetration loss coefficients. These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.

Other clutter values

Open / Terrain

Enclosed receiver loss offset	17	dB
Enclosed receiver loss scaling coefficient	0.25	
Enclosed receiver loss frequency exponent coefficient	1	
Receiver height		m

Sparse forest

Enclosed receiver loss offset	0	dB
Enclosed receiver loss scaling coefficient	0.82	
Enclosed receiver loss frequency exponent coefficient	0.65	
Receiver height		m

Medium density urban (High buildings)

Enclosed receiver loss offset	0	dB
Enclosed receiver loss scaling coefficient	0.89	
Enclosed receiver loss frequency exponent coefficient	0.65	
Receiver height		m

Enclosed receiver loss offset

The initial entry loss into an obstacle within the clutter class, expressed as an offset in dB, which is added to the path loss grid. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Enclosed receiver loss scaling coefficient

Represents additional signal loss as a function of the distance travelled in an obstacle within the clutter class. Higher values increase path loss. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Enclosed receiver loss frequency exponent coefficient

Reflects additional loss inside an obstacle within the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Receiver height

Receiver height value used when receiver lands on this clutter class. If this value is not defined, the main receiver height value defined in the prediction model is used.

Clutter Classes default values

	Enclosed receiver loss offset	Enclosed receiver loss scaling coefficient	Enclosed receiver loss frequency exponent coefficient
Open / Terrain	17	0.25	1
Grassland	0	0.82	0.65
Sparse forest	0	0.82	0.65
Medium dense forest	0	0.89	0.65
Very dense forest	0	0.95	0.65
Low density urban (Low buildings)	0	0.89	0.65
Low density urban (High buildings)	0	0.89	0.65
Medium density urban (Low buildings)	0	0.89	0.65
Medium density urban (High buildings)	0	0.89	0.65
High density urban (Low buildings)	0	0.89	0.65
High density urban (High buildings)	0	0.89	0.65
High density urban (Very high buildings)	0	0.89	0.65
Building blocks	0	0.89	0.65
Transportation	0	0.89	0.65
Agriculture	0	0.89	0.65
Plantation	0	0.89	0.65
Parks	0	0.89	0.65
Airport	0	0.89	0.65
Sea	0	0.89	0.65
Inland water	0	0.89	0.65
Concrete building	5	0.25	1
Glass building	2	0.25	1
Wood building	2	0.25	1

Low loss building	8.5	0.25	1
High loss building	17	0.25	1

3.1.9.3 ITU-R P.1546 Model

This model is designed for wide-area radio propagation prediction based on empirical data and statistical analysis of measured field strengths. It provides path loss estimations over land, sea, and mixed terrain for frequencies ranging from 30 MHz to 3 GHz, using parameters defined in ITU-R Recommendation P.1546. The model is applicable to terrestrial broadcasting, mobile, and public safety networks, including technologies such as 2G, 3G, and 4G, within its frequency range.

The ITU-R P.1546 model accounts for antenna heights, terrain elevation (DTM), land cover types (clutter), and environmental conditions, incorporating corrections for time variability and location-specific effects. It is particularly suitable for modeling line-of-sight (LOS) and non-line-of-sight (NLOS) propagation over long distances where detailed building data is not available.

This model is recommended for national or regional coverage planning, especially in cases where high-resolution DTM (e.g., 30m) and clutter data (e.g., 10m resolution) are available, but building heights and detailed 3D structures are not. Its statistical approach allows for reliable estimations of field strength in both urban and rural environments, without requiring ray-tracing or detailed geometry-based modeling.

For scenarios where accurate building geometry and heights are available and a higher modeling frequency range (up to 6 GHz) is required, the CEC ITU-R 3GPP Model is recommended instead, offering greater precision in dense urban and high-frequency applications.

3.1.9.3.1 ITU R. P1546 Model settings

Model parameters

Configuration name

Radius

km

Receiver height

1.5

m

Time percentage

10

%

Configuration name

Name of the prediction configuration name.

Radius

Maximum prediction radius in kilometers to calculate path loss.

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

Time percentage

How radio signal levels fluctuate over time due to changes in atmospheric conditions, tropospheric refraction, and diffraction. This effect is represented as a statistical factor in the model, which allows predictions for different time percentages. Should not exceed outside the range from 1% to 50%.

Other clutter values

Open / Terrain

Receiver height m

Sparse forest

Receiver height m

Medium density urban (High buildings)

Receiver height m

Agriculture

Receiver height m

Receiver height

Receiver height value used when receiver lands on this clutter class. If this value is not defined, the main receiver height value defined in the prediction model is used.

3.1.9.4 ITU-R P525 Model

Line of Sight model is typically used for mmWave band frequencies within the 6 GHz – 100 GHz frequency range and provides results only for line-of-sight areas.

3.1.9.4.1 ITU-R P525 Model settings

 Prediction models

Model parameters

Configuration name	
Radius	km
Receiver height	1.5 m
Effective earth radius	8500 km
Offset coefficient	37 dB
Distance coefficient	20
Frequency coefficient	20

Configuration name

Name of the prediction configuration name.

Radius

Maximum prediction radius in kilometers to calculate path loss.

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

Effective earth radius

Earth radius in kilometers, used for the calculations.

Offset coefficient

Represents the offset in decibels added to the path loss grid. The default value is 37 dB.

Distance coefficient

Defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

Frequency coefficient

Indicates the slope determined by the frequency value, with a default value of 20.

The screenshot shows a mobile application interface titled "Other clutter values" with a back arrow in the top right corner. Below the title, there are five distinct sections, each representing a clutter class. Each section contains a label for the clutter class and a text input field for "Receiver height" followed by a unit "m". The sections are: "Open / Terrain" (light green border), "Sparse forest" (green border), "Medium density urban (High buildings)" (orange border), "Agriculture" (yellow border), and "Plantation" (teal border).

Clutter Class	Receiver height (m)
Open / Terrain	
Sparse forest	
Medium density urban (High buildings)	
Agriculture	
Plantation	

Receiver height

Receiver height value used when receiver lands on this clutter class. If this value is not defined, the main receiver height value defined in the prediction model is used.

3.1.9.5 UniMacro Model

This model is designed for deterministic tracking of the main, strongest radio ray in Line of Sight (LOS) areas, while propagation modeling in Obstructed Line of Sight (OLOS) and Non-Line of Sight (NLOS) areas uses empirically determined parameters defined in ITU-R and 3GPP recommendations. It also models the scattering of other rays around the receiver. The model applies empirically validated values for the 400 MHz to 3 GHz frequency range and is suitable for modeling all cellular mobile and public safety networks, including 2G, 3G, 4G, and 5G, within that frequency range.

This model is recommended for wide-area propagation and coverage modeling when building data and their heights are unavailable, and only DTM and clutter data at a resolution of 10 meters or lower are accessible.

For accurate wide-area propagation and coverage modeling where building data and their heights are available, the CEC ITU-R 3GPP Model is recommended, offering a broader application frequency range of 100 MHz to 6 GHz.

3.1.9.5.1 UniMacro Model settings

<

Prediction models

Model parameters

Configuration name

Radius

km

Receiver height

1.5

m

Effective earth radius

8500

km

Offset coefficient

37

dB

Distance coefficient near

20

Distance coefficient far

30

Frequency coefficient

20

A0

36

A1

32

A2

-12

A3

0.1

Configuration name
Name of the prediction configuration name.

Radius
Maximum prediction radius in kilometers to calculate path loss.

Receiver height
Receiver height above the receiver reference height selected in the workspace settings.

Effective earth radius
Earth radius in kilometers, used for the calculations.

Offset coefficient

Represents the offset in decibels added to the path loss grid. The default value is 37 dB.

Distance coefficient near

Defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

Distance coefficient far

Represents the slope based on breakpoint distance between the cell and the receiver location. The default value is 30.

Frequency coefficient

Indicates the slope determined by the frequency value, with a default value of 20.

A0

Constant offset in dB this value simply added to loss grid. Adjusting this value, you can minimize mean error. It regulates the absolute level of the loss curve. Default value 36.

A1

Distance influence coefficient. Physically it represents loss dependant on distance such as atmospheric (dust, hydrometeors, etc...) losses. It regulates slope of the curve. Default value 32.

A2

Transmitter height influence coefficient. It is related to errors in DTM, real Earth curvature, etc. It regulates loss curve vertical position like the A0, but with respect to antenna height. Default value -12.

A3

Okumura-Hata type of multiplying factor for $\log(h_M)\log(d)$. Default value 0.1.

The Clutter Class option defines several predefined clutter categories, each with unique values for diffraction loss, clutter loss, penetration loss, and receiver loss coefficients. These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.

Other clutter values

Open / Terrain

Nominal distance

1

m

Diffraction loss coefficient

1

Enclosed receiver loss offset

17

dB

Enclosed receiver loss scaling coefficient

0.25

Enclosed receiver loss frequency exponent coefficient

1

Receiver point loss offset

0

dB

Receiver height

m

Sparse forest

Nominal distance

27

m

Diffraction loss coefficient

0.5

Enclosed receiver loss offset

0

dB

Enclosed receiver loss scaling coefficient

0.82

Enclosed receiver loss frequency exponent coefficient

0.65

Receiver point loss offset

0

dB

Receiver height

m

Nominal distance

The average distance between objects within the clutter class, ranging from 1 to 100 meters.

Diffraction loss coefficient

A multiplier used in diffraction calculations. Lower values result in reduced diffraction loss, while higher values increase it. Typically, this coefficient is higher for buildings compared to forests or other clutter types. Multiplier for diffraction calculations. If value is lower, diffraction will be lower, if higher – then diffraction will be higher. Usually, for buildings clutter class this parameter is higher than forest or other clutter classes.

Enclosed receiver loss offset

The initial entry loss into an obstacle within the clutter class, expressed as an offset in dB, which is added

to the path loss grid. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Enclosed receiver loss scaling coefficient

Represents additional signal loss as a function of the distance travelled in an obstacle within the clutter class. Higher values increase path loss. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Enclosed receiver loss frequency exponent coefficient

Reflects additional loss inside an obstacle within the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies. Applies when receiver point ends up enclosed within an obstacle within this clutter class.

Receiver point loss offset

An additional loss offset in dB applied to the path loss grid, representing user equipment (UE) losses.

Receiver height

Receiver height value used when receiver lands on this clutter class. If this value is not defined, the main receiver height value defined in the prediction model is used.

Clutter Classes default values

	Nominal Distance	Diffraction loss coefficient	Enclosed receiver loss offset	Enclosed receiver loss scaling coefficient	Enclosed receiver loss frequency exponent coefficient	Receiver point loss offset
Open / Terrain	1	1	17	0.25	1	0
Grassland	27	0.5	0	0.82	0.65	0
Sparse forest	27	0.5	0	0.82	0.65	0
Medium dense forest	27	0.5	0	0.89	0.65	0
Very dense forest	15	0.5	0	0.95	0.65	0
Low density urban (Low buildings)	27	0.5	0	0.89	0.65	0
Low density urban (High buildings)	27	0.5	0	0.89	0.65	0
Medium density urban (Low buildings)	15	0.6	0	0.89	0.65	0
Medium density urban (High buildings)	15	0.6	0	0.89	0.65	0
High density urban (Low buildings)	10	0.7	0	0.89	0.65	0
High density urban (High buildings)	10	0.7	0	0.89	0.65	0
High density urban (Very high buildings)	10	0.7	0	0.89	0.65	0

Building blocks	15	0.6	0	0.89	0.65	0
Transportation	27	0.5	0	0.89	0.65	0
Agriculture	27	0.5	0	0.89	0.65	0
Plantation	27	0.5	0	0.89	0.65	0
Parks	27	0.5	0	0.89	0.65	0
Airport	27	0.6	0	0.89	0.65	0
Sea	27	0.5	0	0.89	0.65	0
Inland water	27	0.5	0	0.89	0.65	0
Concrete building	1	1	5	0.25	1	0
Glass building	1	1	2	0.25	1	0
Wood building	1	0.6	2	0.25	1	0
Low loss building	1	1	8.5	0.25	1	0
High loss building	1	1	17	0.25	1	0

3.1.9.6 ITU-R P368 Model

ITU-R P.368 is ground-wave propagation of radio signals model. It is specifically designed to estimate the field strength and attenuation of radio waves over the Earth's surface, particularly for frequencies below 30 MHz.

This model is widely used in planning and designing long-distance communication systems, such as maritime, broadcasting, and low-frequency navigation systems, where ground-wave propagation plays a critical role. It accounts for factors such as terrain conductivity, dielectric properties, and surface roughness to deliver accurate predictions of signal behavior.

3.1.9.6.1 ITU-R P368 Model settings


Prediction models

Model parameters

Configuration name

Radius km

Receiver height m

Configuration name

Name of the prediction configuration name.

Radius

Maximum prediction radius in kilometers to calculate path loss.

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

The general model parameters include the radius and receiver height. Additional parameters used for path loss calculations are derived from the clutter classes. Each clutter class has its own unique set of parameters:

Other clutter values

Open / Terrain

Surface refractivity	315	N-Units
Relative permittivity	5	
Surface conductivity	0.001	S/m
Receiver height		m

Sparse forest

Surface refractivity	315	N-Units
Relative permittivity	13	
Surface conductivity	0.004	S/m
Receiver height		m

Medium density urban (High buildings)

Surface refractivity	315	N-Units
Relative permittivity	5	
Surface conductivity	0.001	S/m
Receiver height		m

Surface refractivity

A measure of the refractive index's influence on electromagnetic wave propagation, particularly in the lower atmosphere close to the Earth's surface. It is expressed as a dimensionless value, typically dependent on atmospheric pressure, temperature, and humidity. High surface refractivity can significantly affect radio wave bending and propagation, such as ducting or anomalous refraction.

Relative permittivity

Quantifies a material's ability to permit electric field propagation relative to vacuum. It is a complex quantity with the real part representing energy storage capability and the imaginary part representing energy dissipation within the material.

ITU-R P.368 uses relative permittivity to model how radio waves interact with various surface materials, such as soil, water, or vegetation. These interactions influence reflection, refraction, and absorption phenomena at the surface.

Surface conductivity

Refers to a material's ability to conduct electrical currents across its surface. It is measured in siemens per meter (S/m). Higher conductivity indicates that a surface can easily allow current flow, affecting the reflection and absorption of radio waves.

According to ITU-R P.368, surface conductivity is a critical factor in determining the reflective properties of surfaces, such as dry versus wet soil, or metallic versus dielectric surfaces.

Receiver height

Receiver height value used when receiver lands on this clutter class. If this value is not defined, the main receiver height value defined in the prediction model is used.

Clutter Classes default values

	Surface refractivity	Relative permittivity	Surface conductivity
Open / Terrain	1	1	17
Grassland	27	0.5	0
Sparse forest	27	0.5	0
Medium dense forest	27	0.5	0
Very dense forest	15	0.5	0
Low density urban (Low buildings)	27	0.5	0
Low density urban (High buildings)	27	0.5	0
Medium density urban (Low buildings)	15	0.6	0
Medium density urban (High buildings)	15	0.6	0
High density urban (Low buildings)	10	0.7	0

High density urban (High buildings)	10	0.7	0
High density urban (Very high buildings)	10	0.7	0
Building blocks	15	0.6	0
Transportation	27	0.5	0
Agriculture	27	0.5	0
Plantation	27	0.5	0
Parks	27	0.5	0
Airport	27	0.6	0
Sea	27	0.5	0
Inland water	27	0.5	0
Concrete building	1	1	5
Glass building	1	1	2
Wood building	1	0.6	2
Low loss building	1	1	8.5
High loss building	1	1	17

3.1.9.7 ISO9613 Model

ISO 9613 is an international standard used for predicting outdoor sound propagation, including the assessment of siren noise levels. It provides a structured method for calculating sound attenuation over distance, considering factors such as geometric spreading, atmospheric absorption, ground effects, reflections, and obstacles.

In the context of siren sound prediction, ISO 9613 helps determine the effective coverage area, ensuring that warning signals reach the intended audience with sufficient audibility. This standard is essential for optimizing siren placement, regulatory compliance, and designing effective emergency alert systems.

The primary factors included in the standard are:

1. Geometric Spreading

- This refers to how sound spreads out as it moves away from its source. Sound intensity decreases as the distance from the source increases, following the inverse square law (with spherical spreading) or other forms depending on terrain.
- As the distance from the sound source increases, the intensity of sound diminishes, which is considered in the calculation of sound levels at various receiver points.

$$A_{div}=20*LOG_{10}(d)+11$$

2. Atmospheric Absorption

- Sound is absorbed by the atmosphere as it travels, particularly at higher frequencies. The absorption depends on factors like temperature, humidity, and air pressure, and is often more significant over longer distances.
- Atmospheric absorption reduces the intensity of sound as it propagates, especially at higher frequencies. The standard takes into account these effects to calculate the reduction in sound level.

3. Obstacles

- Physical barriers such as walls, hills, and buildings block or scatter sound waves, reducing the sound level that reaches certain areas.
- Obstacles can cause significant sound reduction, especially if they are large or located between the sound source and the receiver. The standard accounts for the shadow zones created by these barriers.

4. Directivity of the Source

- This parameter accounts for the directionality of the sound source, which may not emit sound equally in all directions. Sirens, for example, may have directional characteristics that focus their sound output in certain directions.
- The directivity of the sound source influences how the sound energy is distributed, and therefore, how far and in what pattern the sound propagates.

5. Meteorological Conditions

- Weather conditions such as wind speed, temperature gradients, and humidity can significantly affect sound propagation. For example, sound may travel farther downwind or be absorbed more by humid air.
- These conditions are integrated into the model to adjust the calculations of sound attenuation, ensuring more accurate predictions for different weather scenarios.

3.1.9.7.1 ISO9613 Model settings

 **Prediction models**

Model parameters

Configuration name	
Radius	km
Receiver height	1.5 m
Effective earth radius	8500 km
Distance coefficient	20
Meteorological conditions	3 dB
Temperature	20 C
Humidity	50 %

Configuration name

Name of the prediction configuration name.

Radius

Maximum prediction radius in kilometers to calculate path loss.

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

Effective earth radius

Earth radius in kilometers, used for the calculations.

Distance coefficient

Defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

Meteorological conditions

It is a factor, in decibels, which depends on local meteorological statistics for wind speed and direction, and temperature gradients. Experience indicates that values of Meteorological conditions (C0) in practice are limited to the range from zero to approximately + 5 dB.

Temperature

Air temperature in Celsius.

Humidity

Air humidity in percentage.

Other clutter values

Open / Terrain

Receiver height

m

Sparse forest

Receiver height

m

Medium density urban (High buildings)

Receiver height

m

Agriculture

Receiver height

m

Receiver height

Receiver height value used when receiver lands on this clutter class. If this value is not defined, the main receiver height value defined in the prediction model is used.

3.1.9.8 CEC 3GPP TR Indoor model

In the current version, each transmitter must be defined individually as a separate indoor antenna, including its specific attributes such as transmit power. When importing the building wall layout from a CAD drawing, it is converted into both Clutter Heights and a Clutter Class raster. You can assign Clutter Class 0 to represent open space and Clutter Class 1 for obstacles. Additional clutter classes can be defined based on the wall material properties during the CAD-to-clutter conversion process. These classifications are then specified within the Geodata Set, enabling more precise indoor propagation modeling.

3.1.9.8.1 CEC 3GPP TR Model settings

 Prediction models

Model parameters

Configuration name	
Radius	km
Receiver height	1.5 m
Effective earth radius	8500 km
Offset coefficient	37 dB
Distance coefficient	20
Distance coefficient obstructed	30
Frequency coefficient	20

Configuration name

Name of the prediction configuration name.

Radius

Maximum prediction radius in kilometers to calculate path loss.

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

Effective earth radius

Earth radius in kilometers, used for the calculations.

Offset coefficient

Represents the offset in decibels added to the path loss grid. The default value is 37 dB.

Distance coefficient

Defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

Distance coefficient obstructed

Represents the slope based on the obstructed distance between the cell and the receiver location. The default value is 20.

Frequency coefficient

Indicates the slope determined by the frequency value, with a default value of 20.

Other clutter values

Open / Terrain

Penetration loss offset

17

dB

Penetration loss scaling coefficient

0.25

Penetration loss frequency exponent coefficient

1

Receiver height

m

Sparse forest

Penetration loss offset

0

dB

Penetration loss scaling coefficient

0.82

Penetration loss frequency exponent coefficient

0.65

Receiver height

m

Medium density urban (High buildings)

Penetration loss offset

0

dB

Penetration loss scaling coefficient

0.89

Penetration loss frequency exponent coefficient

0.65

Receiver height

m

Penetration loss offset

The initial entry loss applied when crossing an obstacle within the clutter class, expressed as an offset in dB, which is added to the path loss grid.

Penetration loss scaling coefficient

Represents additional signal loss as a function of the distance travelled in an obstacle within the clutter class. Higher values increase path loss.

Penetration loss frequency exponent coefficient

Reflects additional loss inside an obstacle within the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.

Receiver height

Receiver height value used when receiver lands on this clutter class. If this value is not defined, the main receiver height value defined in the prediction model is used.

Note: penetration losses stack when multiple obstacles are crossed.

Clutter Classes default values

	Penetration receiver loss offset	Penetration receiver loss scaling coefficient	Penetration receiver loss frequency exponent coefficient
Open / Terrain	17	0.25	1
Grassland	0	0.82	0.65
Sparse forest	0	0.82	0.65
Medium dense forest	0	0.89	0.65
Very dense forest	0	0.95	0.65
Low density urban (Low buildings)	0	0.89	0.65
Low density urban (High buildings)	0	0.89	0.65
Medium density urban (Low buildings)	0	0.89	0.65
Medium density urban (High buildings)	0	0.89	0.65
High density urban (Low buildings)	0	0.89	0.65
High density urban (High buildings)	0	0.89	0.65
High density urban (Very high buildings)	0	0.89	0.65
Building blocks	0	0.89	0.65
Transportation	0	0.89	0.65
Agriculture	0	0.89	0.65
Plantation	0	0.89	0.65
Parks	0	0.89	0.65
Airport	0	0.89	0.65
Sea	0	0.89	0.65
Inland water	0	0.89	0.65
Concrete building	5	0.25	1
Glass building	2	0.25	1

Wood building	2	0.25	1
Low loss building	8.5	0.25	1
High loss building	17	0.25	1

3.1.10 Settings

Click this button  to open Settings tool.

Settings

Application

Active module

RCP

▼

Keep prediction results open between sessions

☒

Load geodata when opening workspace

☒

Show only feature layers related to active module

☒

Appearance

Theme

Light

▼

Toolbar location

split

▼

Show seconds on prediction history log timestamps

☒

Show workspace preview thumbnails

☒

Show mini calculation task window for newly created tasks

☒

Calculations

Calculation result raster interpolation mode

Nearest

▼

Zoom to layer extent when opening prediction results

☐

Automatically apply snapped feature in Profile & Quick RF prediction widgets

☒

Measurements

Units

Meters

▼

Application

Active module

The currently active CE Express module. Switches the toolset to the required module. Module list consists of currently licensed CE Express modules.

Keep prediction results open between sessions

When enabled - opening a workspace loads open prediction result layers from the last session.

Load geodata when opening workspace

If a geodata set is applied to the workspace, the geodata rasters will be loaded under the geodata section of the layer list, and can be used to visualize the geodata that is used for calculations.

Show only feature layers related to active module

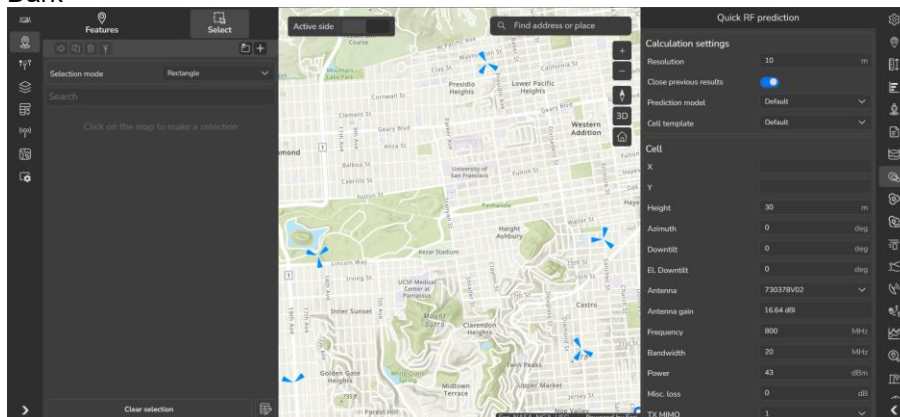
When enabled – hides features types not related to active module in tools like layer list, features, networks, etc.

Appearance

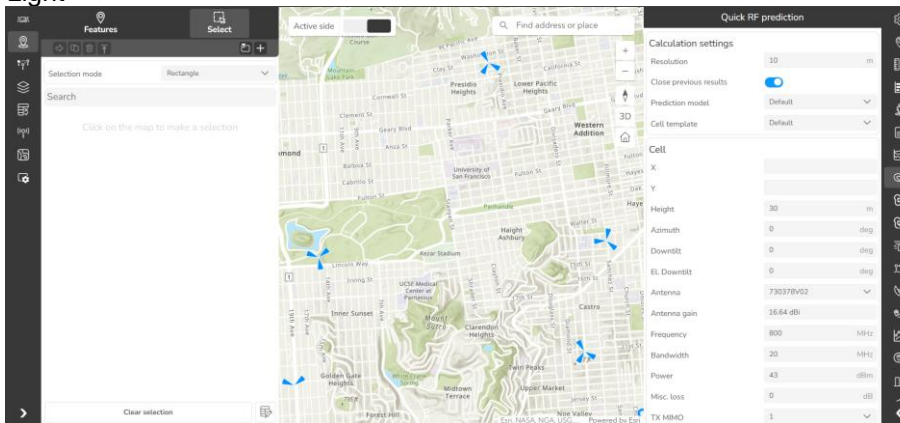
Theme

Change application theme:

- Dark



- Light



Toolbar location

Change the location of the tools:

- Split
- Left
- Right

Show seconds on prediction history log timestamps

May be enabled or disabled.

Show workspace preview thumbnails

May be enabled or disabled.

Show mini calculation task window for newly created tasks

When enabled, a small calculation window appears upon creating a calculation task. The window shows the calculation log while the task is executing, then switches to showing the result list once the task is finished. Results may also be opened from this window.

Calculations**Calculation result raster interpolation mode**

Allows you to select the interpolation mode for calculation result raster layers.

Available options:

- Nearest (default)
- Cubic

Automatically apply first feature templates when calculation widgets are first opened

When enabled – automatically sets the feature template parameters on calculation widgets to the first (recently edited) one.

Automatically apply snapped feature in Profile & Quick RF prediction tools

When enabled – while using Profile or Quick RF prediction tools and snapping to a feature, the parameters from this feature will be automatically applied without having to select it from the snapped features dropdown list. If disabled, only the location will be snapped to.

Measurements**Units**

Select the displayed distance units for the measurements tool.

3.1.11 Identify

Click this button to open Identify tool.

Shows the signal values at an exact location for loaded predictions in the Map view. The dialog is empty if there are no loaded prediction rasters in the Layers tool.

Identify

Identify settings

Real time

Coordinates

X

Y

No layers to show

If the Layer tool has loaded prediction rasters,

Prediction Results

RF prediction 4G Field Strength 1, dBm

this will be shown in the Identify tool.

Identify

Identify settings

Real time

Coordinates

X

Y

Prediction Results

RF prediction 4G 1800 Field Strength 1, dBm

Click on the map to preview the signal value.

Identify

Identify settings
Real time ☐

Coordinates
X -122.40501130636198
Y 37.768033926941925

Prediction Results
RF prediction 4G Field Strength 1, dBm -119

Identify settings

Real time

When enabled – you do not need to click on the map to identify the value of the raster, the value is automatically updated as you hover over different pixels of the calculation result.

Coordinates

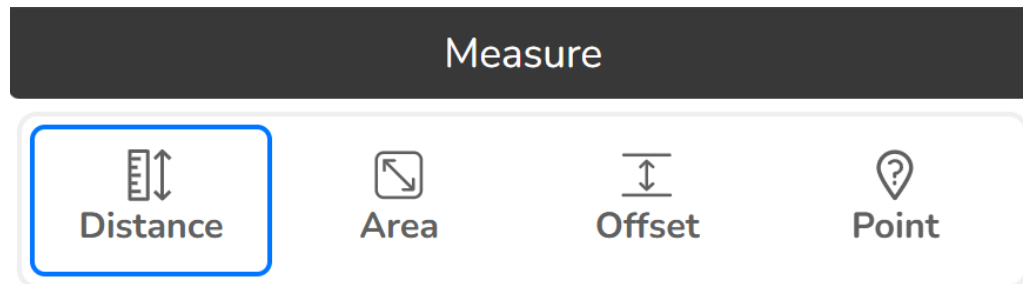
To display the coordinates, click on your desired location on the map.

Coordinates
X -122.43305484725133
Y 37.777472735581505

3.1.12 Measurement tool

Click this button  to open Measurement tool tool.

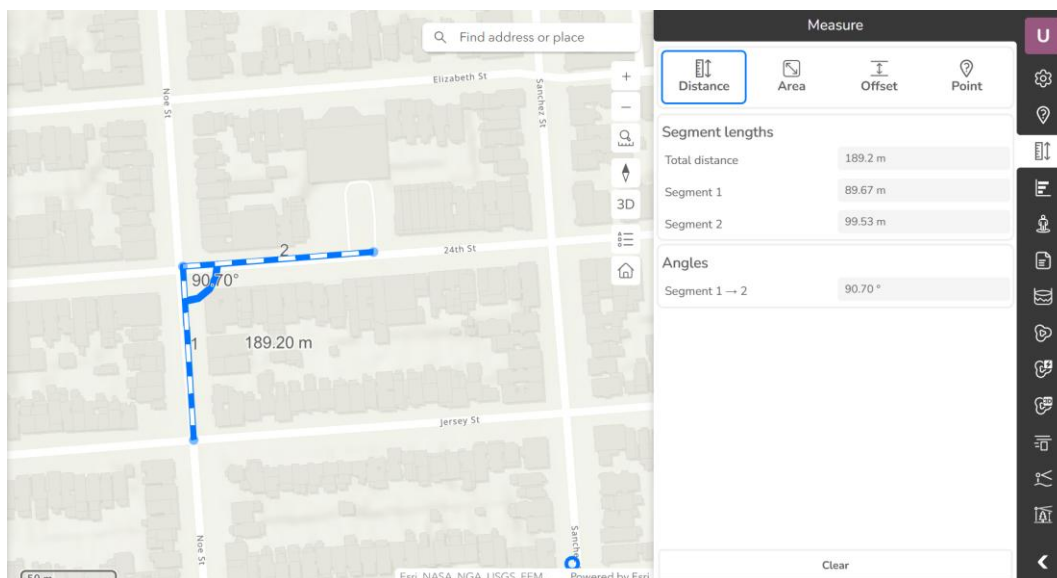
The Measurement Tool allows users to calculate spatial properties. It provides three measurement modes: Distance, Area, and Offset.



Click on the map to start measuring

3.1.12.1 Distance

Measure the distance between two or more points.



Click once to finish line segment. Double click to finish drawing.

Segment lengths

Total distance

Total length of all line segments.

Segment X

Shows the length of the line segment.

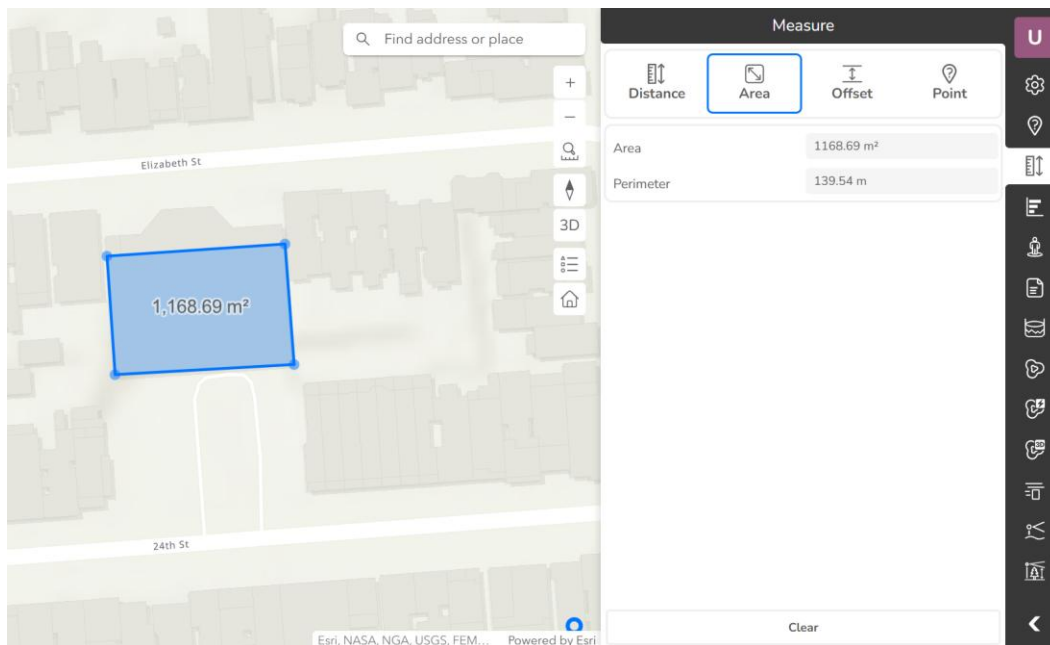
Angles

Segment 1 → 2

Angles between line segments.

3.1.12.2 Area

Calculates the surface area and perimeter of the area.

**Area**

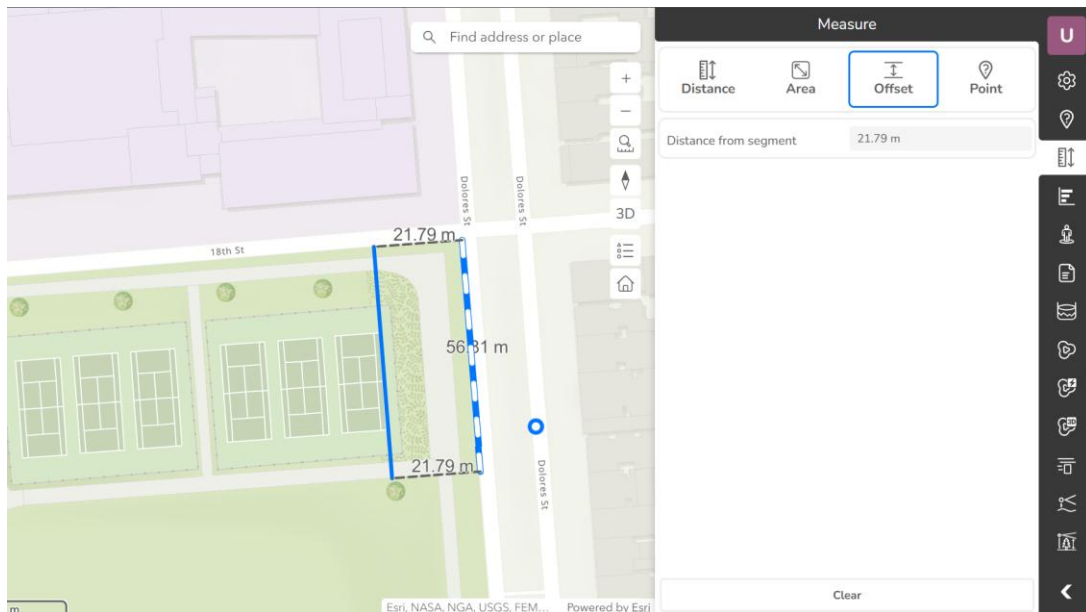
Total enclosed surface.

Perimeter

Total boundary length.

3.1.12.3 Offset

Helps find the shortest distance from a line to a point on the map.

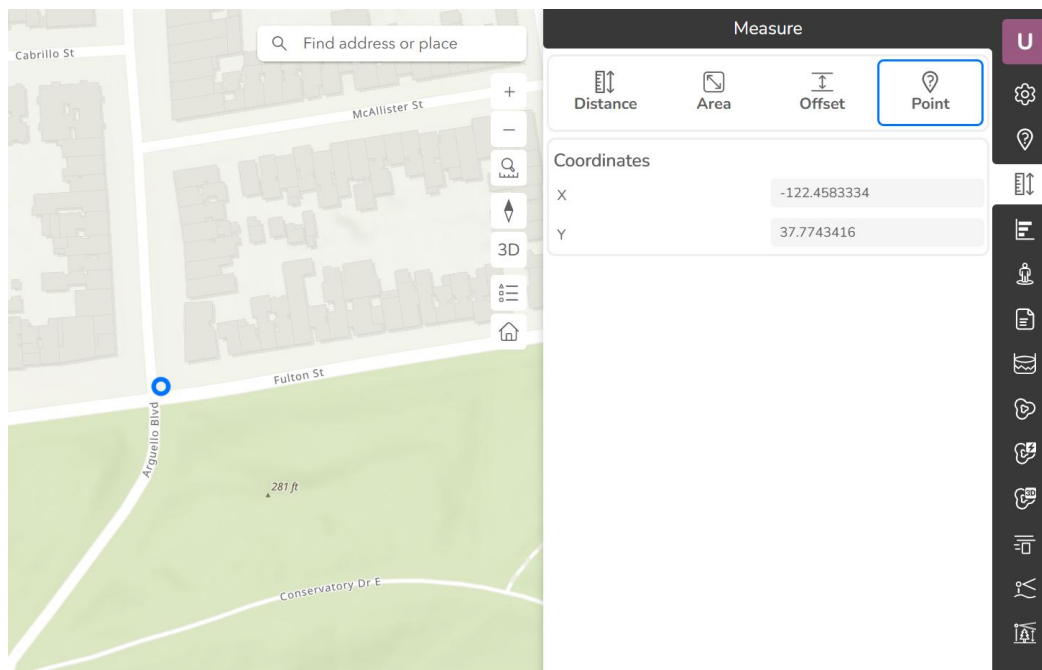


Click once to start drawing the reference line, then once again – to finish. Then, click on the map to select the offset point, towards which you would like to find the distance.

Distance from segment

Distance from the line segment to the offset point.

3.1.12.4 Point



Coordinates

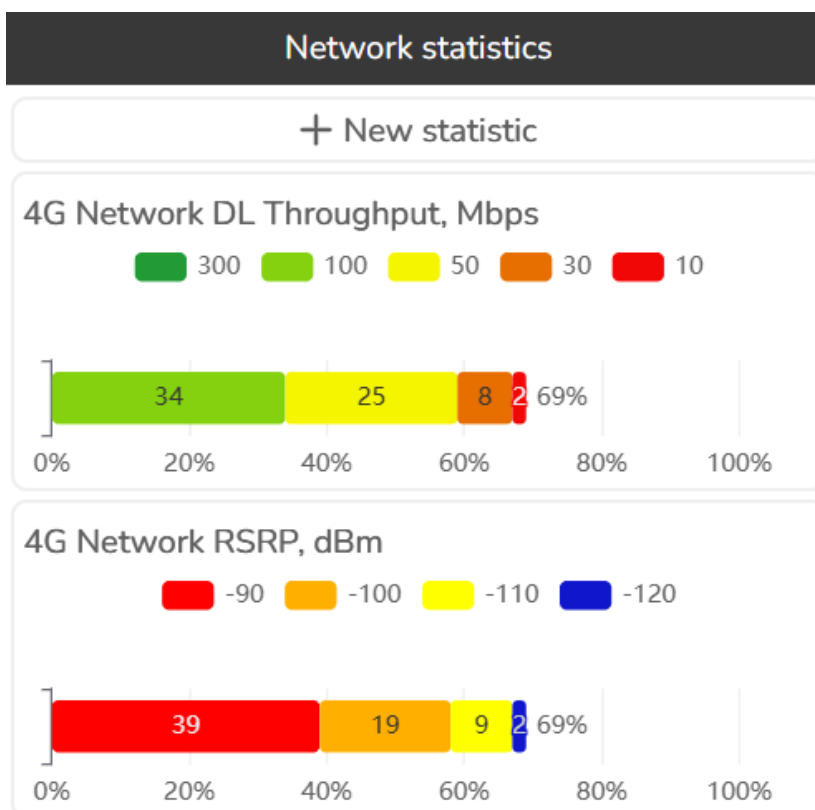
To display the coordinates, click on your desired location on the map.

3.1.13 Network statistics

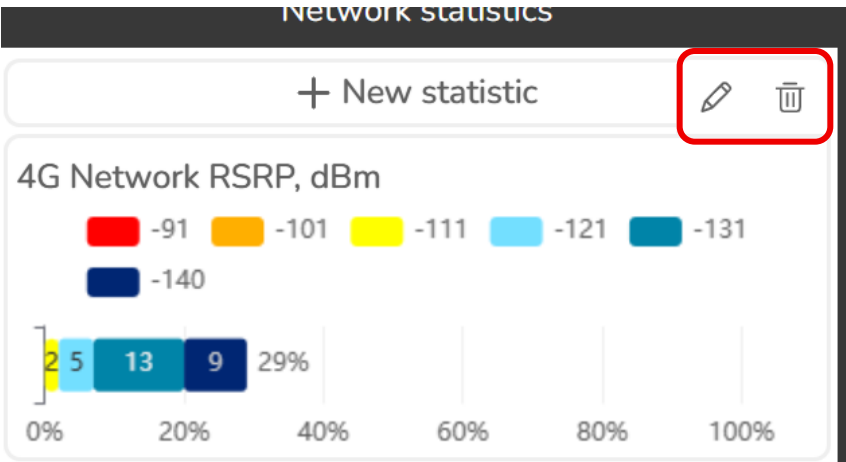


Click this button to open Network statistics tool.

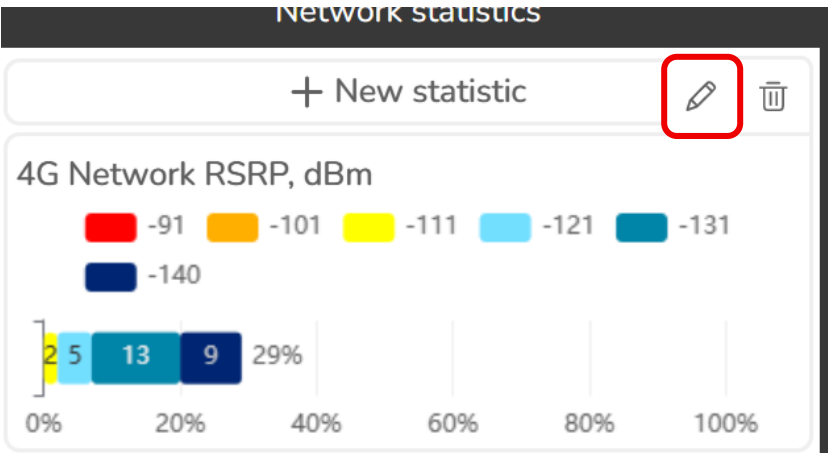
Network Statistics is a tool that calculates the total coverage of a polygon based on its overall coverage (signal strength, dl throughput, etc.). The resulting statistics include the total coverage and individual coverage of each polygon segment.



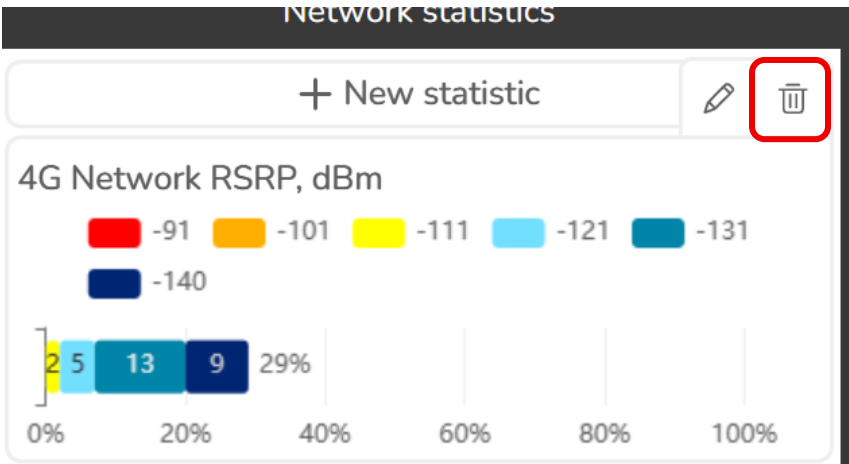
Upon hovering the mouse over a statistic item, options for it appear.



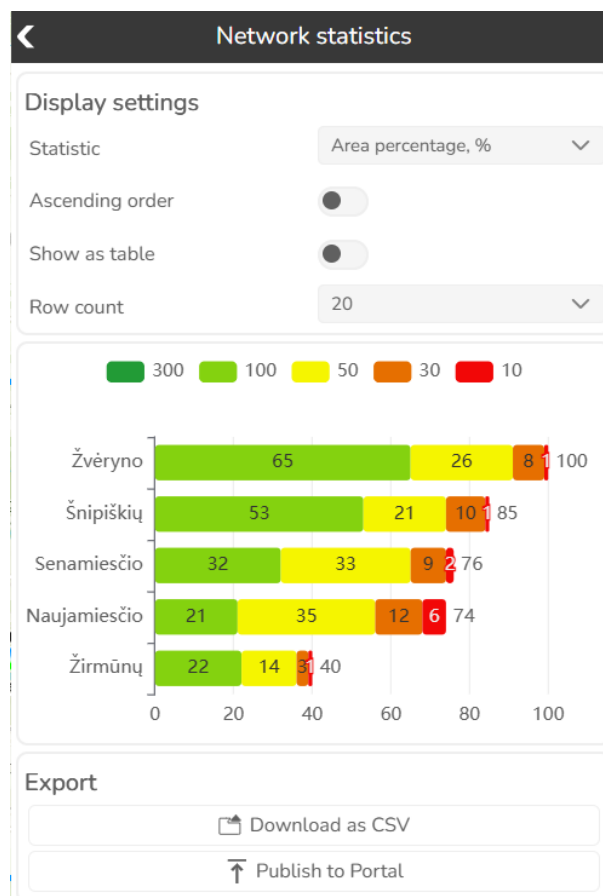
Edit statistic



Delete statistic



A statistics item may be clicked to reveal detailed statistic.



Display settings

Statistic

- Area percentage, % – percentage of area covered by each color band. If the statistic is based on a population point layer, then it's the percentage of population.
- Area, km² – area in km² covered by each color band.
- Population – population covered by each color band, when polygon population field is set.
- Point count - only appears on statistics based on a population point layer. Displays the number of weighted points covered by each color band.

Ascending order

Sort in descending/ascending order.

Show as table

View the results in tabular view.

Row count

Defines the maximum number of rows displayed.

Export

Download as CSV

Download statistic results as csv file.

Publish to Portal

Publish statistic polygons as an ArcGIS layer. The output polygons will have the statistic results assigned to them as attributes

3.1.13.1 Add statistic

Used for adding new statistics items to the main screen.

Network statistics

Statistic settings

Statistic name: New statistic

Raster layer: RF prediction 4G LTE-800 ...

Network: KLP_LIG-Su saitu

Band 1: >= -140

Band 2: >= -131

Band 3: >= -121

Band 4: >= -111

Band 5: >= -101

Band 6: >= -91

Polygon layer: Regions

Name field: sen_pav

Population field (optional): stat_01_viso

Polygon count: 21

Point layer (optional): Adresai Vilnius

Weight field (optional): none

Accept Cancel

Statistic name

The name of the Statistic.

Raster layer

The calculation result layer that will be used as the source for statistics item.

Band

The color of each segment in the statistics. The color bands are automatically retrieved from the raster layer.

Polygon layer

Workspace 'extra layer' (polygon feature layer) that is used to divide statistics item into areas by polygon.

Name Field

Defines the names of territories that will be displayed in the results.

Population field (optional)

If selected, adds a population calculation to statistics item.

Point layer (optional)

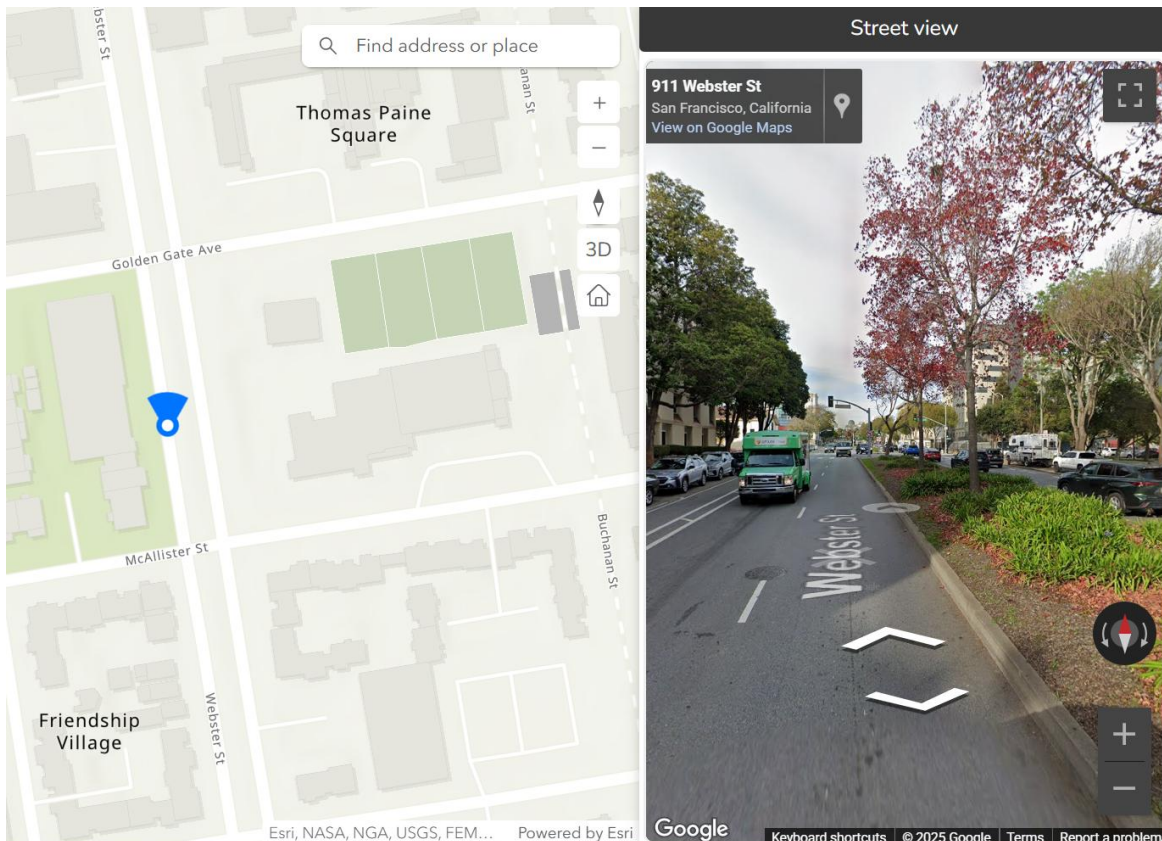
Workspace 'extra layer' (point feature layer). If selected, the statistics will be calculated not by total polygon area coverage, but by point coverage (% and #) within each polygon.

3.1.14 Street view



Click this button to open Street view tool.

This interface links a 2D map with a street-level view. The map shows spatial context and landmarks. The street view shows the real-world appearance of the same location. Both panels update together when you select a place.



3.1.15 Feature report



Click this button to open Feature report tool.

This tool allows you to create PDF reports of selected features with attribute tables, as well as a page with a view of the map.

Feature report

Report options

Map page

☒

Download map as image

☐

Preliminary report

☒

Table styling (disable for printing)

☒

Filename

Report

Title

Report

Report info

▼

Table data

▼

EMF options

▼

Generate

Report options

General options related to report

Map page

Whether to add a page with a view of the map to the PDF report output.

Download map as image

If this is enabled, an additional image file of the map will be downloaded upon report generation.

Preliminary report

Whether to add "Preliminary report" notice at the front page of the report.

Table styling (disable for printing)

Enable / disable colors applied to table.

Filename

Report file name.

Title

Title for the front page of the report.

Report info

Author

Organization

Project

Street

Zip code

City

Contact person

Phone

Email

Fax

Report info

Contact data of the report generating person / organization. Displayed in the final report.

Table data

Cells

X

Y

Height, m

Azimuth, deg

Downtilt, deg

El. Downtilt, deg

Frequency, MHz

Power, dBm

Misc. loss, dB

Bandwidth, MHz

Noise figure, dB

Downlink duplex factor

Subcarrier spacing, kHz

Table data
Enable / disable fields that will be added to the feature tables within the report. Only selected features are added to the table data.

EMF options

Cell arrows

Enable ☐

Arrow length 310 m

Cell names ☒

Perimeter circles

Enable ☐

Feature type none

Draw radius line ☐

Radius 1 300 m

Omen points

Name prefix

Place omen points

EMF options

Options pertaining to the EMF module. Only visible when EMF module is active.

Cell arrows**Enable**

Whether to draw the arrows originating from selected cells

Arrow length

Length of arrows in meters

Perimeter circles**Enable**

Whether to draw perimeter circles around selected features

Feature type

Feature type which is used as the origin for perimeter circles

Draw radius line

Whether to draw a line within the circles marking the radius of the circle. A label of the radius length is displayed above the dashed line.

+			
Radius 1	300	m	X
Radius 2	150	m	X
Radius 3	50	m	X

Perimeter radius lengths

Multiple perimeters around each feature may be drawn. This list allows you to define the length of each perimeter radius.

Omen points**Place omen points**

This button places omen points on the intersection points of cell arrow lines and radius circles.

Name prefix

Each placed point is numbered with 1 – n in the name (1, 2, 3, ..., n). This prefix is added to each name. For example, if the prefix is defined as “omen”, the created omen points will be named as “omen 1”, “omen 2”, “omen n”.

Report example:



Report

Vilniaus apskritis

Preliminary document - not for distribution

EPSG: 4326

Extent X max: 26.7593670

Extent Y min: 24.3866560

Extent Y max: 55.5176900

Extent Y min: 54.1276550

Author: CE

Organization: Cellular Expert

Project: CE Report

Author: CE
Organization: Cellular Expert
Project: CE Report
Date: 2025-07-25 11:09

Report
Vilniaus apskritis

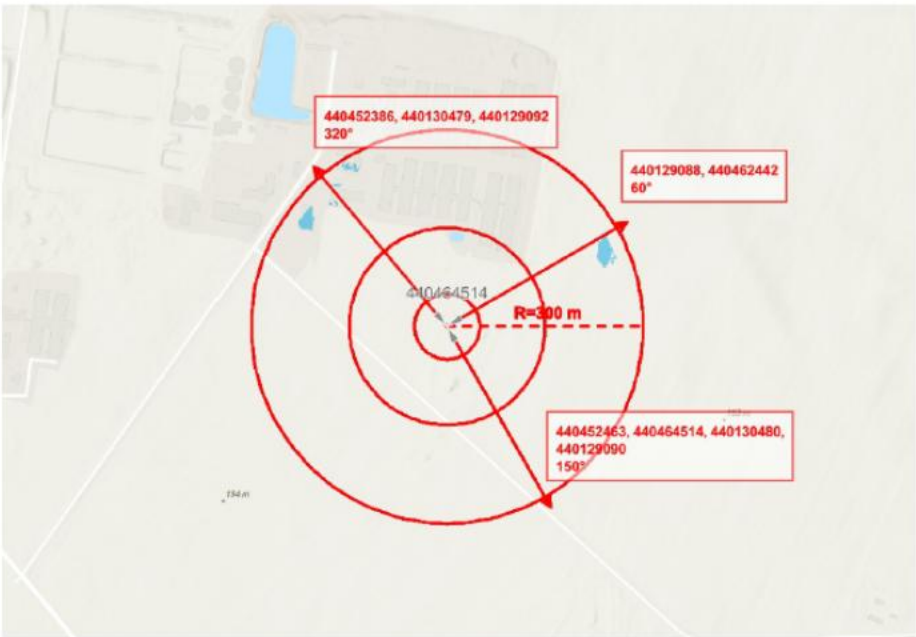


Cells:	440452463	440452386	440129090	440129092	440129088	440130479
ID	563611	563634	564371	564389	564417	564936
X	25.433188	25.433188	25.433188	25.433188	25.433188	25.433188
Y	54.890318	54.890318	54.890318	54.890318	54.890318	54.890318
Height, m	348	348	348	348	348	343
Azimuth, deg	150	320	150	320	60	320
Downtilt, deg	0	0	0	0	0	0
Frequency, MHz	2140	2140	946.6	937.3999999 99999	947.7999999 99999	806
Power, dBm	62.39999999 99999	62.39999999 99999	59.29999999 99999	59.29999999 99999	59.29999999 99999	61.1

Cells:	440130480	440464514	440462442
ID	564937	564961	565208
X	25.433188	25.433188	25.433188
Y	54.890318	54.890318	54.890318
Height, m	343	348	348
Azimuth, deg	150	150	60
Downtilt, deg	0	0	0
Frequency, MHz	806	773	773
Power, dBm	61.1	60.89999999 99999	60.89999999 99999

Author: CE
Organization: Cellular Expert
Project: CE Report
Date: 2025-07-25 11:09

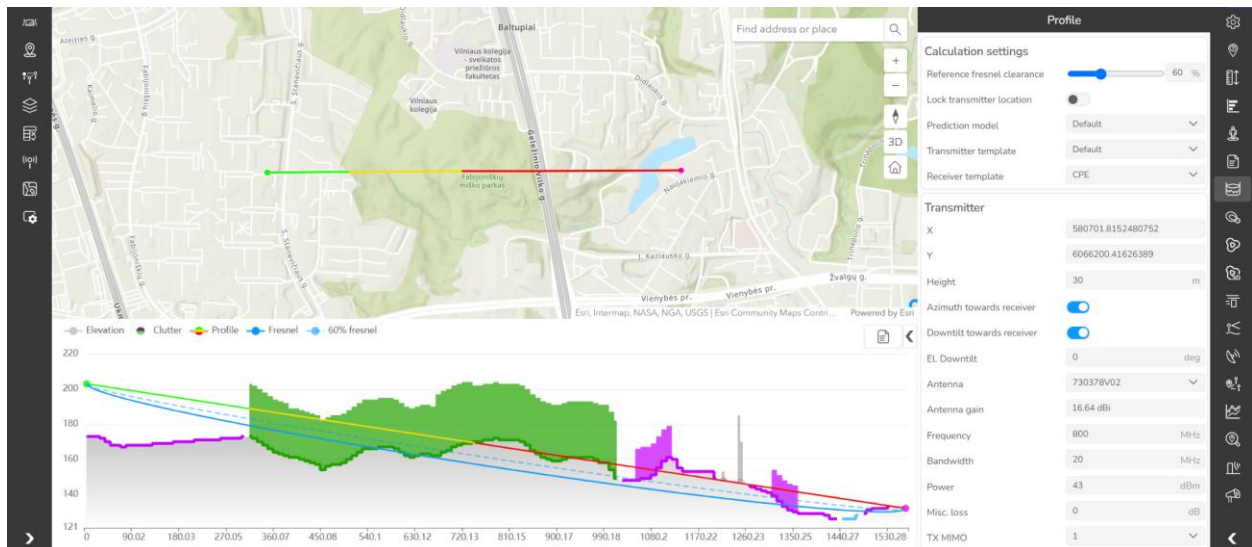
Report
Vilniaus apskritis



3.1.16 Profile

Click this button  to open Profile tool.

This tool generates a detailed profile between two points. The locations From (transmitter) and To (receiver) can be chosen by clicking on the map. The first click defines the transmitter, the second one the receiver.



3.1.16.1 Properties

Profile

Calculation settings

Reference fresnel clearance 60 %

Lock transmitter location ☐

Prediction model 3km radius

Transmitter template 5G template ▼

Receiver template CPE ▼

Calculation settings

Reference Fresnel clearance

Percentage by which the primary Fresnel zones will be scaled up or down thus creating a secondary Fresnel zone. The percentage must be in the range of 1 to 200 %.

Lock transmitter location

Toggling the switch will enable the Locked Transmitter location, which will modify only the receiver’s positioning when drawing the profile on the map.

Prediction model

Prediction model list.

Transmitter template

The template that is used for transmitter’s default values.

Receiver template

The template that is used for receiver’s default values.

Transmitter

X

Y

Height

30

m

Azimuth towards receiver

Downtilt towards receiver

El. Downtilt

0

deg

Antenna

Typical Omni TR 36 942

Antenna gain

0 dBi

Frequency

26000

MHz

Bandwidth

200

MHz

Power

35

dBm

Misc. loss

0

dBm

TX MIMO

32

RX MIMO

32

Subcarrier spacing

60

kHz

Transmitter

Snapped feature

A feature from which the profile will be drawn. The parameters of the feature will be taken into calculation if the feature is snapped to by the profile tool.

X

X coordinate of the transmitter.

Y

Y coordinate of the transmitter.

Height

Height above the ground in meters.

Azimuth towards receiver

Enabled by default. When enabled, the transmitter's azimuth automatically set towards the receiver. Disabling this option allows the user to enter a static azimuth value for the transmitter.

Downtilt towards receiver

Enabled by default. When enabled, the transmitter's tilt is automatically set towards the receiver. Disabling this option allows the user to enter a static tilt value for the transmitter.

EI. Downtilt

Electrical downtilt value for the transmitter, in degrees.

Antenna

Antenna that will be used in the prediction calculations.

Frequency

The frequency value in MHz.

Bandwidth

Value in MHz. Required for 4G and 5G technologies. For other technologies define the value as 0.015.

Power

Transmitter power in dBm. If **calculate EIRP** is set to false in the workspace settings, the power value is assumed to be EIRP.

Misc. loss

Miscellaneous loss value in dB.

TX MIMO

Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

RX MIMO

Receiver antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

Subcarrier spacing

Value in kHz. Required for 4G and 5G technologies. For other technologies define value 15.

Receiver

X	
Y	
Height	2 m
Azimuth towards transmitter	☒
Downtilt towards transmitter	☒
Antenna	10P-2L8M-B5-V2_Port 3 +45...
Antenna gain	16.41 dBi
Power	0 dBm
Misc. loss	0 dBm

Receiver**X**

X coordinate of the receiver.

Y

Y coordinate of the receiver.

Height

Receiver height above the receiver reference height selected in the workspace settings.

Azimuth towards transmitter

Enabled by default. When enabled, the receiver's azimuth automatically set towards the transmitter. Disabling this option allows the user to enter a static azimuth value for the receiver.

Downtilt towards transmitter

Enabled by default. When enabled, the receiver's tilt is automatically set towards the transmitter. Disabling this option allows the user to enter a static tilt value for the receiver.

Antenna

Antenna that will be used in the prediction calculations.

Power

A power that the receiver possesses.

Misc. loss

Miscellaneous loss value in dB.

3.1.16.2 Draw profile

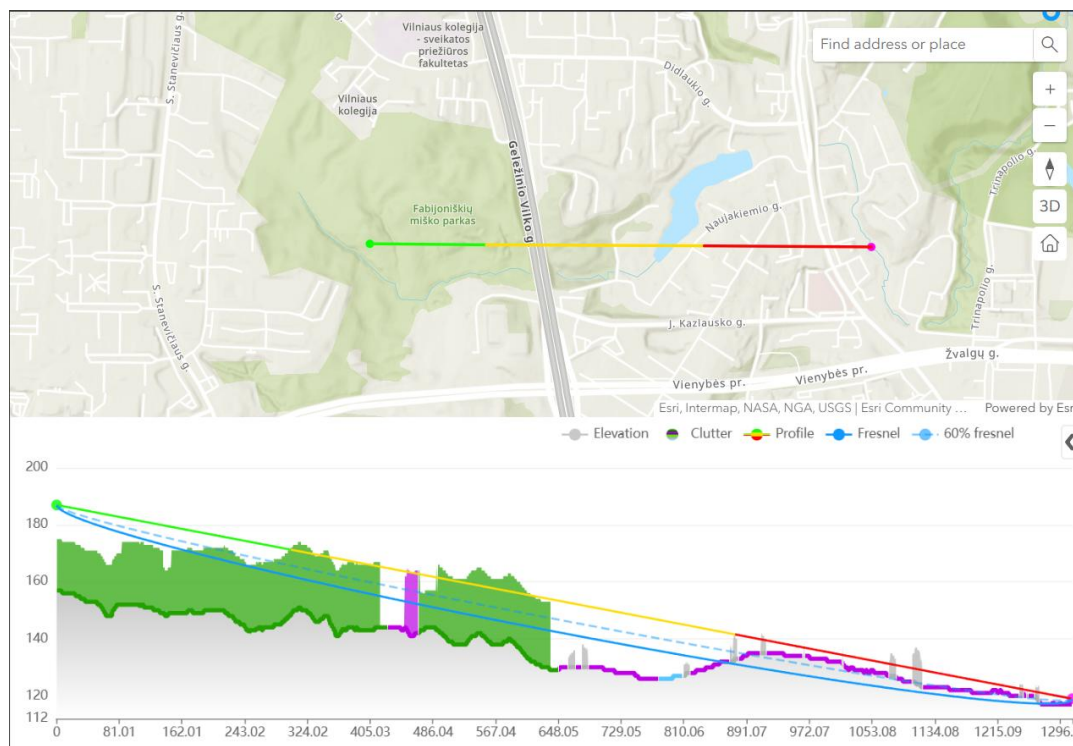
When the Profile tool is selected you will be able to select two points on the map in turn creating a Profile line.

The profile also lets you snap to different network objects (cells, CPE, sites). To snap to an object, use the CTRL key and hover the mouse near the required feature. Snapped features appear in the **Snapped feature** dropdown list under transmitter and receiver sections. Once a snapped feature is selected in the dropdown menu, all parameters of that feature that are necessary to draw a profile will be read and included in the calculations.

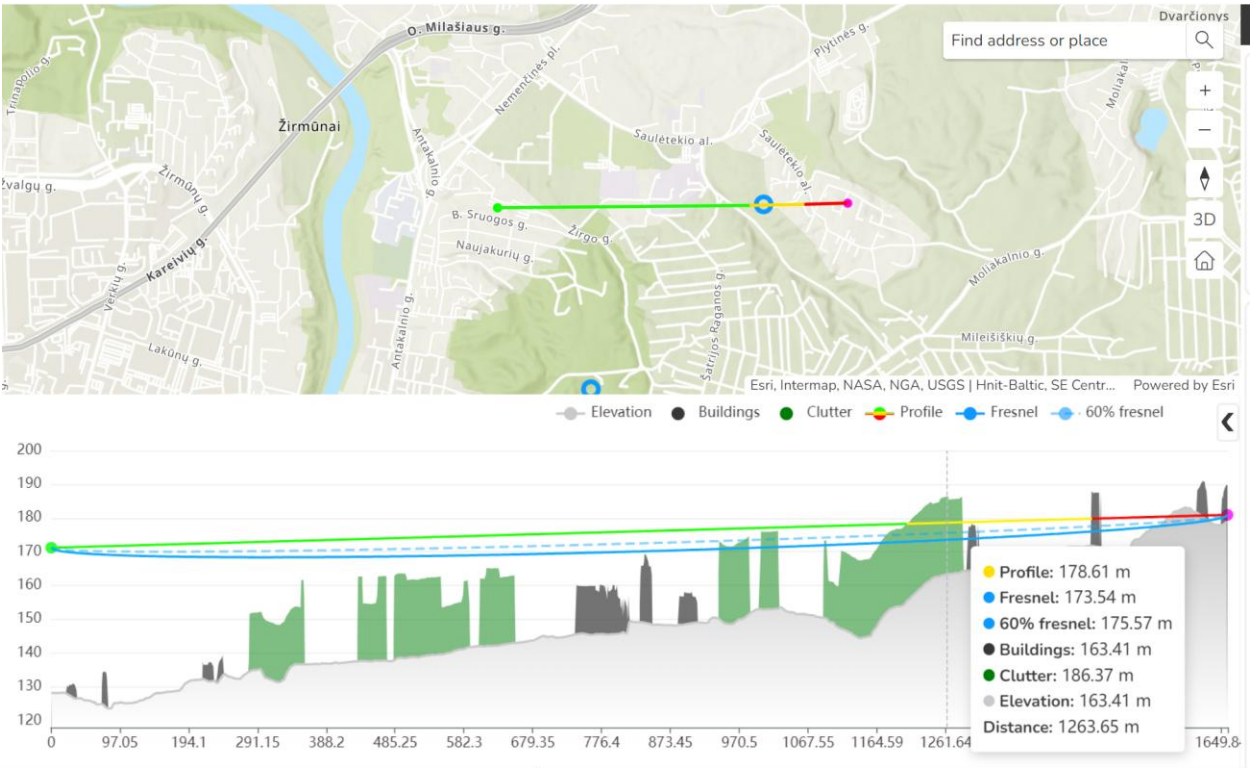
The profile is automatically recalculated when a new parameter is defined.

Upon the selection of a second point, these geometries will be created between the points:

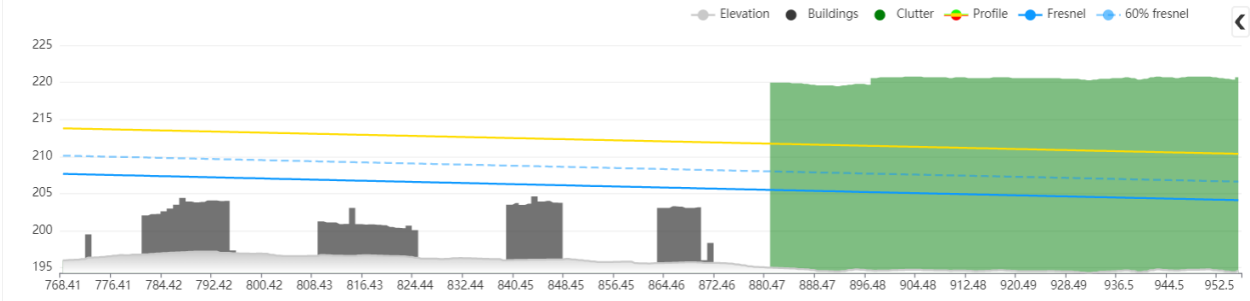
- LOS (green) — the section until the first obstruction in the profile's way
- OLOS (yellow) — the section of profile that is obstructed by clutter
- NLOS (red) — the section of profile that is obstructed by buildings or earth
- Rx (purple) — the receiver point
- Tx (green) — the transmitter point
- Fresnel (blue) — the Fresnel lines



You can inspect the values at particular points by moving the cursor around the plot. The cursor movement on the plot will be projected as a moving point on the map.



Zoom in/out in the profile view.



Click on  button to view the see the Calculation results.

General calculations

Path type

NLOS

Visibility clearance

-38.46 m

Fresnel clearance

-48.22 m

Clearance percentage

-394.01 %

Distance to NLOS

740.14 m

Distance to OLOS

310.06 m

Power budget

Downlink FS

-123.71 dBm

Uplink FS

-130.07 dBm

FWA downlink RSL

-75.82 dBm

FWA uplink RSL

-98.82 dBm

Path loss

Total path loss

135.46 dB

Basic loss

94.84 dB

Terrain / building diffraction loss

27.77 dB

3.1.16.3 Profile report

The input data and calculation results can be automatically transferred into a **Profile Report**. This report will show transmitter/receiver input data, calculation results as well as the Profile plot and map view in which the profile was drawn.

Profile report

Language

English

Landscape orientation

Author

Organization

Project

Calculations page

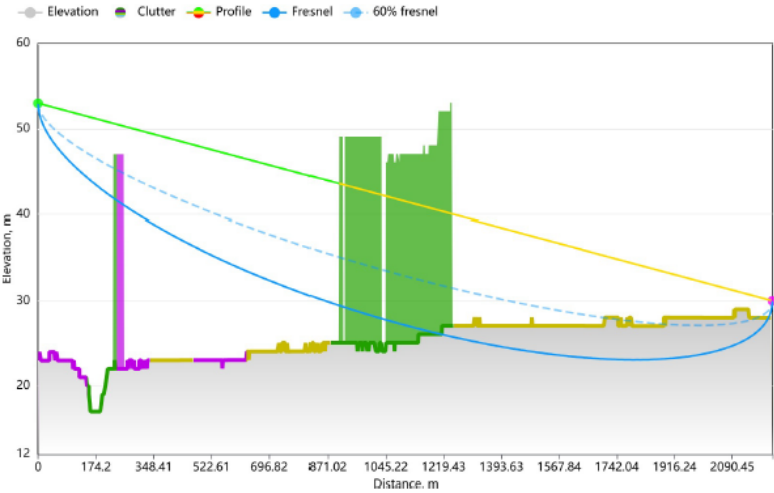
Map page

Table styling

Accept

Cancel

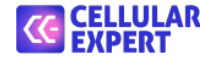
The resulting Profile report will look similar to this example:



Transmitter		Receiver	
Name	-	Name	-
Longitude, latitude	21.2174, 55.7877	Longitude, latitude	21.2527, 55.7875
DMS	55 47 15.6N 021 13 02.7E	DMS	55 47 15.0N 021 15 09.6E
TX Azimuth		RX Azimuth	
TX Tilt		RX Tilt	-0.6 deg
TX to RX Azimuth	90 deg	RX to TX Azimuth	270 deg
TX to RX Tilt	0.6 deg	RX to TX Tilt	-0.6 deg
Height	30 m	Height	2 m
Absolute height	53 m	Absolute height	30 m
Elevation at TX point	23 m	Elevation at RX point	28 m
Antenna model	730378V02	Antenna model	Typical Omni TR 36 942
Frequency	800 Mhz	Power	
Power	43 dBm		
Misc Loss			

Author: -
 Organization: -
 Project: -
 Date: 2025-05-22 10:30

Profile report
 TX: (21.2174, 55.7877) → RX: (21.2527, 55.7875)



Path Loss		General Calculations	
Prediction model name	CEC ITU-R (100MHz - 6GHz)	Visibility Clearance	-13 m
Model configuration	15km AB	Clearance	-27.27 m
Total Path Loss	106.29 dB	Clearance percentage	-91.11 %
Basic Loss	98.46 dB	Distance to NLOS	2211.59 m
Diffraction Loss	0 dB	Distance to OLOS	905.06 m
Clutter Loss	7.83 dB		
Penetration Loss	0 dB		
Penetration Obstacle Loss	0 dB		
RX Clutter Loss	0 dB		

Power Budget	
Downlink FS	-104.17 dBm
Uplink FS	-100.9 dBm
FWA downlink RSL	-56.28 dBm
FWA uplink RSL	-79.28 dBm

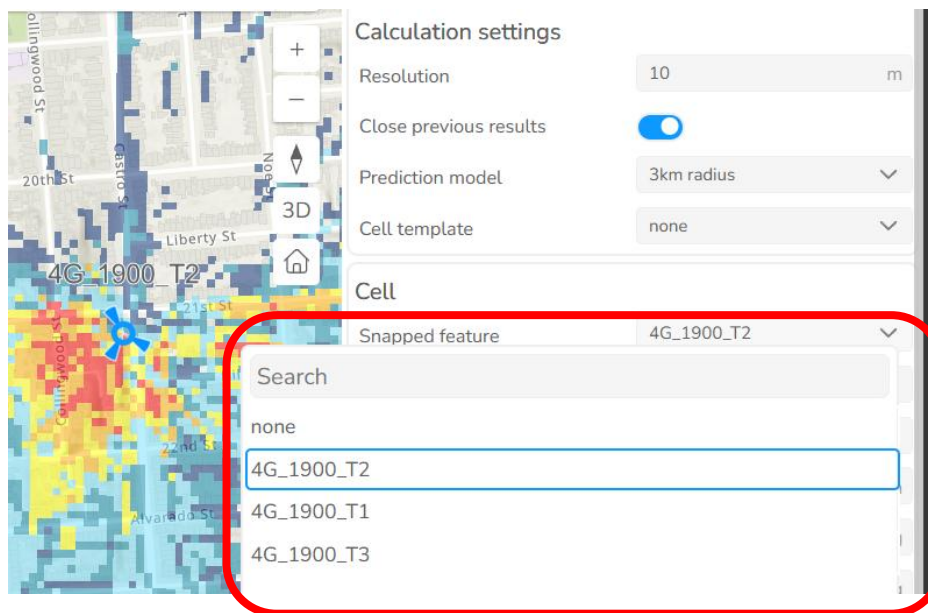
3.1.17 Quick RF Prediction



Click this button to open Quic RF Prediction tool.

The Quick RF Prediction is a tool that lets you select a point on the map and make a prediction without the need to create a cell. Quick prediction also lets you select a cell as the point, meaning that you can also make quick predictions with the created cells. The Quick RF Prediction tool calculates only the field strength coverage.

- To select a cell, hold the CTRL key and click to snap to the required cell. Snapped cells will be visible in the dialog.



- To predict a preferred location without a cell, define the exact coordinates in the X/Y fields or click on the map to fill the coordinate fields automatically.

Quick RF prediction

Calculation settings

Resolution

10

m

Close previous results

☒

Prediction model

3km radius

Cell template

5G template

▼

Cell

Snapped feature

none

▼

X

25.27377393070274

Y

54.70475252774594

Height

30

m

Azimuth

0

deg

Downtilt

0

deg

El. Downtilt

0

deg

Antenna

10P-2L8M-B5-V2_Port 3 +45...

Antenna gain

16.41 dBi

Frequency

3600

MHz

Bandwidth

100

MHz

Power

50

dBm

Misc. loss

0

dBm

TX MIMO

32

▼

Subcarrier spacing

30

kHz

Calculation settings

Resolution

Prediction raster cell size in meters.

Close previous results

If this option is enabled, clicking elsewhere on the map or changing the calculation parameters will close the previous result raster and open the new one. If disabled, the old calculation raster stays displayed on the map.

Prediction model

Prediction models that will be used in the prediction calculations.

Cell template

If a snapped cell does not have all the required parameters or no cell is snapped, the parameters will be taken from the template.

Cell**Snapped feature (optional)**

A cell from which the calculation will be done.

X

X coordinate of the Cell.

Y

Y coordinate of the Cell.

Height

Cell height above the ground in meters.

Azimuth

Cell direction from the North in degrees.

Downtilt

Cells vertical angle offset.

El. Downtilt

Electrical downtilt value for the cell, in degrees.

Antenna

Antenna that will be used in the prediction calculations.

Frequency

Frequency value in MHz.

Bandwidth

Value in MHz. Required for 4G and 5G technologies. For other technologies define the value as 0.015.

Power

Power value in dBm.

Misc. loss

Miscellaneous loss value in dB.

TX mimo

Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

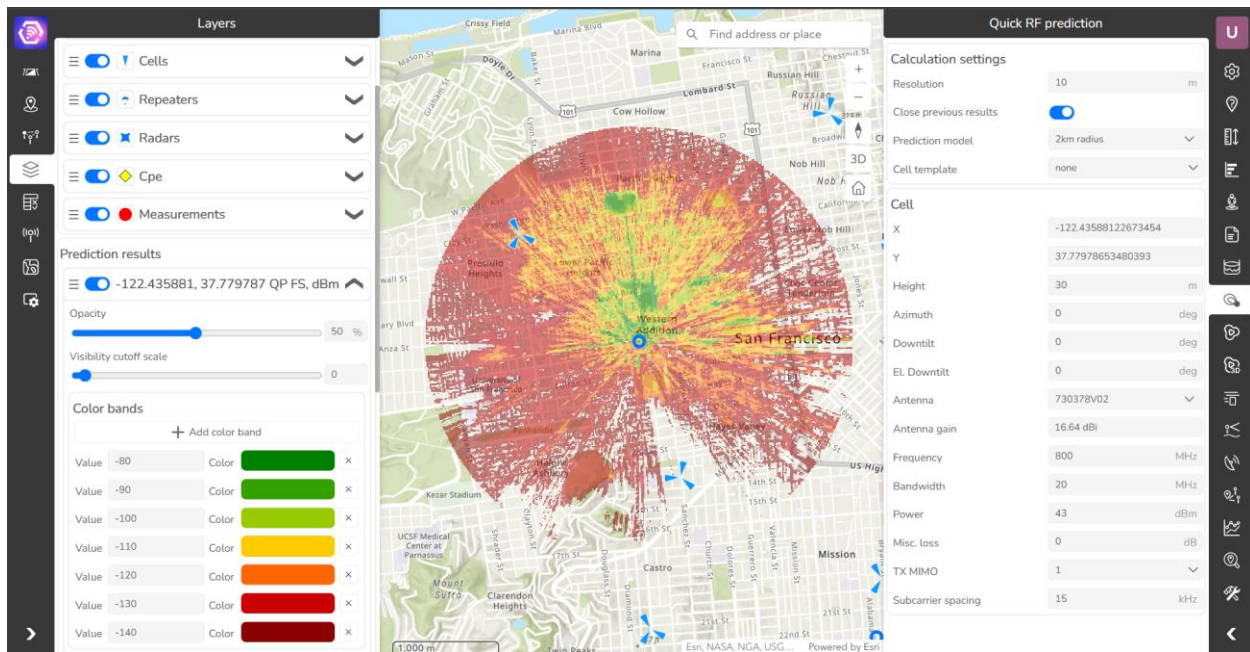
Subcarrier spacing

Value in kHz. Required for 4G and 5G technologies. For other technologies define value 15.

Click on the map to start the calculation. The results will be loaded automatically in the Map view and will be added to the Layer list tool.

Results:

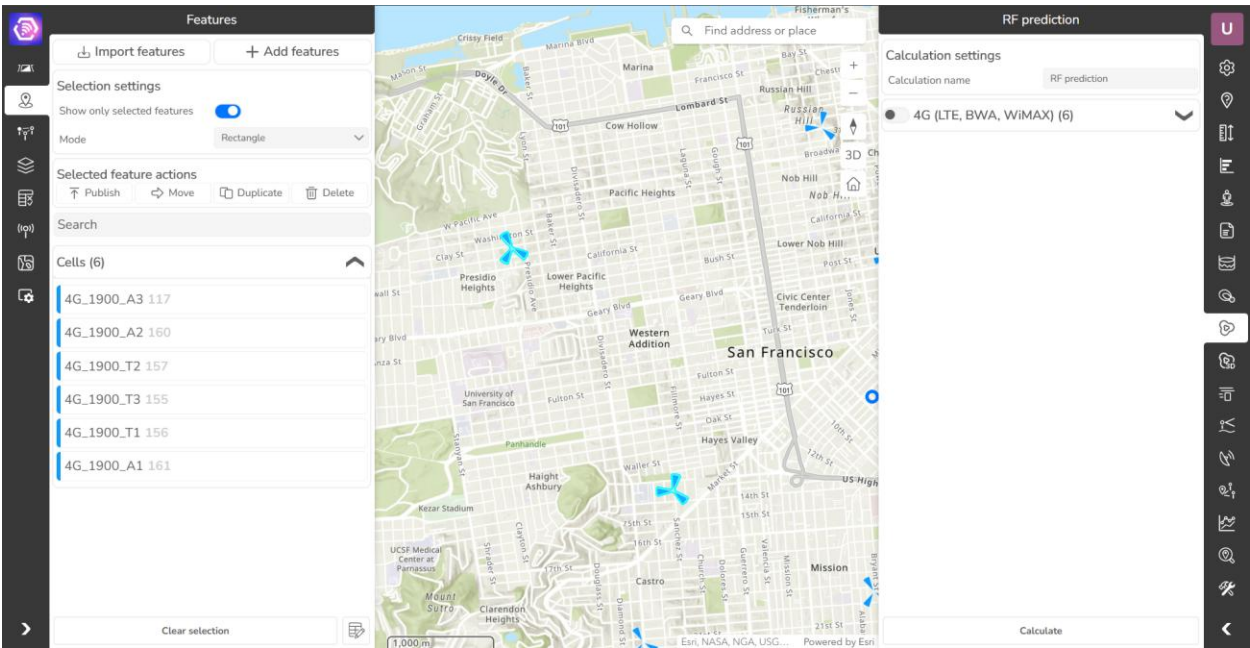
- Field Strength raster in dBm.



3.1.18 RF Prediction

Click this button  to open RF Prediction tool.

This tool allows to perform predictions for cell objects selected on the map. Calculations can be performed for more than one cell at the same time. Use the Features tool  to select cells on the map.



The selected cell count is visible in the RF Prediction dialog.



The cells are split into technologies automatically. The cell's technology is defined by the attributes in the technology field.



To preview the calculation parameters, use the expand option for the technology.

RF prediction

Calculation settings

Calculation nameRF prediction

☒ 4G (LTE, BWA, WiMAX) (6) ^

Resolution10m

Best server count2v

Cell templateDefault (4G 800 Band 20)v

Repeater template4G Templatev

Calculate FS overlap☐

Calculate uplink☐

Calculate broadband coverage☐

Calculate technology totals☐

Calculate neighbours☐

Select the technology with the switch button and press Calculate to initiate the predictions. The prediction will be added to the Prediction History dialog.

3.1.18.1 2G technology (GSM/CDMA-850/TETRA/P-25)

☒ 2G (GSM, CDMA-850, TETRA, P-25) (2)

Resolution	10	m
Best server count	2	▼
Cell template	GSM 900 MHz	▼
Repeater template	2G Template	▼
Calculate interference	☒	
C/I max field margin	20	
C/A max field margin	20	
Calculate uplink	☒	
RX EIRP	24	dBm
Calculate technology totals	☒	
Calculate neighbours	☒	
Minimum FS	-110	dBm
Neighbour minimum coverage	5	%
Maximum neighbour count	16	

Resolution

Prediction raster cell size in meters.

Best server count

There is the possibility to calculate up to 5 best servers. The prediction rasters show the 1st, 2nd, 3rd, and so on strongest signal sources.

Cell template

If the cell is missing the required parameters, the parameters from the template will be used.

Repeater template

If the repeater is missing the required parameters, the parameters from the template will be used.

Calculate interference

The interference for 2G cells will be calculated. Note that for this calculation the C/I and C/A thresholds must be defined.

C/I Max Field Margin, dB

The corresponding channel's signal interference if the first and the second strongest signals differ by less than or equal to 10 dBm.

C/A Max Field Margin, dB

The neighboring channel's signal interference if the first and the second strongest signals differ by less than or equal to 10 dBm.

Calculate uplink

An option to calculate the Uplink signal strength.

RX EIRP, dBm

The power that the receiving antenna can capture from the transmitted signal in dBm.

Calculate technology totals

Calculates the overall field strength and the aggregate downlink throughput for several frequency bands in the same technology.

Calculate neighbours

An option to calculate neighbours for each cell in the technology section

Minimum FS

Field strength threshold in dBm above which the area is considered as "covered" by the cell. The same threshold is used for calculating whether a cell's potential neighboring cell has overlapping areas, i.e. cells only overlap where both of their coverage field strength values are above this threshold.

Neighbour minimum coverage

Coverage area overlap percentage above which a cell is considered a neighbour.

Maximum neighbour count

Maximum number of neighbours a cell can have. If a cell has more potential neighbours than this number, only the ones with the highest overlap are returned.

Available coverage rasters:

- Received signal level raster in dBm, within the channel bandwidth appropriate to this technology. It is possible to calculate separately the rasters for the 1st, 2nd, 3rd, 4th and 5th strongest signal levels, and it depends on the count defined in the Best server count option.
- The best server raster shows the identification of cells generating the strongest signals at each pixel. Raster count depends on the Best server count parameter value specified by the user.
 - 1st best server shows the serving cell with the strongest field strength.
 - 2nd best server shows the second strongest field strength cell identification.
 - 3rd best server shows third shows the strongest field strength cell identification.
 - 4th best server fourth shows the strongest field strength cell identification.
 - 5th best server shows the fifth shows the strongest field strength cell identification.
- C/I rasters show Carrier-to-interference ratio in dB, to account for inter-cell interference from nearby cells that utilise the same carrier.
- C/A raster show Carrier-to-Adjacent interference ratio in dB, to account for inter-cell interference from nearby cells that operate the adjacent carrier (adjacent frequency channel).
- Uplink Field Strength raster shows receiver signal strength in dBm.

3.1.18.2 3G technology (UMTS/CDMA)

☒ 3G (UMTS, CDMA) (1)

Resolution

10

m

Best server count

2

▼

Cell template

3G 900MHz

▼

Repeater template

3G Template

▼

Calculate FS overlap

☒

FS overlap minimum FS

-110

FS overlap maximum field margin

20

Calculate uplink

☒

RX EIRP

24

dBm

Calculate technology totals

☒

Calculate neighbours

☒

Minimum FS

-110

dBm

Neighbour minimum coverage

5

%

Maximum neighbour count

16

Resolution

Prediction raster cell size in meters.

Best server count

There is the possibility to calculate up to 5 best servers. The prediction rasters show the 1st, 2nd, 3rd, and so on strongest signal sources.

Cell template

If the cell is missing the required parameters, the parameters from the template will be used.

Repeater template

If the repeater is missing the required parameters, the parameters from the template will be used.

Calculate FS overlap

This calculation in Cellular Expert calculates the geographical areas where two different cells provide similar signal strength. Specifically, it identifies regions where the absolute difference between the field strengths of Cell 1 and Cell 2 is less than a defined threshold (e.g., 10 dB). This analysis is useful for understanding coverage overlaps and potential handover zones between cells.

FS overlap minimum FS

Defines the lowest acceptable field strength for overlap consideration.

FS overlap max field margin

Threshold under which the cell field strength overlap values are presented in the results.

Calculate uplink

An option to calculate the Uplink signal strength.

RX EIRP, dBm

The power that the receiving antenna can capture from the transmitted signal in dBm.

Calculate technology totals

Calculate technology totals

Calculates the overall field strength and the aggregate downlink throughput for several frequency bands in the same technology.

Calculate neighbours

An option to calculate neighbours for each cell in the technology section

Minimum FS

Field strength threshold in dBm above which the area is considered as “covered” by the cell. The same threshold is used for calculating whether a cell’s potential neighboring cell has overlapping areas, i.e. cells only overlap where both of their coverage field strength values are above this threshold.

Neighbour minimum coverage

Coverage area overlap percentage above which a cell is considered a neighbour.

Maximum neighbour count

Maximum number of neighbours a cell can have. If a cell has more potential neighbours than this number, only the ones with the highest overlap are returned.

Available coverage rasters:

- Received signal level raster in dBm, within the channel bandwidth appropriate to this technology. It is possible to calculate separately the rasters for 1st, 2nd, 3rd, 4th and 5th strongest signal level rasters, and it depends on the count defined in Best server count option.
- The best server raster shows the identification of cells generating the strongest signals at each pixel. Raster count depends on the Best server count parameter value specified by the user.
 - 1st best server shows the serving cell with the strongest field strength.
 - 2nd best server shows the second strongest field strength cell identification.
 - 3rd best server shows third shows the strongest field strength cell identification.
 - 4th best server fourth show the strongest field strength cell identification.
 - 5th best server shows the fifth shows the strongest field strength cell identification.
- Uplink Field Strength raster shows receiver signal strength in dBm.

3.1.18.3 4G technology (LTE/BWA/WiMAX)

☒ 4G (LTE, BWA, WiMAX) (47)

Resolution	10	m
Best server count	2	▼
Cell template	Default	▼
Repeater template	4G Template	▼
Calculate FS overlap	☒	
FS overlap minimum FS	-110	
FS overlap maximum field margin	20	
Calculate uplink	☒	
RX EIRP	24	dBm
BS RX noise floor	6	dB
UL interference ceiling	10	dB
Calculate broadband coverage	☒	
Calculate technology totals	☒	
Calculate neighbours	☒	
Minimum FS	-110	dBm
Neighbour minimum coverage	5	%
Maximum neighbour count	16	

Resolution

Prediction raster cell size in meters.

Best server count

There is the possibility to calculate up to 5 best servers. The prediction rasters show the 1st, 2nd, 3rd, and so on strongest signal sources. We recommend to use at least 3 best servers for more precise RSRQ, RSSI, SINR, and Throughput calculations.

Cell template

If the cell is missing the required parameters, the parameters from the template will be used.

Repeater template

If the repeater is missing the required parameters, the parameters from the template will be used.

Calculate FS overlap

This calculation in Cellular Expert calculates the geographical areas where two different cells provide similar signal strength. Specifically, it identifies regions where the absolute difference between the field strengths of Cell 1 and Cell 2 is less than a defined threshold (e.g., 10 dB). This analysis is useful for understanding coverage overlaps and potential handover zones between cells.

FS overlap minimum FS

Defines the lowest acceptable field strength for overlap consideration.

FS overlap max field margin

Threshold under which the cell field strength overlap values are presented in the results.

Calculate uplink

An option to calculate the Uplink signal strength.

Rx EIRP, dBm

The EIRP value of the receiver in dBm.

BS RX Noise Floor, dB

Best Server receiver's lowest noise level. No values below this level will be included in the calculation.

UL interference ceiling, dB

Uplink's interference highest value. No values above this point will be included in the calculation.

Calculate broadband coverage

An option to calculate the Broadband coverage.

- RSSI;
- RSRQ;
- RS-SINR;
- DL Throughput.

Calculate technology totals

Calculates the overall field strength and the aggregate downlink throughput for several frequency bands in the same technology.

Calculate neighbours

An option to calculate neighbours for each cell in the technology section

Minimum FS

Field strength threshold in dBm above which the area is considered as "covered" by the cell. The same threshold is used for calculating whether a cell's potential neighboring cell has overlapping areas, i.e. cells only overlap where both of their coverage field strength values are above this threshold.

Neighbour minimum coverage

Coverage area overlap percentage above which a cell is considered a neighbour.

Maximum neighbour count

Maximum number of neighbours a cell can have. If a cell has more potential neighbours than this number, only the ones with the highest overlap are returned.

Available coverage rasters:

- Reference Signal Receive Power (RSRP) raster in dBm, as measured for a reference OFDMA sub-carrier bandwidth. It is possible to calculate separately the rasters for the 1st, 2nd, 3rd, 4th and 5th strongest signal levels and it depends on the count defined in the Best server count option.
- The best server raster shows the identification of cells that generate the strongest signals at each pixel. Raster count depends on the Best server count parameter value specified by the user.
 - 1st best server shows the serving cell with the strongest field strength.
 - 2nd best server shows the second strongest field strength cell identification.
 - 3rd best server shows third show strongest field strength cell identification.
 - 4th best server fourth shows the strongest field strength cell identification.
 - 5th best server shows the fifth shows the strongest field strength cell identification.
- Received Signal Strength Indicator (RSSI) estimated values' raster in dBm.
- Reference Signal Receive Quality (RSRQ) estimated values' raster in dB.
- Reference Signal's – Signal to Interference and Noise Ratio (RS-SINR) estimated values' raster in dB.
- Estimated achievable Downlink Throughput values' raster in Mbps.
- Uplink Field strength raster in dBm.
- Uplink SINR raster
- Uplink Throughput

3.1.18.4 5G technology (NR/CBRS)

5G (NR, CBRS) (2)

Resolution

10

m

Best server count

2

Cell template

5G Private (CBRS) 3.5GHz

Repeater template

5G Template

Calculate FS overlap

FS overlap minimum FS

-110

FS overlap maximum field margin

20

Calculate uplink

RX EIRP

24

dBm

BS RX noise floor

6

dB

UL interference ceiling

10

dB

Calculate broadband coverage

Calculate technology totals

Calculate neighbours

Minimum FS

-110

dBm

Neighbour minimum coverage

5

%

Maximum neighbour count

16

Resolution

Prediction raster cell size in meters.

Best server count

There is the possibility to calculate up to 5 best servers. The prediction rasters show the 1st, 2nd, 3rd, and so on strongest signal sources. We recommend to use at least 3 best servers for more precise RSRQ, RSSI, SINR, and Throughput calculations.

Cell template

If the cell is missing the required parameters, the parameters from the template will be used.

Repeater template

If the repeater is missing the required parameters, the parameters from the template will be used.

Calculate FS overlap

This calculation in Cellular Expert calculates the geographical areas where two different cells provide similar signal strength. Specifically, it identifies regions where the absolute difference between the field strengths of Cell 1 and Cell 2 is less than a defined threshold (e.g., 10 dB). This analysis is useful for understanding coverage overlaps and potential handover zones between cells.

FS overlap minimum FS

Defines the lowest acceptable field strength for overlap consideration.

FS overlap max field margin

Threshold under which the cell field strength overlap values are presented in the results.

Calculate uplink

An option to calculate the Uplink signal strength.

Rx EIRP, dBm

The EIRP value of the receiver in dBm.

BS RX Noise Floor, dB

Best Server receiver's lowest noise level. No values below this level will be included in the calculation.

UL interference ceiling, dB

Uplink's interference highest value. No values above this point will be included in the calculation.

Calculate broadband coverage

An option to calculate the Broadband coverage.

- RSSI;
- RSRQ;
- RS-SINR;
- DL Throughput.

Calculate technology totals

Calculates the overall field strength and the aggregate downlink throughput for several frequency bands in the same technology.

Calculate neighbours

An option to calculate neighbours for each cell in the technology section

Minimum FS

Field strength threshold in dBm above which the area is considered as "covered" by the cell. The same threshold is used for calculating whether a cell's potential neighboring cell has overlapping areas, i.e. cells only overlap where both of their coverage field strength values are above this threshold.

Neighbour minimum coverage

Coverage area overlap percentage above which a cell is considered a neighbour.

Maximum neighbour count

Maximum number of neighbours a cell can have. If a cell has more potential neighbours than this number, only the ones with the highest overlap are returned.

Available coverage rasters:

- Equivalent Reference Signal Receive Power (RSRP) raster in dBm, as measured for a reference OFDMA sub-carrier bandwidth. It is possible to calculate separately the rasters for the 1st, 2nd, 3rd, 4th and 5th strongest signal levels and it depends on the count defined in the Best server count option.
- The best server raster shows the identification of cells that generate the strongest signals at each pixel. Raster count depends on the Best server count parameter value specified by the user.
 - 1st best server shows the serving cell with the strongest field strength.
 - 2nd best server shows the second strongest field strength cell identification.
 - 3rd best server shows third shows the strongest field strength cell identification.
 - 4th best server fourth shows show strongest field strength cell identification.
 - 5th best server shows the fifth shows the strongest field strength cell identification.
- Received Signal Strength Indicator (RSSI) estimated values' raster in dBm.
- Reference Signal Receive Quality (RSRQ) estimated values' raster in dB.
- Reference Signal's Signal to Interference and Noise Ratio (RS-SINR) estimated values' raster in dB.
- Estimated achievable Downlink Throughput values' raster in Mbps.
- Uplink Field strength raster in dBm.
- Uplink SINR.
- Uplink Throughput.

3.1.18.5 WiFi technology

WiFi (1)

Resolution

10

m

Best server count

2

Cell template

WiFi 2.4GHz Omni

Repeater template

WiFi

Calculate FS overlap

FS overlap minimum FS

-110

FS overlap maximum field margin

20

Calculate uplink

RX EIRP

24

dBm

BS RX noise floor

6

dB

UL interference ceiling

10

dB

Calculate broadband coverage

Calculate technology totals

Calculate neighbours

Minimum FS

-110

dBm

Neighbour minimum coverage

5

%

Maximum neighbour count

16

Resolution
Prediction raster cell size in meters.

Best server count
There is the possibility to calculate up to 5 best servers. The prediction rasters show the 1st, 2nd, 3rd, and so on strongest signal sources. We recommend to use at least 3 best servers for more precise RSRQ, RSSI, SINR, and Throughput calculations.

Cell template

If the cell is missing the required parameters, the parameters from the template will be used.

Repeater template

If the repeater is missing the required parameters, the parameters from the template will be used.

Calculate FS overlap

This calculation in Cellular Expert calculates the geographical areas where two different cells provide similar signal strength. Specifically, it identifies regions where the absolute difference between the field strengths of Cell 1 and Cell 2 is less than a defined threshold (e.g., 10 dB). This analysis is useful for understanding coverage overlaps and potential handover zones between cells.

FS overlap minimum FS

Defines the lowest acceptable field strength for overlap consideration.

FS overlap max field margin

Threshold under which the cell field strength overlap values are presented in the results.

Calculate uplink

An option to calculate the Uplink signal strength.

Rx EIRP, dBm

The EIRP value of the receiver in dBm.

BS RX Noise Floor, dB

Best Server receiver's lowest noise level. No values below this level will be included in the calculation.

UL interference ceiling, dB

Uplink's interference highest value. No values above this point will be included in the calculation.

Calculate broadband coverage

An option to calculate the Broadband coverage.

- RSSI
- RSRQ
- RS-SINR
- DL Throughput

Calculate technology totals

Calculates the overall field strength and the aggregate downlink throughput for several frequency bands in the same technology.

Calculate neighbours

An option to calculate neighbours for each cell in the technology section

Minimum FS

Field strength threshold in dBm above which the area is considered as "covered" by the cell. The same threshold is used for calculating whether a cell's potential neighboring cell has overlapping areas, i.e. cells only overlap where both of their coverage field strength values are above this threshold.

Neighbour minimum coverage

Coverage area overlap percentage above which a cell is considered a neighbour.

Maximum neighbour count

Maximum number of neighbours a cell can have. If a cell has more potential neighbours than this number, only the ones with the highest overlap are returned.

Available coverage rasters:

- Equivalent Reference Signal Receive Power (RSRP) raster in dBm, as measured for a reference OFDMA sub-carrier bandwidth. It is possible to calculate separately the rasters for the 1st, 2nd, 3rd, 4th and 5th strongest signal levels and it depends on the count defined in the Best server count option.
- The best server raster shows the identification of cells that generate the strongest signals at each pixel. Raster count depends on the Best server count parameter value specified by the user.
 - 1st best server shows the serving cell with the strongest field strength.
 - 2nd best server shows the second strongest field strength cell identification.
 - 3rd best server shows third shows the strongest field strength cell identification.
 - 4th best server fourth shows show strongest field strength cell identification.
 - 5th best server shows the fifth shows the strongest field strength cell identification.
- Received Signal Strength Indicator (RSSI) estimated values' raster in dBm.
- Reference Signal Receive Quality (RSRQ) estimated values' raster in dB.
- Reference Signal's Signal to Interference and Noise Ratio (RS-SINR) estimated values' raster in dB.
- Estimated achievable Downlink Throughput values' raster in Mbps.
- Uplink Field strength raster in dBm.
- Uplink SINR.
- Uplink Throughput.

3.1.19 3D RF Prediction

Click this button  to open 3D RF Prediction tool.

This tool allows to perform 3D predictions for cell objects selected on the map. Calculations can be performed for more than one cell at the same time. Use the Features tool  to select cells on the map.

3D RF prediction

Calculation settings

Calculation name
3D RF prediction

Color bands

+ Add color band

Value	-80	Color		x
Value	-90	Color		x
Value	-100	Color		x
Value	-110	Color		x
Value	-120	Color		x
Value	-130	Color		x
Value	-140	Color		x

☒ 4G (LTE, BWA, WiMAX) (3)

Resolution

10

m

Vertical resolution

10

m

Min Z

33

m

Max Z

64

m

Cell template

Default

▼

Repeater template

4G Template

▼

Color bands

You can set custom colors for your polygon using color bands. After the calculation, you can still change the colors, but you cannot change the values.

Resolution

Prediction raster cell size in meters.

Vertical resolution

Vertical prediction step size in meters.

Min Z

Lowest height level, in meters, at which the prediction is calculated.

Max Z

Highest height level, in meters, up to which the prediction is calculated.

Cell template

If the cell is missing the required parameters, the parameters from the template will be used.

Repeater template

If the repeater is missing the required parameters, the parameters from the template will be used.

3.1.20 Visibility prediction

Click this button  to open Visibility prediction tool.

Visibility calculations refer to the determination of line-of-sight between transmitting and receiving antennas, assessing whether any obstructions might impede direct signal transmission.

Visibility prediction

Calculation settings

Calculation name	Visibility prediction
Resolution	10 m
Max radius	3 km
Receiver height	1.5 m
Effective earth radius	8500 km
Layer to calculate	cells
Template	Default

Selected

cells	12
-------	----

Calculation name

Name of the calculation that will be displayed in the Prediction history.

Resolution

Prediction raster cell size in meters.

Max radius

Maximum calculation radius in kilometers.

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

Effective earth radius

Earth radius was used for the calculations.

Layer to calculate

All selected layers on which predictions can be performed.

Template

A template which corresponds to the selected layers. When the layer changes the templates change as well.

Visibility Prediction is a tool that calculates 4 different results:

- **Clearance, m** – the distance from the profile that is covered or is not covered.
- **Line of Sight (LOS)** – confirms whether visibility exists between the receiver and transmitter with the provided receiver height.
- **Best Server** – the same calculation as for **RF Prediction**.

3.1.21 Antenna visibility prediction

Click this button  to open Antenna visibility prediction tool.

The Antenna Visibility prediction tool is designed to calculate and visualize the feature's Line-of-sight clearance for features based on their azimuth, tilt and selected antenna pattern.

Antenna visibility prediction

Calculation settings

Calculation name	Antenna visibility prediction
Resolution	10 m
Attenuation threshold	30 dB
Cell template	none ▾

Use the selection tool to select cells

Calculation Name

Name of the calculation that will be displayed in Prediction history.

Resolution

Prediction raster cell size in meters.

Cell template

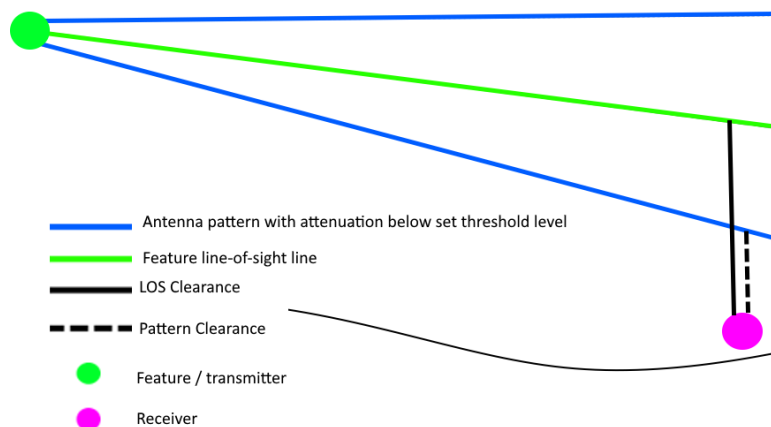
If the cell is missing the required parameters, the parameters from the template will be used.

Attenuation threshold, db

Maximum allowable signal loss for determining effective signal visibility.

Results:

- Line of Sight (LOS) clearance – height difference between the receiver point and feature's straight line-of-sight based on tilt and azimuth.
- Pattern clearance – Height difference between the receiver point and the lowest point of the antenna pattern that's above the set threshold.

**3.1.22 Minimum receiver height**

Click this button to open Minimum receiver height tool.

Calculates the minimum receiver height required to satisfy LOS condition for selected features.

Minimum receiver height

Calculation settings

Calculation name	Minimum receiver height	
Resolution	10	m
Max radius	3	km
Effective earth radius	8500	km
Overlap priority	Lowest (Reach at least 1) ▼	

Use the selection tool to select features

Calculation settings

Calculation name

Name of the calculation that will be displayed in Prediction history.

Resolution

Prediction raster cell size in meters.

Max radius

Maximum calculation radius in kilometers.

Effective earth radius

Earth radius was used for the calculations.

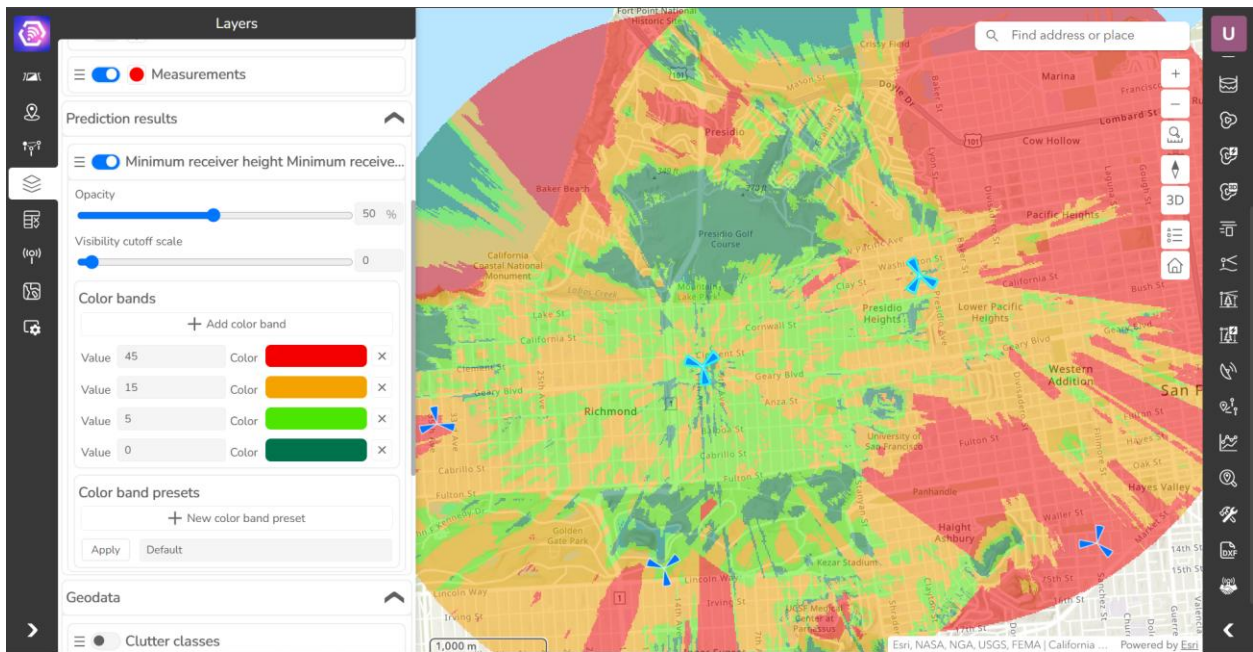
Overlap priority

Value priority in places where multiple feature calculations overlap

- Lowest (Reach at least 1) – minimum value will be used
- Highest (Reach all) – maximum value will be used

Results:

- Minimum receiver height, m - minimum receiver height required to satisfy LOS condition



- **Best Server** – Object id of the feature which has the value priority in each pixel. E.g.: In the case of “lowest” priority setting, pixels have the object id of the feature which has the lowest available minimum receiver height (best visibility).

3.1.23 Quick minimum receiver height



Click this button to open Quick minimum receiver height tool.

Calculates the minimum receiver height required to satisfy LOS condition for a single feature. Feature parameters are defined in the tool input fields, therefore it may be a theoretical feature that does not exist in the database.

Quick minimum receiver height

Calculation settings

Resolution	10	m
Max radius	3	km
Effective earth radius	8500	km
Close previous results	☒	

Feature

X	
Y	
Height	

Calculation settings**Resolution**

Prediction raster cell size in meters.

Max radius

Maximum calculation radius in kilometers.

Effective earth radius

Earth radius was used for the calculations.

Close previous results

If this option is enabled, clicking elsewhere on the map or changing the calculation parameters will close the previous result raster and open the new one. If disabled, the old calculation raster stays displayed on the map.

Feature**X**

X coordinate of the Feature.

Y

Y coordinate of the Feature.

Height

Feature height above the ground in meters.

3.1.24 Radar prediction

Click this button  to open Radar prediction tool.

Radar Prediction is a tool that lets you calculate predictions on radars. Depending on the view angle and the size of the radar, Radar Predictions will show the reach of radar signals.

Radar prediction

Calculation settings

Calculation name	Radar prediction
Resolution	10 m
Best server count	3 ▼
Radar template	RADAR ▼
Target cross-section	20 m ²

Calculation settings

Calculation name

Name of the calculation that will be displayed in Prediction history.

Resolution

Prediction raster cell size in meters.

Best server count

Option to calculate up to 5 best servers.

Radar template

If the radar misses the required parameters, the parameters from the template will be used.

Target cross-section, m²

This parameter describes the target size in square meters.

Selected

Current amount of selected radars.

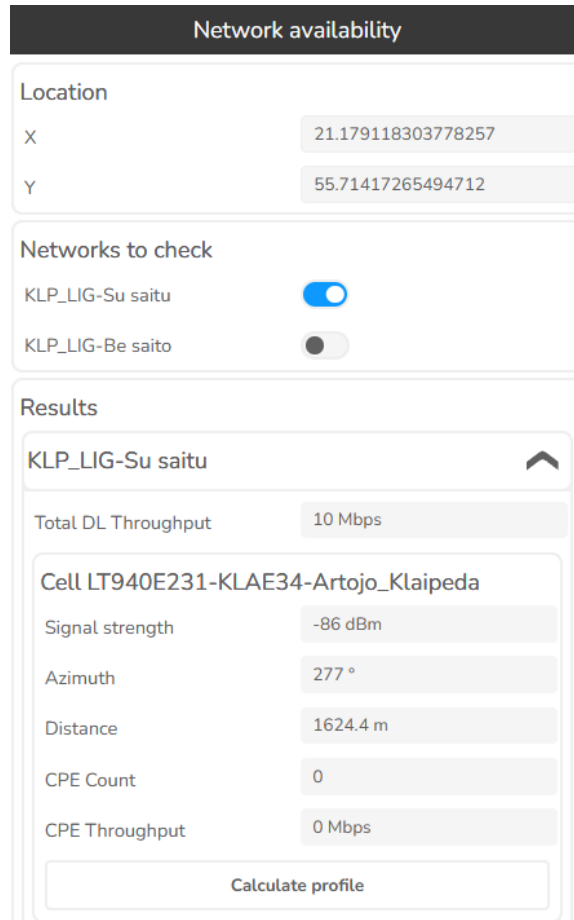
Results:

- Field Strength raster in dBm

3.1.25 Network availability

Click this button  to open Network availability tool.

This tool is designed to check if coverage exists at a chosen location. Define the exact coordinates in the dialog or click on the map to fill the coordinate fields automatically and the tool automatically provides the information about possible networks at this location.



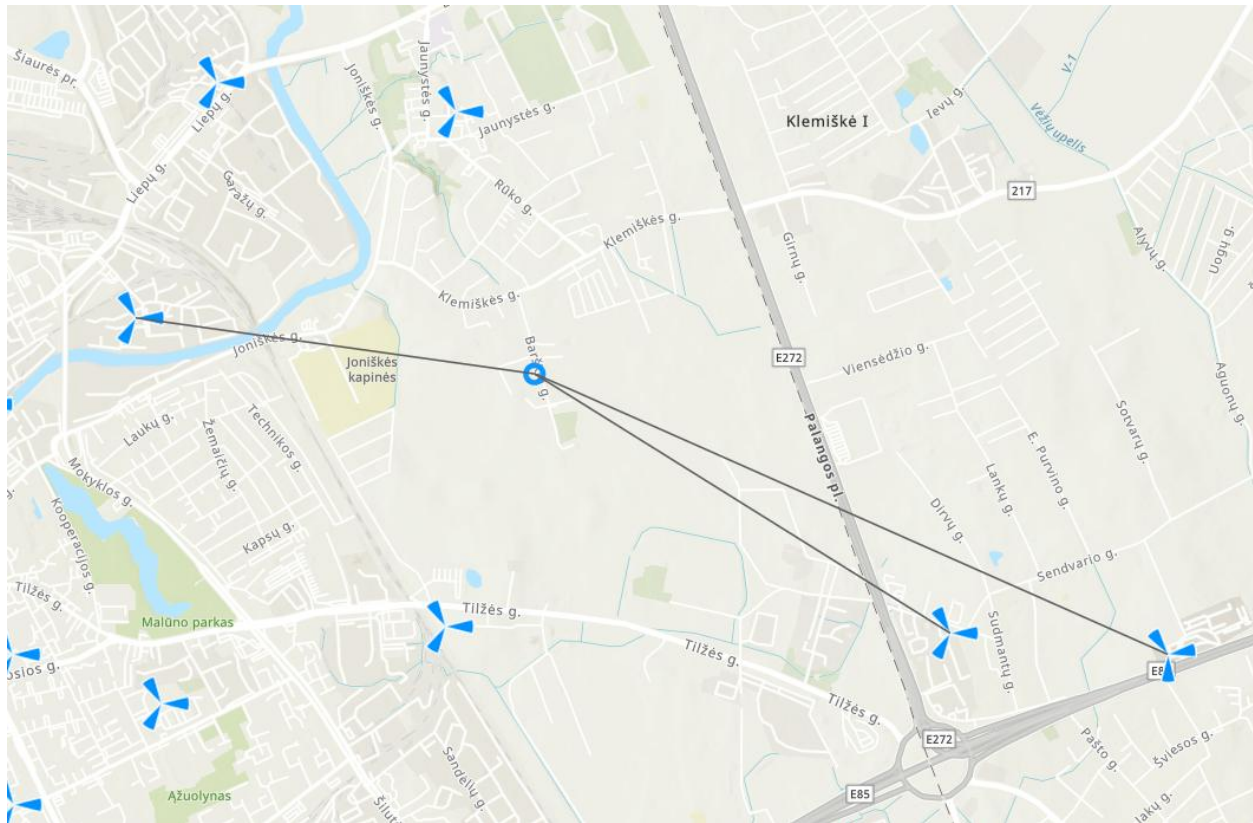
The screenshot shows the 'Network availability' tool interface. It has a dark header with the title 'Network availability'. Below the header, there are three main sections: 'Location', 'Networks to check', and 'Results'. The 'Location' section contains two input fields for 'X' and 'Y' coordinates, with values '21.179118303778257' and '55.71417265494712' respectively. The 'Networks to check' section has two toggle switches: 'KLP_LIG-Su saitu' (turned on) and 'KLP_LIG-Be saitu' (turned off). The 'Results' section displays information for 'KLP_LIG-Su saitu', including 'Total DL Throughput' (10 Mbps), 'Cell LT940E231-KLAE34-Artojo_Klaipeda', 'Signal strength' (-86 dBm), 'Azimuth' (277 °), 'Distance' (1624.4 m), 'CPE Count' (0), and 'CPE Throughput' (0 Mbps). At the bottom of the 'Results' section is a 'Calculate profile' button.

Network availability	
Location	
X	21.179118303778257
Y	55.71417265494712
Networks to check	
KLP_LIG-Su saitu	☒
KLP_LIG-Be saitu	☐
Results	
KLP_LIG-Su saitu	
Total DL Throughput	10 Mbps
Cell LT940E231-KLAE34-Artojo_Klaipeda	
Signal strength	-86 dBm
Azimuth	277 °
Distance	1624.4 m
CPE Count	0
CPE Throughput	0 Mbps
Calculate profile	

The location can be defined manually in the X/Y fields or by clicking on the map. As a result, detailed information about possible connections is displayed:

- Cell name – cell identification.
- Signal strength – signal value in dBm.
- Azimuth – direction from Receiver to Transmitter.
- Distance – distance between Transmitter and Receiver.
- CPE count – cpe count in a cell.
- CPE throughput – total cpe throughput in a cell.
- Calculate profile – option to open a detailed profile between Transmitter and Receiver.

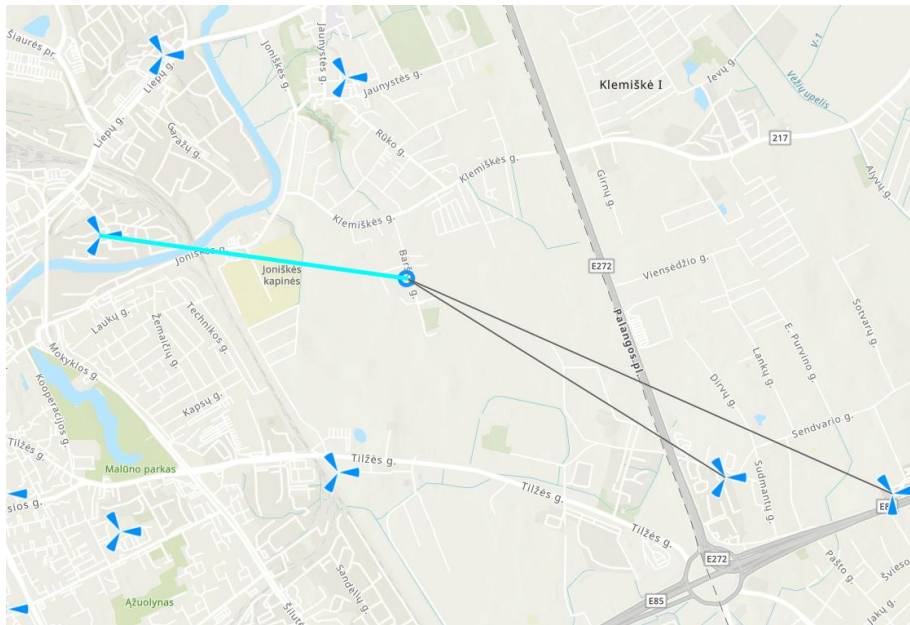
In addition, the lines between Transmitter and Receiver are displayed on the map.



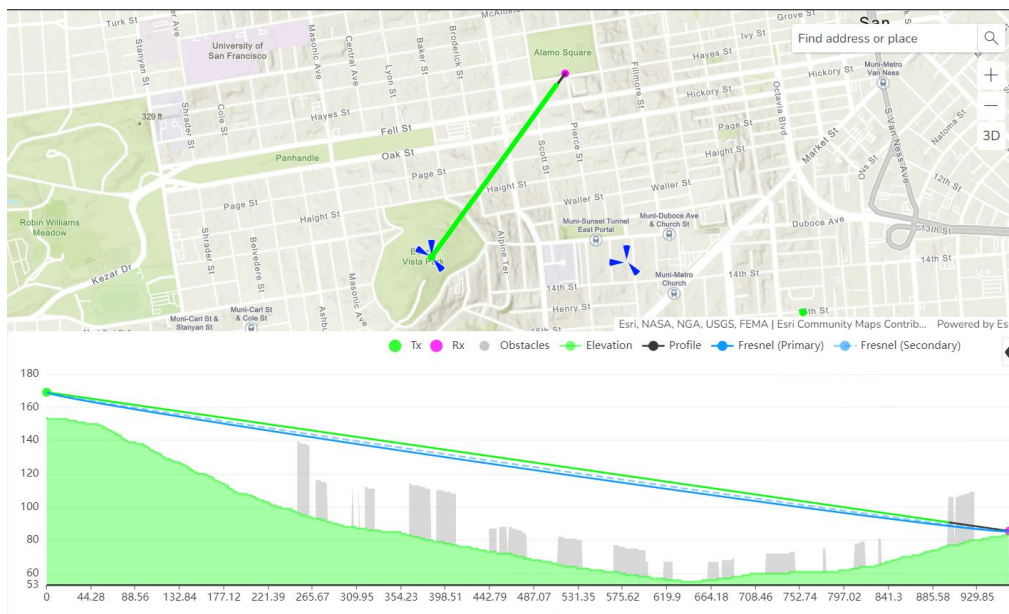
The lines on the map can be visualized in different ways. Move the mouse cursor on the calculate profile button in tabular view.

Cell LT940E231-KLAE34-Artojo_Klaipeda	
Signal strength	-86 dBm
Azimuth	277 °
Distance	1624.4 m
CPE Count	0
CPE Throughput	0 Mbps
Calculate profile	

Then the appropriate line will be visualized differently.



It is possible to draw a profile from the provided results. Press the Calculate profile button to open the profile.



3.1.26 Model Tuning

Click this button  to open Model tuning tool.

Model Tuning is a tool to optimize prediction model parameters based on test points. These test points

must be bound to cells by their cell id.

Model tuning

Calculation settings

Calculation name	Model tuning
Resolution	10 m
Prediction model	none ▼

Use the selection tool to select measurements

Calculation Name

Name of the calculation that will be displayed in the Prediction history.

Resolution

Prediction raster cell size in meters.

Prediction model

Select which model will be tuned.

Use the Features tool to select measurements.

Selected

Measurements273

Cell to use

VLN08C☒

Selected

The count of selected measurement points.

Cell to use

Select the Cell with which you want to do the drive test. At the moment, measuring for a single cell is supported.

Press Calculate to initiate the model tuning. The model tuning task will be added to the Prediction History dialog.

Model tuning

Calculation status

Complete

Created

11/11/2025 11:39

Log

Task is waiting in queue	11:38
Starting task	11:39
Comparing path loss	11:39
Tuning model	11:39
Calculating tuned FS	11:39
Building overviews	11:39
Task completed	11:39

Results

Cell 46193 Untuned Field Strength, ...	Export	Open
Cell 46193 Tuned Field Strength, dB...	Export	Open
Untuned difference points, dBm	Export	Open
Tuned difference points, dBm	Export	Open
Tuning data	Export	Open

Click the *open* button next to the tuning data to open results.

Results

Reselect measurements by visibility

LOS

Reselect

Measurements

1

OLOS

Reselect

Measurements

15

NLOS

Reselect

Measurements

88

Apply selection

Reselect measurements by visibility
Enable the visibility types of your choosing, then click apply selection to select only the measurement points, which fall into your selected visibility categories. Then, you may run the model tuning tool again with the newly selected features. Useful when you want to tune parts of the prediction model related to specific visibility types.

Untuned statistics	
Difference mean	3.39 dB
Difference standard deviation	20.83 dB
Difference standard error	1.51

Tuned statistics	
Difference mean	0.02 dB
Difference standard deviation	13.05 dB
Difference standard error	0.94

The comparison between the untuned and tuned prediction models is presented by highlighting the differences in their mean values, standard deviations, and standard errors.

- **Difference Mean** during model tuning measures how the average prediction accuracy or

performance improves after optimizing model parameters compared to the default settings. For example, tuning might reduce the average prediction error, leading to more accurate coverage or signal estimations.

- **Standard Deviation** reflects the variability in prediction performance. Comparing standard deviations between untuned and tuned models shows whether tuning makes predictions more consistent and reliable. For example, lower standard deviation after tuning indicates the model's predictions are less scattered and more stable.
- **Standard Error** during model tuning quantifies the precision of the mean prediction. A lower standard error in the tuned model suggests that the average prediction is more reliable and less influenced by sample variability, indicating improved model stability and confidence in its performance.

Suggested prediction model values	
Offset coefficient	50.57 dB
Distance coefficient	25
Distance coefficient obstructed	32.61

Model coefficients that are recommended to be changed.

- **Offset coefficient** – represents the offset in decibels added to the path loss grid.
- **Slope coefficient distance** – defines the slope based on the distance between the cell and the receiver location.
- **Slope coefficient distance obstructed** – represents the slope based on the obstructed distance between the cell and the receiver location.

Suggested clutter class values	
Open / Terrain	
Nominal distance	1.26 m
Diffraction loss coefficient	0.9
Enclosed receiver loss offset	10 dB
Enclosed receiver loss scaling coefficient	0.34
Medium dense forest	
Nominal distance	22.88 m
Diffraction loss coefficient	0.5
Enclosed receiver loss offset	0 dB
Enclosed receiver loss scaling coefficient	0.54
Low density urban (Low buildings)	
Nominal distance	17.01 m
Diffraction loss coefficient	0.54
Enclosed receiver loss offset	0 dB
Enclosed receiver loss scaling coefficient	0.91

Clutter classes

Recommended clutter loss values

3.1.27 Optimal placement

Click this button  to open Optimal placement tool.

Optimal placement is a tool that lets the user find optimal positions for a site based on specified parameters.

Optimal placement

Calculation settings

Calculation name	Optimal placement prediction
Prediction model	1km radius
Resolution	10m
CPE template	CPE
Tower height	15m
Frequency	900MHz
TX power	40dBm
TX antenna gain	5dBi
RX antenna gain	0dBi
RX sensitivity	-110dBm
Path loss cutoff	155 dBm

Use the selection tool to select CPE features

Calculation Name
Name of the calculation that will be displayed in the Prediction history.

Prediction Model
The prediction model list.

Resolution
Prediction raster cell size in meters.

CPE templates
If the CPE is missing required parameters, the parameters from the template will be used.

Tower height
The height of the tower that hosts the site.

Frequency
The frequency value in MHz.

TX power
Transmitter power in dBm.

TX antenna gain
Transmitter gain in dBi.

RX antenna gain

Receiver gain in dBi.

RX sensitivity

Receiver gain in dBi.

Path loss cutoff

The path loss is the floor of the prediction.

The prediction produces two rasters:

- Covered Points – defines what number of points are covered in a certain pixel
- Coverage Percentage – the percentage by which the pixel is covered. Meaning that 100% is a fully covered point

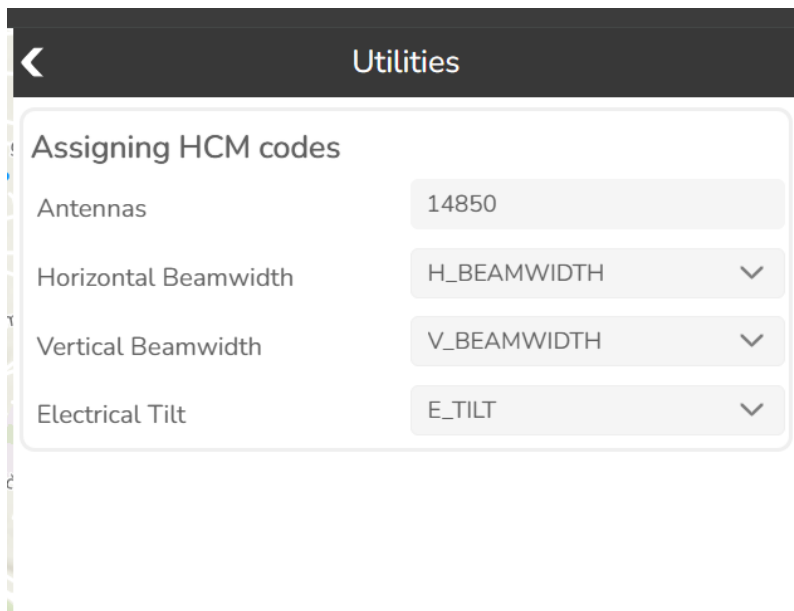
3.1.28 Utilities

Click this button  to open Utilities tool.

3.1.28.1 Beamwidth to HCM code

Used to generate HCM pattern codes from vertical & horizontal beamwidth values. Useful when you do not know the exact antenna model for your cells, and only have beamwidth values assigned to them.

After uploading the CSV file you are prompted to assign the CSV fields to the required parameters:



The screenshot shows a mobile application interface titled 'Utilities'. Below the title bar, there is a section titled 'Assigning HCM codes'. This section contains four rows, each with a label on the left and a corresponding input field on the right:

Parameter	Assigned Value
Antennas	14850
Horizontal Beamwidth	H_BEAMWIDTH
Vertical Beamwidth	V_BEAMWIDTH
Electrical Tilt	E_TILT

3.1.28.2 Feature name to object ID

Used to extract feature object_id values given their names in the csv. Useful when you want to import

features with object_id references to other already existing features. E.g.: assign cell_id field to measurements given cell_name.

3.1.28.3 Assign antennas to cells by name

Takes a .csv file with an antenna name field, and outputs the same file with antenna_id field added. Antenna_id values are filled by finding a matching antenna in the CE Express antenna database and assigning it's ID to the row.

3.1.28.4 SMADEF-XML to frequency plan CSV

Pulls allotment section of SMADEF-XML file and outputs a CSV with a frequency plan carrier list, that may be imported through the link frequency plans tool.

3.1.28.5 Export features to KMZ

Exports selected or all features to a KMZ file.

3.1.29 Points to DXF



Click this button to open Points to DXF tool.

Allows for easy reference point transfer between the CE Express interface and CAD applications.

Points to DXF

DXF options

Marker

Cross point

Marker radius

1

Label

☒

Label line

☒

Label offset

3

Point options

Layer name

Points

Click on the map to create points

DXF options**Marker**

Type of marker to draw

Marker radius

Size of marker

Label

Draw number labels next to the points

Label line

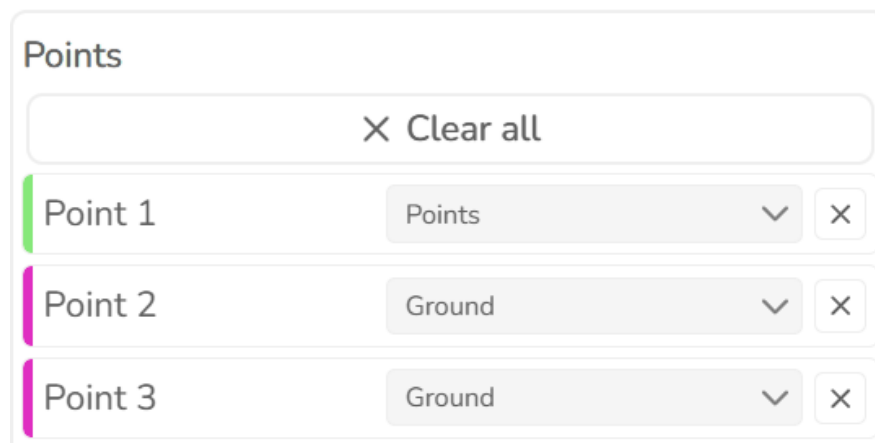
Draw line from the labels to the points

Label offset

How far to place the labels from the points

Point options**Layer name**

Output points are placed into different layers. New points are placed into this layer upon map click. Point layer may be changed after the fact within the line for each point in the UI.



The screenshot shows a UI panel titled "Points". At the top is a button labeled "X Clear all". Below this are three rows, each representing a point. Each row has a colored vertical bar on the left (green for Point 1, purple for Point 2 and Point 3), the point name, a dropdown menu showing the assigned layer, and a small "X" button to remove the point. The layers are "Points" for Point 1 and "Ground" for Point 2 and Point 3.

Point	Layer	Action
Point 1	Points	X
Point 2	Ground	X
Point 3	Ground	X

3.1.30 EMF

Click this button  to open EMF tool.

Used for calculating the electromagnetic field, or non-ionising radiation for regulatory purposes.

EMF

Calculation settings

Calculation name	EMF prediction
Resolution	10 m
Cell template	none ▼
Reflection factor	No reflection ▼
Receiver height	1.5 m

Use the selection tool to select cells

Calculation name

Name of the calculation that will be displayed in the Prediction history.

Resolution

Prediction raster cell size in meters.

Cell template

If the cell is missing the required parameters, the parameters from the template will be used.

Reflection factor

- No reflection – ground reflections aren't considered
- Realistic reflection – reflection coefficient of 2.56
- Full reflection - reflection coefficient of 4

Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

Results:

- EMF Limit usage, % - weighted percentage of frequency band regulatory EMF limit.
- EMF, W/m² – EMF value in mW/m²
- EMF, mV/m – EMF value in mV/m

3.1.31 Audibility

Click this button  to open Audibility tool.

Audibility

Calculation settings

Calculation name	Audibility
Resolution	10 m
Siren template	none ▼

Use the selection tool to select sirens

Calculation name

Name of the calculation that will be displayed in Prediction history.

Resolution

Prediction raster cell size in meters.

Siren Template

Option to use the parameters from the template if the siren misses the required parameters.

Results:

- **Audibility, dB.** The siren sound prediction tool calculates the sound pressure level (SPL) at various locations around a siren using **ISO 9613-2** as the standard for sound propagation. The results are represented in **decibels (dB)**, showing how loud the siren is at different distances and conditions. Each location's sound pressure level is affected by several factors:
 1. **Distance from the Source (Siren)**
 - SPL decreases as the distance from the siren increases due to **geometrical spreading**.
 - The tool provides results that help visualize how sound intensity reduces at different locations.
 2. **Obstacles and Terrain**
 - **Buildings, hills, and other obstacles** can block or reflect sound, reducing the SPL.
 - Flat, open areas will have a higher SPL compared to urban environments with multiple obstacles.
 3. **Atmospheric Conditions**
 - Atmospheric factors like temperature, humidity, and wind influence sound propagation.
 - These effects are taken into account according to ISO 9613-2.
 4. **Frequency-Dependent Attenuation**
 - Higher frequencies (treble) attenuate faster than lower frequencies (bass).
 - The tool might calculate the overall SPL or provide frequency-specific results.
- The **best server raster** shows the identification of siren generating the strongest sound value at each pixel.

3.1.32 Lux calculation

Click this button  to open Lux calculation tool.

Lux calculation

Calculation settings

Calculation name	Lux calculation
Resolution	1 m
Light template	10m Pole ▼

Calculation name

Name of the calculation that will be displayed in the Prediction history.

Resolution

Prediction raster cell size in meters.

Light template

Option to use the parameters from the template if the light misses the required parameters.

Results:

- **Illuminance, centilux (lx * 100)**

3.1.33 Geoclimatic data

Click this button  to open Geoclimatic data tool.

Geoclimatic Data is a tool that lets you adjust the geoclimatic settings that will be used in certain calculations (e.g. Link Prediction).

Geoclimatic data

Gaseous absorption

Temperature

Multipath fading

Rain fading

Statistics

3.1.33.1 Gaseous Absorption

Gaseous absorption define values for dry air pressure and water vapour density. These values can be obtained from predefined geoclimatic data maps. Water vapour density data according to ITU-R P.836-3. It is used in gaseous absorption evaluation.

Gaseous absorption

Dry air pressure

1013

hPa

Water vapour density

7.5

g/m3

Use geoclimatic data

☒

Water vapour density data

SURFWV

Dry Air Pressure

The atmospheric pressure contributed by air that contains no water vapor, typically measured in hectopascals (hPa).

Water Vapour Density

The mass of water vapor present in a unit volume of air, typically measured in grams per cubic meter (g/m³).

Use Geoclimatic Data

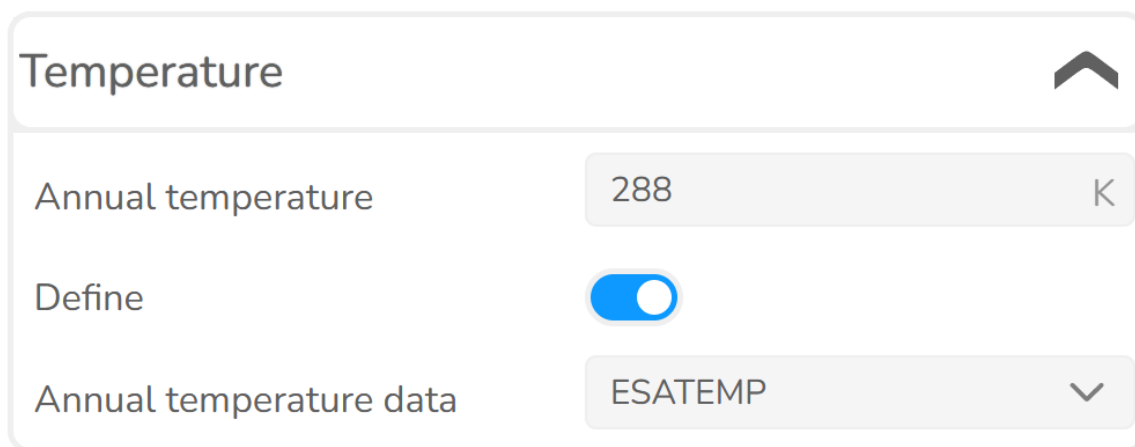
Enable/Disable this Geoclimatic data

Water Vapour Density Data

A dropdown menu that allows the user to select the source or type of data used to determine water vapor density.

3.1.33.2 Temperature

Annual mean surface temperature at 2m above the surface of the Earth according to ITU-R P.1510. The data is used to evaluate the thermal noise of a receiver.

A configuration panel titled "Temperature" with a collapse icon in the top right. It contains three settings: "Annual temperature" set to "288" with a unit "K" dropdown; "Define" with a blue toggle switch turned on; and "Annual temperature data" with a dropdown menu showing "ESATEMP" and a downward arrow.

Temperature	
Annual temperature	288 K
Define	☒
Annual temperature data	ESATEMP

Annual Temperature

Annual Temperature expressed in Kelvins.

Use Geoclimatic Data

Enable/Disable this Geoclimatic data

Annual Temperature Data

A dropdown menu allowing the user to select the dataset from which the annual temperature data should be sourced.

3.1.33.3 Multipath Fading

The Multipath Fading page defines refractivity data and calculation parameters for ITU and Vigants-Barnett methods.

Worst month-to-annual statistics conversion can be performed according to ITU-R P. 530 or ITU-R P. 841 methods.

Refractivity gradient data is based on ITU-R P.453-9. Refractivity gradient data is used for multipath fading analysis.

Multipath fading

ITU method

Use ITU method

ITU-R P.530 version

10

Geoclimatic data use

Use geoclimatic data

Geoclimatic factor

Define value manually

Refractivity gradient not exceeded 1% of a year

-500

N-unit/km

Approximation of terrain roughness

Linear

Percentage of time refractivity gradient in the lowest 100 m is less than units/km

10

C0

1.7

dB

CLat

0

dB

CLon

3

dB

Design

Quick planning

Percentage of time

Small percentages of time

Vigants-Barnett method

Use Vigants-Barnett method

Geoclimatic factor (propagation conditions)

0.5 - good

Worst month conversion

Method

ITU-R P.841

Interf. correction factor for multipath and focusing effects.

0.01

ITU Method
Enable/Disable the ITU Method

ITU-R P.530 version
A dropdown menu allowing the user to select the ITU-R P.530 version.

Use Geoclimatic Data

Enable/Disable this Geoclimatic data.

Geoclimatic Factor (Define value manually)

A field to input or calculate a factor used in propagation modeling, affecting the prediction of multipath fading.

Geoclimatic Factor (Calculate)

Calculate the geoclimatic factor used in propagation modeling, affecting the prediction of multipath fading.

- **Approximation of terrain roughness** - Options to select how terrain roughness is modeled in the calculations - as a constant value or as a linear approximation.
- **Percentage of time that refractivity gradient in the lowest 100 m is less than 100 units/km** - An input field for the percentage of time when the refractivity gradient at low altitudes meets a specific threshold.
- **Co, dB / Lat, dB / Lon, dB** - Fields to input correction values in decibels for specific geographic coordinates, possibly related to the orientation or position of the transmitter/receiver.

Percentage of Time / Design

A dropdown menus to choose the granularity of the time percentage for which the calculations are relevant, and the level of detail required for planning.

Use Viggants-Barnett Method

Enabled/Disabled Viggants-Barnett Method.

Worst month conversion

A section to select the ITU recommendation (either ITU-R P.530 or ITU-R P.841) for converting worst-month statistics to annual statistics.

Interf. correction factor for multipath and focusing effects

Input for the percentage of time to apply an interference correction factor to account for multipath and focusing effects.

3.1.33.4 Rain Fading

The rain fading page defines rain regions (ITU and Crane) for rain rate statistics and calculation methods (ITU and Crane). The rain rate exceedance parameter for 0.01% of the time can be set manually or automatically according to the rain zone.

Rain fading

Rain fall data

Geoclimatic data use

Use geoclimatic data ☒

ESA rain rate data ESARAIN_V5

Fading method

ITU method

Use ITU method ☒

ITU-R P.530 version none

Crane method

Use Crane method ☒

End-to-end reliability

Multi-hop Accumulation

Two-way Averaging

Geoclimatic data - ESA rain rate data / ESA rain rate

Options to select the dataset (such as "ESARAIN_V5") and the specific field within that dataset (like "MT") for rain rate data, used for calculating rain attenuation.

Fading method - ITU method / ITU-R P.530 version

A checkbox to select the ITU method for rain fading calculations and a dropdown to choose the version of the ITU-R P.530 recommendation being applied.

Crane

This is a model selection option related to the Crane model for calculating rain attenuation.

End-to-end reliability method - Multi-hop / Two-way

Selection options for the method of calculating the reliability of a signal in a communication path that may involve multiple hops or two-way transmission.

3.1.33.5 Statistics

These settings are used to handle statistical conversions for telecommunication planning, ensuring that systems are designed to cope with the worst-case scenarios based on historical data and predictive models.

Statistics

ITU-R P.841 conversion

$P_w = Q * P$
P - average annual time percentage of excess
P_w - average annual worst-month time percentage of excess
Q - conversion factor

Conversion factor

Define

☐

Q1 (1...12)

2.85

Beta

0.13

Define

Options to either define a fixed conversion factor or to calculate it based on additional inputs.

Q1 (1...12)

An input field likely used to define a conversion factor or another parameter for a specific month or set of months.

Beta

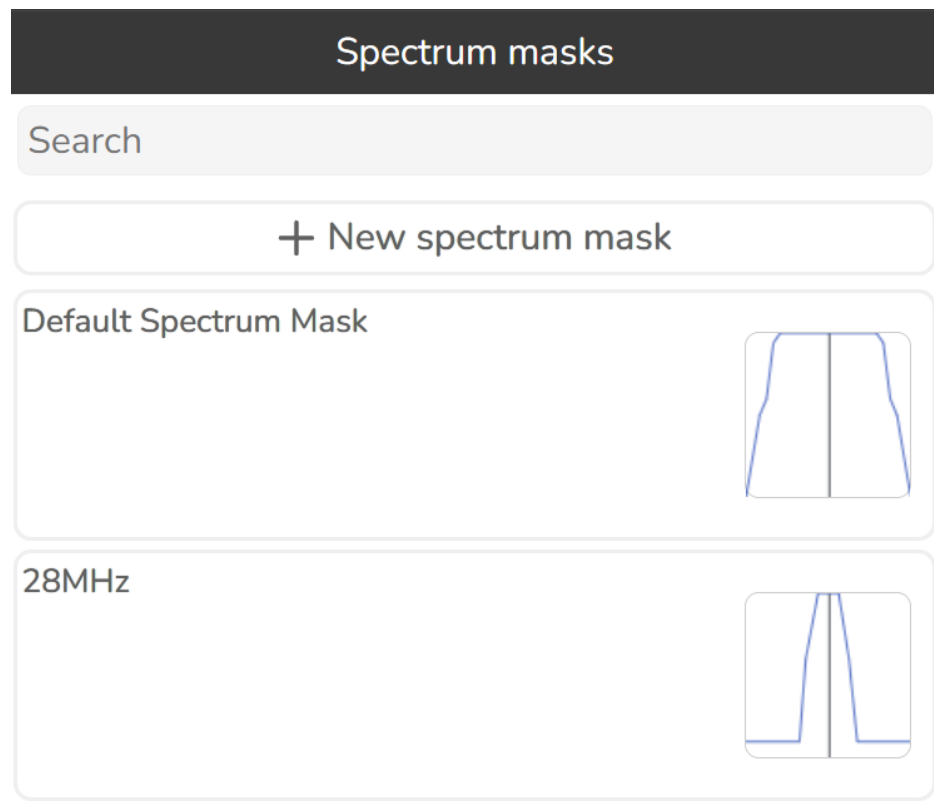
An input field for the beta parameter, which may be part of the statistical model or conversion formula within the ITU-R P.841 recommendation.

3.1.34 Spectrum masks



Click this button to open Spectrum masks tool.

The tool enables you to create spectrum masks that will be necessary to be used in calculations.



Move the mouse cursor over the spectrum mask and delete it using *Delete* button.



By selecting a spectrum mask you can view its properties and change them. Changes in the mask point values will be reflected in the graph.

3.1.34.1 Add spectrum mask

To create a spectrum mask, press the *New spectrum mask* button.

Spectrum masks

Mask attributes

Name New mask

+ Add point

Spectrum mask

Attenuation, dB

Mask attributes**Mask Name**

Spectrum mask identification.

Add point

Add spectrum mask point (frequency and attenuation).

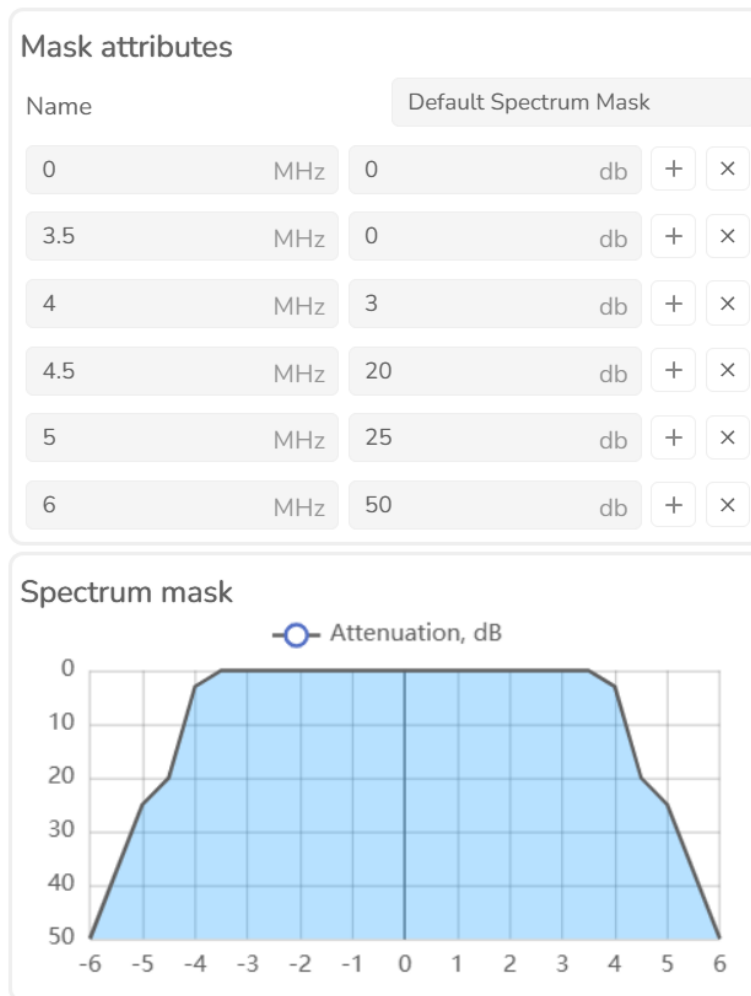
Adds a new mask point to the spectrum mask



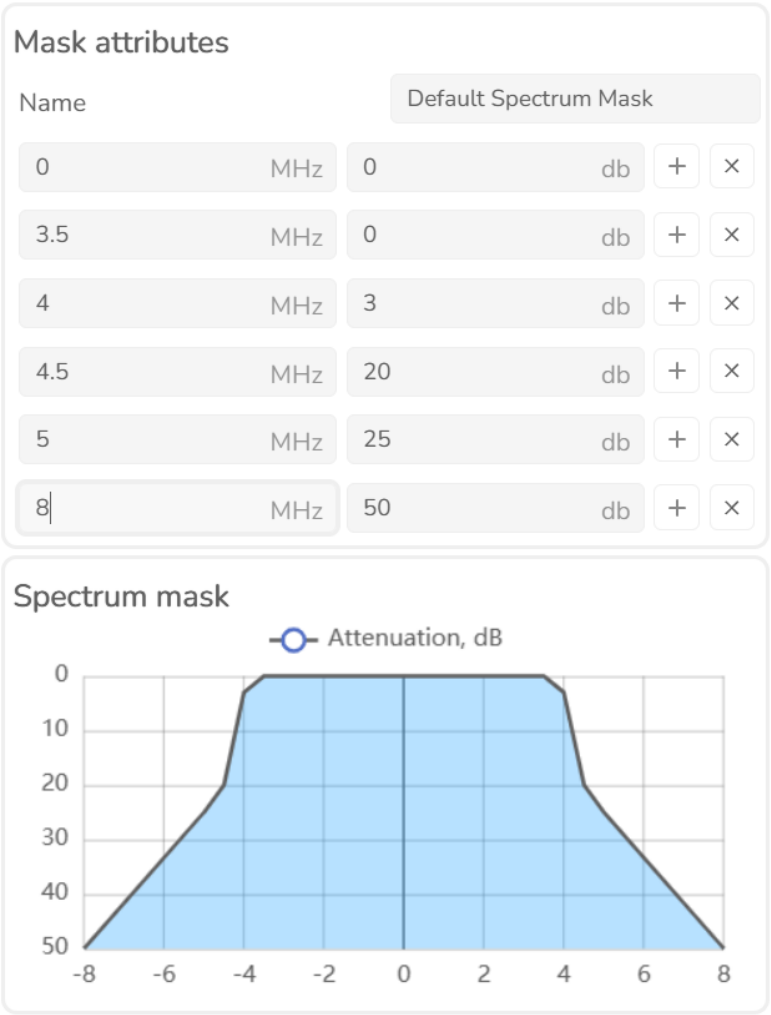
Deletes mask point.

**Spectrum mask**

When you create a spectrum mask, you can see its visual representation as well as the values of each spectrum mask point (frequency and attenuation)



You can change any values of these points and those changes will be reflected in the graph.



3.1.35 Radios

Click this button  to open Radios tool.

The tool enables you to create and preview radios that will be necessary to create a link and later be used in calculations.

Radios

Search

+ New radio

Ceragon

Model

FibeAir 56MHz

ID

2

Power

20

Bandwidth

56

Model

FibeAir 5G

ID

5

Power

0

Bandwidth

28

Default

Model

SHF

ID

25

Power

18

Bandwidth

10

Move the mouse cursor over the radio and delete radio using *Delete* button.

Ceragon

Model

FibeAir 56MHz

ID

2

Power

20

Bandwidth

56

By clicking an already existing radio, you can edit its properties and save them.

Radios

General

Model

FibeAir 56MHz

Manufacturer

Ceragon

Bandwidth, MHz

56

MTBF, h

300000

MTTR, h

24

Dispersive Fade Margin

50

Emission Designator

56M0FMK

Receiver

BER 10-3 Threshold, dBm

-83

BER 10-6 Threshold, dBm

-80

Noise Floor, dB

-96.5

Noise Figure, dB

3.6

Receiver Noise, dBm

-92.9

Residual BER

0

XPIF, dB

10

Accept

Cancel

3.1.35.1 Add new radio

To create a radio press the *New radio* button.

General

Model

Manufacturer

Bandwidth, MHz

MTBF, h

MTTR, h

Dispersive Fade Margin

Emission Designator

General

Model

Radio identification or name.

Manufacturer

Radio manufacturer (company or entity).

Bandwidth, MHz

Range of frequencies within which the radio operates, value in MHz.

MTBF, h

Mean time between failures: average time (in hours) between successive failures of the radio system during normal operation, indicating reliability.

MTTR, h

Mean time to repair: average time (in hours) required to repair the radio system after a failure occurs, indicating maintainability.

Dispersive Fade Margin, dB

Additional signal strength required to overcome dispersion and fading effects in the radio communication system, measured in dB.

Emission Designator

Code describing the characteristics of the radio emission, including modulation type, signal nature, and bandwidth.

Receiver

BER 10-3 Threshold, dBm

BER 10-6 Threshold, dBm

Noise Floor, dB

Noise Figure, dB

Receiver Noise, dBm

Residual BER

XPIF, dB

Manufacturer's guaranteed
minimum XPD, dB

Carrier-to-interference ratio for
reference BER, dB

Receiver

BER 10-3 Threshold, dBm

Receiver sensitivity threshold for BER = 10-6.

BER 10-6 Threshold, dBm

Receiver sensitivity threshold for BER = 10-3.

Noise Floor, dB

Refers to the minimum power level of unwanted noise or interference at the receiver at the base station.

Noise Figure, dB

Measure of the degradation of the signal-to-noise ratio (SNR) as a signal passes through a radio component or system. Value in dB. Required for 4G and 5G technologies.

Receiver Noise, dBm

The sum of Noise Floor and Noise Figure.

Residual BER

The remaining bit error rate after all error correction methods have been applied.

XPIF, dB

Cross-polarization interference factor: interference caused by signal components with orthogonal polarization states, indicating the impact of cross-polarization on system performance, expressed in decibels (dB). The default value is 20. The value should be higher than 0 for rain ITU fading calculations to work.

Manufacturer's guaranteed minimum XPD, dB

The manufacturer's guaranteed minimum cross-polar discrimination, expressed in decibels (dB). The default value is 30.

Carrier-to-interference ratio for a reference BER, dB

Carrier-to-interference ratio for a reference bit error rate, expressed in decibels (dB). The default value is 35.

Transmitter

Power, dBm

ATPC Range, dB

Transmitter**Power, dBm**

Power value in dBm.

Automatic Transfer Power Control Range (ATPC), dB

The range over which the radio can automatically adjust its transmission power, measured in dB.

Link modulations

Add modulation

none

BPSK

Name

BPSK

Sensitivity

-94

SNR

4

Downlink throughput

0

Mbps

The modulations are used in MW link calculations and specifically can be defined for Radios.

Modulations

Add modulation

A new modulation with default values can be initialized.

3.1.36 Frequency plans



Click this button to open Frequency plans tool.

The tool enables you to create frequency plans that will be necessary to create a link and later be used in calculations.

Frequency Plans

Search

+ New frequency plan

FR 10GHz

FR 15GHz

FR 3.5GHz Simplex

SHF

Move the mouse cursor over the frequency plan and delete it using *Delete* button.

+ New frequency plan

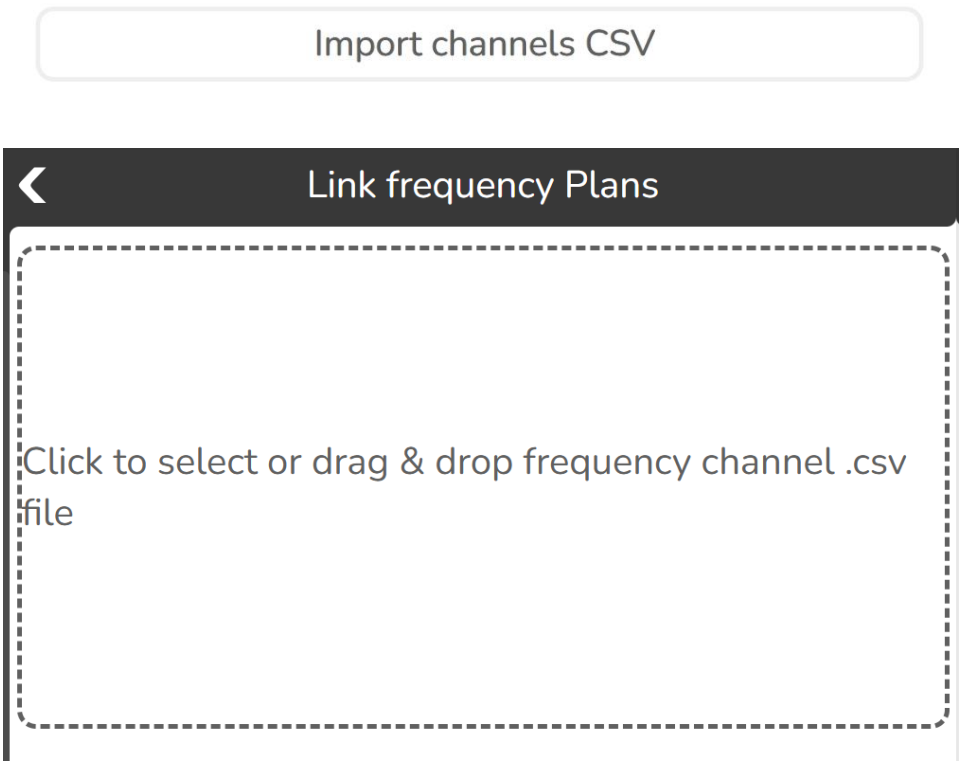
FR 10GHz

3.1.36.1 Import frequency plan

To import a frequency plan press the *New frequency plan* button

+ New frequency plan

and then select *Import channels CVS*.



To start the import procedure, select or drag and drop the frequency channel csv file. Here is an example:

	A	B
1	carrierID	frequency
2	1	4400
3	2	4420
4	3	4440
5	4	4460
6	5	4480
7	1'	4800
8	2'	4820
9	3'	4840
10	4'	4860
11	5'	4880

3.1.36.2 Add frequency plan manually

To add manually a frequency plan press the *New frequency plan* button

+ New frequency plan

and then select *Create manually*.

Create manually

Frequency Plans

General

Name

Duplex ☒

Generate channels

Low frequency MHz

High frequency MHz

Carrier spacing MHz

Duplex spacing MHz

Index of first channel

Generate channels

Channels

+ Add channel

Accept Cancel

General

Name

Frequency Plan identification

Duplex

Specifies whether the signal can be transmitted and received simultaneously. Changing the duplex mode will affect the carrier selection.

Generate channels

Low/high frequency

Frequencies in MHz.

Carrier spacing

The frequency separation between adjacent carrier frequencies in a communication system, ensuring non-interference between carriers.

Duplex spacing

Frequency separation between the transmit and receive bands in a two-way communication system.

Index of first channel

The numbering start point for generated channels.

Generate channels

Creates a list of channels based on the inputs.

Channels

Add channel

Allows manual addition of individual channels to the frequency plan.

Channels

1	10	MHz	+	×
1'	20	MHz	+	×
2	20	MHz	+	×
2'	30	MHz	+	×
3	30	MHz	+	×
3'	40	MHz	+	×

Channel

Channel identifier.

Frequency

Channel's frequency value in MHz.

Add channel using *Add channel below* button



Delete channel using *Delete channel* button



3.1.37 Link prediction

Click this button  to open Link prediction tool.

Link prediction

Calculation settings

Calculation name	Link prediction
Link template	Default 

Calculation settings

Calculation Name

Link Prediction identification.

Link Template

The template will fill all empty or not specified fields with default values that are not necessary for predictions.

Interference

Calculate interference	☐
Use interference radius	☐
Interference radius	1000
Use interference limits	☐
Interference creation limit	-1700
Interference analysis limit	-1500
Tx/Rx filter discrimination	1000

Interference

Calculate Interference

If checked will calculate the interference of multiple links.

Use interference radius

If enabled, also allows to specify a certain radius from either link endpoint. Points outside this range will not be included in the calculation, speeding up the calculation process. If disabled all links will be used in the calculations.

Interference creation limit

The power threshold below which no interference for links will be calculated.

Interference analysis limit

The power threshold below which no links with lesser interference will be included in the analysis.

Tx/Rx filter discrimination

The ability of filters in a duplex system to effectively separate and prevent interference between Tx and Rx frequencies. No dBm greater than this will be accounted for in the calculation.

Field Margin

Include Field Margin in Power Budget ☐

Thermal Fade Margin Based on receiver level 10... ▼

Field Margin

Include Field Margin in Power Budget

If checked, it will take into account the specified field margin for Power Budget calculations.

Thermal Fade Margin

The threshold below which thermal fade will not be calculated

Tropospheric scatter

Calculate tropospheric scatter ☐

Troposcatter time percentage 99.99 ▼

Tropospheric scatter

Calculate Tropospheric Scatter

Enable/Disable this option.

Troposcatter time percentage

Time Percentage in %.

ESR multipath

Method

ITU

▼

Statistics

Worst-month

▼

SESR multipath

Method

Vigants-Barnett

▼

Statistics

Worst-month

▼

UATR multipath

Method

Crane

▼

Statistics

Annual

▼

Total multipath

Method

Vigants-Barnett

▼

Total rain

Method

Crane

▼

Method

The standard used to calculate the performance metric.

Statistics

A selection indicating if the metric is calculated based on yearly data or the data from the month with the poorest performance.

3.1.37.1 Link prediction results

After calculations, you will be able to view Profile, Interference, Performance and Capacity calculation results.

Results

Search

Links

link1 A->B

Carrier

(1') 2 MHz

Diversity improvement

1

Protection improvement

4166.333354476914

Profile

Interference

Performance

Capacity

Generate report

link1 B->A

link2 A->B

link2 B->A

Interfering Links

Carrier

Select different carriers to view the results for each one of them.

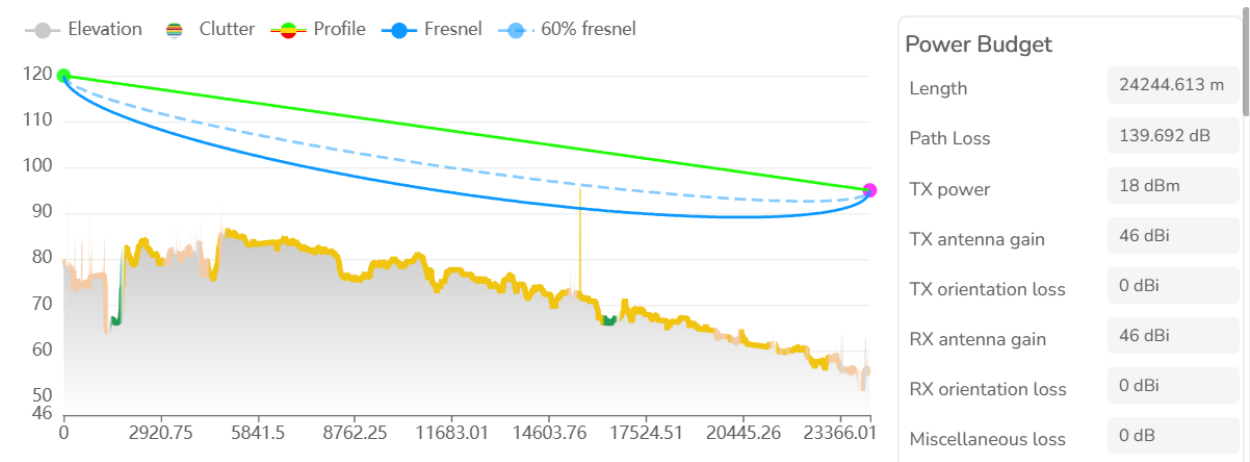
Diversity improvement

To alleviate the effect of multipath fading, various propagation diversity techniques are employed.

Protection improvement

The protection improvement factor is a ratio between unprotected and protected unavailability.

Profile - The Link Profile behaves in virtually the same way as a regular profile.



Power Budget - the calculation of the balance between transmitted power, power losses in the system, and receiver sensitivity in a communication system.

Path Loss - the reduction in signal strength as it travels from the transmitter to the receiver.

Interference - the undesired impact of one signal on another, leading to potential signal degradation.

Link prediction results					
Interference from (interfering links)			Interference to (interfered links)		
Link name	Channel	Distance, km	Link name	Channel	Distance, km
Il kl1 - I Tragbar1	11000 (1')	17.57	X gr5 - Il kl3	11000 (1')	16.64
X gr5 - Il kl3	11000 (1')	33.06	Il kl1 - I Tragbar1	11000 (1')	31.81
XX gr1 - X gr5	11000 (1')	29.64	Il gr4 - Il kl2	11000 (1')	11.06
X gr5 - Il gr4	11000 (1')	19.98	X gr5 - Il kl2	11000 (1')	16.64
X gr5 - Il kl2	11000 (1')	24.05	Il gr4 - Il gr3	11000 (1')	11.06
Il gr3 - Il kl1	11000 (1')	11.9	Il gr3 - Il kl1	11000 (1')	21.5

Performance - assessments of key metrics like data rate, error rates, and signal-to-noise ratio, crucial for evaluating and optimizing a communication system's efficiency.

Link prediction results				
	Propagation reliability		Total reliability	
	Total Outage Annual	Total Outage W.Month	Multipath ITU Annual	Multipath
Unavailability, %	0.00175	0.00703	0.00000	0.00000
Availability, %	99.99825	99.99297	100.00000	100.0000
Objective, %	99.99800		99.99900	
Unavailability time	0:09:13	0:03:08	0:00:00	0:00:00
Availability time	8765:50:47	743:56:52	8766:00:00	744:00:00

3.1.37.2 Interfering links

In this tab, you will be able to view interference Power Budget, Path Loss, Profile, and Spectrum Mask results in an unordered fashion.

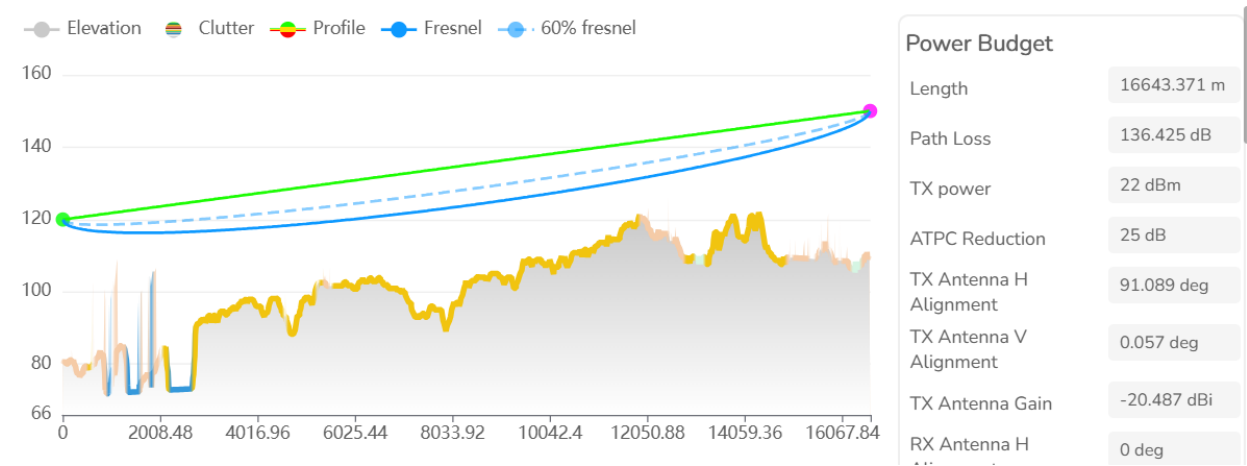
Interfering Links

link1 (A) 2 MHz -> link2 (B) 2 MHz

Profile

Spectrum mask

Profile - The Link Profile behaves in virtually the same way as a regular profile.

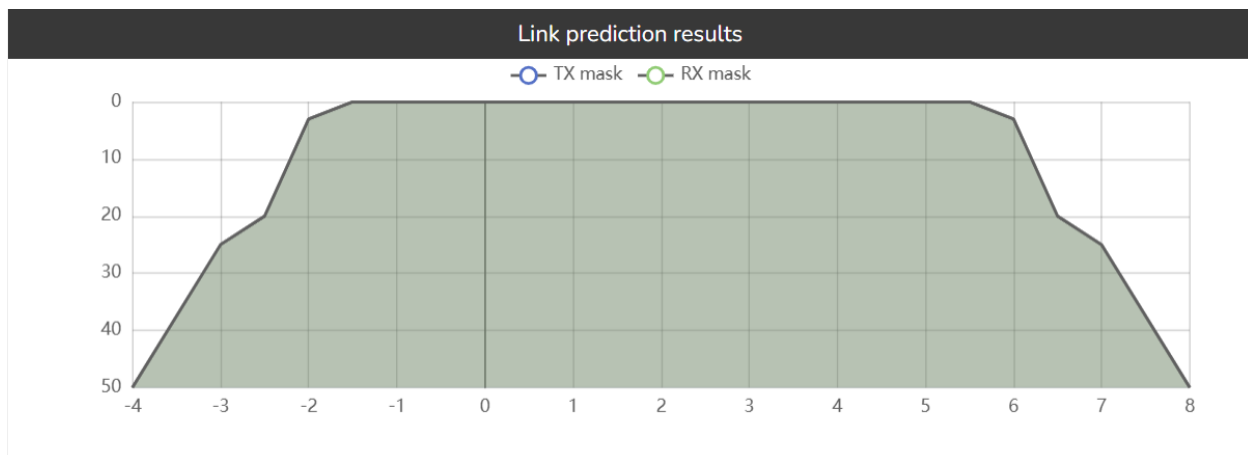


Power Budget - the calculation of the balance between transmitted power, power losses in the system,

and receiver sensitivity in a communication system.

Path Loss - the reduction in signal strength as it travels from the transmitter to the receiver.

Spectrum Mask - the method used to assess the usage efficiency of a frequency spectrum in a telecommunications system, involving the measurement of how much and how effectively different frequencies are being utilized.



3.1.37.3 Generate report

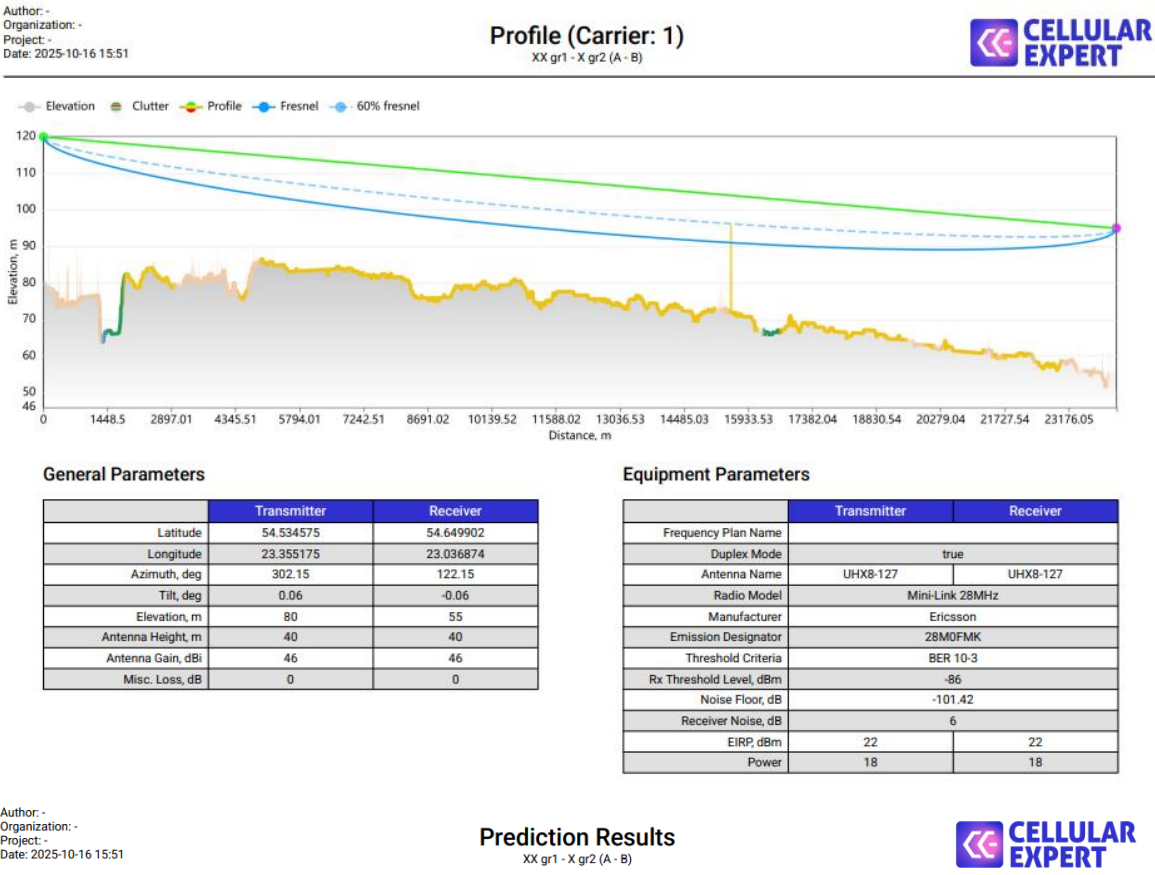
The calculation results can be automatically transferred into a Link Prediction Report. This report will show profile, prediction parameter and results, performance and propagation reliability. The report can be exported in PDF format.

The "Profile report" dialog box contains the following settings:

- Language: English (dropdown menu)
- Landscape orientation: ☒ (toggle)
- Author: (text field)
- Organization: (text field)
- Project: (text field)
- Calculations page: ☒ (toggle)
- Map page: ☒ (toggle)
- Table styling: ☒ (toggle)

At the bottom, there are two buttons: "Accept" and "Cancel".

Report example:



Author: -
Organization: -
Project: -
Date: 2025-10-16 15:51

Prediction Results
XX gr1 - X gr2 (A - B)



Total Path Loss

Frequency, MHz	10000
Carrier ID	1
K Factor	1.33
Polarization	Horizontal
Path Length, m	24244.61
Clearance, m	-4.91
Clearance, %	62.01
Clearance Distance, m	15553.6
Visibility Clearance, m	8.01
OLOS, m	24244.61
NLOS, m	24244.61
Path Loss, dB	139.69

Geoclimatic Calculations

ITU Climatic Factor	2.68e-5
ITU C0 Factor	1.7
V Barnett Climatic Factor	0.5
Fade Occurrence factor, %	1.87e+0
Refractivity grad., N-units/km	-152.34
Refractivity Gradient, %	10
Rain Rate R0.01, mm/h	32.36

Power Budget

Field Margin, dB	0
Received Signal Level, dBm	-29.69
Thermal Fade Margin, dB	56.31
Composite Fade Margin, dB	48.34
Dispersive Fade Margin, dB	50
Signal to Noise Ratio, dB	65.73
Bit Error Rate	0.00e+0
Total Interference, dBm	0
Actual Tx Antenna Gain, dBi	46
Tx Orientation Loss, dBi	0
Actual Rx Antenna Gain, dBi	46
Rx Orientation Loss, dBi	0

Author: -
Organization: -
Project: -
Date: 2025-10-16 15:51

Performance (Carrier: 1)

XX gr1 - X gr2 (A - B)



Propagation Reliability

	Multipath ITU Annual	Multipath ITU W.Month	Multipath V-Barnett Annual	Multipath V-Barnett W.Month	Multipath XPD W.Month	Rain Crane Annual	Rain Crane W.Month	Rain ITU Annual	Rain ITU W.Month	Rain XPD W.Month	Total Outage Annual	Total Outage Worst
Unavailability, %	0.00000	0.00000	0.00002	0.00011	0.00002	0.00000	0.00000	0.00175	0.00701	0.00000	0.00175	0.00703
Availability, %	100.00000	100.00000	99.99998	99.99989	99.99998	100.00000	100.00000	99.99825	99.99299	100.00000	99.99825	99.99297
Objective, %	99.99900		99.99900			99.99900		99.99900			99.99800	
Unavailability Time	0:00:00	0:00:00	0:00:05	0:00:03	0:00:01	0:00:00	0:00:00	0:09:13	0:03:08	0:00:00	0:09:13	0:03:08
Availability Time	8766:00:00	744:00:00	8765:59:55	743:59:57	743:59:59	8766:00:00	744:00:00	8765:50:47	743:56:52	744:00:00	8765:50:47	743:56:52
Objective Time	8765:54:44		8765:54:44			8765:54:44		8765:54:44			0:00:00	

Total Reliability

	Equipment Protected	Equipment Unprotected	Propagation Protected	Propagation Unprotected	Total Protected	Total Unprotected
Unavailability, %	0.00000	0.00800	0.00177	0.00177	0.00177	0.00977
Availability, %	100.00000	99.99200	99.99823	99.99823	99.99823	99.99023
Objective, %	99.99900	99.99900	99.99800	99.99800	99.99700	99.99700
Unavailability Time	0:00:00	0:42:04	0:09:18	0:09:18	0:09:18	0:51:23
Availability Time	8766:00:00	8765:17:56	8765:50:42	8765:50:42	8765:50:42	8765:08:37
Objective Time	8765:54:44	8765:54:44	8765:49:29	8765:49:29	8765:44:13	8765:44:13

3.1.38 Automatic frequency planning



Click this button to open Automatic frequency planning tool.

The automatic frequency planning tool for each selected link suggests the best channel from the chosen frequency plan and the selected channel list. It considers inter-link interference, evaluates signal-to-interference ratio (SIR) across available carriers, and aims to maximize link performance while minimizing co-channel interference. The tool accounts for duplex link pairing, node-layer priority, and site location constraints (e.g., neighboring links and links sharing the same site).

Automatic frequency planning

Automatic frequency planning

Calculation name Automatic frequency planning

Link template Default

Frequency plan

Frequency plan none

Recalculate already assigned frequencies ☐

Use the selection tool to select links

Automatic frequency planning

Calculation name

Name of the calculation that will be displayed in Prediction history.

Link template

Template to be used for filling up missing parameters of the link.

Frequency plan

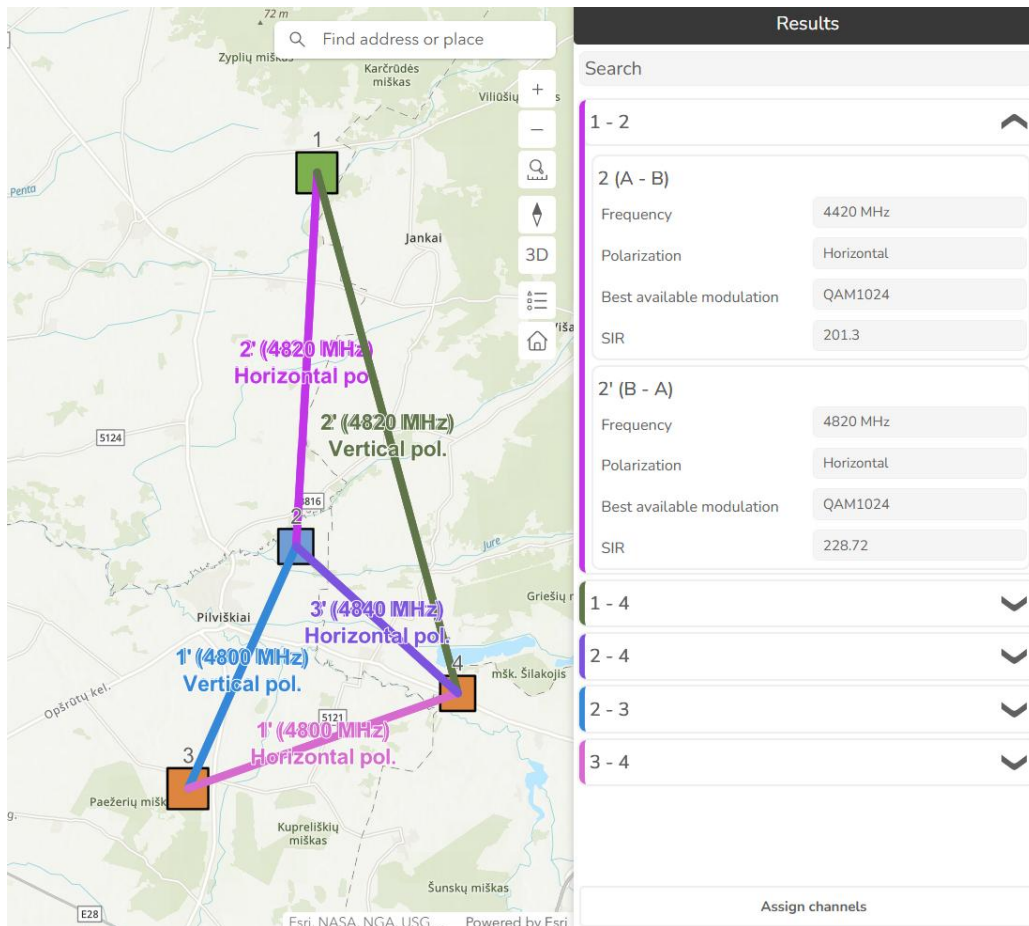
Frequency plan

Frequency plan that is applied to the links.

Recalculate already planned frequencies

Where applicable, the tool respects existing channel assignments. If frequency recalculation is explicitly enabled, i.e., the Recalculate already planned frequencies option is checked, the frequencies are reassigned for all links regardless of whether they have channel assignments.

The resulting plan includes the assigned frequency, polarization, best available modulation and SIR score.



Assign channels

Assign suggested channel & polarization values to link objects.

3.1.39 Link HCM-FS prediction

Click this button  to open Link HCM-FS prediction tool.

This calculates interference between selected links and HCM-FS records available in the database.

Calculation settings

Calculation name
HCM-FS calculations

Link template
Default

Calculation settings

Calculation name

Name of the calculation.

Link template

Template to be used for filling up missing parameters of the link.

Trigger options

Limit TD	☒
TD, dB >	0.1
Limit Interference	☒
Interference level, dBW >	-400
Distance PTX to RX [km] <	1.99

Trigger options

Trigger options are used to limit the size of calculation report. If all trigger options are disabled, calculation report may become very huge since all calculation cases will be logged into the report.

Limit TD

If enabled, calculation case is logged into the report only if threshold degradation is greater than defined.

TD, dB >

Value of threshold degradation used as a trigger for logging calculation case into the report.

Limit Interference

If enabled, calculation case is logged into the report only if interference level is greater than defined. Can be used along with TD limit.

Interference level, dBW

Value of interference level used as a trigger for logging calculation case into the report.

Distance PTX to RX [km]

Distance between passive transmitter and receiver. In the case of passive receiver, calculation will be performed only if distance between passive transmitter and receiver is less than defined. Decision whether to include this calculation into the report will additionally depend on the rest of Trigger options.

3.1.39.1 Link HCM-FS prediction results

Results

Lazdijai

Maximum outgoing TD

0 dB

Maximum incomming TD

0 dB

Contains errors

No

Show Incoming interference

Show Outgoing interference

Report lists all the selected links and following parameters:

- **Maximum outgoing TD, dB.** Maximum interference (threshold degradation) to a single HCM-FS receiver caused by the link.
- **Maximum incoming TD, dB.** Maximum interference (threshold degradation) from HCM-FS transmitter to the receiver of the link.

Show Incoming interference

Show incoming interference.

Displays list of HCM-FS transmitters calculated against receivers of the link.

Link HCM-FS prediction results						Close
List of interfering transmitters						
Show profile	Site	Type	Link freq.	HCM station name	Frequency	
Profile		RX	38552.5	Lazdijai (B)	38552.5	
Profile		RX	38552.5	Lazdijai (B)	38552.5	
Profile		RX	38552.5	Lazdijai (B)	38552.5	

List shows site of the affected link, HCM-FS parameters and interference calculation result.

3.1.40 Mesh topology builder

The mesh topology builder tool optimizes the process of making a connection between 2 desired mesh nodes.

Mesh topology builder

Calculation settings

Resolution

10

m

Mesh node template

Mesh_template

▼

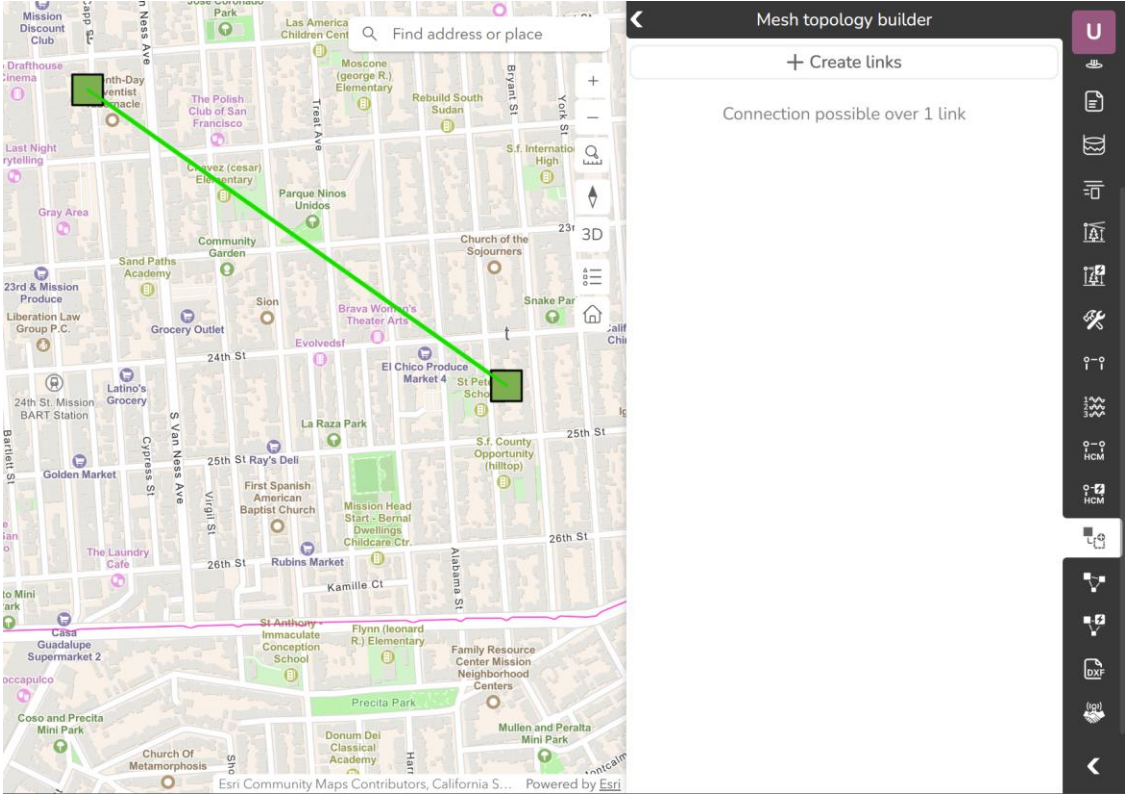
Link template

Link 23 GHz Duplex

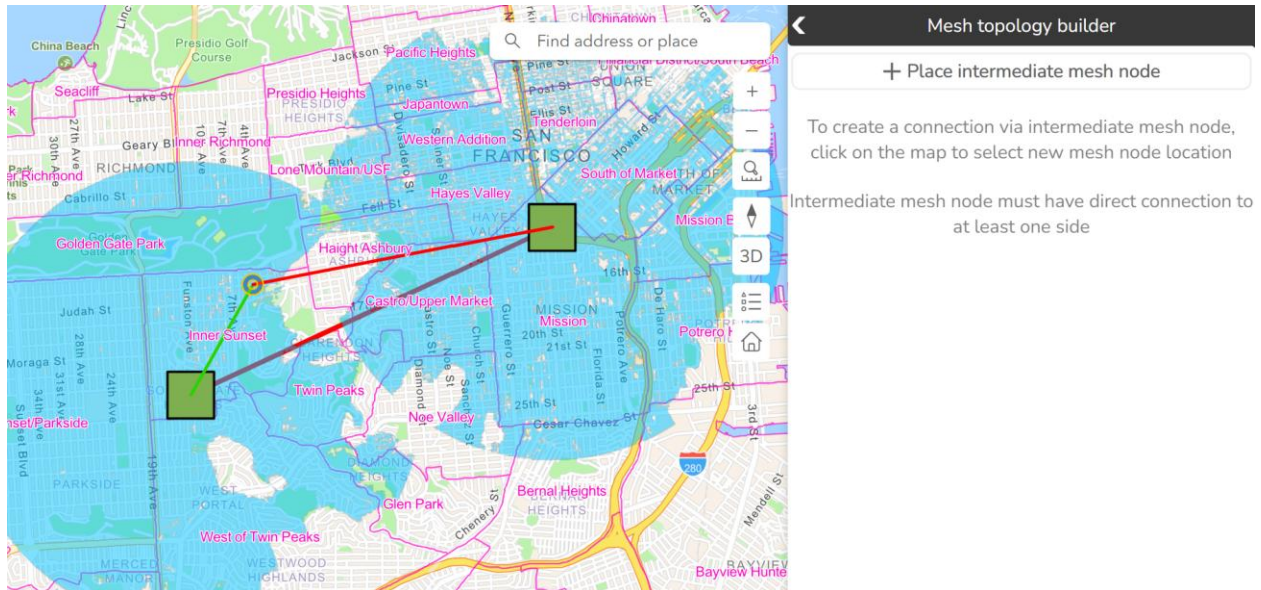
▼

Please select 2 Mesh Nodes you want to connect

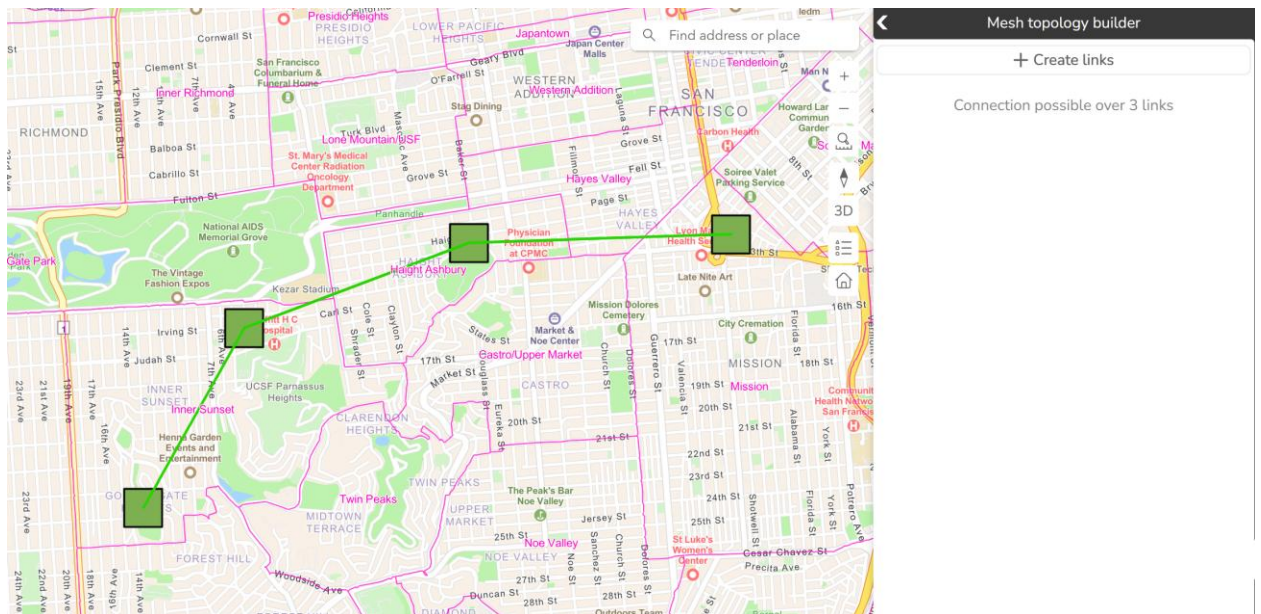
The tool first analyzes the possible link between the 2 selected nodes, and if a direct connection is possible, it prompts you to create the link object between them.:



If a direct connection is not possible, the tool will automatically run an optimal placement prediction between the 2 nodes and show you the resulting raster. You are then prompted to select a location for an intermediate mesh node with the help of the displayed optimal placement layer, and quick mesh connectivity lines for the location you choose. (SS)



Once a new intermediate node location is chosen, you can click *Place intermediate mesh node* to create it, at which point a new process is started between the new node and the unconnected link from it. If all connections are possible, the tool prompts you to *create links* for the multi-node path, at which point the originally selected mesh nodes are connected through a series of intermediate mesh nodes.



3.1.41 Mesh connectivity



Click this button to open Mesh connectivity tool.

This tool enables rapid and accurate assessment of wireless peer-to-peer link quality during mesh network planning and optimization.

Mesh connectivity

Calculation settings

Calculation name Mesh connectivity

Mesh node template Default

Use the selection tool to select mesh nodes

Calculation name

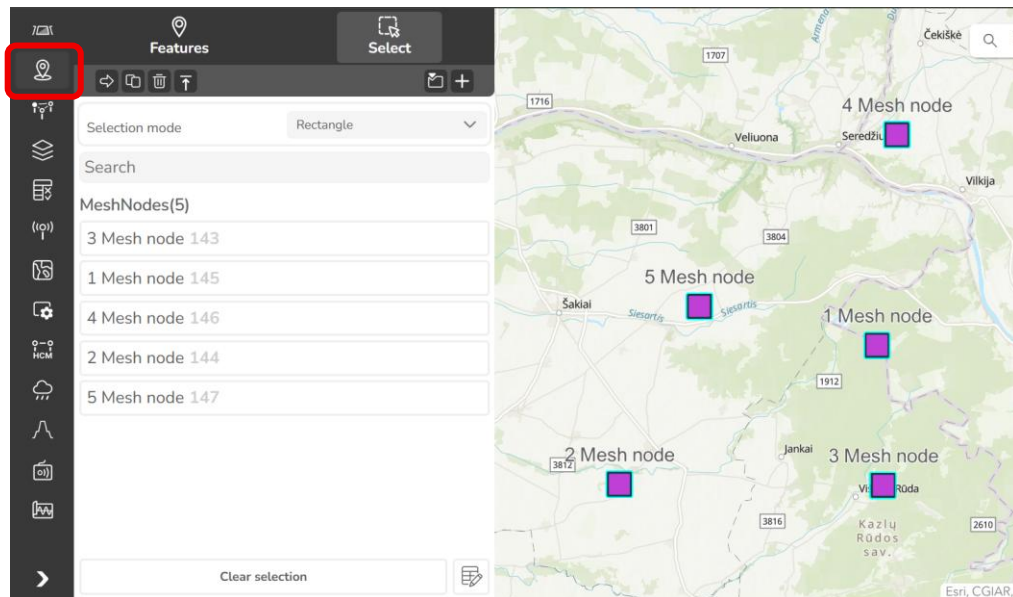
Name of the calculation that will be displayed in the Prediction history.

Mesh node template

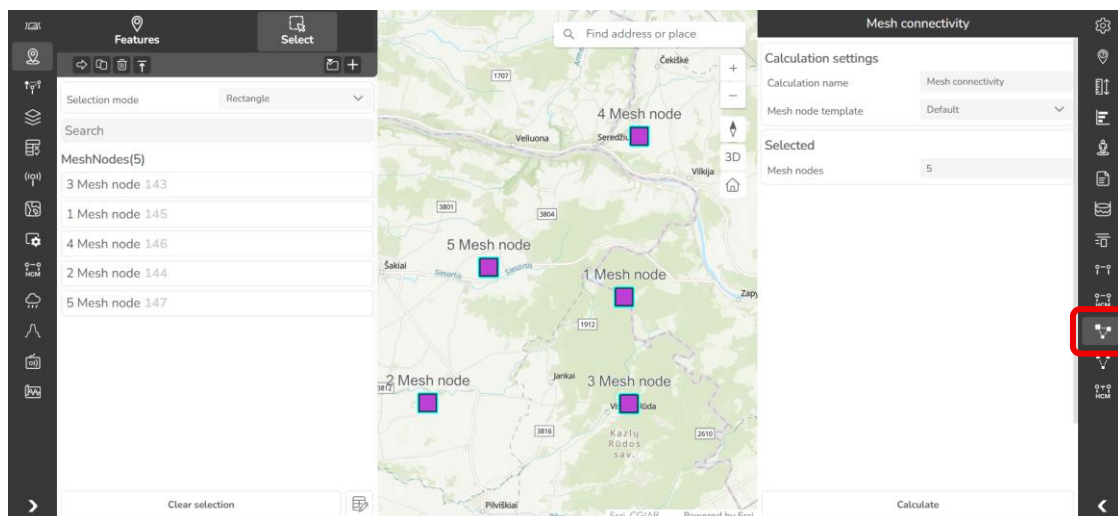
Mesh node template if selected features won't have all necessary parameters.

How It Works:

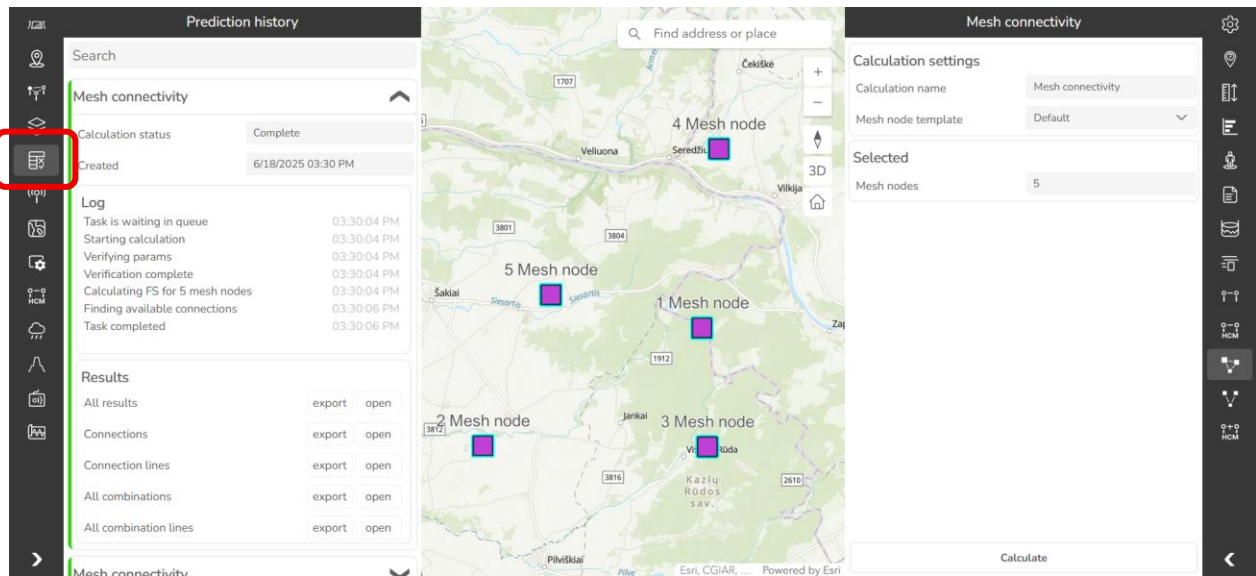
- Select one or more Mesh Nodes in the workspace.



- Open Mesh Connectivity tool.

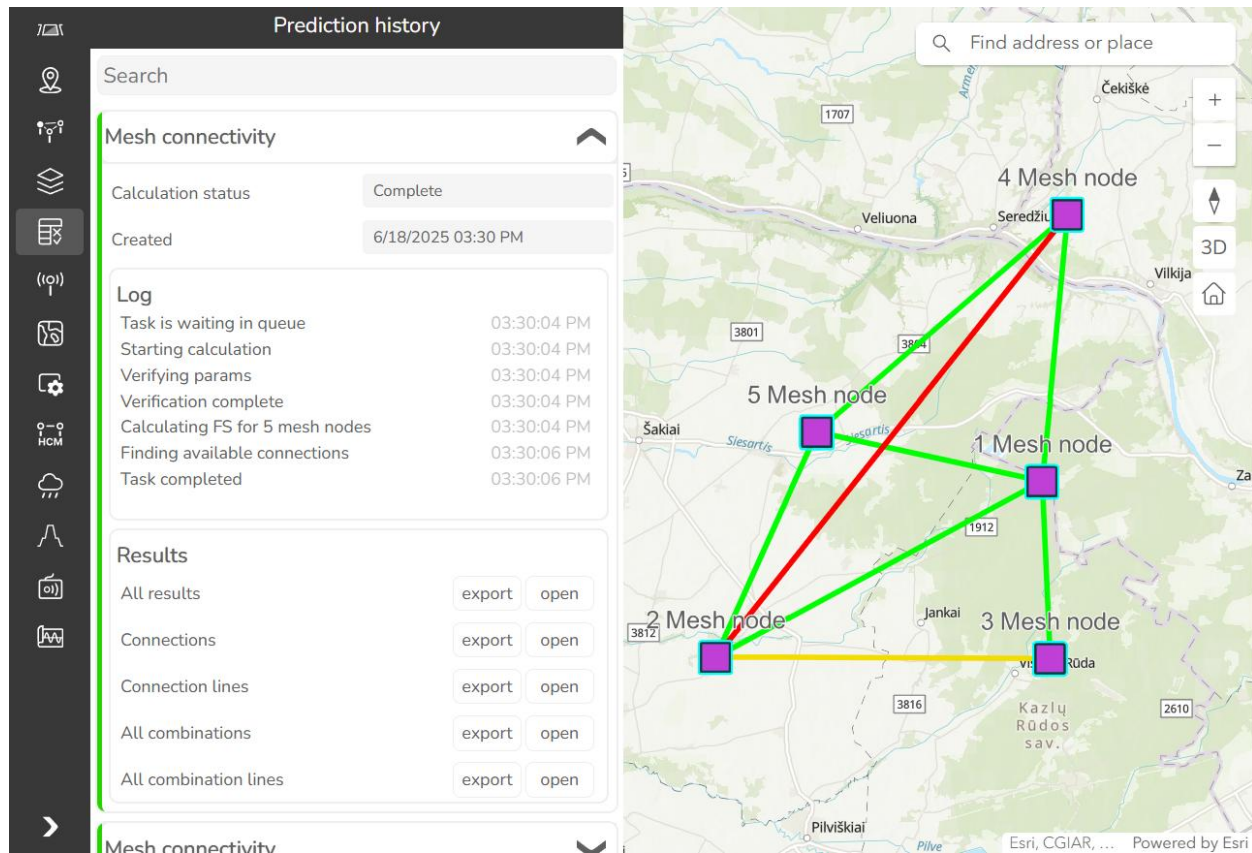


- Define:
 - Calculation name
 - Mesh node template
- Press Calculate button to start simulation.
- The tool calculates the RSL between each node pair and compares it against the configured RSL sensitivity threshold defined for the Mesh Node object.
- Open Prediction history tool.



Connectivity status is then visually rendered using a clear, color-coded scheme:

-  Green : Bi-directional connectivity – both nodes meet or exceed the RSL sensitivity for each other.
-  Yellow: Uni-directional connectivity – only one node meets the RSL threshold for the other.
-  Red: No connectivity – RSL falls below the sensitivity threshold in both directions



Results:

- **All results** – It shows all possible combinations of Mesh network connections.
- **Connections** – opens new dock on the right where you can create links.
- **Connection lines** – Shows Mesh connections where the tool will create Link object

To create a Links between Mesh Nodes, define Mesh connectivity results. Once result is selected, it will add possible options to create a links between Mesh Nodes

Results

Link creation

Link template

none

Delete existing links☒

Create only enabled links☒

Create links

Assigned connections

Search

☒ Mesh1 - Mesh2

☒ Mesh1 - Mesh5

☒ Mesh1 - Mesh4

Links creation

Delete existing links

If selected, existing links that have been created from mesh connectivity results will be deleted.

Create only enabled links

If checked, the links will be created only from the checked connectivity lines in the suggested list.

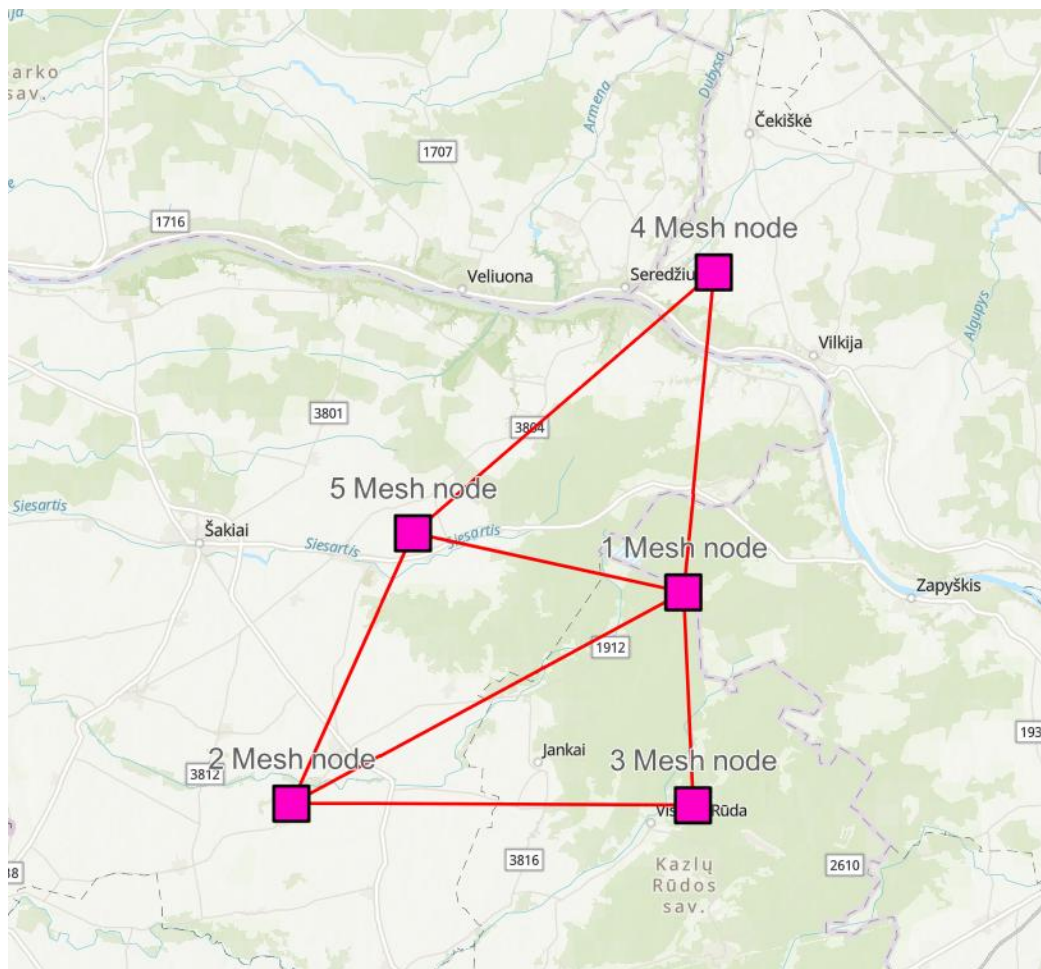
Enable/Disable Links from the suggested list

☒ Mesh1 - Mesh2

☒ Mesh1 - Mesh5

☒ Mesh1 - Mesh4

Once parameters are set, press Create links button to create Links between Mesh objects.



3.1.42 Quick mesh connectivity

Click this button  to open Quick mesh connectivity tool.

To enhance the efficiency of early-stage planning and dynamic mesh layout, the **Quick Mesh Connectivity** tool offers a fast and intuitive way to assess connectivity potential.

This lightweight yet powerful feature enables users to **instantly evaluate whether a Mesh Node can connect to an existing mesh network** based on either its current position or a proposed location.

Whether you are planning the deployment of a single node or simulating a mobile mesh scenario, **Quick Mesh Connectivity** streamlines the decision-making process by providing immediate, actionable feedback.

Quick mesh connectivity

Calculation settings

Mesh node template

Default

Temporary mesh node template

Default

Temporary mesh node

X

Y

Height

40

m

Antenna

Typical Omni TR 36 942

Prediction model

Default

Frequency

5000

MHz

Power

67

dBm

Misc. loss

5.5

dB

Sensitivity

-101

dBm

Calculation settings

Mesh node template

The Mesh Node Object Template is automatically utilized whenever a Mesh object lacks the required parameters needed to complete connectivity calculations. This ensures that all essential data is available for accurate simulation and evaluation, even if the original Mesh object is incomplete.

Temporary mesh node template

The **Temporary Mesh Node Object Template** is automatically utilized whenever a newly proposed Mesh object lacks the required parameters needed to complete connectivity calculations. This ensures that all essential data is available for accurate simulation and evaluation, even if the original Mesh object is incomplete.

Temporary mesh node

X

Coordinate in the projected coordinate system.

Y

Coordinate in the projected coordinate system.

Height

Mesh node object's height above the terrain.

Antenna

Antenna which will be used for Mesh node.

Prediction model

Prediction model used to calculate a path loss between Mesh Nodes.

Frequency

Frequency in MHz for proposed Mesh Node.

Power

Tx power in dBm for proposed Mesh Node.

Misc. Loss

Total losses in dB for proposed Mesh Node.

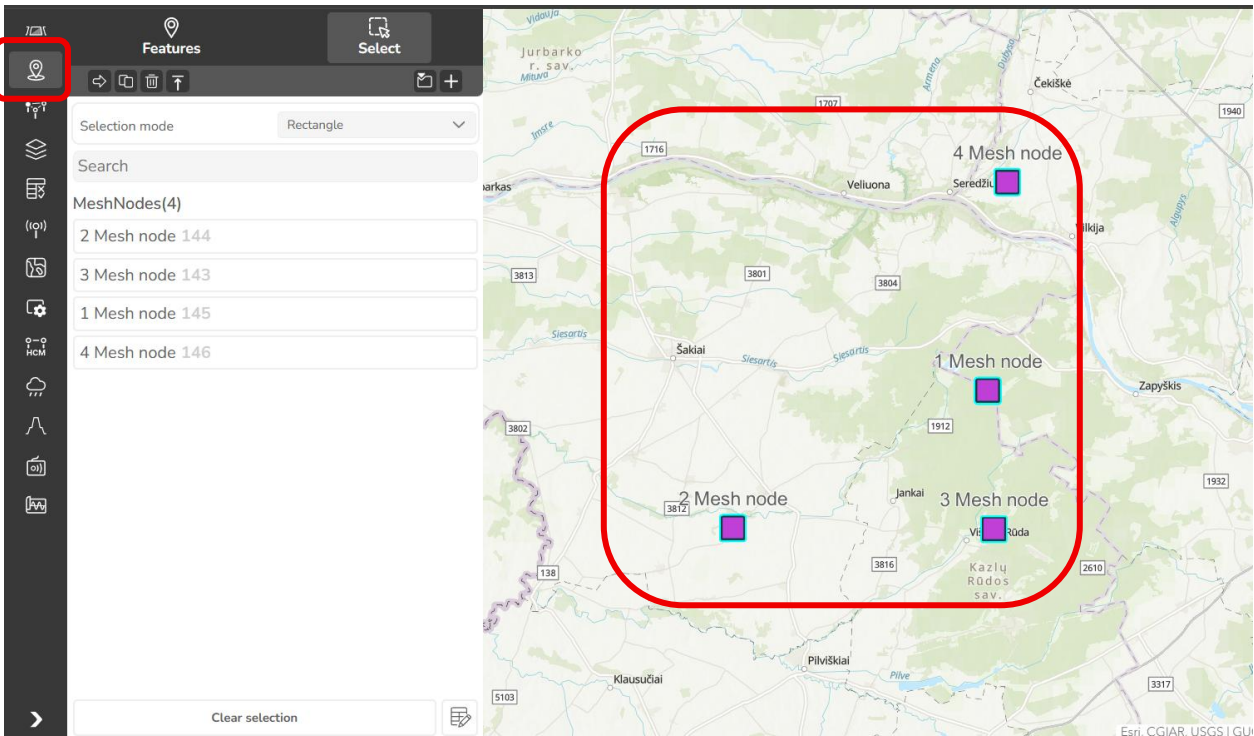
Sensitivity

Receiving Signal Level threshold in dBm at Mesh Node.

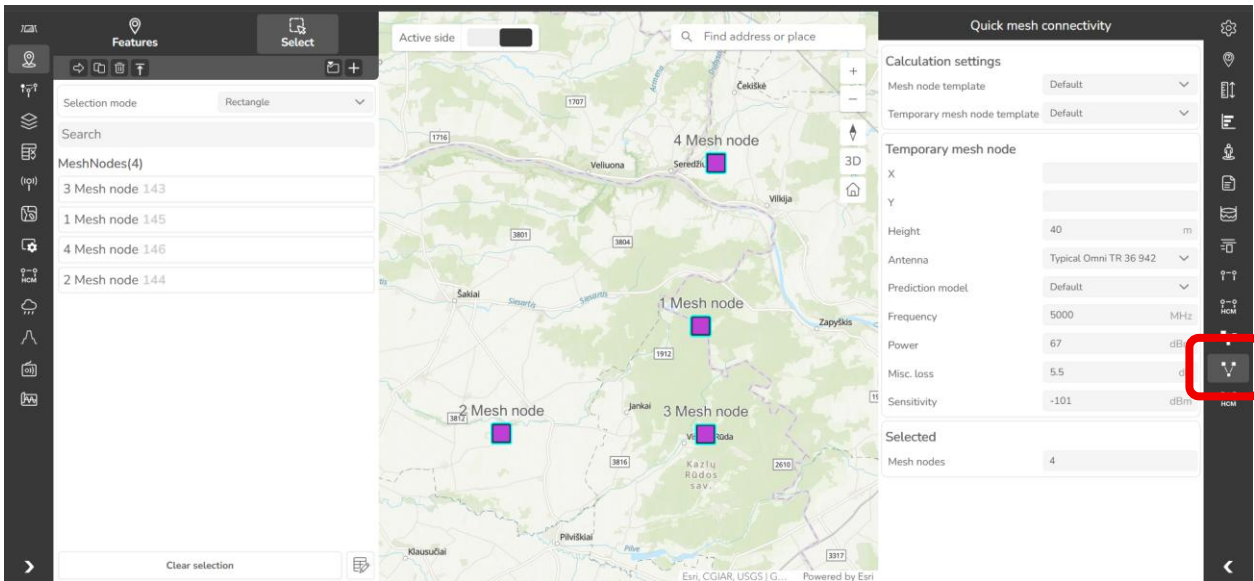
3.1.42.1 Calculate Quick mesh connectivity

- **Select Existing Mesh Nodes**

On the map interface, select the Mesh Nodes that should be included in the connectivity calculation.



- **Open the Quick Mesh Connectivity Tool**



- **Configure Node Parameters**
Input the required parameters for the proposed Mesh Node (e.g., transmission power, antenna type, height).

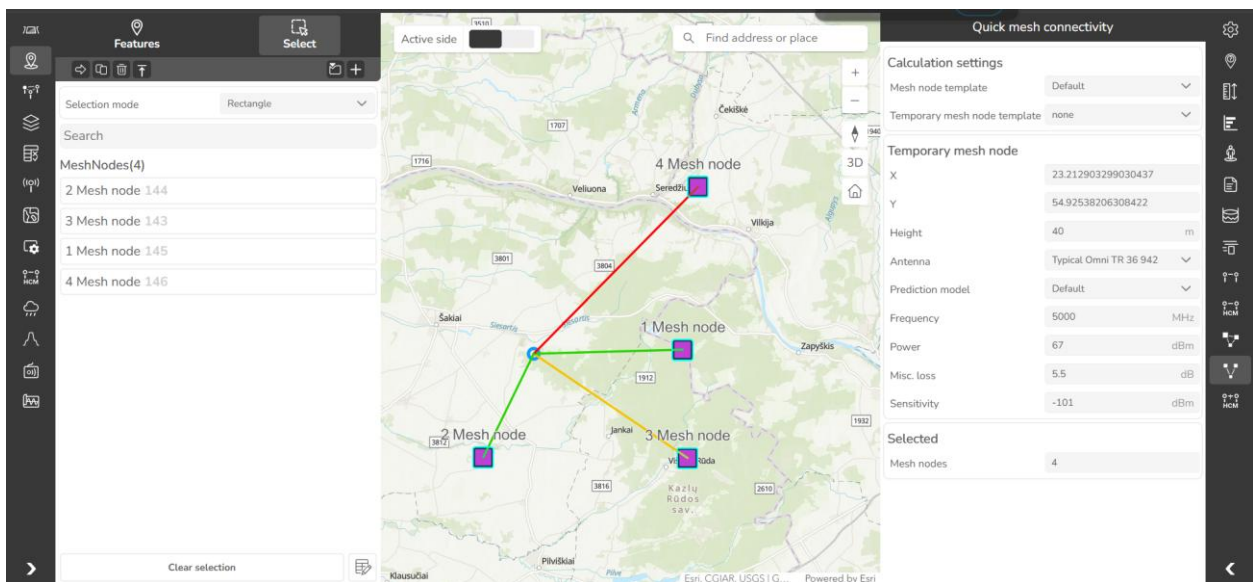
- **Define the Proposed Node Location**

Choose the location of the new Mesh Node either by clicking directly on the map or by entering specific coordinates.

- **View the Results**

Once the analysis is complete, the results will appear on the map:

- **Green** indicates areas with successful two-way connectivity.
- **Yellow** shows one-way connectivity.
- **Red** marks areas with no connectivity.



3.1.43 Quick HCM-FS prediction



Click this button to open Quick HCM-FS prediction tool.

This performs calculations between two HCM-FS files.

Calculation settings

Calculation name

Quick HCM-FS calculations

Calculation name

Name of the calculation.

Trigger options

Limit TD

☒

TD, dB >

0.1

Limit Interference

☒

Interference level, dBW >

-400

Distance PTX to RX [km] <

1.99

Trigger options are used to limit the size of calculation report. If all trigger options are disabled, calculation report may become very huge since all calculation cases will be logged into the report.

Limit TD

If enabled, calculation case is logged into the report only if threshold degradation is greater than defined.

TD, dB >

Value of threshold degradation used as a trigger for logging calculation case into the report.

Limit Interference

If enabled, calculation case is logged into the report only if interference level is greater than defined. Can be used along with TD limit.

Interference level, dBW

Value of interference level used as a trigger for logging calculation case into the report.

Distance PTX to RX [km]

Distance between passive transmitter and receiver. In the case of passive receiver, calculation will be performed only if distance between passive transmitter and receiver is less than defined. Decision whether include this calculation into the report will additionally depend on the rest of Trigger options.

Please note that all calculation cases resulted in errors will be logged into the report regardless of Trigger options.

Reference file

Reference file

File against which test calculations are performed.

Test file

Test file

File that is tested against reference file.

There is no substantial difference between test and reference file. Calculations will be performed in the same way but the calculation report always lists records of test file.

When calculation report is opened it shows all the records from test file along with maximum value of threshold degradation.

Results

Processing log

Verifying parameters

14:14:35

Verification complete

14:14:35

Calculations started

14:14:35

Calculations finished

14:14:35

Task completed

14:14:36

Type

TX

Name

Frequency

10.602632G

Reference

Original status

C

Max TD

0

Type

Type of station of the test record

Name

Name of station of the test record

Frequency

Combines frequency and frequency unit of the test record

Reference

Coordination reference of the test record

Original status

Status of coordination of the test record

Max TD

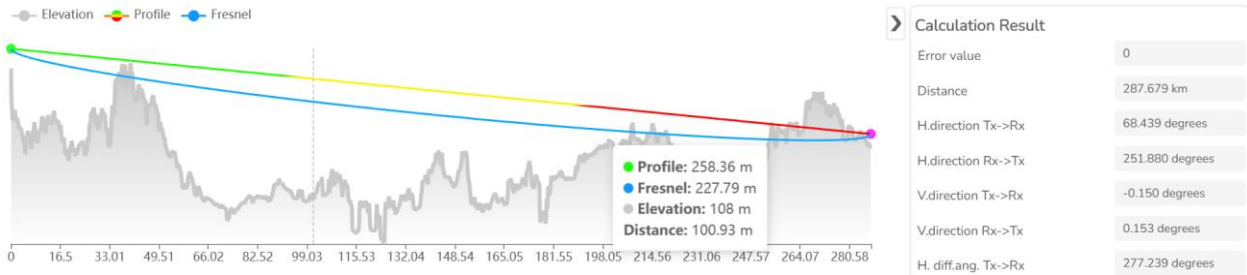
Maximum threshold degradation. In case of test TX record, it shows maximum threshold degradation caused by the transmitter to a receiver of the reference file. In case of test RX record, it shows threshold degradation of this particular receiver. For PTX or PRX test record, it shows maximum threshold degradation of either test or reference receiver.

Show interference list

Show interference list - shows list of calculated stations against particular test record.

List of interfering stations						
Show profile	Type	Name	Frequency	Coord.	Status	C
Profile	PRX	2 TARPINE_TEST_P...	38.55250G	C		l

Clicking on *Profile* button allows user to perform detailed HCM calculations.



3.1.44 HCM requests

Click this button  to open HCM requests tool.

This is the main component used for HCM calculations. Main functionality:

- To list and process all the incoming and outgoing coordination requests
- Create new incoming coordination request
- Create outgoing coordination request

3.1.44.1 List coordination requests

HCM-FS requests

Own requests

Foreign requests

Search

+ New own request

R028

Country

LVA

Code

N

Writing date

30072025

Status

Sent

R027

Country

LVA

Code

N

Writing date

30072025

Status

Sent

Own requests

This lists all outgoing coordination requests (sent to other countries).

+ New own request

Starts the process of creating a new outgoing request

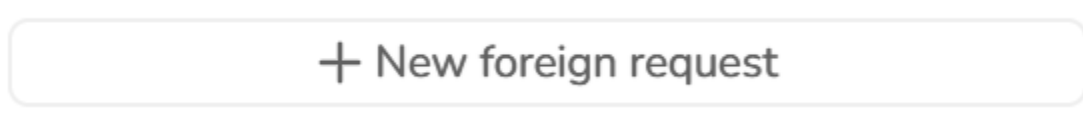
The list of Own requests contains following parameters:

- Destination country
- File contents code (from file header)

- Date of coordination request (from file header)
- You assigned reference number (used for your convenience)
- Status of coordination request

Foreign requests

This lists all incoming coordination requests (received from other countries).



Starts the process of importing a new incoming request

The list of Foreign requests contains following parameters:

- Country of the origin
- File contents code (from file header)
- Date of coordination request (from file header)
- You assigned reference number (used for your convenience)
- Status of coordination request

Each coordination request contains detailed information about the request and/or the answer.

Request

Download request file

View content of request file

View associated records

View processing results

Recalculate

Country from	POL
Country to	LTU
Registration number	R024
Registration date	2025-07-25
Responsible person	
Phone	
Telefax	
Email	
Number of records	2
Writing date	07042025
Status	Processing ended

Country from

Country of origin

Country to

Destination country

Registration number

Your assigned registration number used for your convenience (optional)

Registration date

Your date of registration of the coordination request, used for your convenience (optional)

Responsible person

Name of the responsible person (from HCM request file header)

Phone

Phone number from HCM request file header

Telefax

Telefax number from HCM request file header

Email

Email from HCM request file header

Number of records

Number of records from HCM request file header

Writing date

Writing date from HCM request file header

Status

Current status of the coordination request. Different statuses are available:

- Processing.
- Processing ended.
- Processing error. This shows that processing resulted in error.
- Ready to send. This shows that outgoing request is ready to be sent. Usually, user may need to enter additional parameters required for coordination request or edit defaults before finishing coordination request.
- Finished

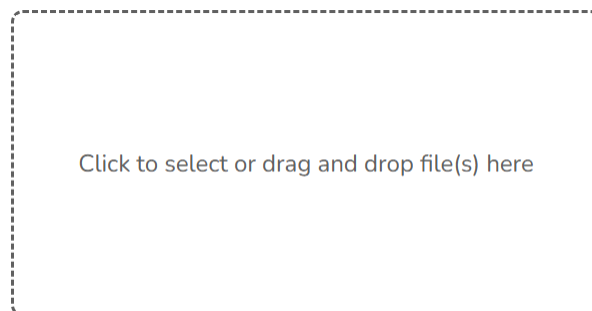
Each request contains a toolbar of the following tools:

- Download request file. downloads original coordination request file.
- View request file. Shows content of the original request file.
- View associated records. Shows current records associated with the request file. Usually, when incoming request is processed, all records are imported into the internal database but with time they may be deleted by another request, or changed. The same is valid for the records of outgoing requests – they may later be changed or deleted. This functionality shows current parameters of the imported (or exported) data.
- View processing results. Opens calculation results that have been performed for the selected coordination request.
- Recalculate. Restarts calculations for the selected coordination request.

Answer to coordination request contains the same parameters as the request.

3.1.44.2 Create new incoming coordination request

New incoming (foreign) coordination request is created by clicking *New foreign request* in HCM requests. Initially this displays area where received coordination file can be dropped.



When coordination file is loaded map displays the records of the file with additional information, such as content of HCM file header, registration data, and Trigger options.



Your data	
Calculation Name	Process POL 07042025 HCM re
Registration Number	
Registration Date	yyyy-mm-dd 

Your data

Calculation Name

Name of the calculation that will be displayed in Prediction history.

Registration Number

You can assign registration number for the coordination request. Optional.

Registration Date

You can assign here registration date for the coordination request. Optional.

File Content	
File Number	1
File Contents	Requests
File Contents Code	N
Country	POL
Responsible Person	
Phone	
Telefax	
Email	
Number of Records	2
Writing Date	07042025

File content section contains header information of the file as it is defined in Annex 2B of HCM-Agreement.

Trigger options	
Limit TD	☐
Limit Interference	☐
Distance PTX to RX [km] <	2

Trigger options

Limit TD

If enabled, calculation case is logged into the report only if threshold degradation is greater than defined.

TD, dB >

Value of threshold degradation used as a trigger for logging calculation case into the report.

Limit Interference

If enabled, calculation case is logged into the report only if interference level is greater than defined. Can be used along with TD limit.

Interference level, dBW

Value of interference level used as a trigger for logging calculation case into the report.

Distance PTX to RX [km]

Distance between passive transmitter and receiver. In the case of passive receiver, calculation will be

performed only if distance between passive transmitter and receiver is less than defined. Decision whether to include this calculation into the report will additionally depend on the rest of Trigger options. Please note that all calculation cases resulted in errors will be logged into the report regardless of Trigger options.

For file with contents code N, following steps are performed:

- Coordination reference data of submitted HCM records are tested to make sure that no records with such references already exist in the database.
- New record of coordination request is created and all HCM records of the request file are imported into the database.
- Own records are selected that fulfill following conditions: have coordination status C, D, E, F, G, H, P received by the administration which new entries are tested against.
- Calculations are performed: new coordination request data against selected own HCM records, taking into account Trigger options.

For file with contents code A, the following steps are performed:

- New coordination request record (answer) is created.
- Appropriate HCM records in the database are updated (that match coordination reference). Only following parameters are updated: coordination status, date of achieving coordination, and remarks.

For file with contents code D, the following steps are performed:

- Received HCM records are tested to contain coordination statuses W and R only.
- Search for records in the database that have the same coordination reference.
- Delete found records in the database.
- Create new coordination request record (answer).

For file with contents code O, the following steps are performed:

- New coordination request record is created.
- Own records for the database are selected that have coordination following status with the country of the coordination request: C, E, F, G, H, P
- File content is compared against the selected records and comparison report is created.

3.1.44.3 Display processing results

View processing results displays calculations performed on an incoming request file of type N or O.

File containing new entries (N)

Processing log

Task is waiting in queue14:45:20
Starting calculation14:45:25
Verifying parameters14:45:25
Verification complete14:45:26
Calculations started14:45:26

TypeTX

NameS [REDACTED] A

Frequency58 [REDACTED] 00M

ReferencePOL20 [REDACTED] 01

Original statusB

Max TD0

Proposed Coord. StatusC

Remarks

Type

Type of station of the test record

Name

Name of station of the test record

Frequency

Combines frequency and frequency unit of the test record

Reference

Coordination reference of the test record

Original status

Status of coordination of the test record

Max TD

Maximum threshold degradation. In case of test TX record, it shows maximum threshold degradation cause by the transmitter to a receiver of the reference file. In case of test RX record, it shows threshold degradation of this particular receiver. For PTX or PRX test record, it shows maximum threshold degradation of either test or reference receiver.

Proposed Coord. Status

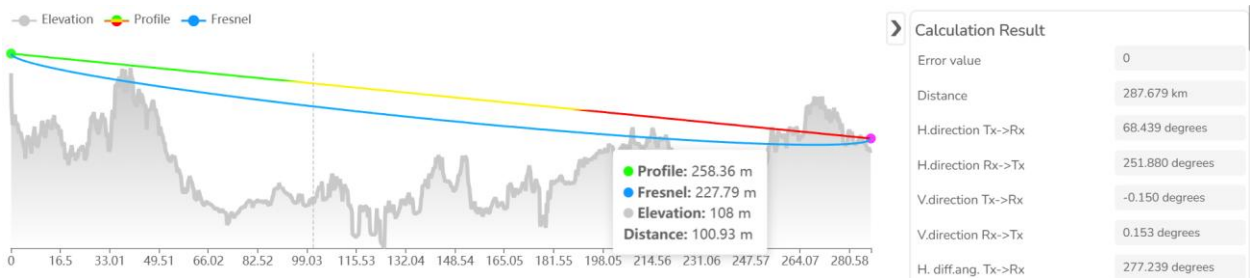
If Max TD is lower than 1 dB and original coordination status is B, software automatically sets proposed coordination status to C. This value will be inserted into the answer file.

Remarks

User can write separate remarks for each HCM record. This information will be added to the answer file.

Show interference list

Shows list of calculated stations against particulat test record.



Profile and calculation values are extracted from HCM calculation report file.

File containing overall list (O)

Results

Processing log

Task is waiting in queue

13:24:43

Starting calculation

13:24:46

Verifying parameters

13:24:46

Verification complete

13:24:46

Comparison started

13:24:46

Comparison finished

13:24:47

Differences in records

RX

22.5505 G

POL

2

RX

22.5645 G

POL

2

Differences in records

Shows records that have at least one different parameter value. By clicking on the record, detailed comparison widget is displayed:

HCM comparison results		
PO [REDACTED]		
Parameter name	Database value	Received value
Geographical co-ordinates	0 [REDACTED] 9	0 [REDACTED] 59
Height of station above sea level	220	221
Error No errors		

Parameter name

Name of the parameter defined in Annex 2B of HCM-Agreement.

Database value

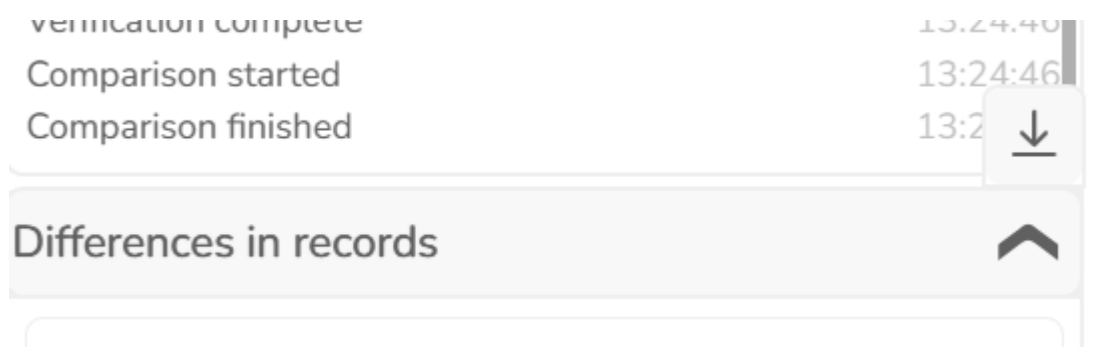
Value of the record stored in the database.

Received value

Value of the record in the received request file.

Error

Contains error message if error occurs while processing the record.

**Download report as CSV file**

Downloads report containing all the records of this section

Missing in database					^
RX	23.12	PO	[REDACTED]	02	
TX	22.11	PO	[REDACTED]	01	

Section “Missing in database” lists all the records that are present in the received file but missing in the database. By clicking on the record, a detailed comparison widget is displayed with a few additional parameters of the record:

HCM comparison results					
POL [REDACTED] 2					
Type	Frequency	Station name	Coordinates	Channel spacing	
RX	23.12 G	[REDACTED]	02 [REDACTED] 4	3.5	

Differences in records					↓
Missing in database					^

Download report as CSV file

Downloads report containing all the records of this section

Missing in received list				
RX	22.55	POL2	[REDACTED]	2
TX	23.56	POL2	[REDACTED]	1

Section “Missing in received list” displays all the records that are present in the database but missing in the received file. By clicking on the record, a detailed comparison widget is displayed with a few additional parameters of the record:

HCM comparison results				
POL [REDACTED] 2				
Type	Frequency	Station name	Coordinates	Channel spacing
RX	22.55 G	[REDACTED]	02 [REDACTED] 9	7

Missing in database				
Missing in received list				

Download report as CSV file

Downloads report containing all the records of this section

3.1.44.4 Create new outgoing coordination request

New outgoing (own) coordination request is created by clicking *New own request* (in HCM requests). This opens a new widget:

New own HCM-FS request

Request type

N - New entries

▼

Link template

Default

▼

Send to country

LVA

▼

Use the selection tool to select links

Request type N

For new coordination request (New entries) user must select at least one link that is going to be sent for coordination to a country set under “Send to country” and link template, in case one is used.

By clicking Create button new outgoing coordination request is created and displayed in the list of Own requests, containing status “Ready to send”. The request contains entries of selected link, converted to HCM Annex 2B format. HCM entries are stored in the database.

Request type D

This creates coordination request containing a list of deleted HCM records. The records to be deleted must be selected by the user. If the selected HCM records have been coordinated by different administrations, several coordination requests will be created.

Request type O

This creates a coordination request containing an overall list of coordinated HCM records with the defined target country. Only records with coordination status C, E, F, G, H, and P will be included.

Created coordination request

Request

View associated records

Country from

LTU

Country to

LVA

Registration number

Registration date

dd/mm/yyyy 

Responsible person

Phone

Telefax

Email

Number of records

4

Writing date

16102025

Status

Ready to send

Finish

Country from

Country of origin. It is defined in the settings of the user.

Country to

Target country, to which coordination request is to be sent.

Registration number

Optional identification of the coordination request.

Registration date

Optional registration date of the request.

Responsible person

Name, as defined in Annex 2B of HCM-Agreement. This field is automatically filled with information from user settings.

Phone

Phone number, as defined in Annex 2B of HCM-Agreement. This field is automatically filled with information from user settings.

Telefax

Telefax number, as defined in Annex 2B of HCM-Agreement. There is no such parameter in the user settings therefore it is not filled in automatically.

Email

Email, as defined in Annex 2B of HCM-Agreement. This field is automatically filled with information from user settings.

Number of records

Defined in Annex 2B of HCM-Agreement. This field is automatically calculated.

Writing date

Writing date, as defined in Annex 2B of HCM-Agreement. Date of the creation of coordination request is automatically written to this field.

Status

Status of coordination request

Send email

If enabled, email is automatically sent to the contact person of the target country. For this to work, the following conditions have to be fulfilled:

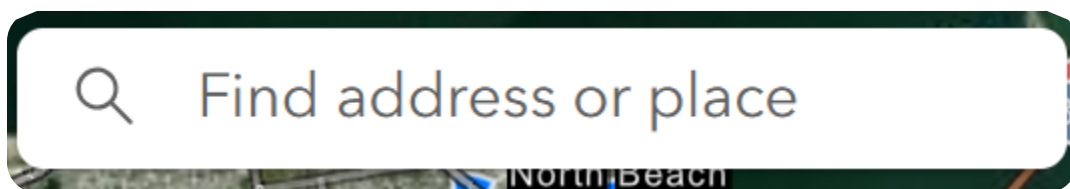
- SMTP server defined in Express settings.
 - Email of the contact person of the target country set in the table hcm_countries.
- If this option is not enabled, coordination request file is prepared but not sent automatically. User must send it manually.

Finish

It updates coordination request with the data and creates content of Annex 2B data exchange file. If "Send email" is enabled, coordination request is automatically sent to the target administration, otherwise prepared Annex 2B exchange file can be downloaded using "Download request file" and sent manually by the user.

3.2 Map

3.2.1 Search bar



Allows searching for an address, place, or location.

3.2.2 Zoom in



Zooms in on the map.

3.2.3 Zoom out



Zooms out on the map.

3.2.4 Fine zoom



When enabled, zoom steps are not locked to standardized scale levels. This allows to zoom more finely when more precision is needed.

3.2.5 Compass



Shows the current map orientation and allows resetting north-up.

3.2.6 3D/2D Button



Switches between 2D and 3D map view.

3.2.7 Legend



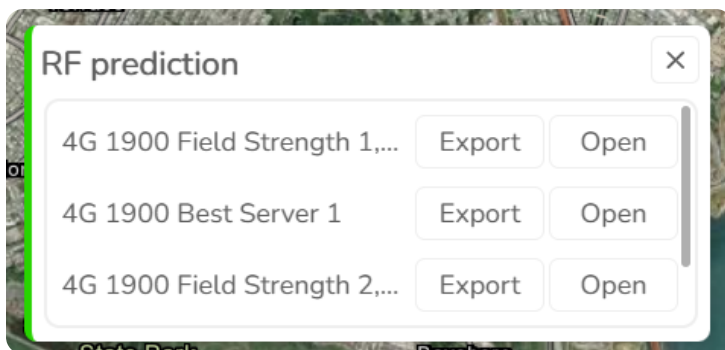
Opens the legend panel showing all visible layers and their symbols.

3.2.8 Home



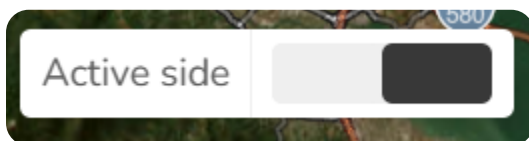
Returns the map to the workspace extent.

3.2.9 Mini calculation task window



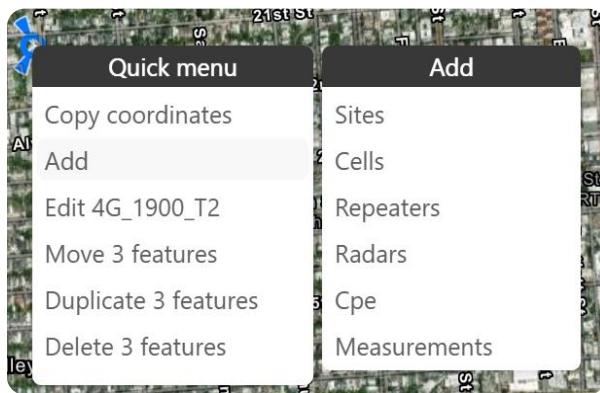
Mini calculation task window displays the result of the most recent calculation. Unlike the full Prediction History tool, this window does not list all past calculations, it only shows the latest created task.

3.2.10 Active side switch



Shows up when two tools, which have map clicking enabled, are opened at the same time. You may then switch between left and right sides, to select which tool is active. For example: Features tool on the left, and Profile tool on the right. When the active side switch is flipped to the left, clicking on the map will start feature selection, rather than draw a profile.

3.2.11 Quick menu



The quick menu opens when middle-clicking on the map. If the click is on a feature, it provides actions such as copying coordinates, adding, editing, moving, duplicating, or deleting the feature. If the click is not on a feature, it shows only copying coordinates and adding new items.

3.3 Data management and visualization section

Available tools for this section:

- Workspace
- Features
- Networks
- Layers
- Prediction history
- Antennas

The dialogs are pinned and can not be moved to another location. If one tool is active and visible in the window and a second one is turned on, the first tool is closed and the new one is opened.

3.4 Calculation section

Available tools for this section:

- Settings
- Identify
- Measurement tools
- Statistics
- Profile
- Quick RF prediction
- RF Prediction
- Visibility prediction
- Antenna visibility prediction
- Radar prediction
- Link prediction
- Network availability
- Model tuning

- Optimal placement
- EMF

The dialogs are pinned and can not be moved to another location. If one tool is active and visible in the window and a second one is turned on, the first tool is closed and the new one is opened.

There are additional tools that open on the right side and can be accessed from the Features tool:

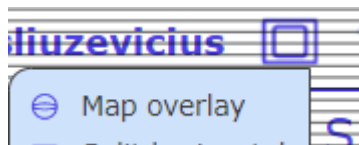
- Move
- Duplicate
- Delete
- Import
- Add

3.5 Map/table view modes

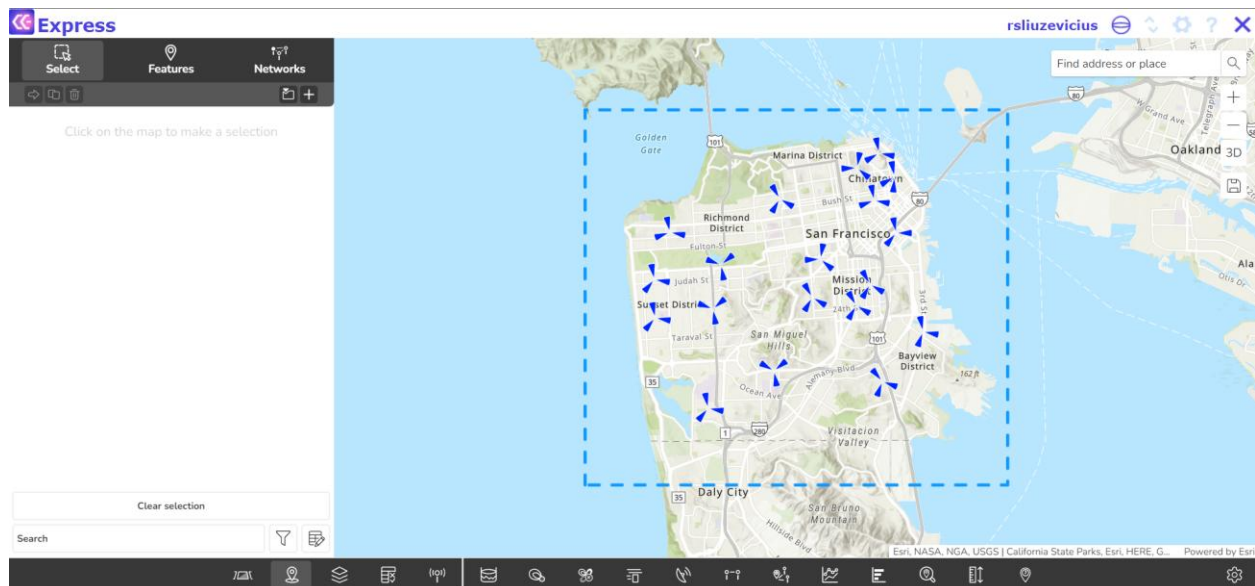
In this section, users can adjust the data view. For example, turn on the Map view, Table view, or turn on both of them, etc.

3.5.1 Map view

Map view widget.

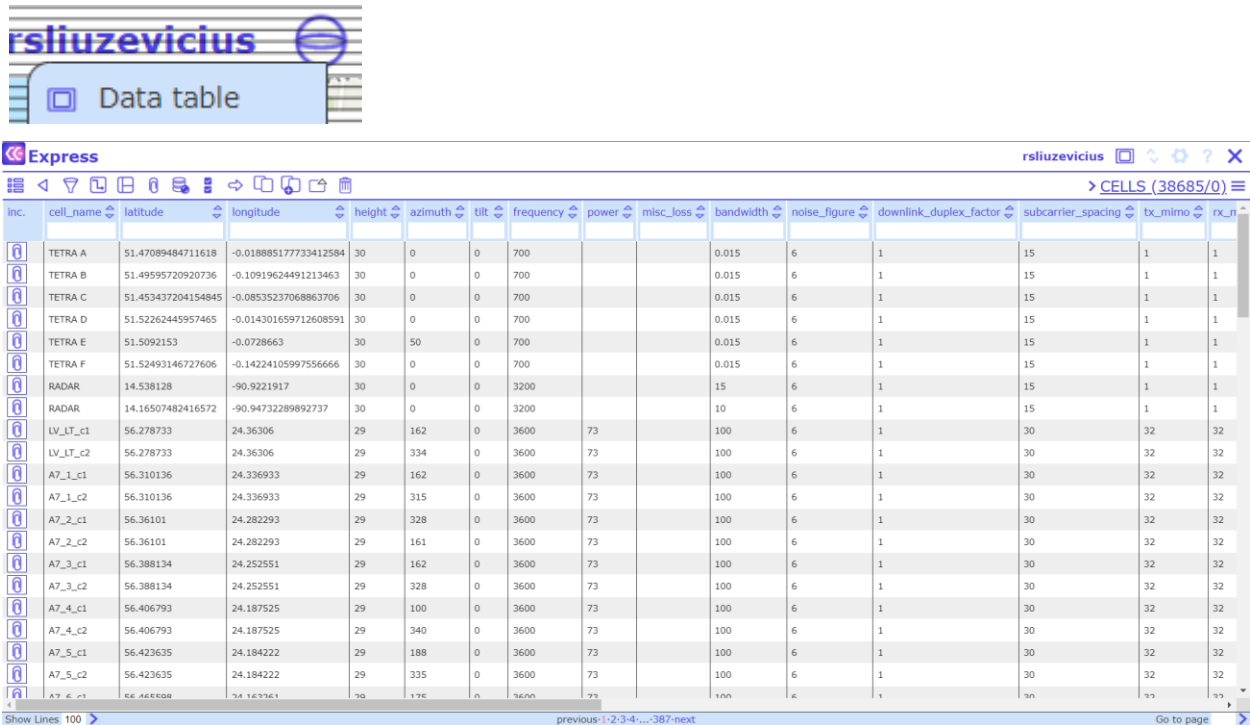


Turns on the map view.



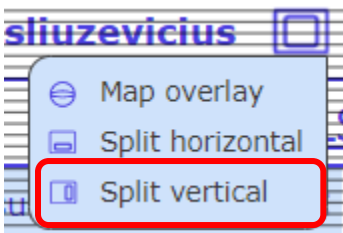
3.5.2 Display data table full screen

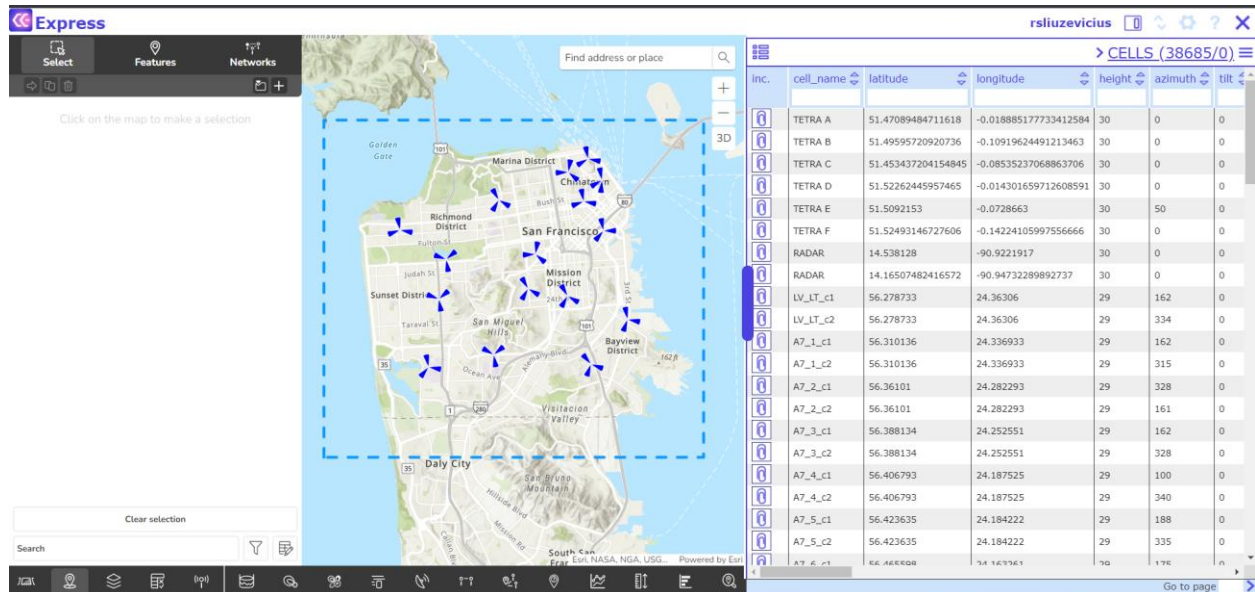
Turns on the Network Data Management view in full screen.



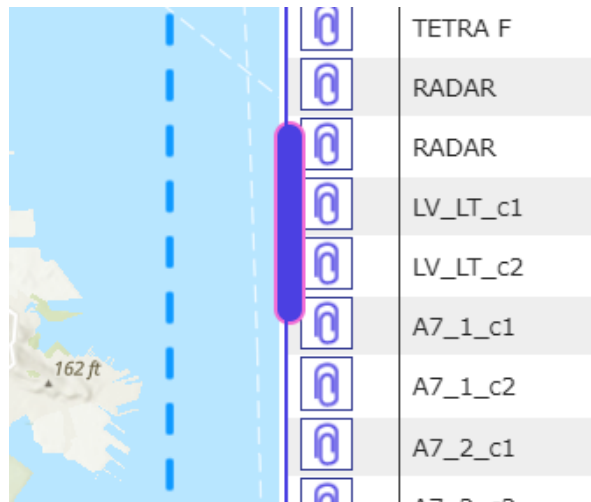
3.5.3 Split window vertically

The screen can be divided into Map view and Table view. In the previous options, the tables were placed on top of the Map view. This means that the tables cover the widgets.



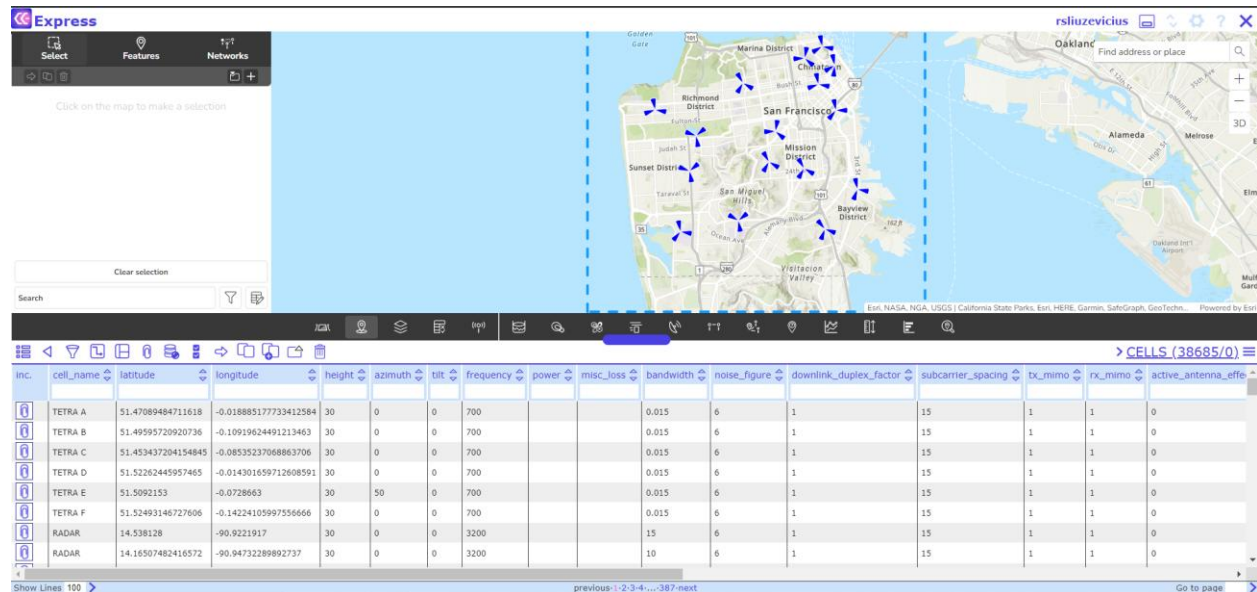
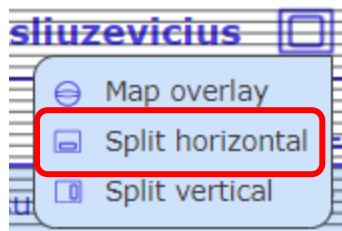


Users can adjust the Table and Map view proportions by using the slider function: Move the mouse cursor on top of the blue line, click on it and change the proportions.

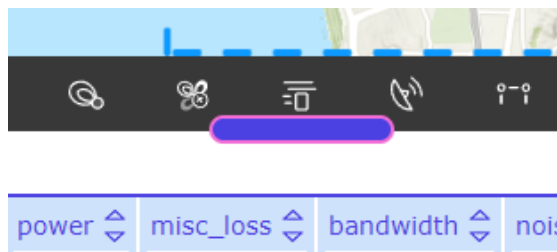


3.5.4 Split window horizontally

The screen can be divided into Map view and Table view. In the previous options, the tables were placed on top of the Map view. This means that the tables cover the widgets



Users can adjust the Table and Map view proportions by using the slider function: Move the mouse cursor on top of the blue line, click on it and change the proportions.



4. Database structure

4.1 Sites

Describes the tower's location and its identification. An object is not used for the calculations. It is possible to add additional fields, but here are the main fields used for this object.

site_name

Object identification.

Y

Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

X

Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

workspace_id

ID for the workspace. This field value is used to filter objects based on a chosen workspace. The parameter is filled automatically.

height

Height of the site in meters.

4.2 Cells

Describes the sector equipment, parameters, and cell logical information in one table. An object is used for point-to-area calculation, which describes the technology, frequency, power, and other information.

cell_name

Object identification.

Y

Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

X

Decimal degrees type coordinate in the WGS 1984 geographical coordinate system.

height

Cell height above the ground in meters.

azimuth

Cell direction from the North in degrees.

tilt

Mechanical tilt value, negative numbers tilt up.

frequency

Frequency value in MHz.

power

Power value in dBm.

misc_loss

Miscellaneous loss value in dB.

bandwidth

Value in MHz. Required for 4G and 5G technologies. For other technologies define the value as 0.015.

noise_figure

Value in dB. Required for 4G and 5G technologies.

downlink_duplex_factor

Value range from 0 to 1. Required for 4G and 5G technologies, and used for Downlink Throughput calculations.

subcarrier_spacing

Value in kHz. Required for 4G and 5G technologies. For other technologies define value 15.

tx_mimo

Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

rx_mimo

Receiver antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

active_antenna_effect

This parameter is dedicated to smart antenna modelling. The default value is 0, but if massive MIMO is used, a smart antenna effect can be included to lower the interference and boost the throughput. Recommended values:

- For MIMO 32x32 – value 6.
- For MIMO 64x64 – value 9.

cell_load

This parameter is described in percentage and varies from 0 to 100. It describes the load of a cell. The cell load affects RSSI, RSRQ, DL Throughput calculations. For example, if the cell load is higher, the DL Throughput is lower.

workspace_id

ID for the workspace. The field value is used to filter objects based on a chosen workspace. The parameter is filled automatically.

color_index

Describes the cell visualization. Available values:

- None – blue color.
- 1 – red color.
- 2 – light green color.
- 3 – dark green color.
- 4 – light blue color.
- 5 – dark blue color.
- 6 – purple color.

technology

Describes the cell technology. Possible values are 2G, 3G, 4G, and 5G.

prediction_model_configuration_id

Defines the prediction model configuration to be used in the model table. The objectID value is taken from the prediction model table.

frequency_group

Used to divide calculations into parts. If the selection range includes two or more different frequency group values, the cells won't be predicted together.

antenna_id

Describes the antenna ID from the antennas table. The objectID is used.

carriers

Describes the carrier values used for 2G calculations: C/I interference and C/A interference. The values are written in brackets, [...]. If more than one value is defined, the values are separated by a comma. If there is no carrier information, the brackets are left empty [].

site_id

Site identification. This field value is optional.

duplex_mode

Lets the user select what type of duplexing technique he wants to assign to the cell.

electrical_tilt

Electrical tilt value, negative numbers tilt up

status

free-form text.

type

free-form text.

4.3 Repeaters

repeater_name

Object identification.

Y

Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

X

Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

height

Repeater height above the ground in meters.

azimuth

Cell direction from the North in degrees.

tilt

Mechanical tilt value, negative numbers tilt up.

frequency

Frequency value in MHz.

threshold 1-3

The minimum field strength in dB at repeater location at which power of the corresponding index (1-3) will be applied. Values should be in ascending order. If a higher threshold is satisfied, the power corresponding to it will be used.

power 1-3

Power value in dBm.

misc_loss

Miscellaneous loss value in dB. Value is optional.

bandwidth

Value in MHz. Required for 4G and 5G technologies. For other technologies define the value as 0.015.

subcarrier_spacing

Value in kHz. Required for 4G and 5G technologies. For other technologies define value 15.

tx_mimo

Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

rx_mimo

Receiver antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

technology

Describes cell technology. Possible values are 2G, 3G, 4G, and 5G.

prediction_model_id

Sets the model type to be used:

- Value 1 – ITU R. P452
- Value 2 – UniMacro

prediction_model_configuration_id

Defines the prediction model configuration to be used in the model table. The objectID value is taken from the prediction model table.

frequency_group

Used to divide calculations into parts. If the selection range includes two or more different frequency group values, the cells won't be predicted together.

antenna_id

Describes the antenna ID from the antennas table. The objectID is used.

workspace_id

ID for the workspace. The field value is used to filter objects based on a chosen workspace. The parameter is filled automatically.

electrical_tilt

Electrical tilt value, negative numbers tilt up

4.4 Radars

Radar_name

Object identification.

Y

Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

X

Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

Height

Radar height above the ground in meters

Tilt

Mechanical tilt value, negative numbers tilt up.

Frequency

Frequency value in MHz.

Power

Power value in dBm. Value is optional.

Misc_loss

Miscellaneous loss value in dB. Value is optional.

View_angle

Radar radiation vertical height in degrees centered at the tilt value.

Prediction_model_id

Sets the model type to be used:

- Value 1 – ITU R. P452
- Value 2 – UniMacro

Prediction_model_configuration_id

Defines the prediction model configuration to be used in the model table. The objectID value is taken from the prediction model table.

Workspace_id

ID for the workspace. The field value is used to filter objects based on a chosen workspace. The parameter is filled automatically.

4.5 CPE

Cpe_name

Object identification.

Y

Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

X

Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

Height

CPE height above the ground in meters

Azimuth

Cell direction from the North in degrees.

Antenna_id

Describes the antenna ID from the antennas table. The objectID is used.

Cell_id

Cell identification.

Throughput

Throughput sold to customer (used in network availability calculation)

Power

Power value in dBm.

Misc_loss

Miscellaneous loss value in dB.

Status

free-form text.

Notes

free-form text.

Workspace_id

ID for the workspace. The field value is used to filter objects based on a chosen workspace. The parameter is filled automatically.

4.6 Measurements

Fs

Measured field strength.

Y

Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

X

Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

Cell_id

Cell identification.

Workspace_id

ID for the workspace. The field value is used to filter objects based on a chosen workspace. The parameter is filled automatically.

4.7 Workspace

Workspace_name

Workspace identification.

Extent_xmin

Minimum workspace extent longitude

Extent_ymin

Minimum workspace extent latitude

Extent_xmax

Maximum workspace extent longitude

Extent_ymax

Maximum workspace extent latitude

Extra_layers

JSON array containing links to ArcGIS layer services.

e.g.: ["url1", "url2"]

Use_clutter_loss

Possible values:

- t – clutter losses included in the prediction model.
- f – clutter losses not included in the prediction model.

Calculate_eirp

True – EIRP is calculated with the formula power - misc_loss + antenna gain.

False – power value is used as EIRP.

Geodata_set_id

Path to folder containing geodata rasters: elevation.tif, building_height.tif (optional), clutter_classes.tif (optional), clutter_height.tif (optional).

5. CE Express API

The (currently incomplete) documentation for the CE Express API endpoints can be found here:

https://cecom2.cellular-expert.com/ce_express_api_documentation/

6. Network Data Management

The Network Data Management view is divided into four sections:

- Tools for data management.
- User administration tasks.
- Table for data editing.
- View options.

Express

rsluizevicius

inc.

1. Data management

gnt

azimuth

tilt

frequency

power

misc_loss

bandwidth

noise_figure

do

TETRA A	51.47089484711618	-0.018885177733412584	30	0	0	700		0.015	6	1	15
TETRA B	51.49595720920736	-0.10919624491213463	30	0	0	700		0.015	6	1	15
TETRA C	51.453437204154845	-0.08535237068863706	30	0	0	700		0.015	6	1	15
TETRA D	51.52262445957465	-0.014301659712608591	30	0	0	700		0.015	6	1	15
TETRA E	51.5092153	-0.0728663	30	50	0	700		0.015	6	1	15
TETRA F	51.52493146727606	-0.14224105997556666	30	0	0	700		0.015	6	1	15
RADAR	14.538128	-90.92221917	30	0	0	3200		15	6	1	15
RADAR	14.16507482416572	-90.94732289892737	30	0	0	3200		10	6	1	15
LV_LT_c1	56.278733	24.36306	29	162	0	3600	73	100	6	1	30
LV_LT_c2	56.278733	24.36306	29	334	0	3600	73	100	6	1	30
A7_1_c1	56.310136	24.336933	29	162	0	3600	73	100	6	1	30
A7_1_c2	56.310136	24.336933	29	315	0	3600	73	100	6	1	30
A7_2_c1	56.36101	24.282293	29	328	0	3600	73	100	6	1	30
A7_2_c2	56.36101	24.282293	29	161	0	3600	73	100	6	1	30
A7_3_c1	56.388134	24.252551	29	162	0	3600	73	100	6	1	30
A7_3_c2	56.388134	24.252551	29	328	0	3600	73	100	6	1	30
A7_4_c1	56.406793	24.187525	29	100	0	3600	73	100	6	1	30
A7_4_c2	56.406793	24.187525	29	340	0	3600	73	100	6	1	30

Show Lines 100

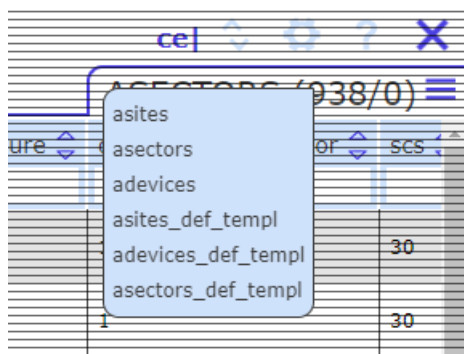
previous: 1-2-3-4...387-next

Go to page

2. View options and User administration

3. Table

To navigate through the tables, click on the table name at the top right corner.



To select a record, click on a record with the left mouse button. Selected records are colored in blue:

Express CELLS (4/206073)

inc.	cell_name	x	y	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect
5G cell A	18.97380976701136	72.81372290611266	10	20	0	3500				100	6	1	30	32	32	9
5G cell B	18.97380976701136	72.81372290611266	10	140	0	3500				100	6	1	30	32	32	9
5G cell C	18.97380976701136	72.81372290611266	10	240	0	3500				100	6	1	30	32	32	9
5G cell A1	18.965292058441445	72.80658823013263	10	310	0	3500				100	6	1	30	32	32	9
5G cell B1	18.965292058441445	72.80658823013263	10	120	0	3500				100	6	1	30	32	32	9
5G cell C1	18.965292058441445	72.80658823013263	10	220	0	3500				100	6	1	30	32	32	9
5G cell A2	18.963044609524736	72.81359952449802	10	340	0	3500				100	6	1	30	32	32	9
5G cell B2	18.963044609524736	72.81359952449802	10	90	0	3500				100	6	1	30	32	32	9
5G cell C2	18.963044609524736	72.81359952449802	10	185	0	3500				100	6	1	30	32	32	9

6.1 Data management tools

The Data management tools are found at the top of the screen:



6.1.1 Table view



Load data from the previously configured server location and open the Table view. Here, several functions can be performed, e.g., sorting or filtering data.

6.1.2 Back



Go one step back and show previous data.

6.1.3 Sieve



Sieve the data in the currently opened table for one or more filter terms.

6.1.4 Multiple table view



View and edit all tables from one site simultaneously in one window.

6.1.5 View attachments



View the list of attachments associated with a given record. To do so, select a record of choice, then click on the View attachments button and a dialog will open with the respective attachments listed. To view an attachment in a separate browser tab simply click on the attachment of choice

6.1.6 Synchronize changes



Synchronize the previously selected server database with changes performed in the Table view.

6.1.7 *Select/unselect all*



Select/unselect all currently shown database records.

6.1.8 *Show selected/Show all*



Shows all currently selected database records or all the records.

6.1.9 *Move record*



Move an object from one site to another. To do so, select an object and click on the Move Record button. Then select the site you want to move the object to and click the move button again. Note that attachments are not moved.

6.1.10 *Copy record*



Duplicate a database record from the Table view. Select the record to be duplicated. Then click the Copy Record button and the duplicated record appears in pink color in the Table view as a new record. Note that if the selected record to be duplicated has child objects, Copy Record will duplicate the entire record incl. child objects. Attachments are not duplicated.

6.1.11 *Add new record*



Add a new record to the Table view. Note that the newly added record will appear in orange writing.

6.1.12 *CE API*



Call CE API tool. Note: The administrator has to configure the tool.

6.1.13 Export selected



Export selected items.

6.1.14 Delete record



Remove a selected record from the current view. Note that removed records will be hidden from the user. Only administrators can delete a removed record from the database permanently (or restore it).

6.1.15 Remove selected



Removes selected objects.

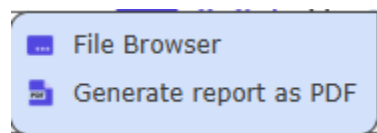
6.2 View options and User administration



6.2.1 Import Export



Opens the menu for exploring and importing data:

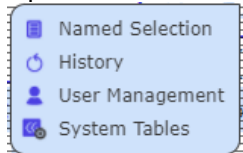


See section 6 for details.

6.2.2 Settings

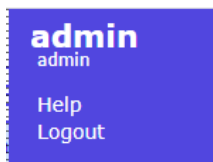


Opens the menu for administrator functions:



Please view Administration Guide for details.

6.2.3 User name/ Help/ Logout



Help - Displays the User Guide for quick help.
Logout – Logoff the user from CE Inventory3D

6.3 Table

The table comprises all network data. The data can be managed with the Data Management tools (see above).

6.4 View Options

Click on one of the buttons to choose a view of the data table aligned with the map (section 3.2).

7. Database organization

7.1 Page setup and navigation

7.1.1 Organization of single and multiple tables

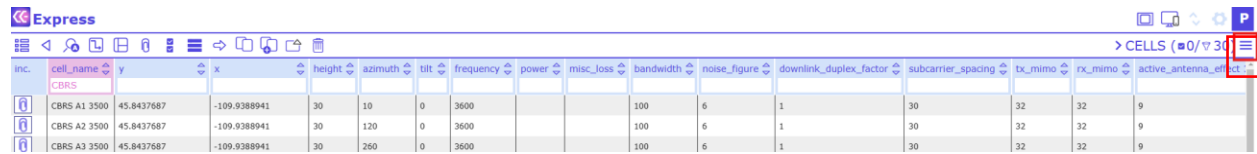
inc	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect
SG cell A	18.97380976701136	72.81372290611266	10	20	0	3500				100	6	1	30	32	32	9
SG cell B	18.97380976701136	72.81372290611266	10	140	0	3500				100	6	1	30	32	32	9
SG cell C	18.97380976701136	72.81372290611266	10	240	0	3500				100	6	1	30	32	32	9
SG cell A1	18.965292058441445	72.80658823013263	10	310	0	3500				100	6	1	30	32	32	9
SG cell B1	18.965292058441445	72.80658823013263	10	120	0	3500				100	6	1	30	32	32	9
SG cell C1	18.965292058441445	72.80658823013263	10	220	0	3500				100	6	1	30	32	32	9
SG cell A2	18.963044609524736	72.81359952449802	10	340	0	3500				100	6	1	30	32	32	9
SG cell B2	18.963044609524736	72.81359952449802	10	90	0	3500				100	6	1	30	32	32	9
SG cell C2	18.963044609524736	72.81359952449802	10	185	0	3500				100	6	1	30	32	32	9
SG cell A3	18.970243441704998	72.80440491199482	10	10	0	3500				100	6	1	30	32	32	9
SG cell B3	18.970243441704998	72.80440491199482	10	120	0	3500				100	6	1	30	32	32	9
SG cell C3	18.970243441704998	72.80440491199482	10	220	0	3500				100	6	1	30	32	32	9
SG cell A4	18.968399373014837	72.81080734491289	10	20	0	3500				100	6	1	30	32	32	9
SG cell B4	18.968399373014837	72.81080734491289	10	120	0	3500				100	6	1	30	32	32	9
SG cell C4	18.968399373014837	72.81080734491289	10	270	0	3500				100	6	1	30	32	32	9
CBRS A1 3500	45.8437687	-109.9388941	30	10	0	3600				100	6	1	30	32	32	9
CBRS A2 3500	45.8437687	-109.9388941	30	120	0	3600				100	6	1	30	32	32	9
CBRS A3 3500	45.8437687	-109.9388941	30	260	0	3600				100	6	1	30	32	32	9
CBRS B1 3500	45.79516579402228	-108.54887500305232	30	340	0	3600				100	6	1	30	32	32	9
CBRS B2 3500	45.79516579402228	-108.54887500305232	30	110	0	3600				100	6	1	30	32	32	9
CBRS B3 3500	45.79516579402228	-108.54887500305232	30	220	0	3600				100	6	1	30	32	32	9
CBRS C1 3500	45.732923908309125	-108.61178889770542	30	340	0	3600				100	6	1	30	32	32	9
CBRS C2 3500	45.732923908309125	-108.61178889770542	30	100	0	3600				100	6	1	30	32	32	9
CBRS C3 3500	45.732923908309125	-108.61178889770542	30	250	0	3600				100	6	1	30	32	32	9
CBRS D1 3500	45.83162838070681	-108.46845164794999	30	340	0	3600				100	6	1	30	32	32	9
CBRS D2 3500	45.83162838070681	-108.46845164794999	30	110	0	3600				100	6	1	30	32	32	9
CBRS D3 3500	45.83162838070681	-108.46845164794999	30	220	0	3600				100	6	1	30	32	32	9
CBRS E1 3500	45.899763394763575	-108.38386550445658	30	340	0	3600				100	6	1	30	32	32	9
CBRS E2 3500	45.899763394763575	-108.38386550445658	30	110	0	3600				100	6	1	30	32	32	9

The number of records displayed per page can be set on the bottom left corner. Furthermore, you may scroll through the pages using the commands in the bottom middle and bottom right corner. The table name (here: cells), the total quantity of entries (here: 20607), and the quantity of currently selected entries (here: 0) are shown on the top right corner. Note that the number of filtered records will be shown, if the user applies a filter:

inc	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect
CBRS																
CBRS A1 3500	45.8437687	-109.9388941	30	10	0	3600				100	6	1	30	32	32	9
CBRS A2 3500	45.8437687	-109.9388941	30	120	0	3600				100	6	1	30	32	32	9
CBRS A3 3500	45.8437687	-109.9388941	30	260	0	3600				100	6	1	30	32	32	9

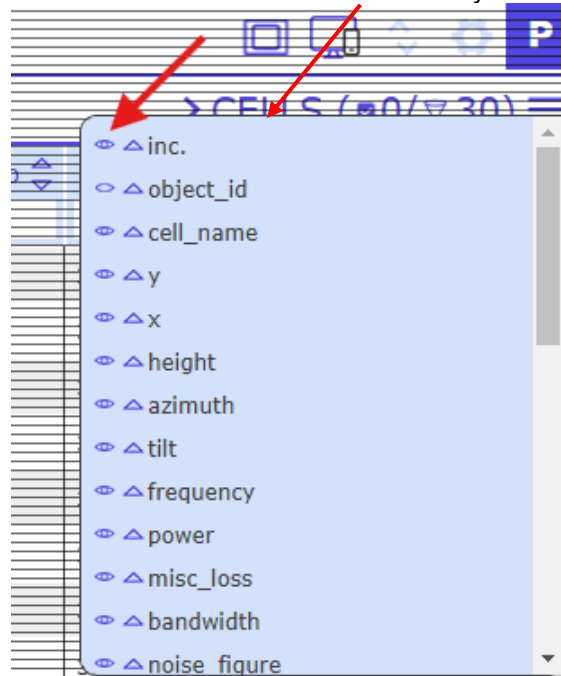
7.1.2 Organization of columns

The column order can be adjusted by clicking on  in the top right corner.

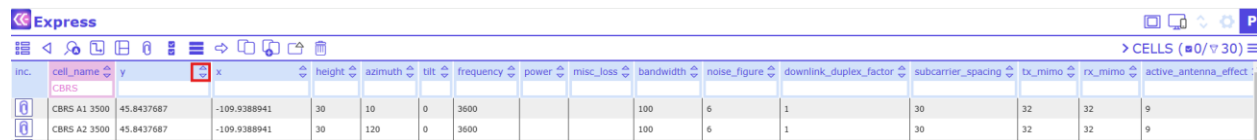


inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect
	CBRS															
	CBRS A1 3500	45.8437687	-109.9388941	30	10	0	3600			100	6	1	30	32	32	9
	CBRS A2 3500	45.8437687	-109.9388941	30	120	0	3600			100	6	1	30	32	32	9
	CBRS A3 3500	45.8437687	-109.9388941	30	260	0	3600			100	6	1	30	32	32	9

A menu opens in which the position of the columns can be sorted by clicking the arrow symbols. Individual columns are switched visible/invisible by clicking on the eye symbols.



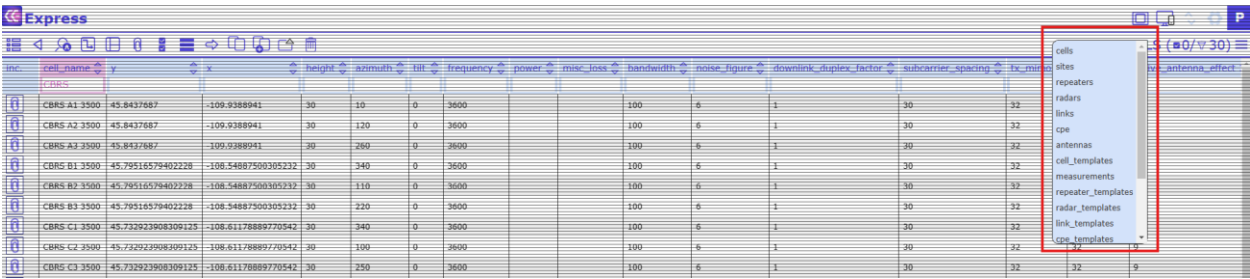
The columns can be sorted alphabetically from A>Z or from Z>A using the up and down arrow symbols within the column headers:



inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect
	CBRS															
	CBRS A1 3500	45.8437687	-109.9388941	30	10	0	3600			100	6	1	30	32	32	9
	CBRS A2 3500	45.8437687	-109.9388941	30	120	0	3600			100	6	1	30	32	32	9

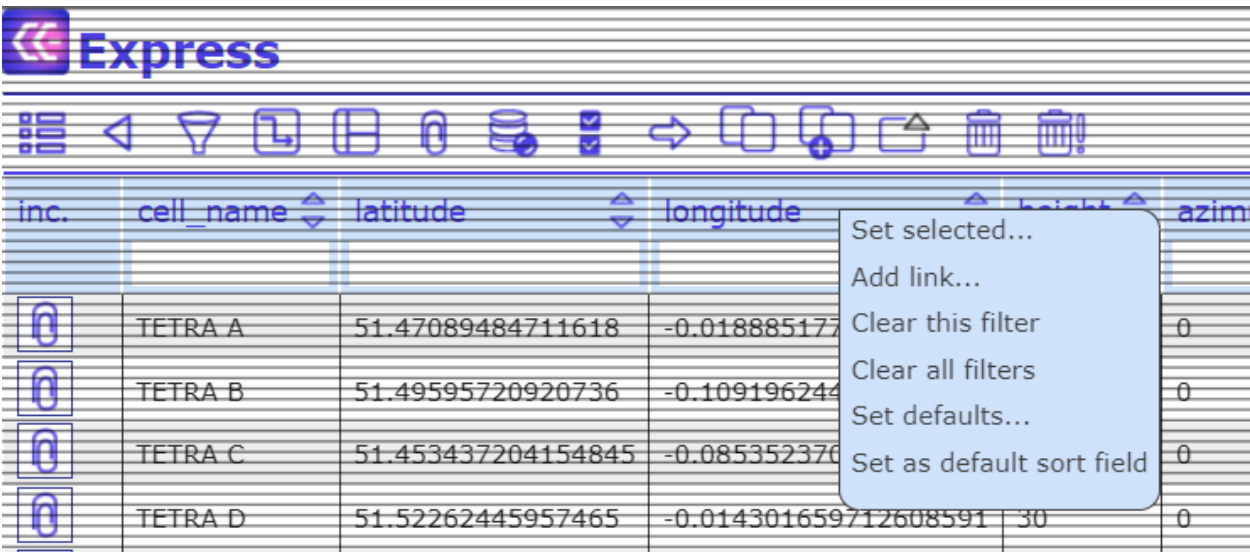
7.1.3 Navigation across tables

There are 2 ways to change from one table to another. Either click on the table name in the top right (here: cells) and a menu opens with all available tables.

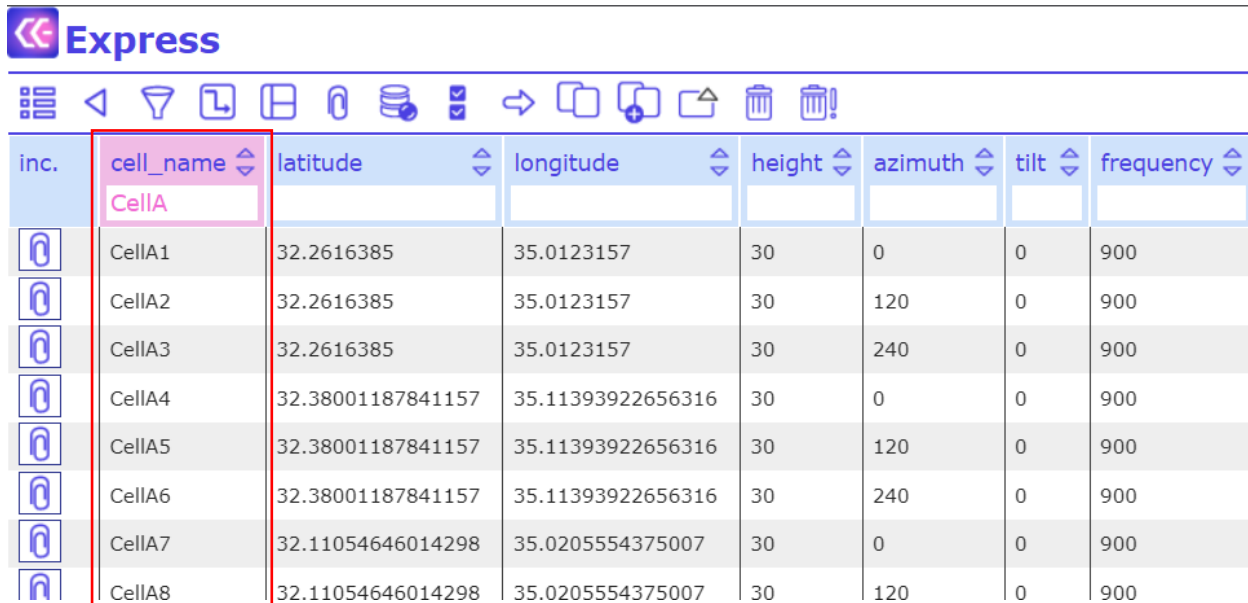


7.2 Filtering, sorting, editing, and linking database records

The database entries can be sorted according to their individual values. To do so, click on the column header and a dropdown menu opens with the following functions: “Set selected”, “Add link”, ”Clear this filter”, “Clear all filters”, “Set defaults” and “Set as default sort field”:



Alternatively, and more convenient for simple searches, columns can be filtered using the “quick filter” fields below the column names:



inc.	cell_name	latitude	longitude	height	azimuth	tilt	frequency
	CellA						
	CellA1	32.2616385	35.0123157	30	0	0	900
	CellA2	32.2616385	35.0123157	30	120	0	900
	CellA3	32.2616385	35.0123157	30	240	0	900
	CellA4	32.38001187841157	35.11393922656316	30	0	0	900
	CellA5	32.38001187841157	35.11393922656316	30	120	0	900
	CellA6	32.38001187841157	35.11393922656316	30	240	0	900
	CellA7	32.11054646014298	35.0205554375007	30	0	0	900
	CellA8	32.11054646014298	35.0205554375007	30	120	0	900

Users can use star symbol “*” as wildcard. Possible filtering options:

* - any non-null value

!* - null value

[text] - anything that contains the [text] value at the beginning. The same as [text]*

[text]*

*[text]

[text]*[text]

![text] - anything but the defined [text] value

"[text]" - for exact match of value

Filtering options for numeric values:

=n - exact match of n

<>n - anything else but n

<n - less than n

>n - greater than n

<=n - less than or equal to n

>=n - greater than or equal to n

The Filter function can be used on several columns simultaneously. The headers of columns with active Filter function are marked in pink color:

Express

Icons: List, Back, Search, Copy, Paste, Print, Link, Filter, Sort, Undo, Redo, Save, Delete, Refresh.

inc.	cell_name	y	x	height	azimuth	tilt	frequency	power
	A7_				>19			
	A7_1_c1	56.310136	24.336933	29	162	0	3600	73
	A7_1_c2	56.310136	24.336933	29	315	0	3600	73
	A7_2_c1	56.36101	24.282293	29	328	0	3600	73
	A7_2_c2	56.36101	24.282293	29	161	0	3600	73
	A7_3_c1	56.388134	24.252551	29	162	0	3600	73

Filters can be cleared by clicking on Clear all filters.

7.2.1 Set selected

This feature allows bulk editing of several records at once: Select the respective records, click on the column header and choose “Set selected” from the dropdown menu. A dialog opens with a text field. Fill in the term you want the selected database records to be changed to. Then click on “Change”. Synchronize the edited records with the database using the Synchronize changes Tool.

Express

Icons: List, Back, Search, Copy, Paste, Print, Link, Filter, Sort, Undo, Redo, Save, Delete, Refresh.

> CELLS (#5/7 23)

inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect	cell_load	work
A7_1					>19		3600	73		100	6	1	30	32	32	9	50	6
A7_3							3600	73		100	6	1	30	32	32	9	50	6
A7_2_c1	56.36101	24.282293	29	328	0	3600	73		100	6	1	30	32	32	9	50	6	
A7_2_c2	56.36101	24.282293	29	161	0	3600	73		100	6	1	30	32	32	9	50	6	
A7_3_c1	56.388134	24.252551	29	162	0	3600	73		100	6	1	30	32	32	9	50	6	
A7_3_c2	56.388134	24.252551	29	328	0	3600	73		100	6	1	30	32	32	9	50	6	
A7_4_c1	56.406793	24.187525	29	100	0	3600	73		100	6	1	30	32	32	9	50	6	

1. Select the records

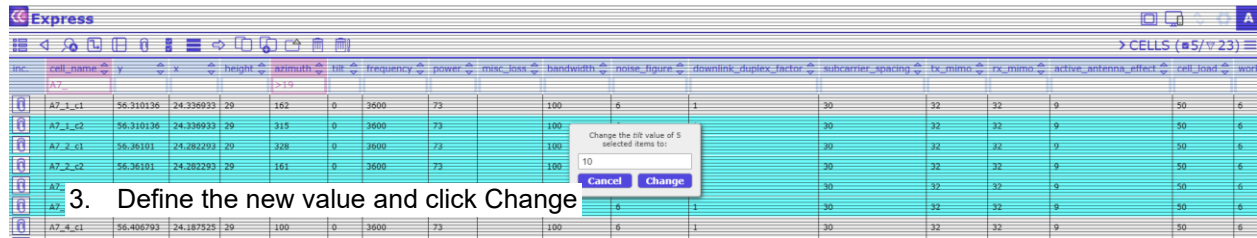
Express

Icons: List, Back, Search, Copy, Paste, Print, Link, Filter, Sort, Undo, Redo, Save, Delete, Refresh.

> CELLS (#5/7 23)

inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect	cell_load	work
A7_1					>19		3600	73		100	6	1	30	32	32	9	50	6
A7_3							3600	73		100	6	1	30	32	32	9	50	6
A7_2_c1	56.36101	24.282293	29	328	0	3600	73		100	6	1	30	32	32	9	50	6	
A7_2_c2	56.36101	24.282293	29	161	0	3600	73		100	6	1	30	32	32	9	50	6	
A7_3_c1	56.388134	24.252551	29	162	0	3600	73		100	6	1	30	32	32	9	50	6	
A7_3_c2	56.388134	24.252551	29	328	0	3600	73		100	6	1	30	32	32	9	50	6	
A7_4_c1	56.406793	24.187525	29	100	0	3600	73		100	6	1	30	32	32	9	50	6	

2. Click on the required column header (here:tilt) and choose Set selected

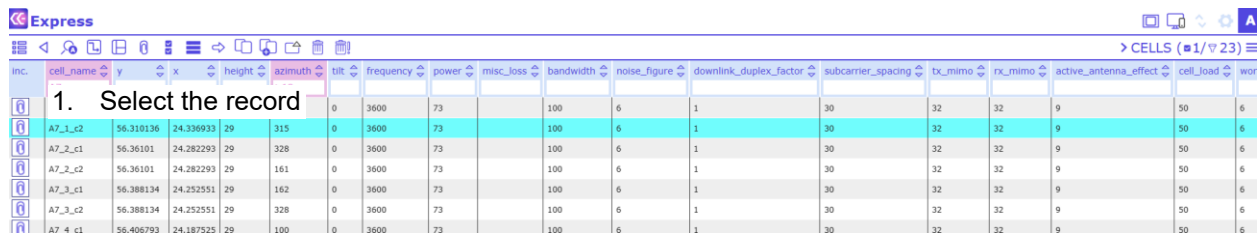


3. Define the new value and click Change

7.2.2 Add link

Each database record may comprise reference links to other database records. This means that a link can be added to reach another database record.

Select the respective record, click on the column header and choose “Add link” from the dropdown menu. In the pop up window, click on Add new reference. A list of component types opens. Select the one you want to link to. Choose the component and confirm the selection. A new link has been created.

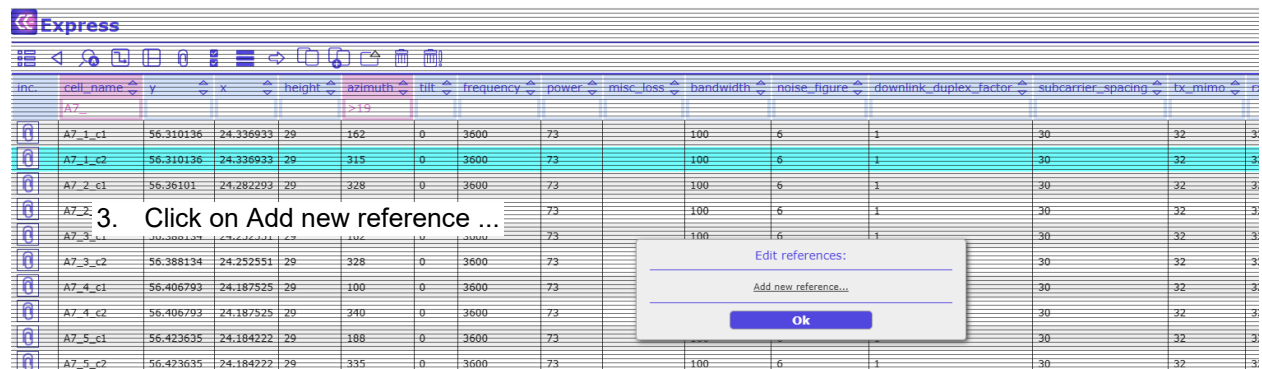


1. Select the record



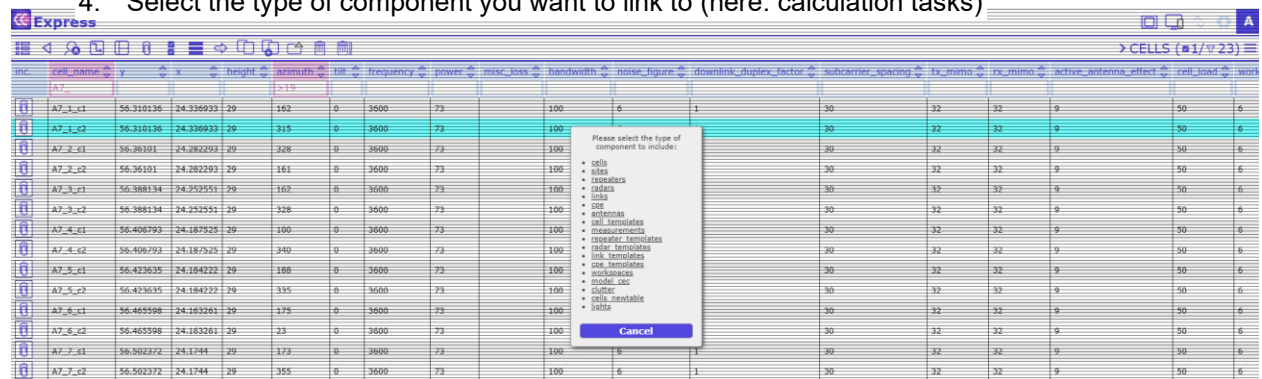
2. Click on the required column header (here: misc_loss) and choose Add link

3. Click on Add new reference ...



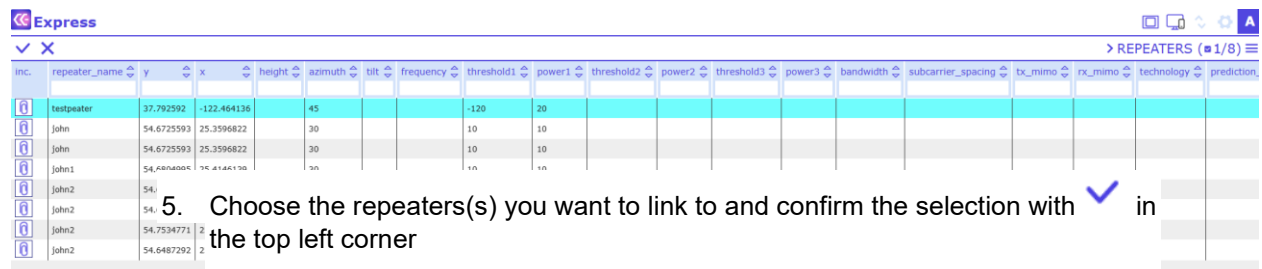
inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect	cell_load	work
A7					>19													
A7_1_c1	56.310136	24.336933	29	162	0	3600	73			100	6	1	30	32	32	9	50	6
A7_1_c2	56.310136	24.336933	29	315	0	3600	73			100	6	1	30	32	32	9	50	6
A7_2_c1	56.36101	24.282293	29	328	0	3600	73			100	6	1	30	32	32	9	50	6
A7_2_c2	56.36101	24.282293	29	161	0	3600	73			100	6	1	30	32	32	9	50	6
A7_3_c1	56.388134	24.252551	29	162	0	3600	73			100	6	1	30	32	32	9	50	6
A7_3_c2	56.388134	24.252551	29	328	0	3600	73			100	6	1	30	32	32	9	50	6
A7_4_c1	56.406793	24.187525	29	100	0	3600	73			100	6	1	30	32	32	9	50	6
A7_4_c2	56.406793	24.187525	29	340	0	3600	73			100	6	1	30	32	32	9	50	6
A7_5_c1	56.423635	24.184222	29	188	0	3600	73			100	6	1	30	32	32	9	50	6
A7_5_c2	56.423635	24.184222	29	335	0	3600	73			100	6	1	30	32	32	9	50	6

4. Select the type of component you want to link to (here: calculation tasks)



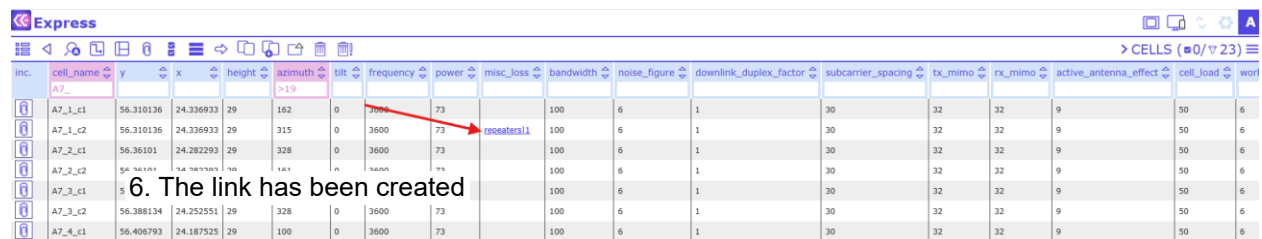
inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect	cell_load	work
A7					>19													
A7_1_c1	56.310136	24.336933	29	162	0	3600	73			100	6	1	30	32	32	9	50	6
A7_1_c2	56.310136	24.336933	29	315	0	3600	73			100	6	1	30	32	32	9	50	6
A7_2_c1	56.36101	24.282293	29	328	0	3600	73			100	6	1	30	32	32	9	50	6
A7_2_c2	56.36101	24.282293	29	161	0	3600	73			100	6	1	30	32	32	9	50	6
A7_3_c1	56.388134	24.252551	29	162	0	3600	73			100	6	1	30	32	32	9	50	6
A7_3_c2	56.388134	24.252551	29	328	0	3600	73			100	6	1	30	32	32	9	50	6
A7_4_c1	56.406793	24.187525	29	100	0	3600	73			100	6	1	30	32	32	9	50	6
A7_4_c2	56.406793	24.187525	29	340	0	3600	73			100	6	1	30	32	32	9	50	6
A7_5_c1	56.423635	24.184222	29	188	0	3600	73			100	6	1	30	32	32	9	50	6
A7_5_c2	56.423635	24.184222	29	335	0	3600	73			100	6	1	30	32	32	9	50	6
A7_6_c1	56.465598	24.163261	29	175	0	3600	73			100	6	1	30	32	32	9	50	6
A7_6_c2	56.465598	24.163261	29	23	0	3600	73			100	6	1	30	32	32	9	50	6
A7_7_c1	56.502372	24.1744	29	173	0	3600	73			100	6	1	30	32	32	9	50	6
A7_7_c2	56.502372	24.1744	29	355	0	3600	73			100	6	1	30	32	32	9	50	6

5. Choose the repeaters(s) you want to link to and confirm the selection with ✓ in the top left corner



inc.	repeater_name	y	x	height	azimuth	tilt	frequency	threshold1	power1	threshold2	power2	threshold3	power3	bandwidth	subcarrier_spacing	tx_mimo	rx_mimo	technology	prediction
	testpeater	37.792592	-122.464136		45			-120	20										
	John	54.6725593	25.3596822		30			10	10										
	John	54.6725593	25.3596822		30			10	10										
	John1	54.648808	25.4161196		30			10	10										
	John2	54																	
	John2	54																	
	John2	54.7534771	2																
	John2	54.6487292	2																

6. The link has been created



inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect	cell_load	work
A7					>19													
A7_1_c1	56.310136	24.336933	29	162	0	3600	73			100	6	1	30	32	32	9	50	6
A7_1_c2	56.310136	24.336933	29	315	0	3600	73			100	6	1	30	32	32	9	50	6
A7_2_c1	56.36101	24.282293	29	328	0	3600	73			100	6	1	30	32	32	9	50	6
A7_2_c2	56.36101	24.282293	29	161	0	3600	73			100	6	1	30	32	32	9	50	6
A7_3_c1	56.388134	24.252551	29	162	0	3600	73			100	6	1	30	32	32	9	50	6
A7_3_c2	56.388134	24.252551	29	328	0	3600	73			100	6	1	30	32	32	9	50	6
A7_4_c1	56.406793	24.187525	29	100	0	3600	73			100	6	1	30	32	32	9	50	6

7.3 Adding, viewing, downloading and deleting attachments

Each database entry can be connected with any type of attachment, for example, a picture or a text file. Entries with connected attachments have a blue check symbol next to the paper clip symbol in the column “Inc.”, while entries without attachment have only the paper clip.

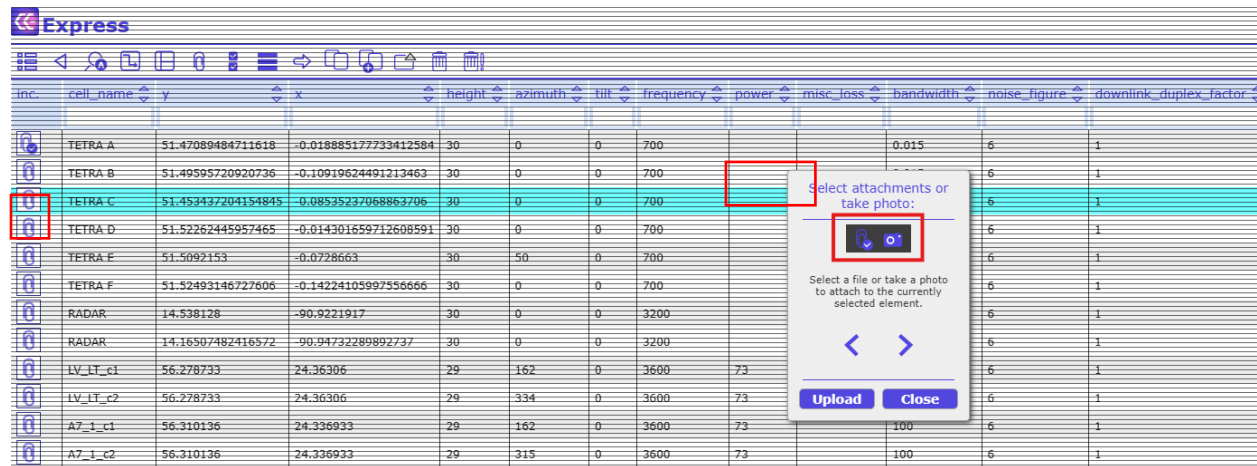
Express

Navigation icons: list, back, search, print, save, link, check, menu, undo, redo, copy, paste, delete, refresh.

inc.	cell_name	y	x	height	azimuth	tilt	frequency
	TETRA A	51.47089484711618	-0.018885177733412584	30	0	0	700
	TETRA B	51.49595720920736	-0.10919624491213463	30	0	0	700
	TETRA C	51.453437204154845	-0.08535237068863706	30	0	0	700
	TETRA D	51.52262445957465	-0.014301659712608591	30	0	0	700
	TETRA E	51.5092153	-0.0728663	30	50	0	700
	TETRA F	51.52493146727606	-0.14224105997556666	30	0	0	700
	RADAR	14.538128	-90.9221917	30	0	0	3200
	RADAR	14.16507482416572	-90.94732289892737	30	0	0	3200
	LV_LT_c1	56.278733	24.36306	29	162	0	3600
	LV_LT_c2	56.278733	24.36306	29	334	0	3600

Adding attachments

To connect a database entry with an attachment, click on the paper clip symbol in the column “Inc.”. A dialog opens that allows you to browse your computer for the attachment of your choice or to take a photo.

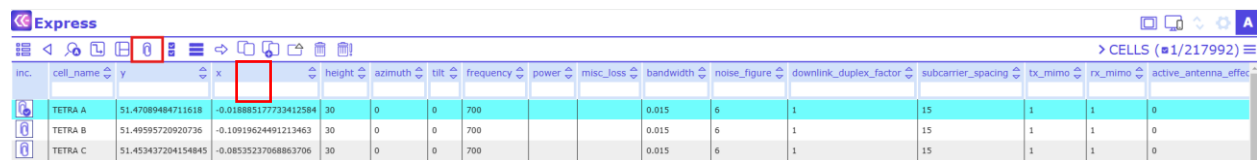


The screenshot shows the Cellular Expert Express interface with a table of cell data. A red box highlights the attachment icon in the toolbar. A modal dialog titled "Select attachments or take photo:" is open, showing options to select a file or take a photo.

inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor
	TETRA A	51.47089484711618	-0.018885177733412584	30	0	0	700			0.015	6	1
	TETRA B	51.49595720920736	-0.10919624491213463	30	0	0	700			0.015	6	1
	TETRA C	51.453437204154845	-0.08535237068863706	30	0	0	700			0.015	6	1
	TETRA D	51.52262445957465	-0.014301659712608591	30	0	0	700			0.015	6	1
	TETRA E	51.5092153	-0.0728663	30	50	0	700			0.015	6	1
	TETRA F	51.52493146727606	-0.14224105997556666	30	0	0	700			0.015	6	1
	RADAR	14.538128	-90.9221917	30	0	0	3200			0.015	6	1
	RADAR	14.16507482416572	-90.94732289892737	30	0	0	3200			0.015	6	1
	LV_LT_c1	56.278733	24.36306	29	162	0	3600	73		0.015	6	1
	LV_LT_c2	56.278733	24.36306	29	334	0	3600	73		0.015	6	1
	A7_1_c1	56.310136	24.336933	29	162	0	3600	73		0.015	6	1
	A7_1_c2	56.310136	24.336933	29	315	0	3600	73		0.015	6	1

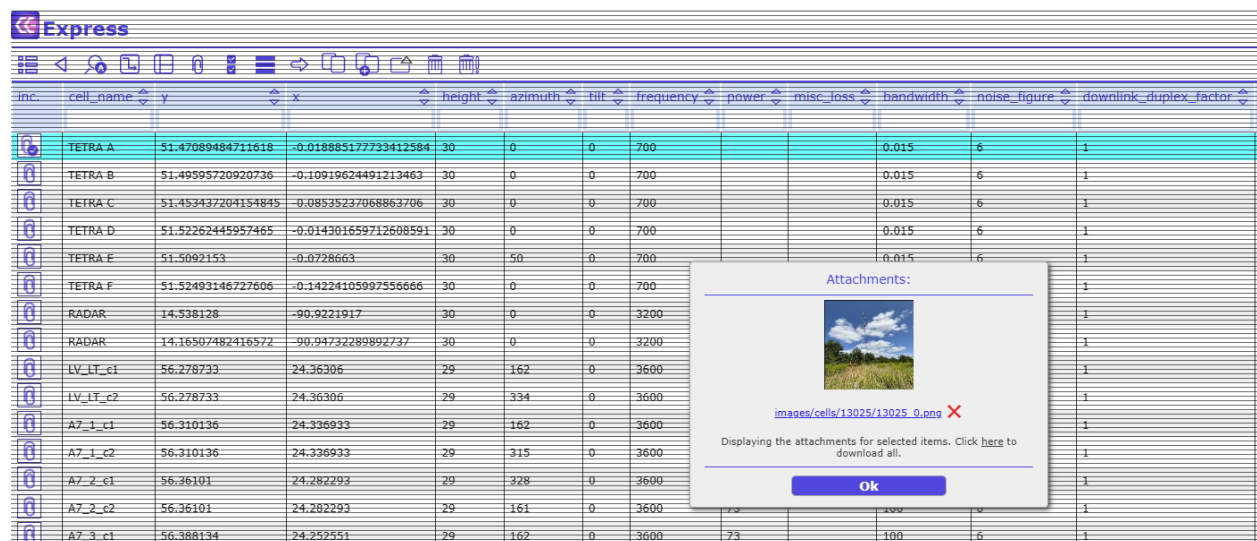
Viewing, downloading and deleting attachments:

To view the list of attachments connected to an entry, select the respective entry and click on  in the toolbar:



The screenshot shows the Cellular Expert Express interface with a table of cell data. A red box highlights the attachment icon in the toolbar. A modal dialog titled "Attachments:" is open, showing a list of attachments with a thumbnail image.

inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effec
	TETRA A	51.47089484711618	-0.018885177733412584	30	0	0	700			0.015	6	1	15	1	1	0
	TETRA B	51.49595720920736	-0.10919624491213463	30	0	0	700			0.015	6	1	15	1	1	0
	TETRA C	51.453437204154845	-0.08535237068863706	30	0	0	700			0.015	6	1	15	1	1	0



The screenshot shows the Cellular Expert Express interface with a table of cell data. A red box highlights the attachment icon in the toolbar. A modal dialog titled "Attachments:" is open, showing a list of attachments with a thumbnail image.

inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor
	TETRA A	51.47089484711618	-0.018885177733412584	30	0	0	700			0.015	6	1
	TETRA B	51.49595720920736	-0.10919624491213463	30	0	0	700			0.015	6	1
	TETRA C	51.453437204154845	-0.08535237068863706	30	0	0	700			0.015	6	1
	TETRA D	51.52262445957465	-0.014301659712608591	30	0	0	700			0.015	6	1
	TETRA E	51.5092153	-0.0728663	30	50	0	700			0.015	6	1
	TETRA F	51.52493146727606	-0.14224105997556666	30	0	0	700			0.015	6	1
	RADAR	14.538128	-90.9221917	30	0	0	3200			0.015	6	1
	RADAR	14.16507482416572	-90.94732289892737	30	0	0	3200			0.015	6	1
	LV_LT_c1	56.278733	24.36306	29	162	0	3600			0.015	6	1
	LV_LT_c2	56.278733	24.36306	29	334	0	3600			0.015	6	1
	A7_1_c1	56.310136	24.336933	29	162	0	3600			0.015	6	1
	A7_1_c2	56.310136	24.336933	29	315	0	3600			0.015	6	1
	A7_2_c1	56.36101	24.282293	29	328	0	3600			0.015	6	1
	A7_2_c2	56.36101	24.282293	29	161	0	3600			0.015	6	1
	A7_3_c1	56.388134	24.252551	29	162	0	3600	73		0.015	6	1

In the attachments the listed images are shown as thumbs. Select an attachment from the list and it opens in a separate browser tab.

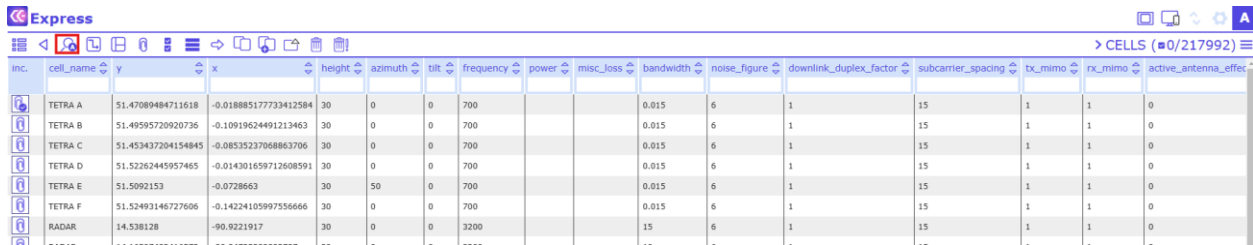
Another way to view an attachment list is to move the pointer over the check symbol in the column “Inc.” and to click the right mouse button.

The selected attachments can also be downloaded.

To delete an attachment click on the red cross symbol.

7.4 Sieve

It is possible to display only selected data from an open table. Click on the button  in the toolbar.



inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effet
	TETRA A	51.47089484711618	-0.018885177723412584	30	0	0	700			0.015	6	1	15	1	1	0
	TETRA B	51.49595720920736	-0.10919624491213463	30	0	0	700			0.015	6	1	15	1	1	0
	TETRA C	51.453437204154845	-0.08535237068863706	30	0	0	700			0.015	6	1	15	1	1	0
	TETRA D	51.52262445957465	-0.014301659712608591	30	0	0	700			0.015	6	1	15	1	1	0
	TETRA E	51.5092153	-0.0728663	30	50	0	700			0.015	6	1	15	1	1	0
	TETRA F	51.52493146727606	-0.14224105997556666	30	0	0	700			0.015	6	1	15	1	1	0
	RADAR	14.538128	-90.9221917	30	0	0	3200			15	6	1	15	1	1	0

And a dialog will open:



Sieve Items

Active:

Add rule
Clear all

Enter one or more search
queries to limit available data.

Ok Cancel

Note that by clicking “Add rule” you may sieve with two or more filter terms that are connected via the operation “OR”, not “AND”. Thus, the result of two filters will show all items that comprise either the first or the second search term in their attributes.

For example, do you want to sieve the table asites for records starting with defined characters? Then the Sieve tool will help.

Sieve Items

Active:

2G

MISS

Add rule

Clear all

Enter one or more search
querry to limit available data.

Ok

Cancel

Click Add rule in order to add another search term, or click Clear all to remove all active searches.

To start the sieve process, click OK.

Results

Express

Active sieves: LV_LT, MISS

CELLS (#0/6)

inc.	cell_name	y	x	height	azimuth	tilt	frequency	power	misc_loss	bandwidth	noise_figure	downlink_duplex_factor	subcarrier_spacing	tx_mimo	rx_mimo	active_antenna_effect
	LV_LT_c1	56.278733	24.36306	29	162	0	3600	73		100	6	1	30	32	32	9
	LV_LT_c2	56.278733	24.36306	29	334	0	3600	73		100	6	1	30	32	32	9
	Mission Bay 5Ghz AP	38.09231567	-92.77516174	3.048	0		5745	21		20	6	0.75	15	4	4	0
	MISS_1900_P3	37.7489267919896	-122.50112432445887	20	100	0	1900	15	0	20	6	0.75	15	4	4	0
	MISS_1900_P1	37.7489267919896	-122.50112432445887	20	349	0	1900	15	0	20	6	0.75	15	4	4	0
	MISS_1900_P2	37.7489267919896	-122.50112432445887	20	220	0	1900	15	0	20	6	0.75	15	4	4	0

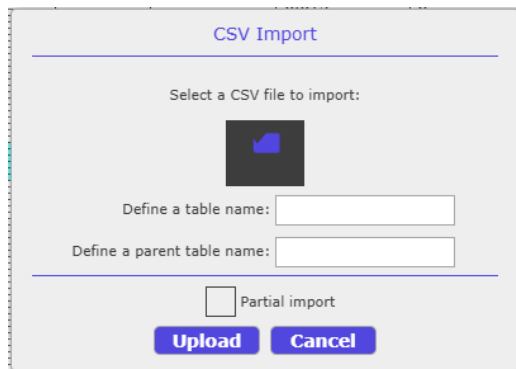
7.5 CE API

It is possible to call other tools configured by the administrator. For every table a different tool can be configured. To start tool selection, click on the button  in the toolbar.

7.6 Import CSV

This tool is integrated into the “CE API” tool. It is possible to import data from .csv files. Note(!): Consult with the administrator before import.

“Import CSV” opens a new window:



CSV Import

Select a CSV file to import:

Define a table name:

Define a parent table name:

☐ Partial import

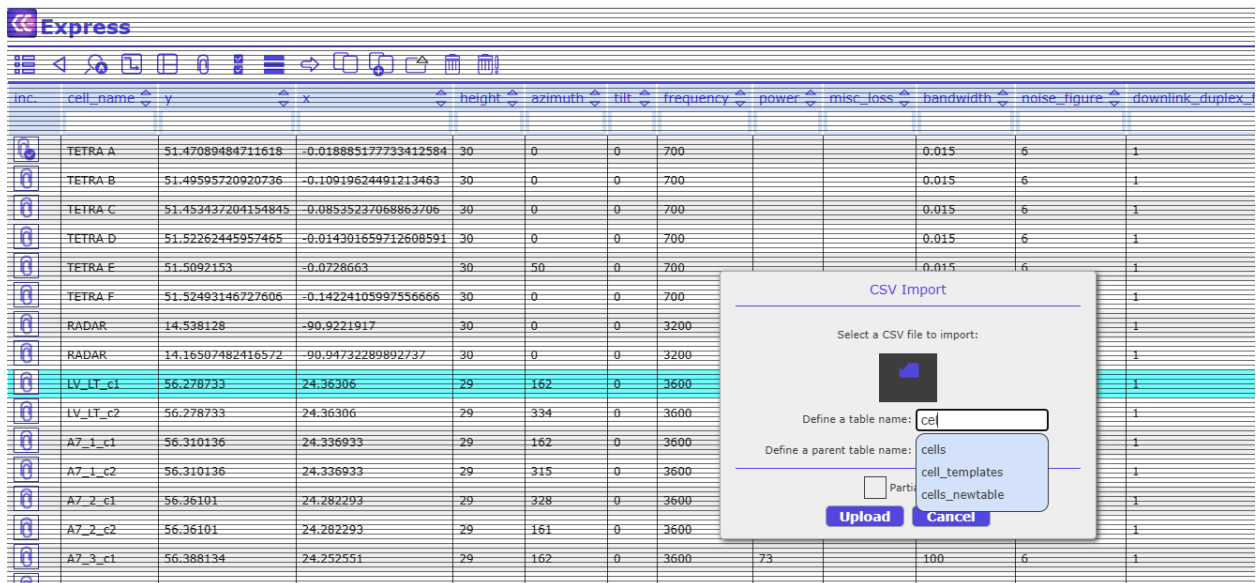
Upload **Cancel**

Data can be imported either as a new table (level 0), or added as a child table to a parent table (level 1). When adding a new table, it is required to define the table name. When adding a child table to a parent table, it is required to enter the parent table name.

Requirements:

1. The data in the CSV file should be separated by the symbol “;”
2. Avoid space and special characters in the table name
3. Level 0 table – CSV file must comprise a column object_id
4. Level 1 table – CSV file must comprise a column object_id and a column parent_id, with the parent_id value equal to the object_id of the parent table

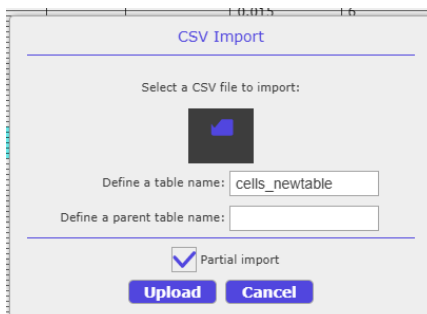
When adding a child table to a parent table, choose the parent table from the dropdown menu:



The screenshot shows the Cellular Expert Express interface with a table of data. The table has columns: inc, cell_name, y, x, height, azimuth, tilt, frequency, power, misc_loss, bandwidth, noise_figure, and downlink_duplex. The data includes rows for TETRA A through F, RADAR, and LV_1T_c1 through LV_1T_c2. A CSV Import dialog box is overlaid on the table, showing the 'Define a parent table name' dropdown menu with options: cells, cell_templates, cells_newtable, and cells_newtable.

Partial Import

The Partial Import feature allows to add records to an already existing table. If the box "Partial import" is checked, the new records will be added to the defined table:



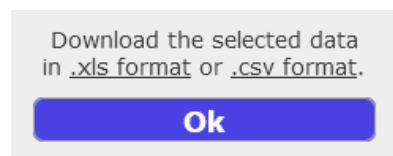
8. Exploring data

8.1 Export selected

Data subsets can be exported as .xls or .csv files. Select the database records of choice, then click Export

Selected  button:

A popup window opens.



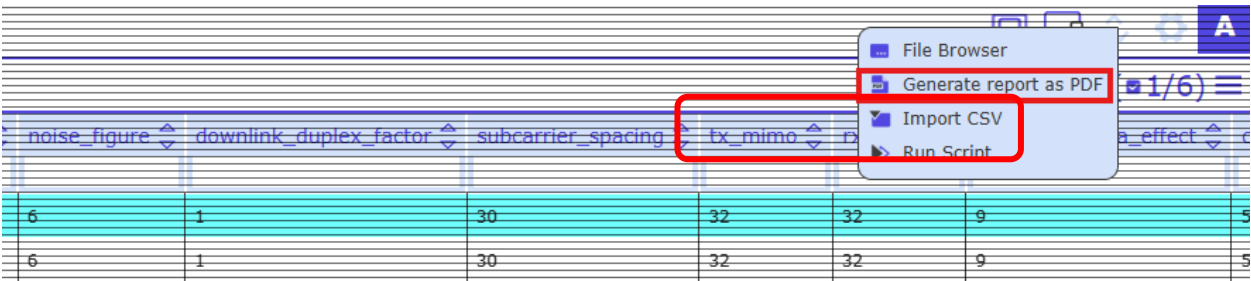
Click on the links to choose between .xls or.csv file formats:

Note that hidden columns are not exported.

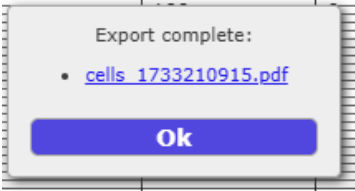
8.2 PDF report

A PDF report summarizes information associated with one or more objects. For generating a report, first

select the objects of choice. Then open the Import Export menu with  and choose "Generate report as PDF"



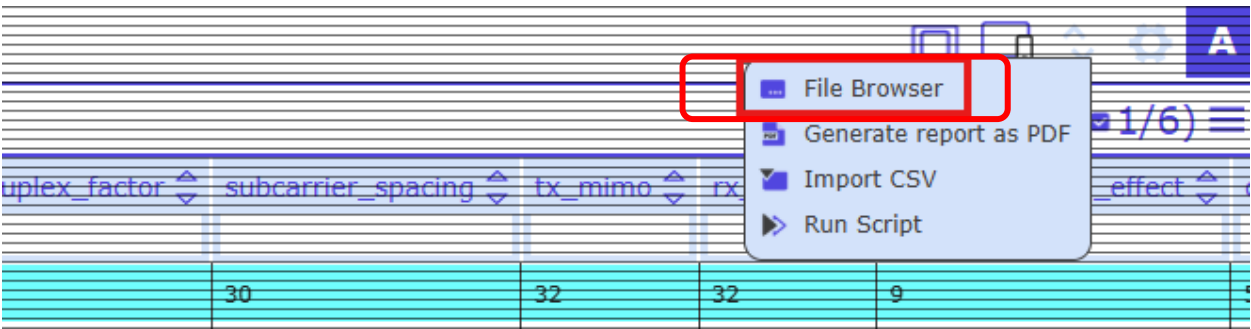
A new dialog appears. Open the PDF report by clicking “here”:



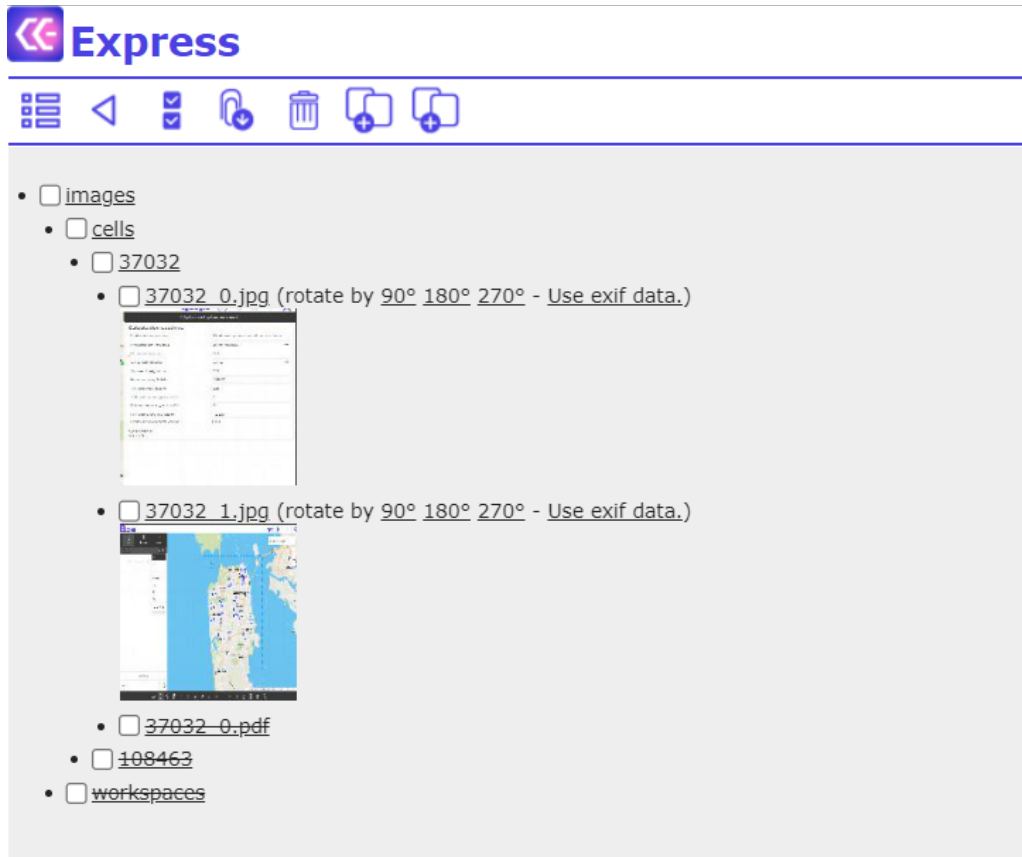
Note that the administrator has to prepare the template for the PDF report. Please inform the administrator in case of problems with generating PDF reports.

8.3 File browser

Open the Data Export menu with  and choose “File Browser”:



View attached files, download or delete files:



Images are shown as thumbs. Note that it is possible to rotate the images.

File Browser Toolbar:



Open the Data View Panel



Go one step back



Select / unselect all files and folders



Download selected files

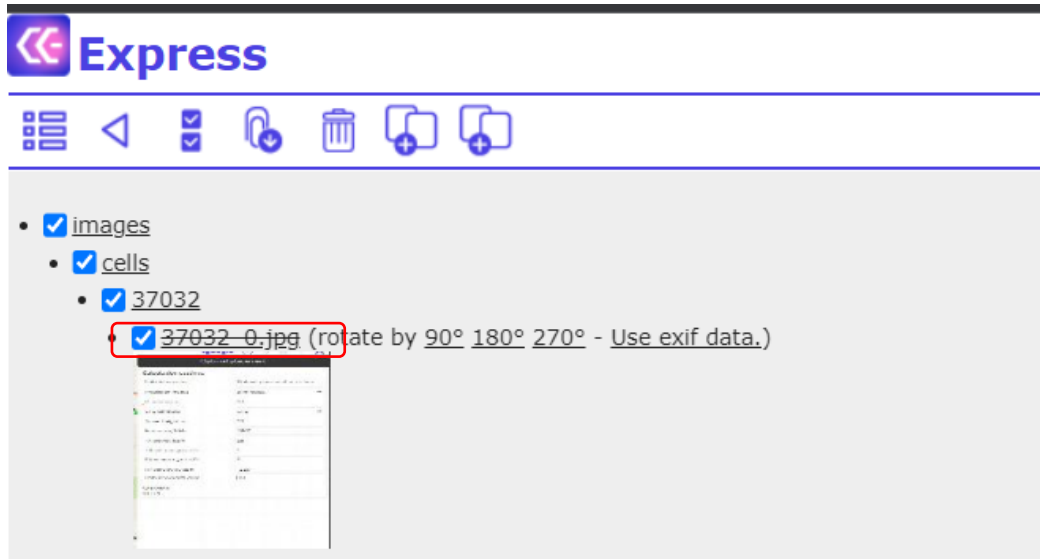


Delete selected files

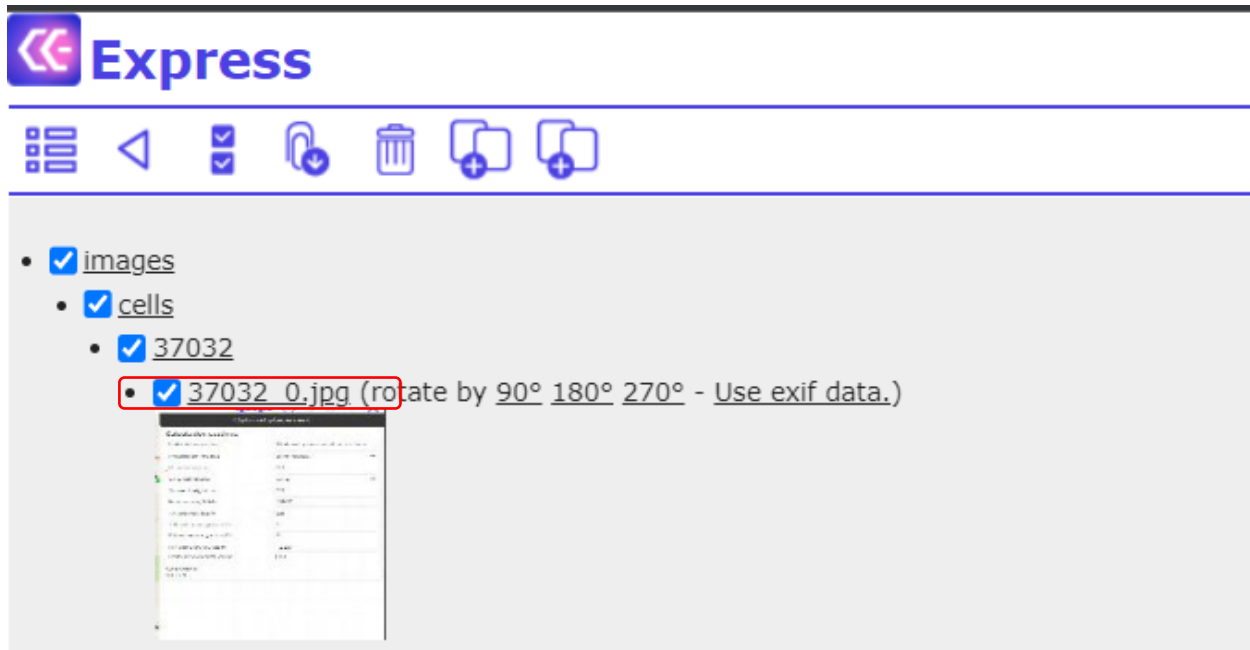


Restore selected files

When an image is deleted, said image is marked as strikethrough on the File Server:



You can restore the image by selecting the strikethrough object and clicking  :

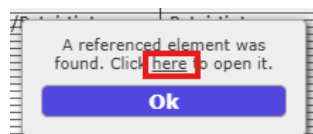


8.4 Quick references

Quick references are defined by the administrator and allow users to open a reference object in a separate browser tab. If Quick references are enabled, the respective column is marked with “*”, for example “siteid”:

inc.	parent_id	latitude	longitude	sectorheight	azimuth	power	miscloss	tilt	antenna	siteid	mimogain	bandwidth	noise
1	1	37.78987	-122.4012	1	0				*	*			
1	1	41.757869	-72.829897	33.53	30	40	0	0	SBNHH-1D65B_Port 1 +45_00DT_0704*	*	0	10	6
1	1	41.757869	-72.829897	33.53	150	40	0	10	SBNHH-1D65B_Port 1 +45_00DT_0704*	*	0	10	6
1	1	41.757869	-72.829897	33.53	30	40	0	10	calculationtasks1	*	0	10	6
1	1	41.757869	-72.829897	33.53	150	40	0	10	SBNHH-1D65B_Port 1 +45_00DT_0704*	*	0	10	6
1	1	41.757869	-72.829897	33.53	30	40	0	0	SBNHH-1D65B_Port 3 +45_00DT_1910*	*	0	10	6
1	1	41.757869	-72.829897	33.53	150	40	0	0	SBNHH-1D65B_Port 3 +45_00DT_1910*	*	0	10	6
1	1	37.75821822	-122.4254333	10	358	40	0	0	SBNHH-1D65B_Port 1 +45_00DT_0704*	2G_4*	0	0	0
1	1	37.75821822	-122.4254333	10	180	40	0	0	SBNHH-1D65B_Port 1 +45_00DT_0704*	2G_4*	0	0	0

To open a reference link, select it in edit mode (right click on it) and click “here” in the opened dialog.



The referenced object is opened in a separate browser tab.

8.5 Default editing and manual editing

If the administrator has defined default values (Administrator Guide: "Set Defaults"), editing column information may include selecting values from a drop-down menu. If the menu does not comprise the desired value, users may manually edit the information or clear the value: